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(12) **United States Patent**
Josse

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(45) **Date of Patent:** **Aug. 12, 2025**

(54) **TISSUE RETRACTOR, RETRACTION
MODULES, AND ASSOCIATED METHODS**

(56) **References Cited**

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(71) Applicant: **Warsaw Orthopedic, Inc.**, Warsaw, IN
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(72) Inventor: **Loic Josse**, Palm Beach Gardens, FL
(US)

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(73) Assignee: **WARSAW ORTHOPEDIC, INC.**,
Warsaw, IN (US)

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 835 days.

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(22) Filed: **Mar. 1, 2022**

(65) **Prior Publication Data**

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Primary Examiner — Eduardo C Robert

Assistant Examiner — Christina NegrelliRodriguez

(74) *Attorney, Agent, or Firm* — FOX ROTHCHILD
LLP

Related U.S. Application Data

(63) Continuation-in-part of application No. 17/336,860,
filed on Jun. 2, 2021, now Pat. No. 12,096,923, which
(Continued)

(51) **Int. Cl.**

A61B 17/02 (2006.01)

A61B 17/00 (2006.01)

(52) **U.S. Cl.**

CPC **A61B 17/0206** (2013.01); **A61B**
2017/00473 (2013.01)

(58) **Field of Classification Search**

CPC ... A61B 17/02; A61B 17/025; A61B 17/0206;
A61B 17/0218; A61B 17/0293; A61B
2017/00367; A61B 2017/0046

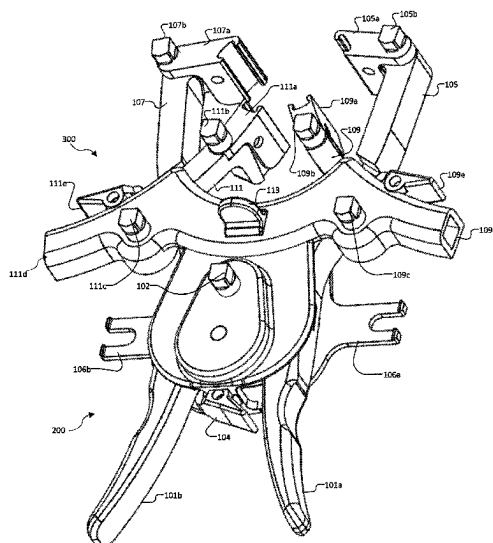
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(57)

ABSTRACT

A modular retractor may include a first body portion that
houses a distraction mechanism for opening and closing a
first arm and a second arm. A first pivoting member may be
coupled to a distal end of the first arm and a second pivoting
member may be coupled to a distal end of the second arm,
for example. A first blade attachment mechanism may be
coupled to the first pivoting member and a second blade
attachment mechanism may be coupled to the second piv-
oting member. The first and second blade attachment mecha-
nisms may be configured to couple to first and second blades
and be independently inclinable. Various embodiments may
include at least one connection point for connecting to at
least one additional retractor module. Various embodiments
may include at least one quick connect coupler for connect-
ing to a snap on table mount.

22 Claims, 163 Drawing Sheets



Related U.S. Application Data

is a continuation-in-part of application No. 16/926, 173, filed on Jul. 10, 2020, now Pat. No. 11,224,415.

(60) Provisional application No. 63/254,929, filed on Oct. 12, 2021.

(58) Field of Classification Search

USPC 600/201, 214, 219
See application file for complete search history.

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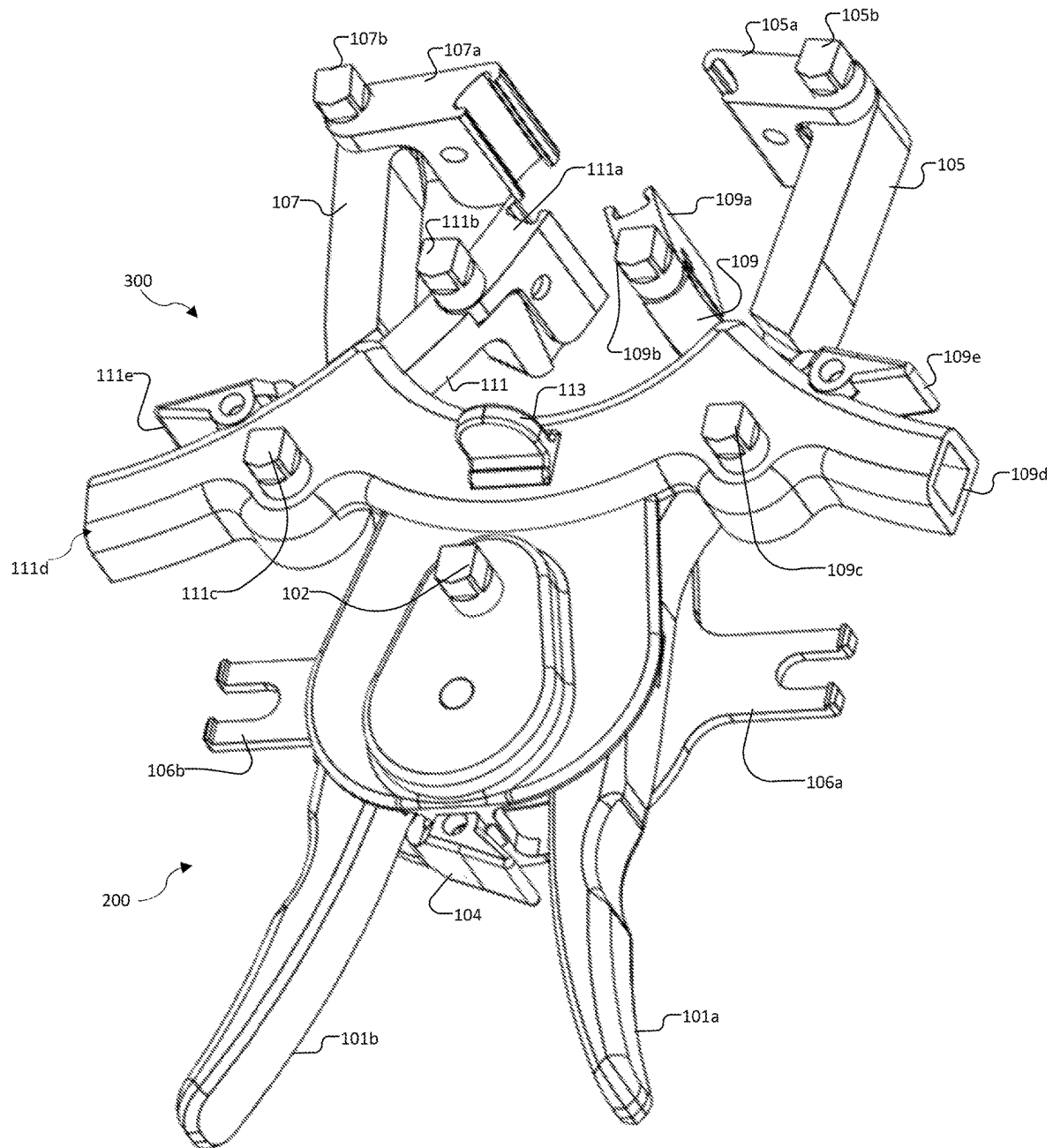
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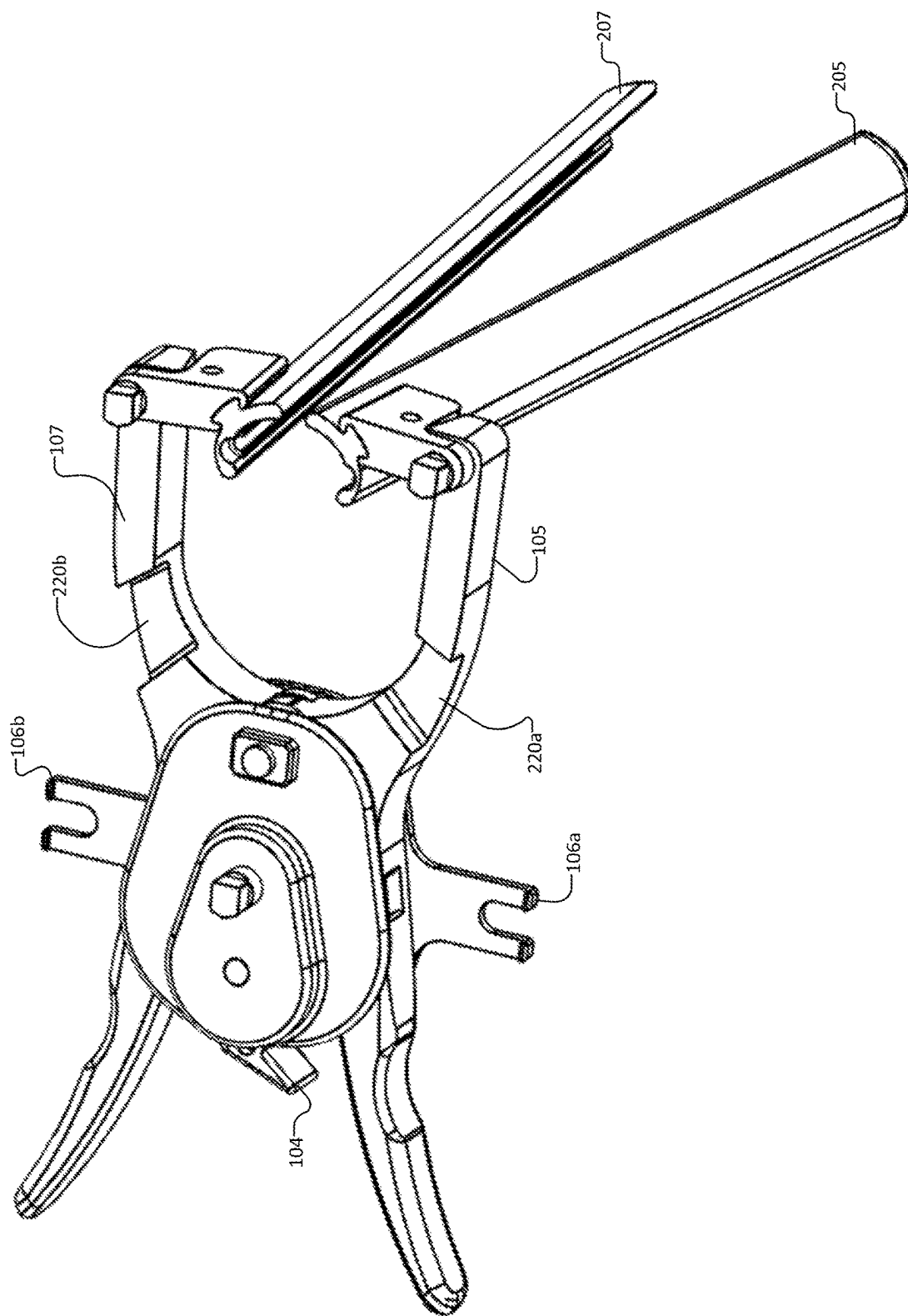
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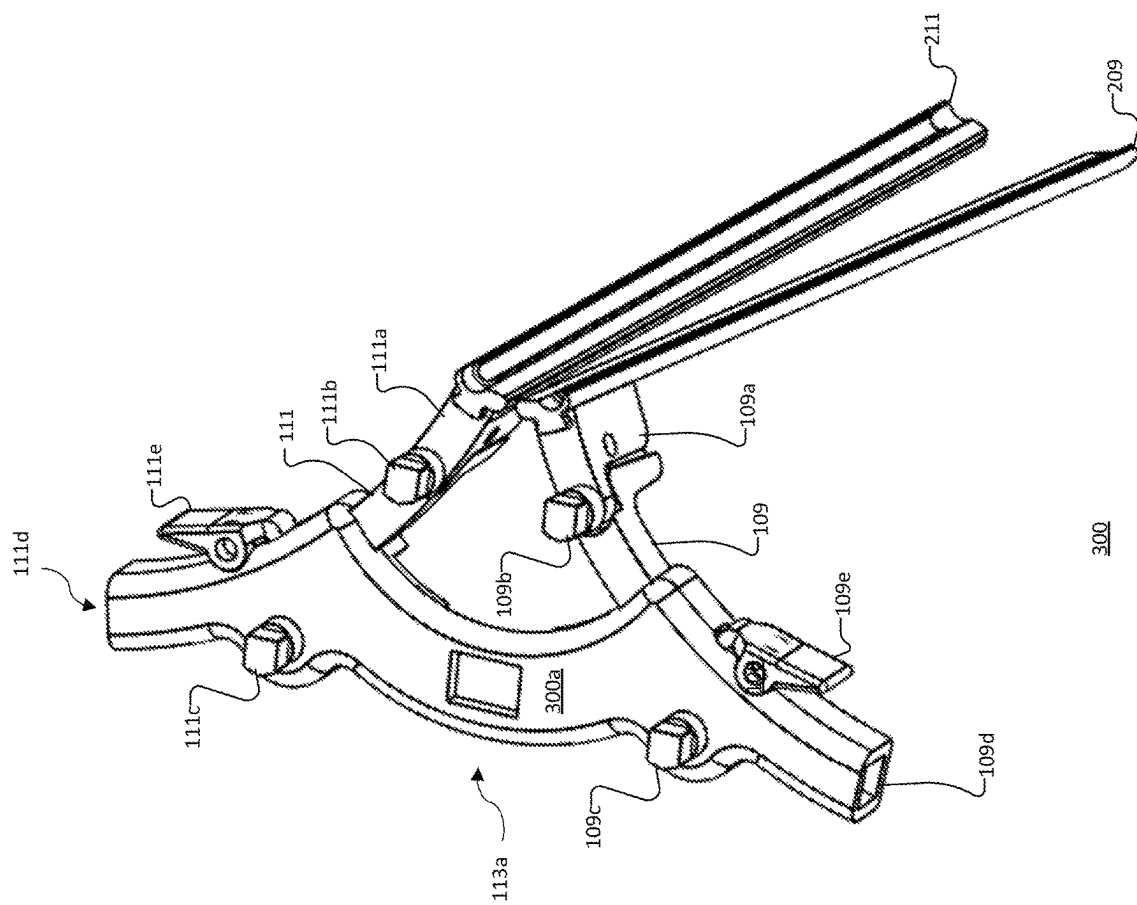
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FIG. 1



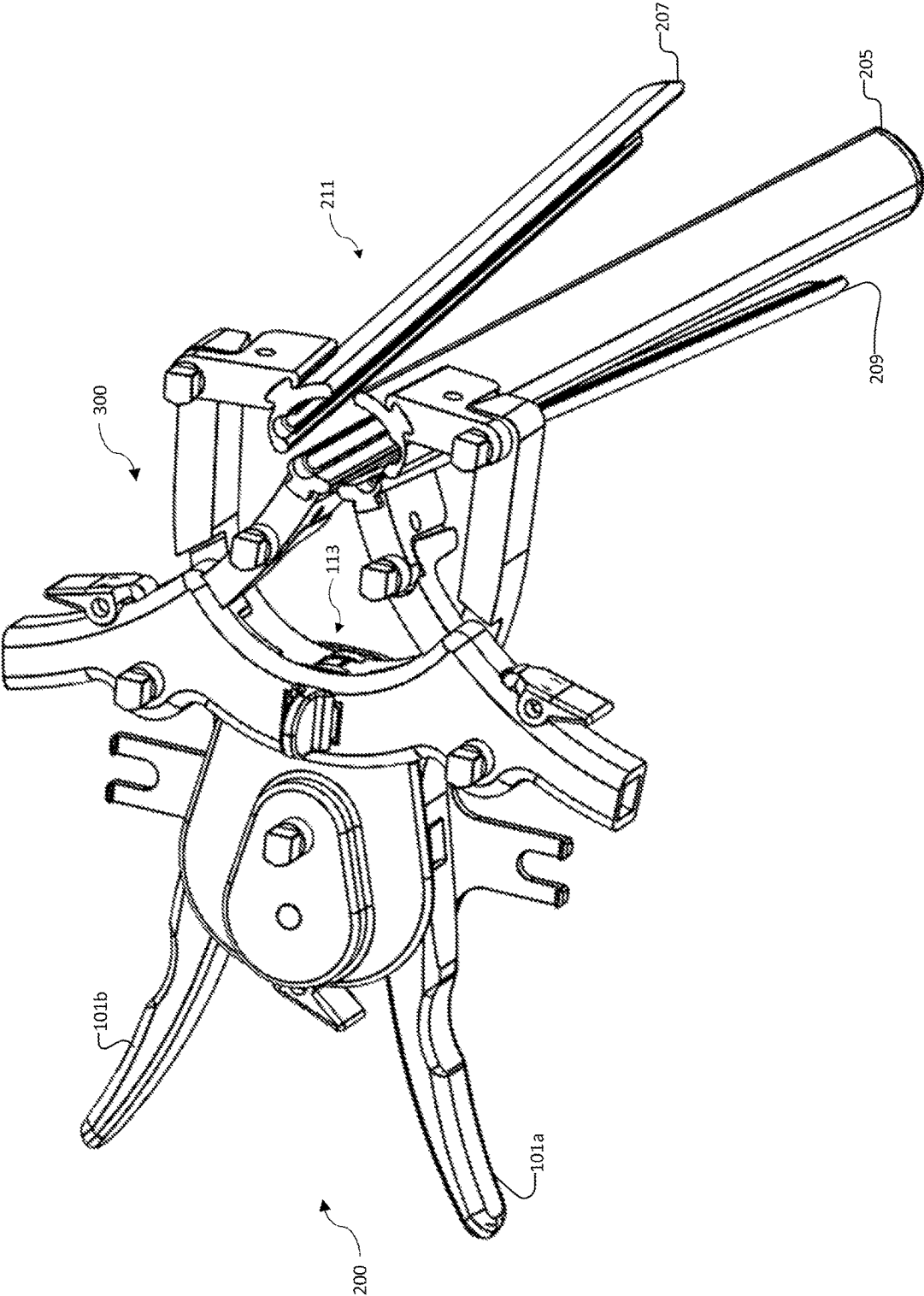
200

FIG. 2



300

FIG. 3



100
FIG. 4

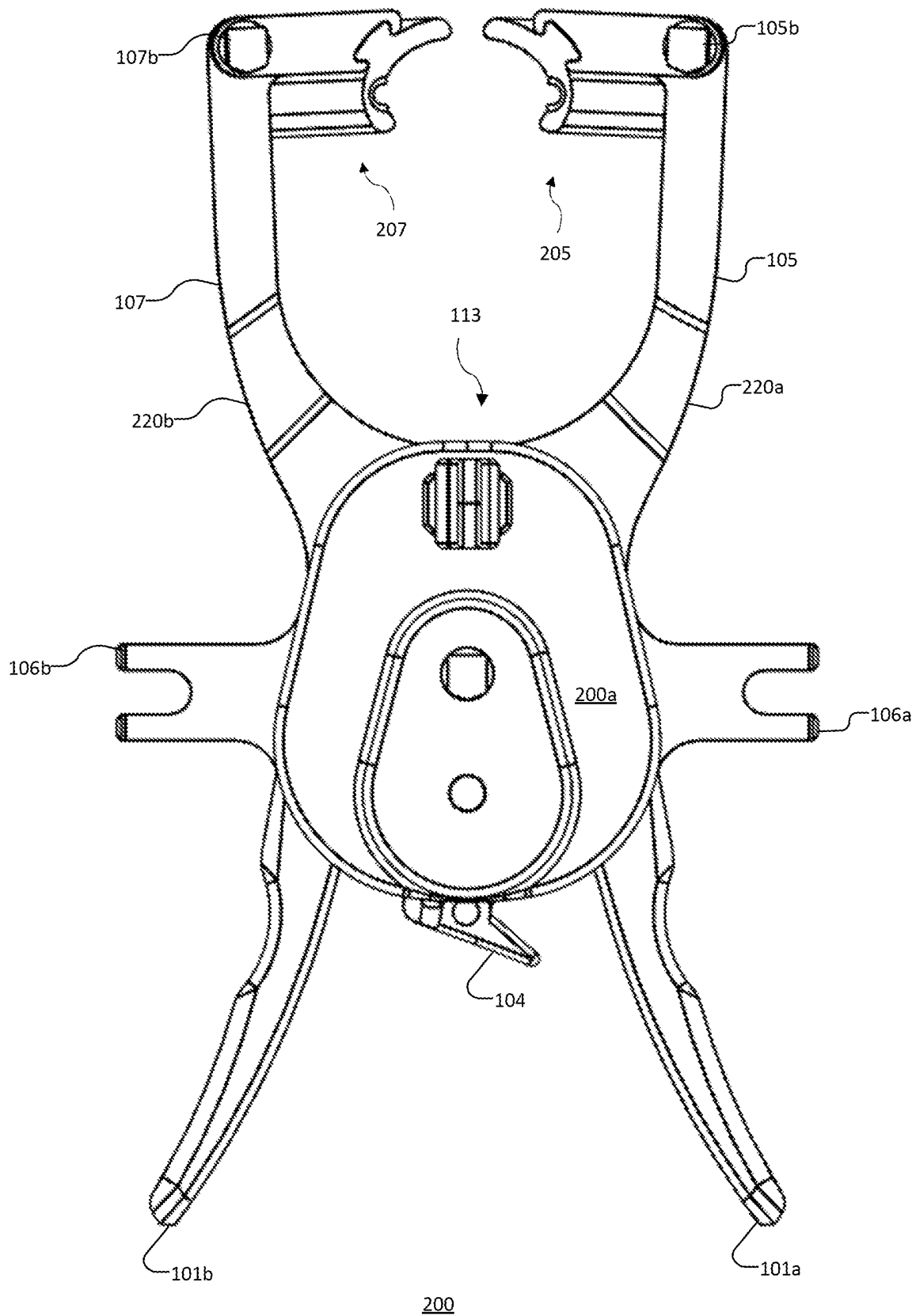
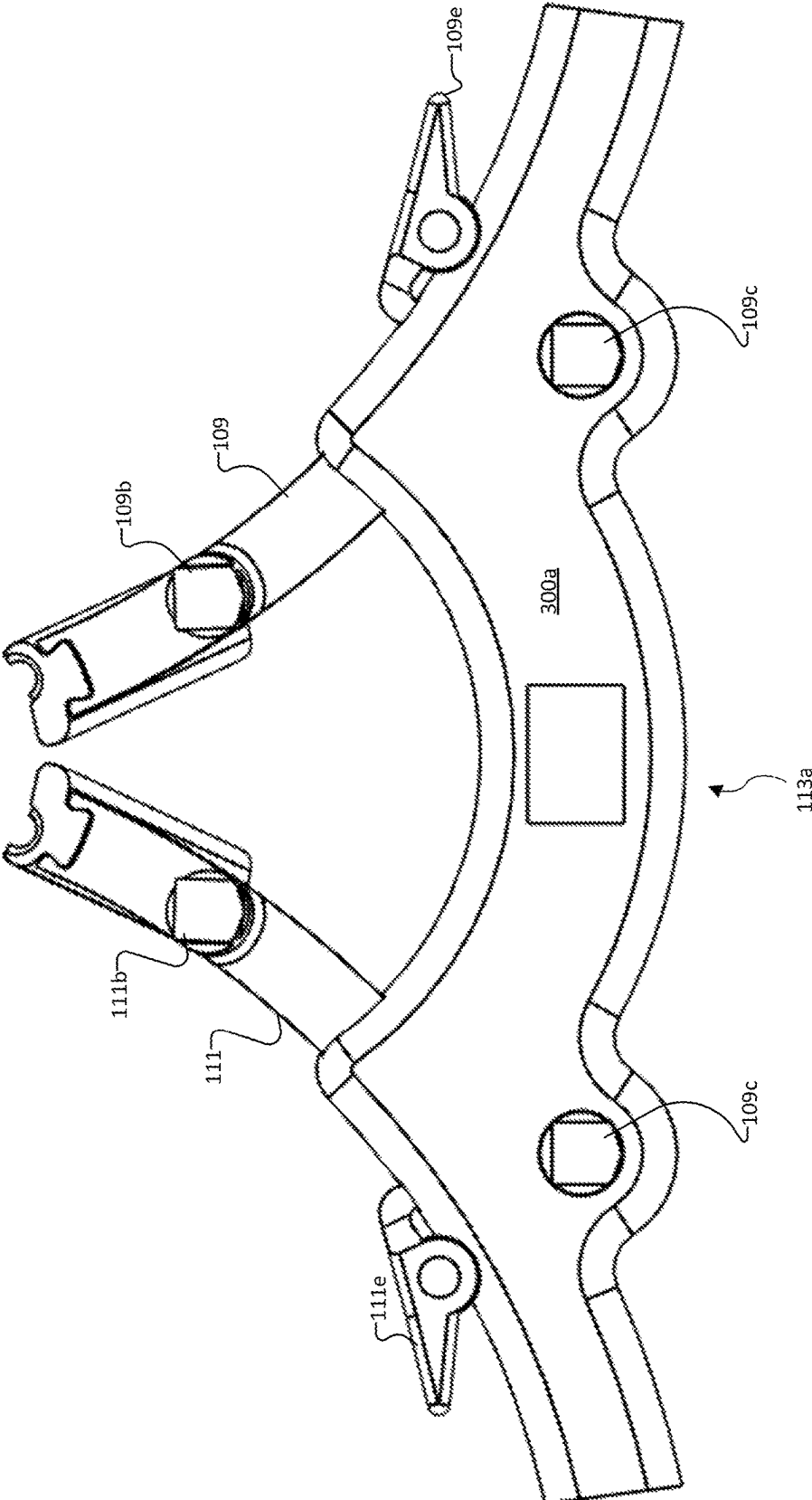


FIG. 5



300

FIG. 6

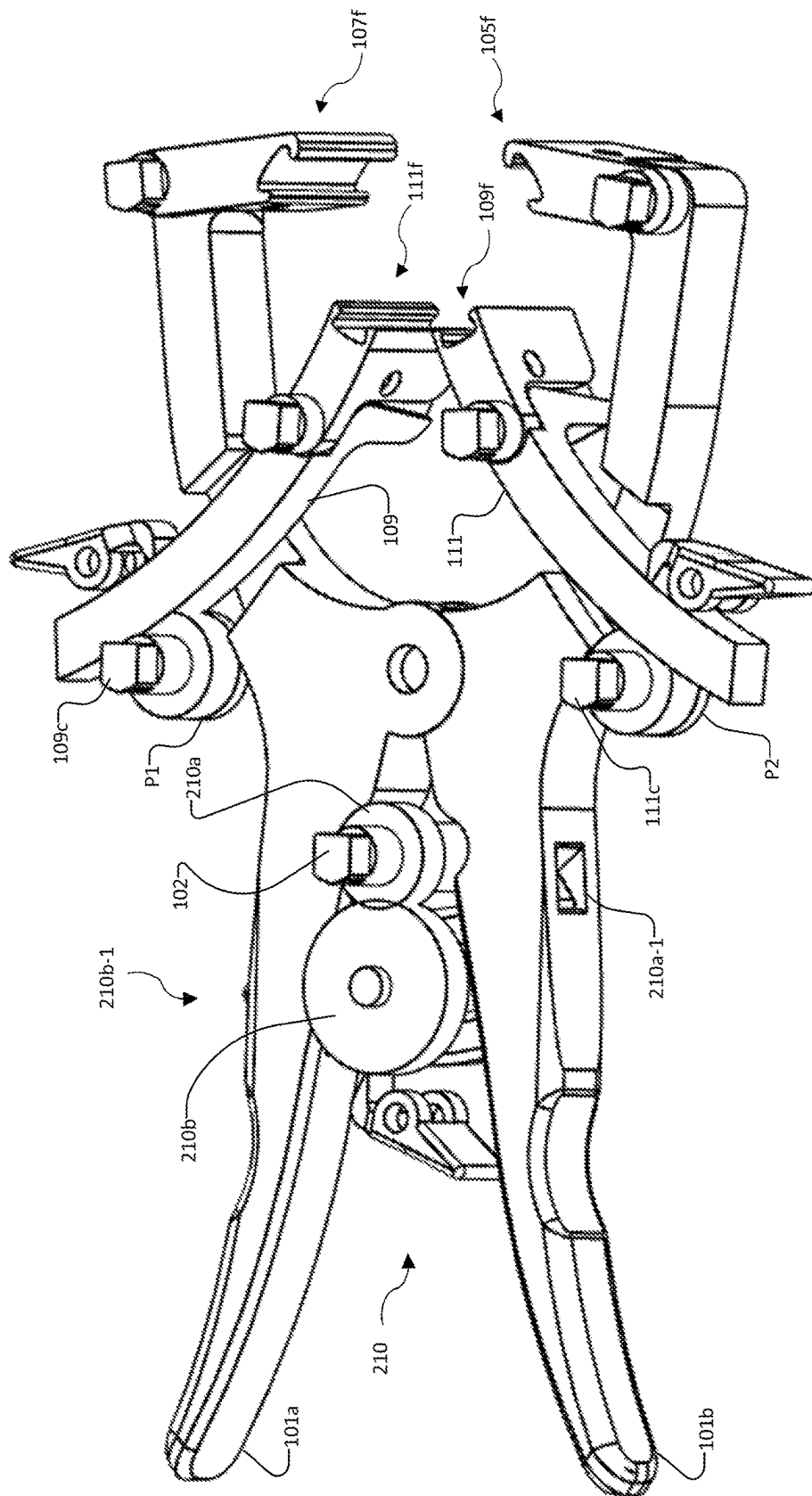


FIG. 7

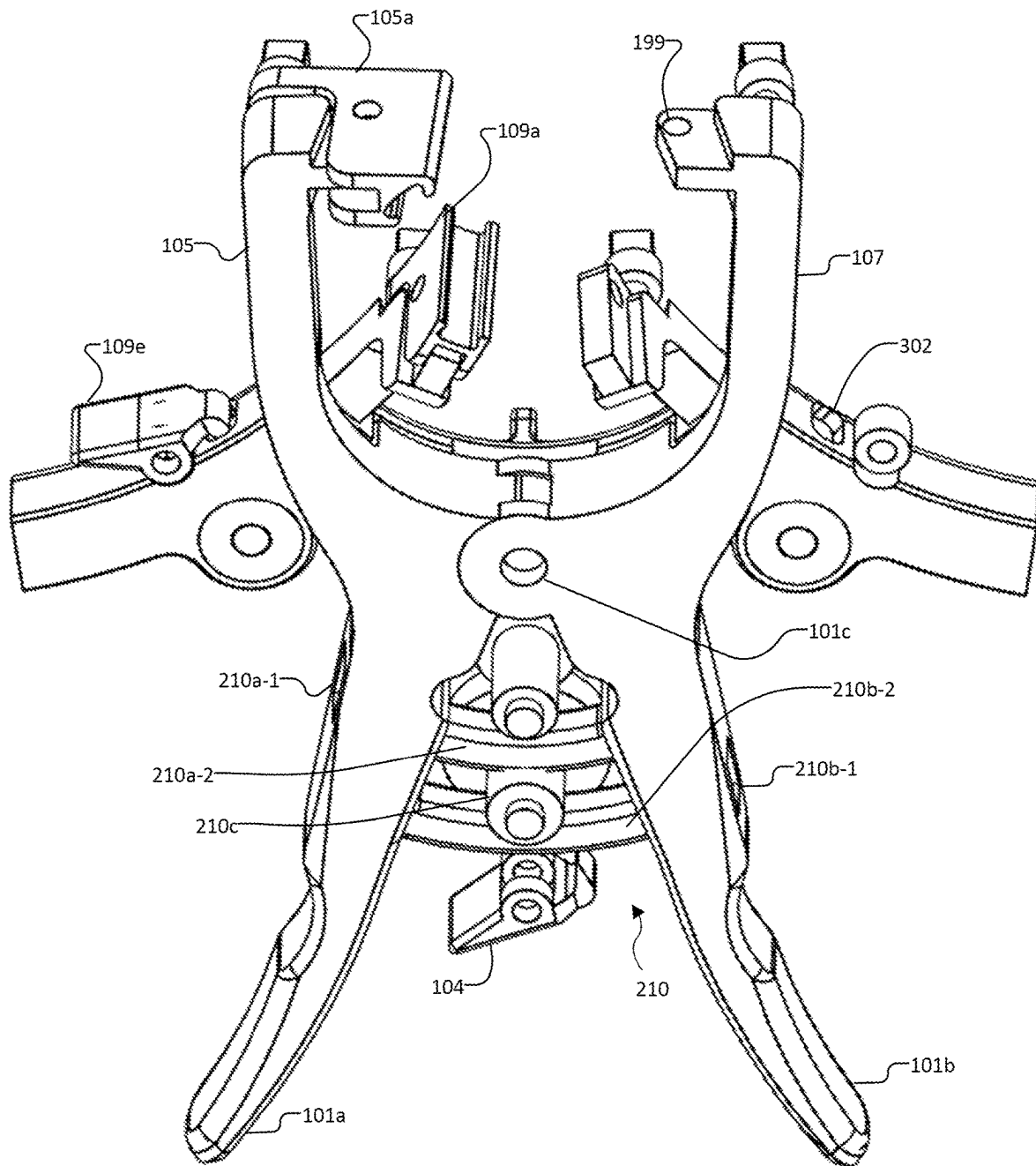
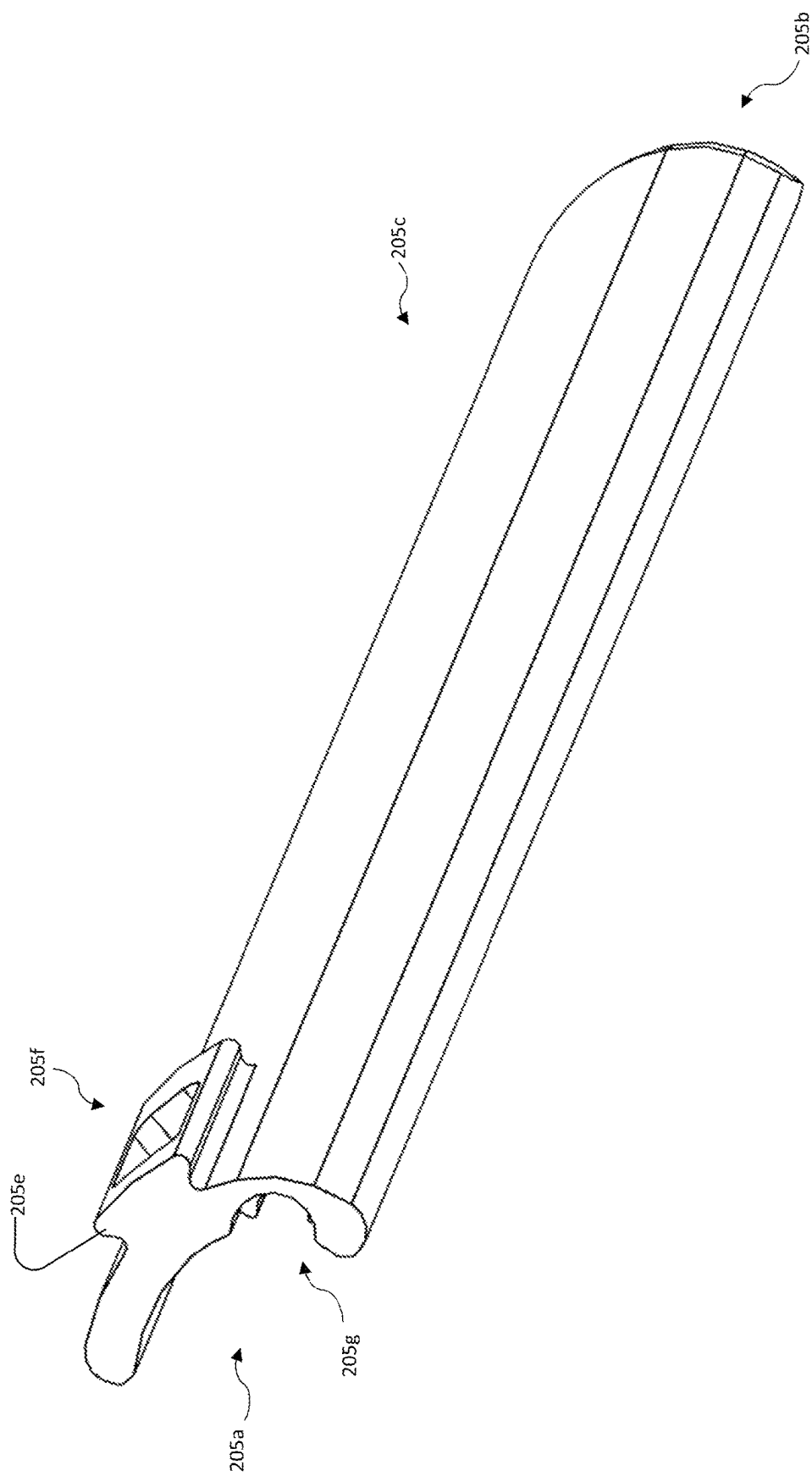


FIG. 8



205

FIG. 9

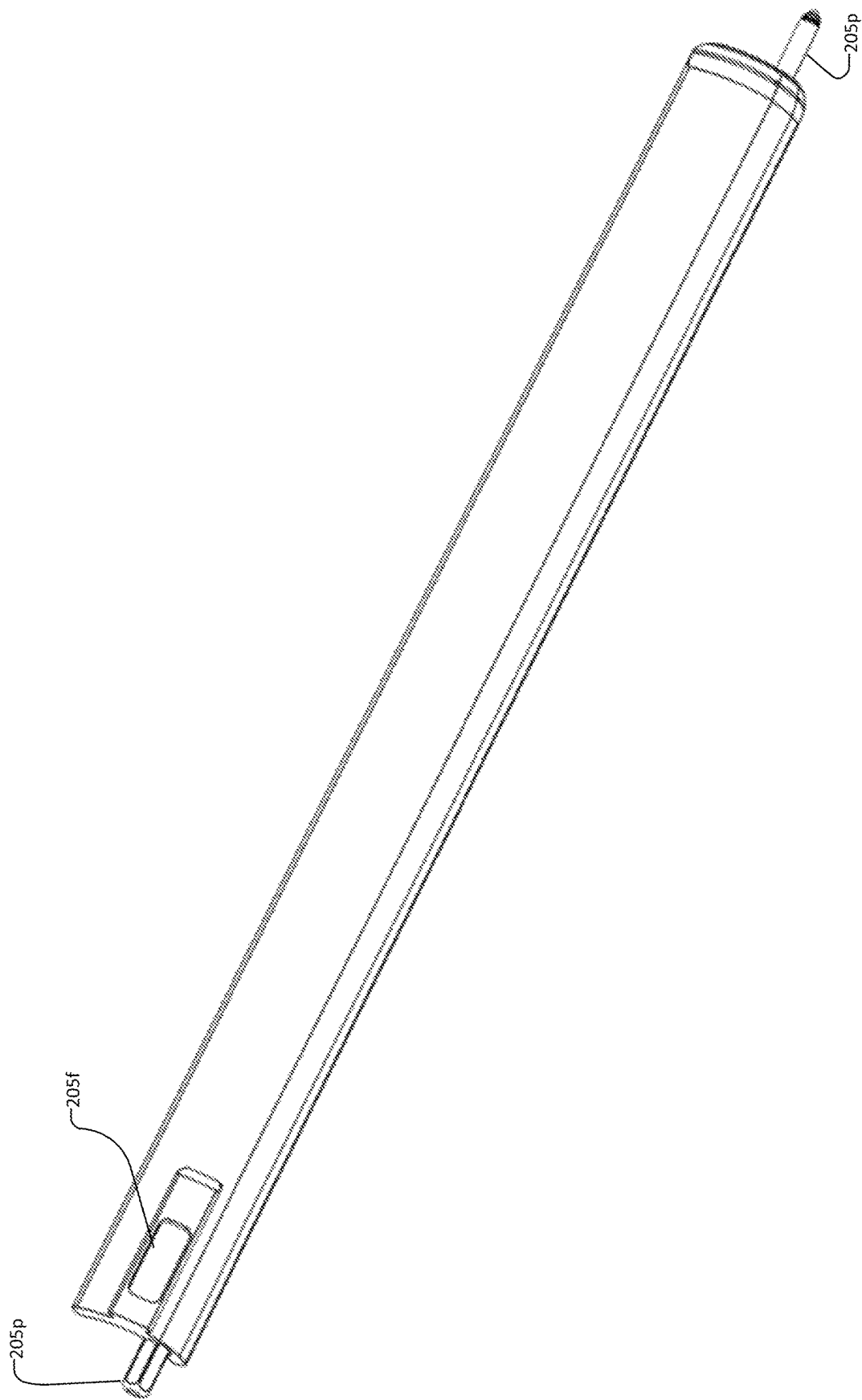
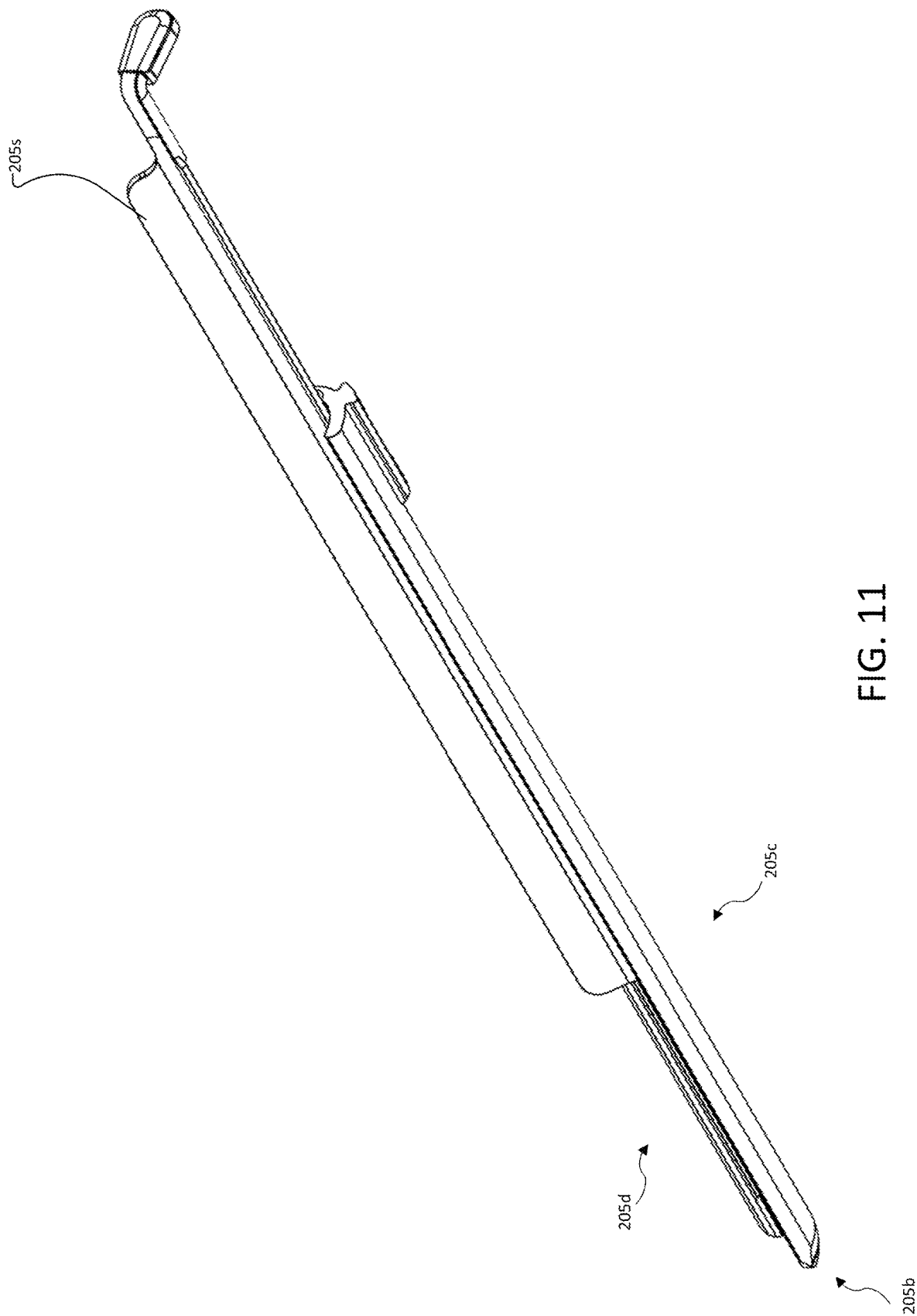
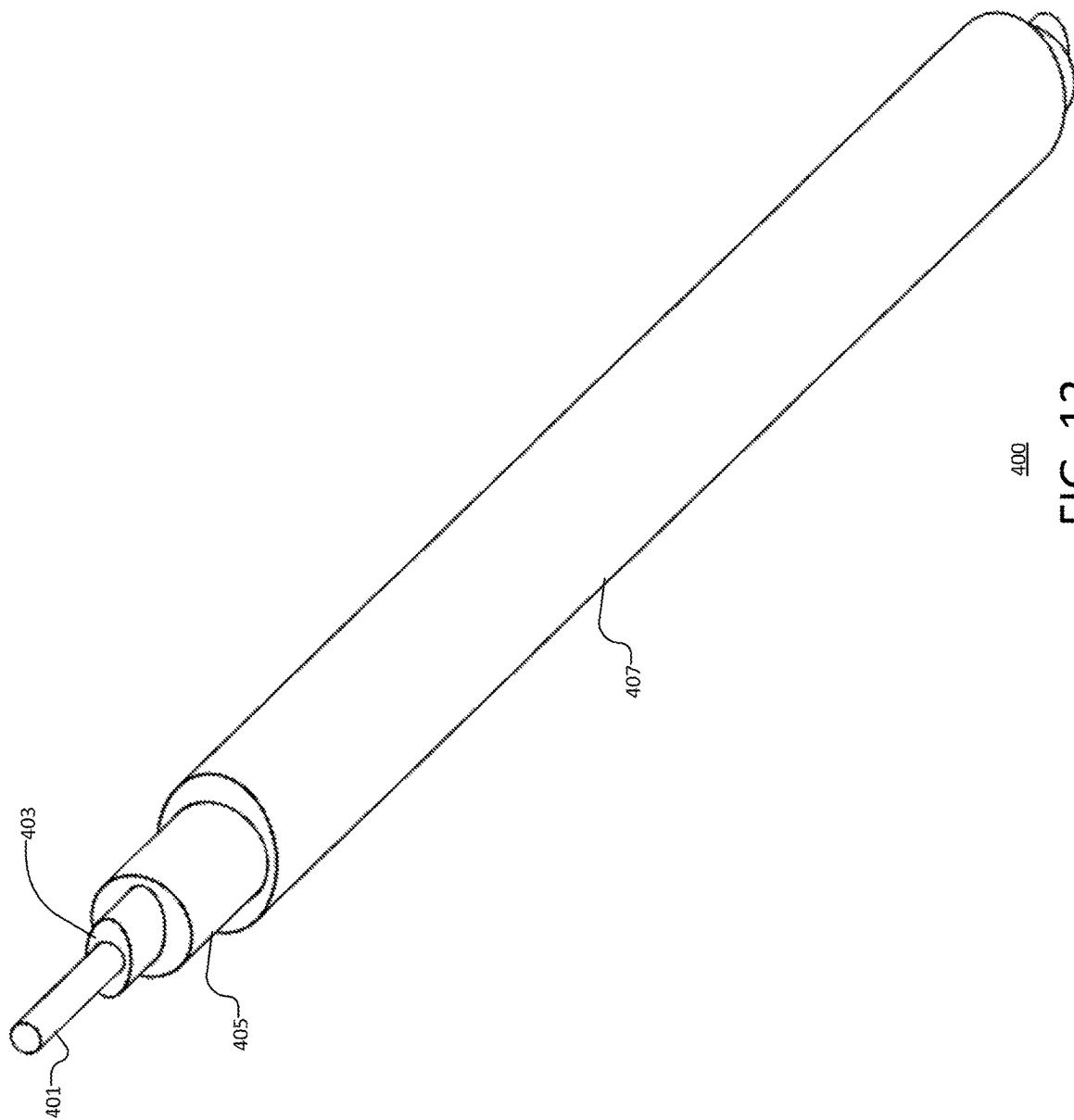


FIG. 10





400
FIG. 12

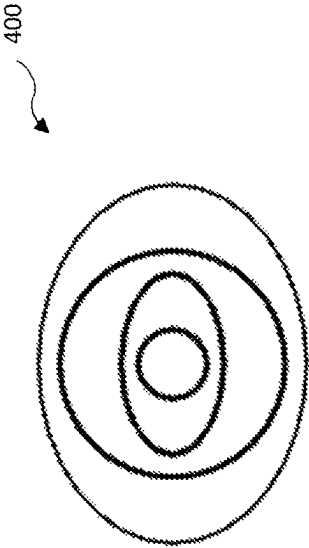


FIG. 13A

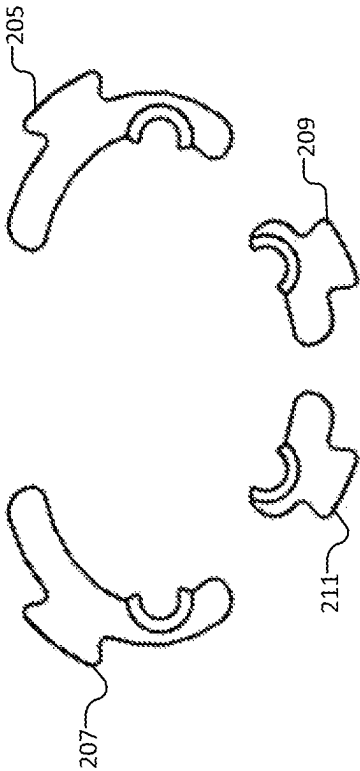


FIG. 13B

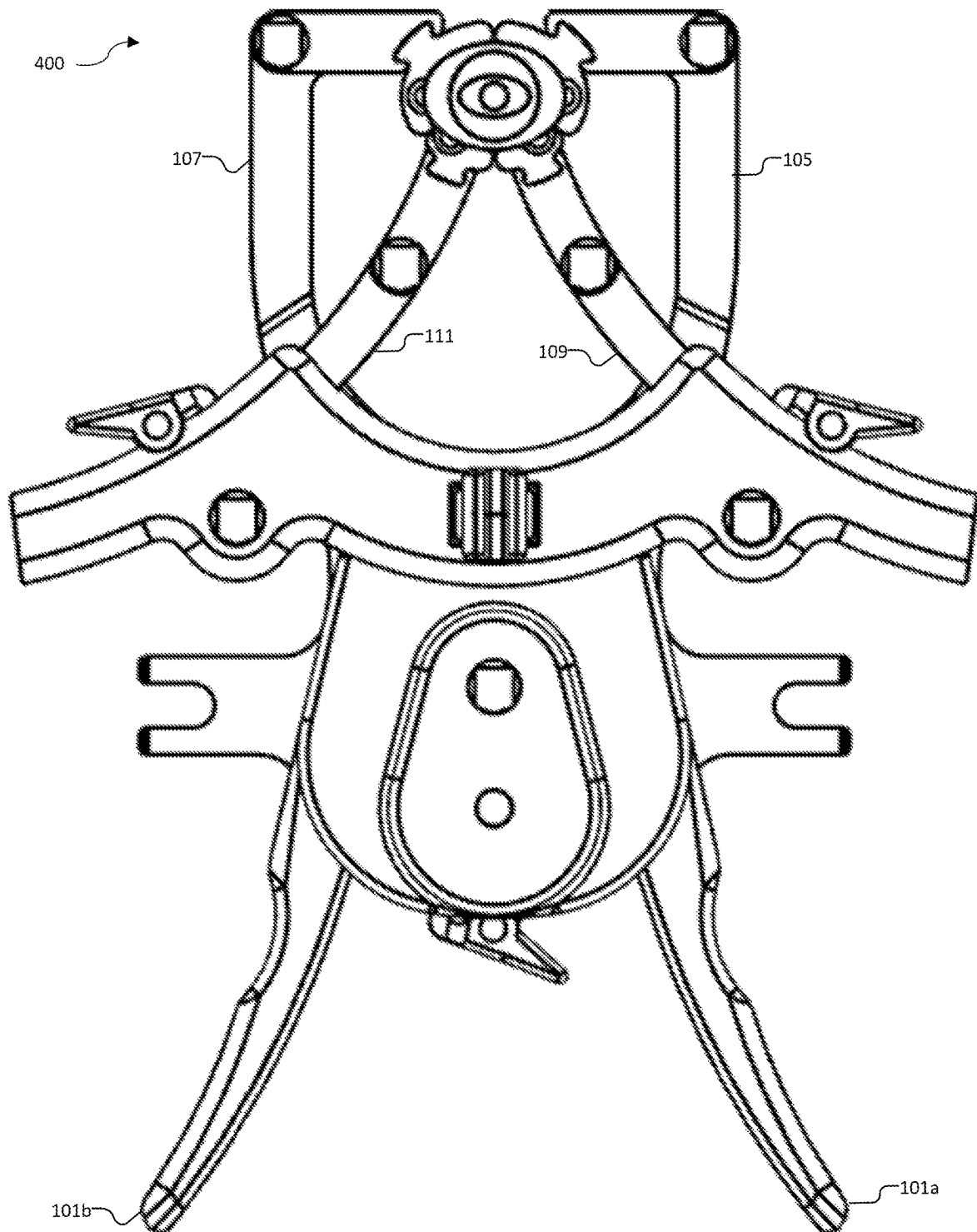


FIG. 14

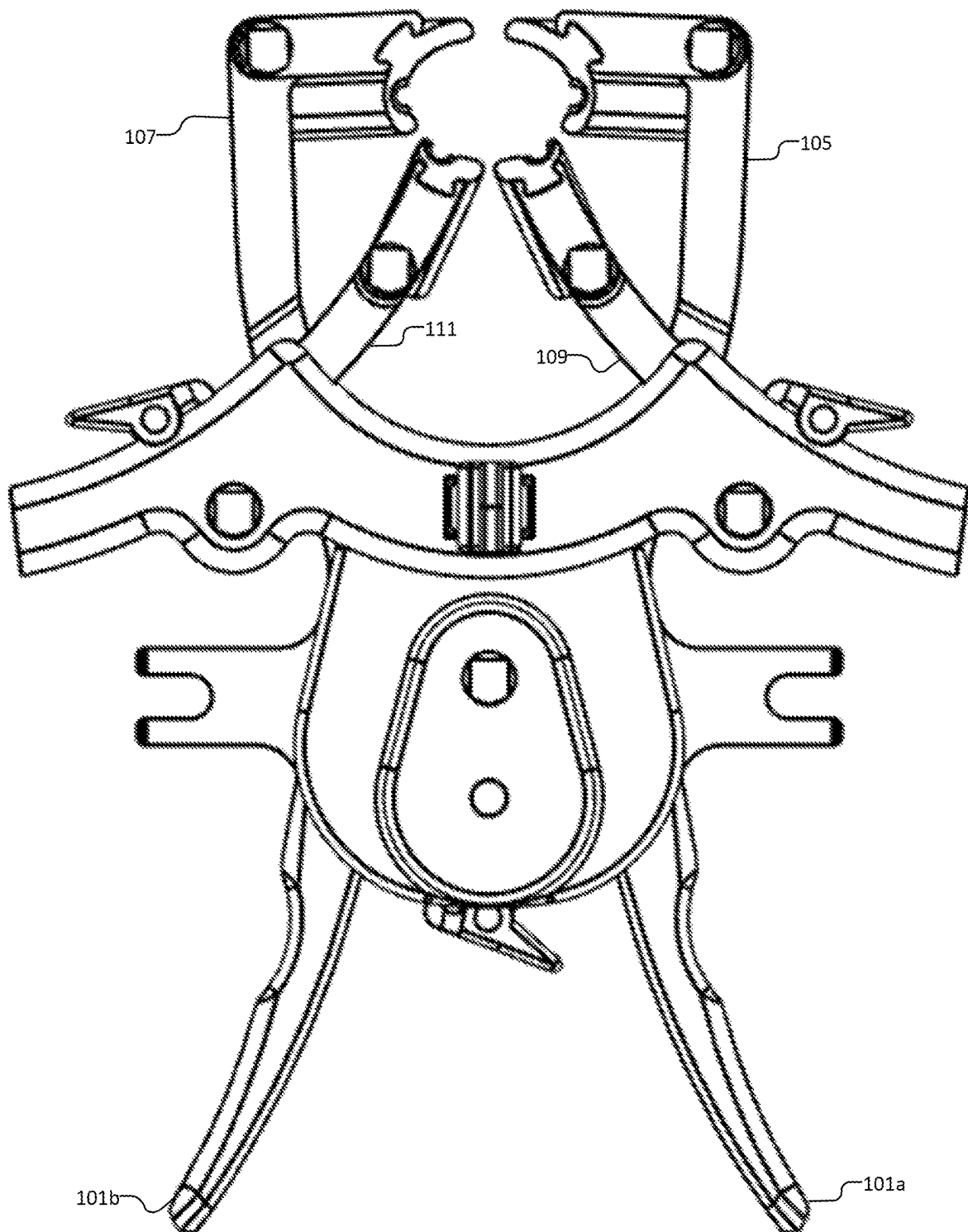


FIG. 15

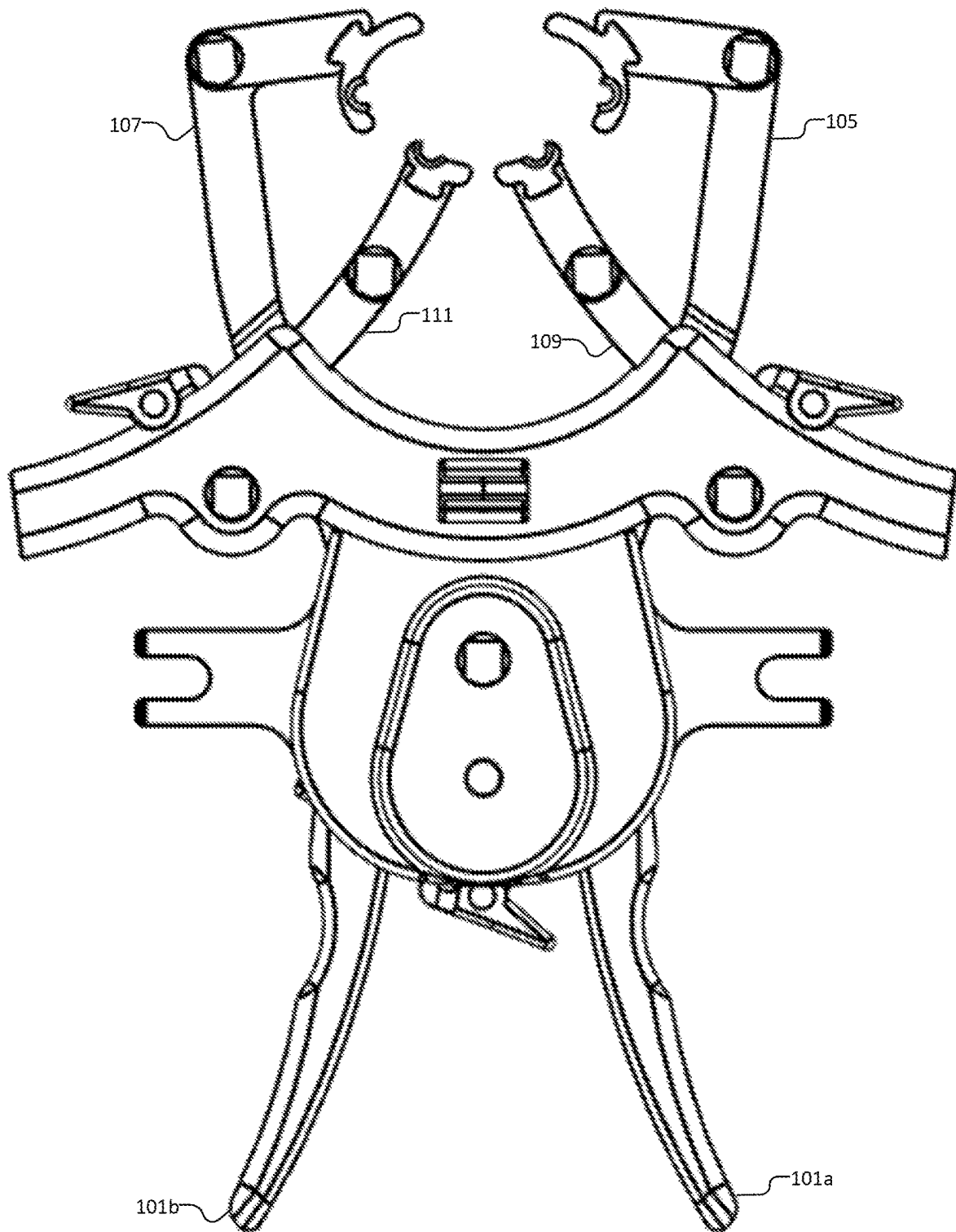


FIG. 16

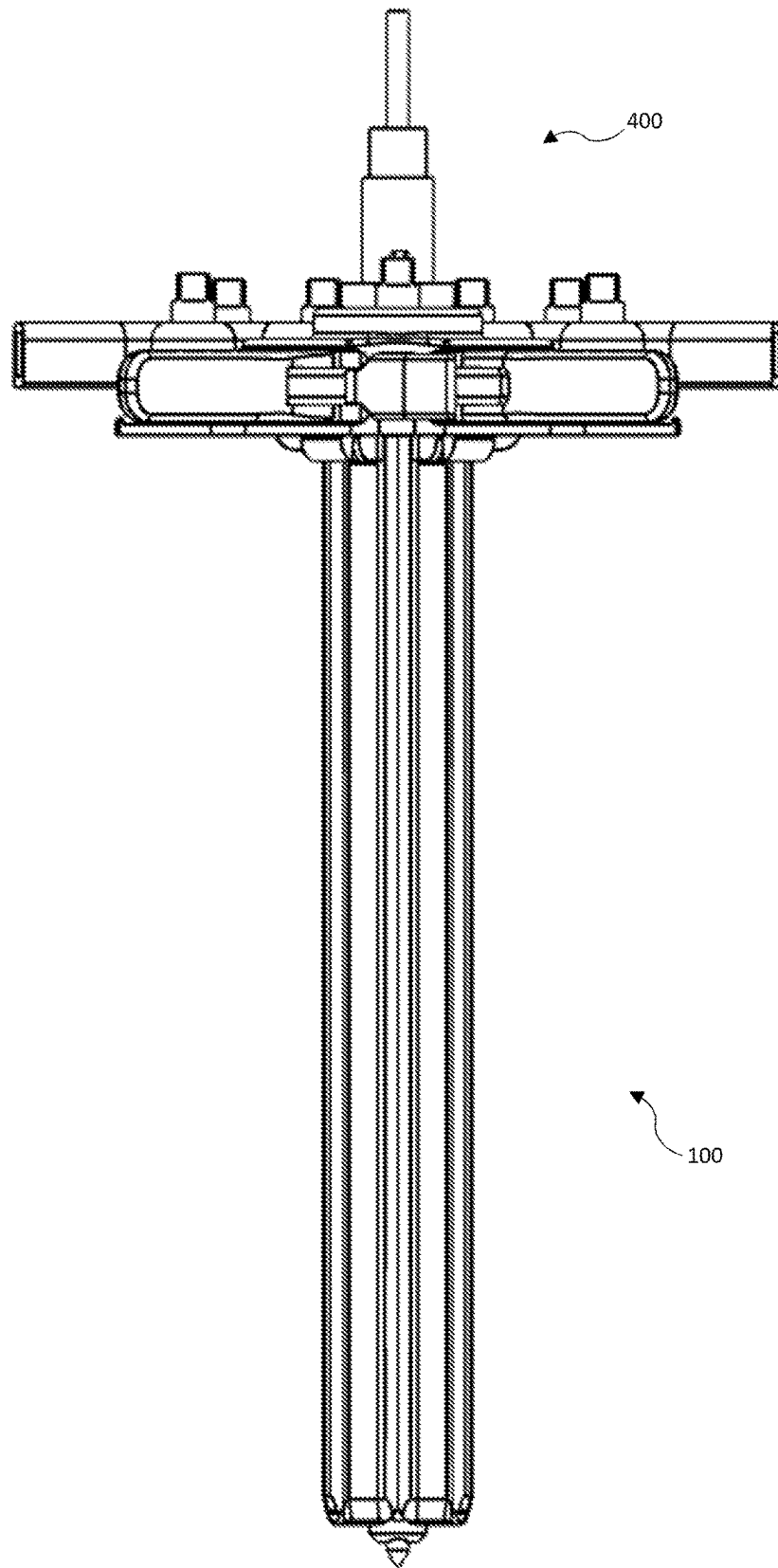


FIG. 17

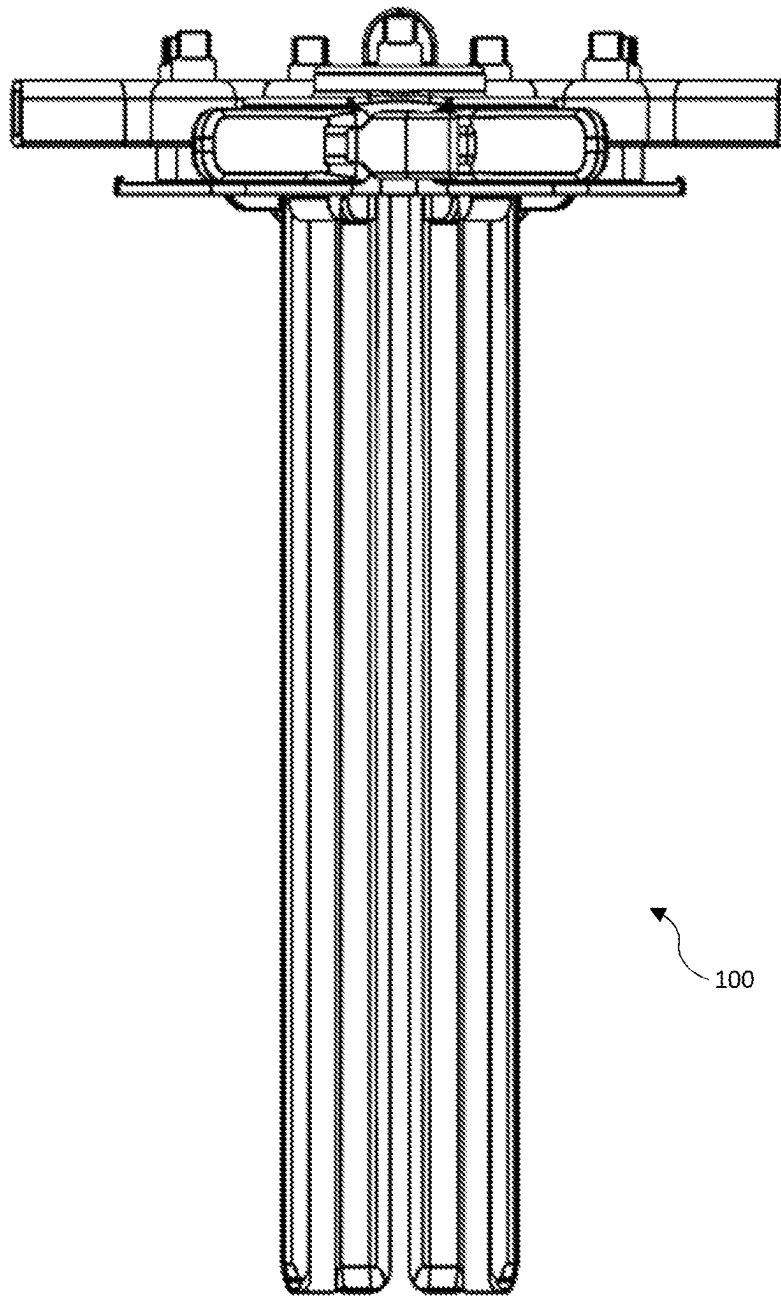


FIG. 18

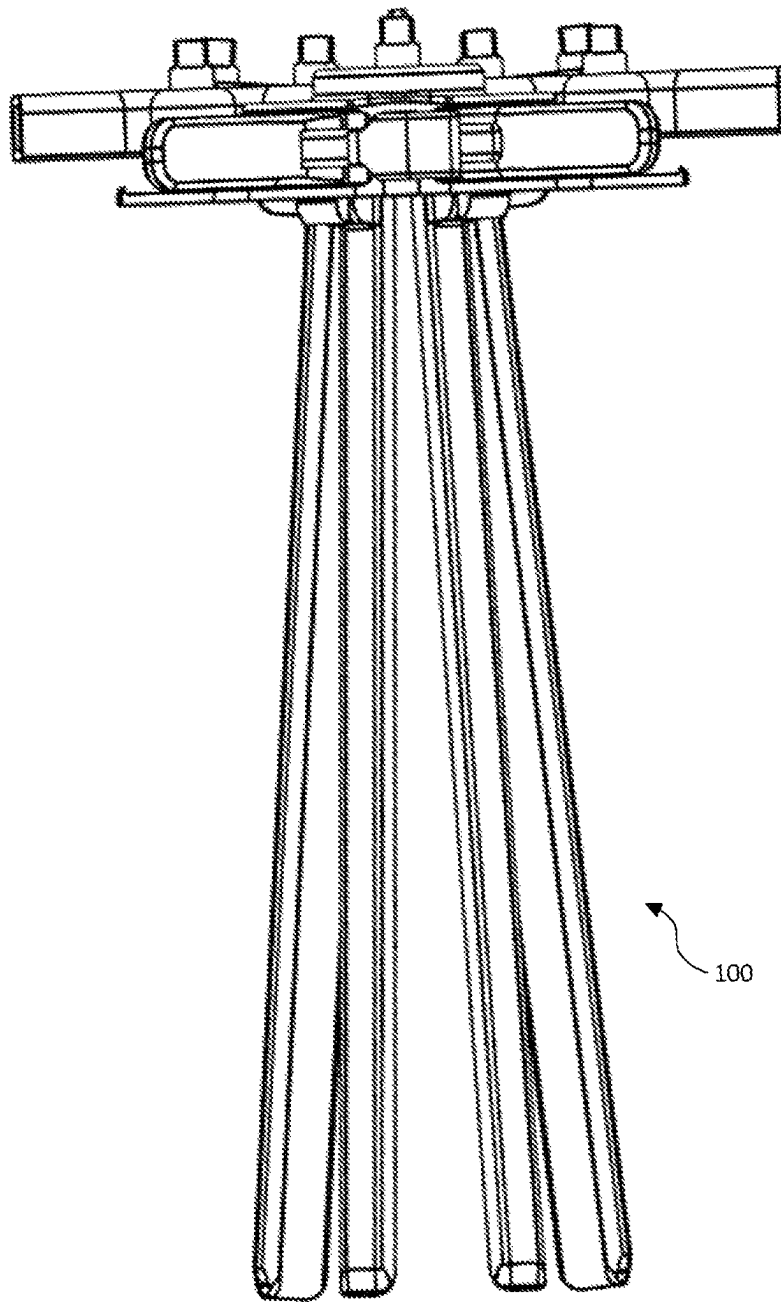


FIG. 19

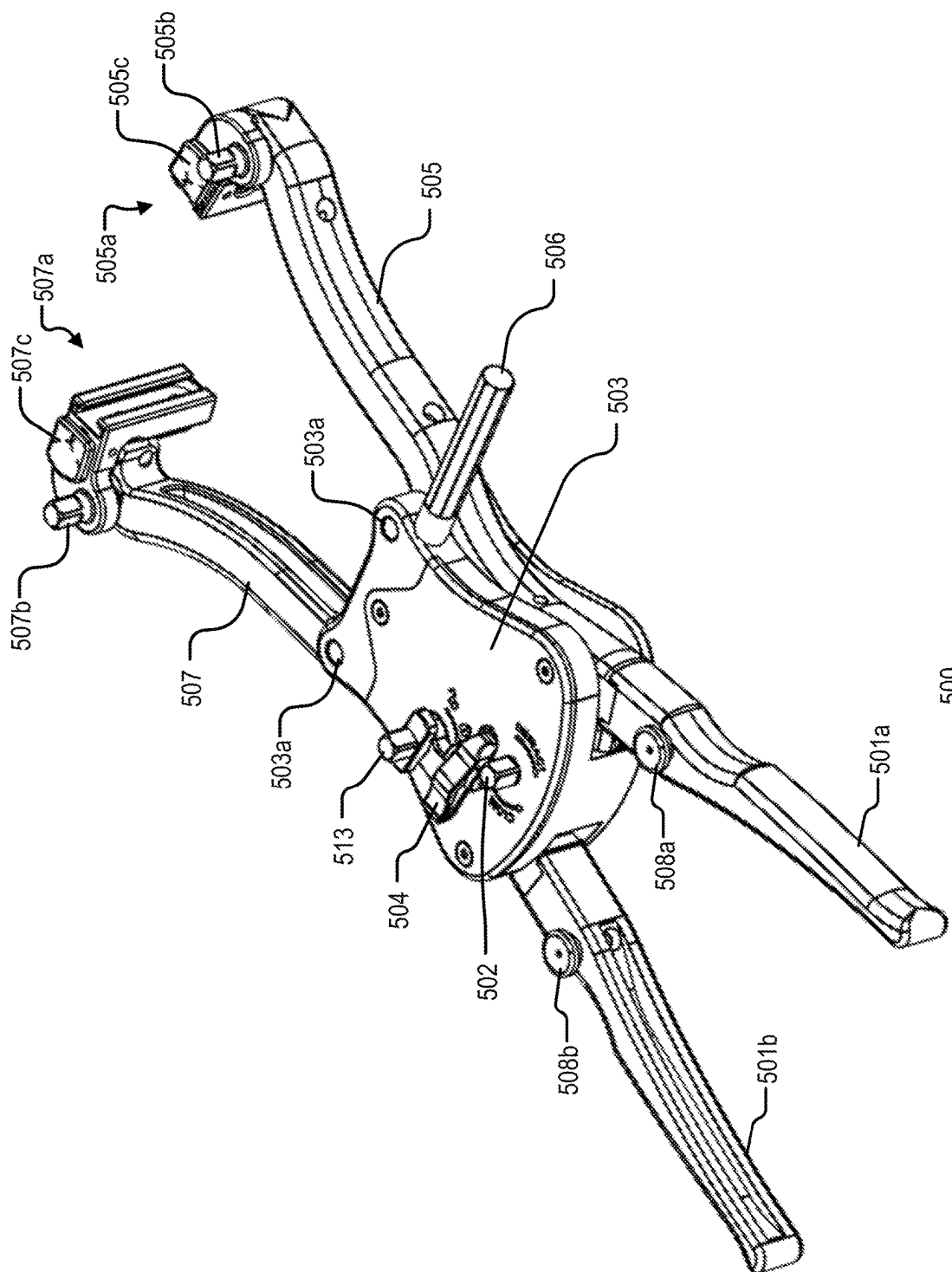


FIG. 20

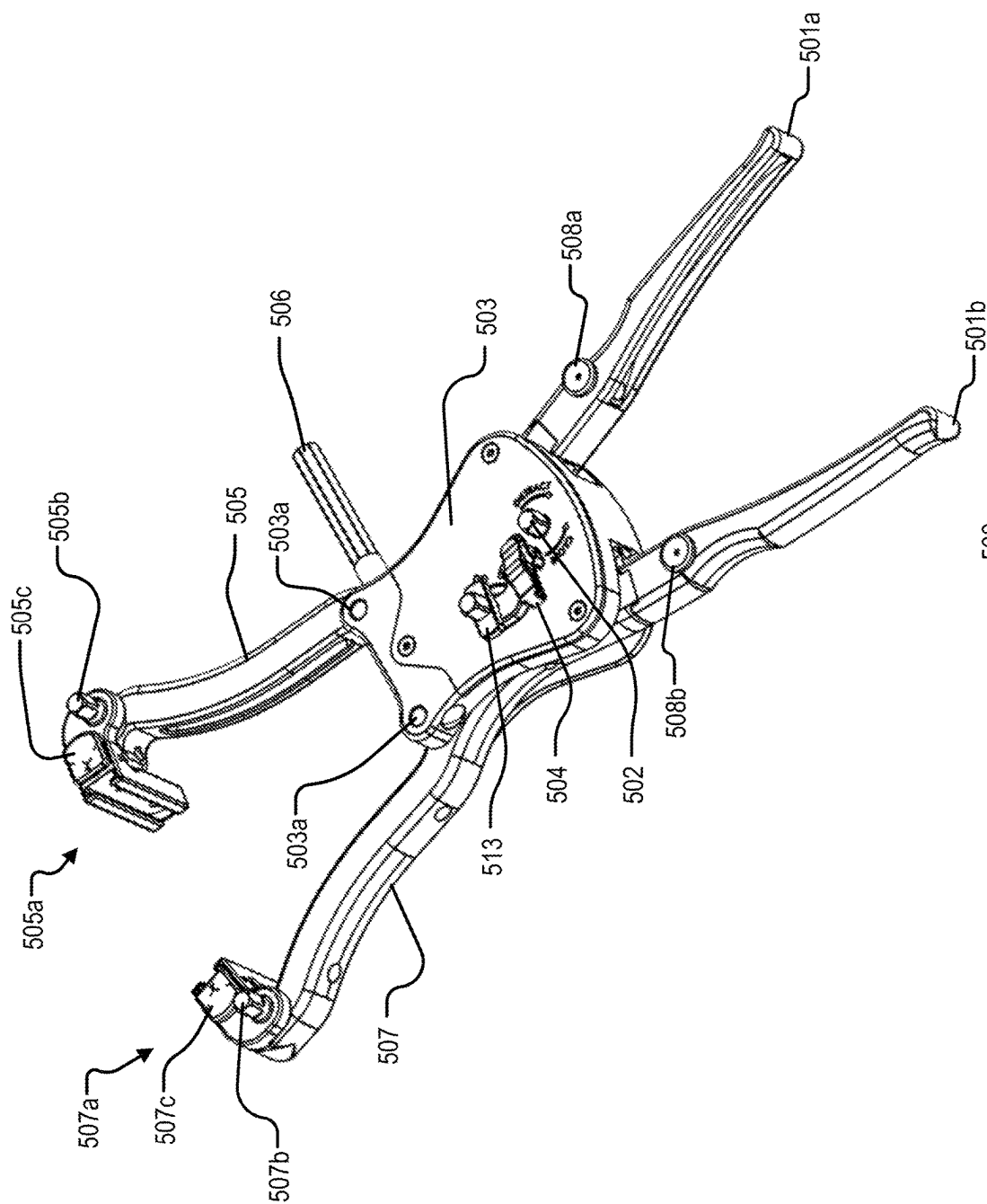


FIG. 21

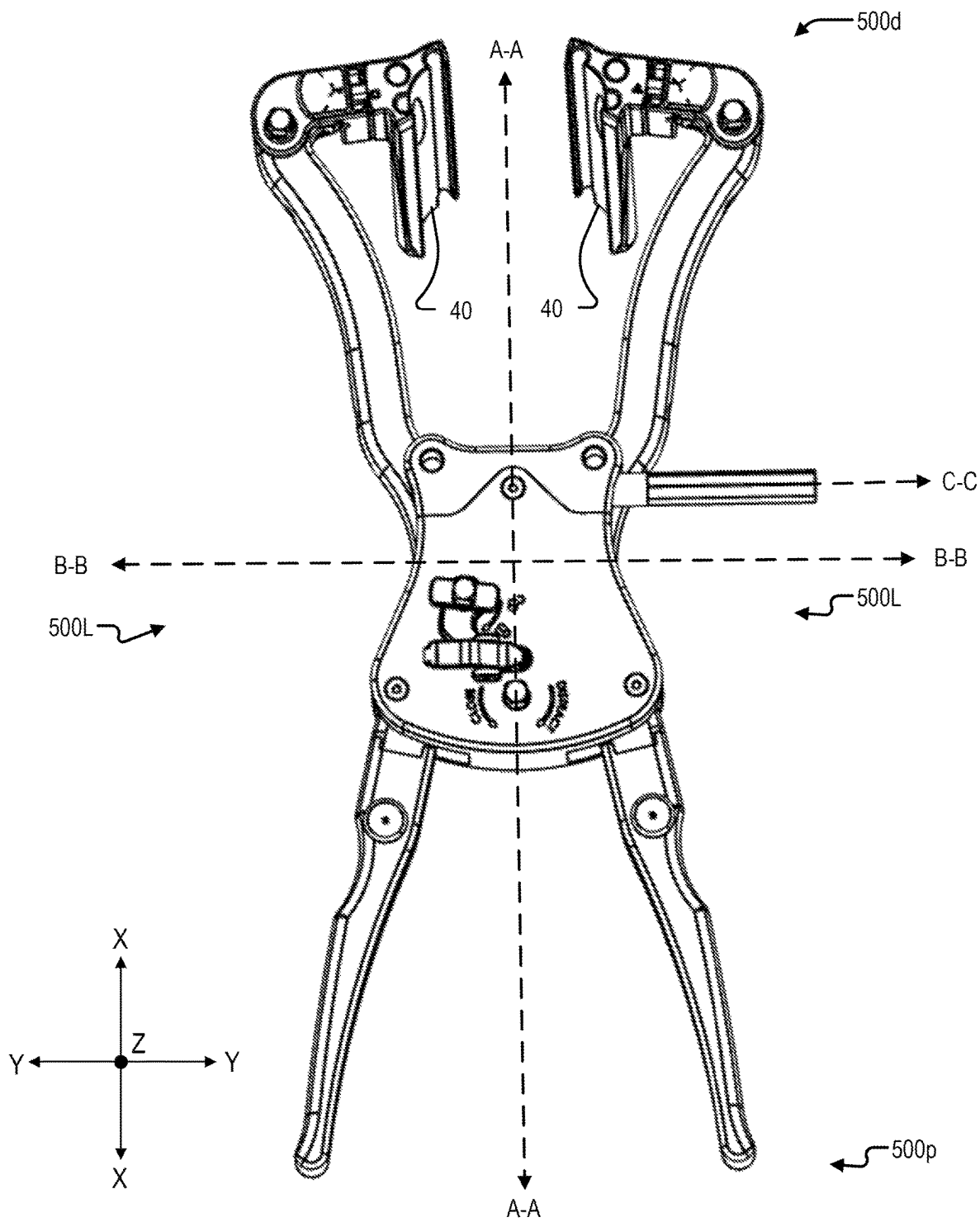
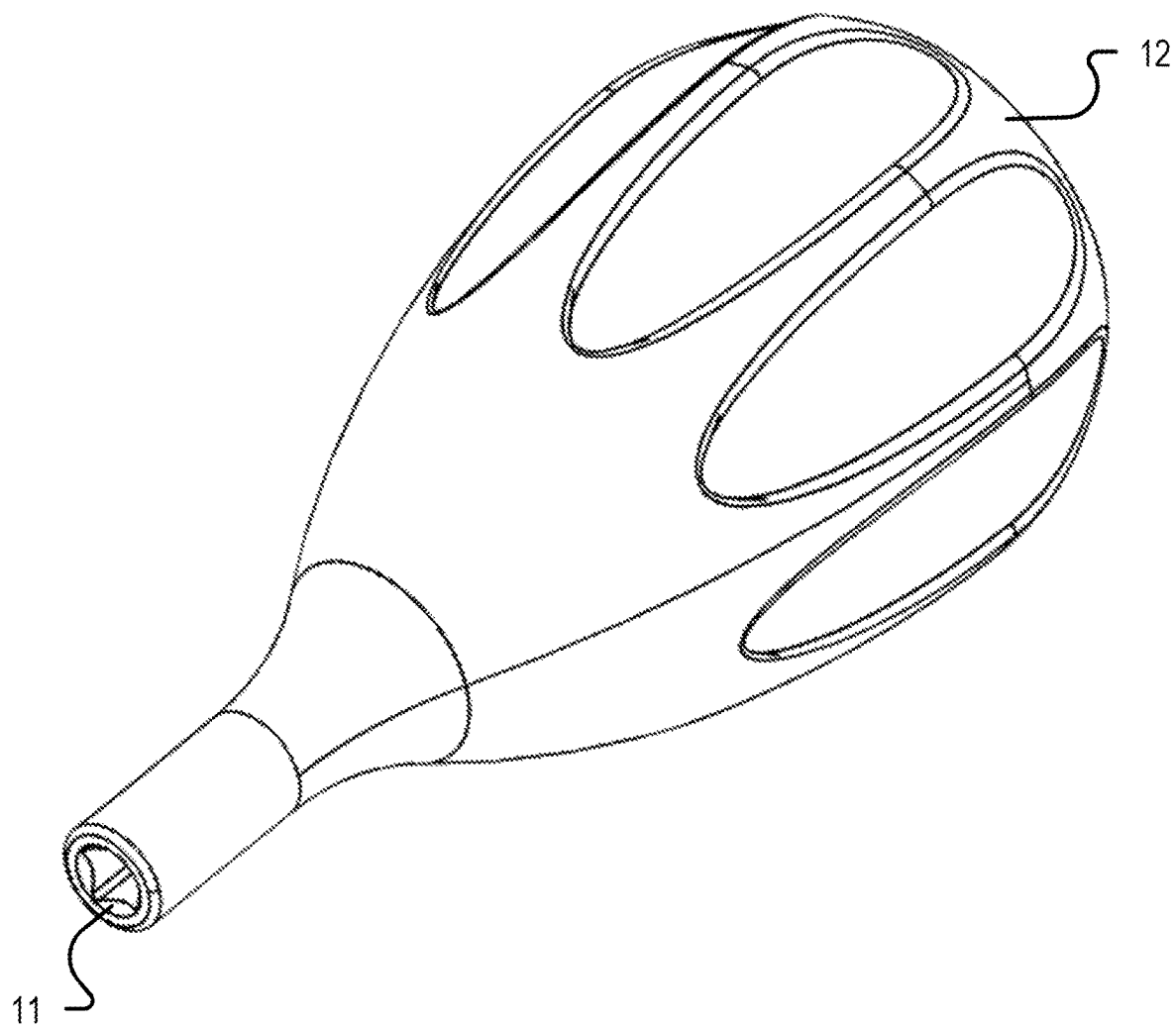


FIG. 22



10

FIG. 23

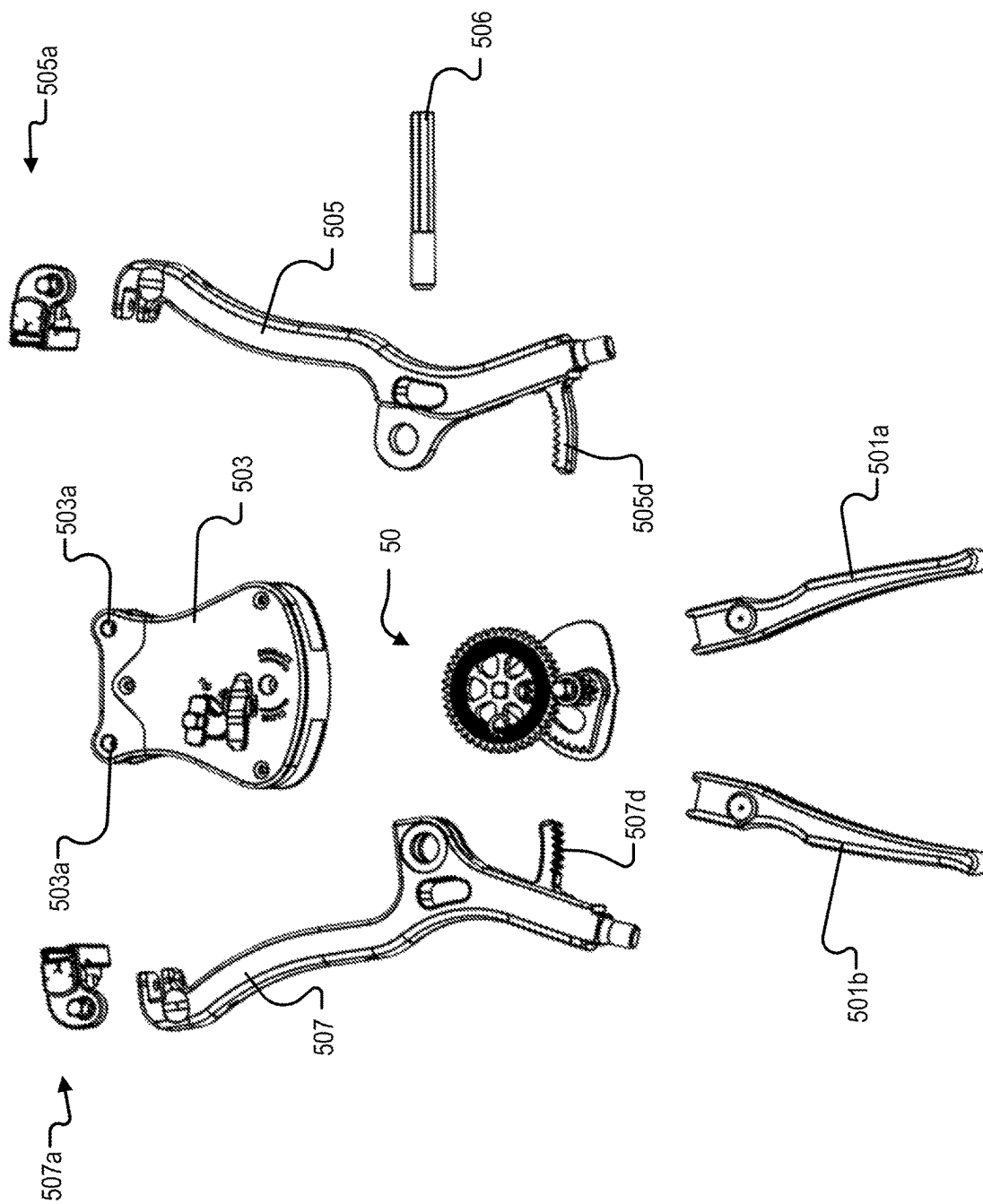


FIG. 24

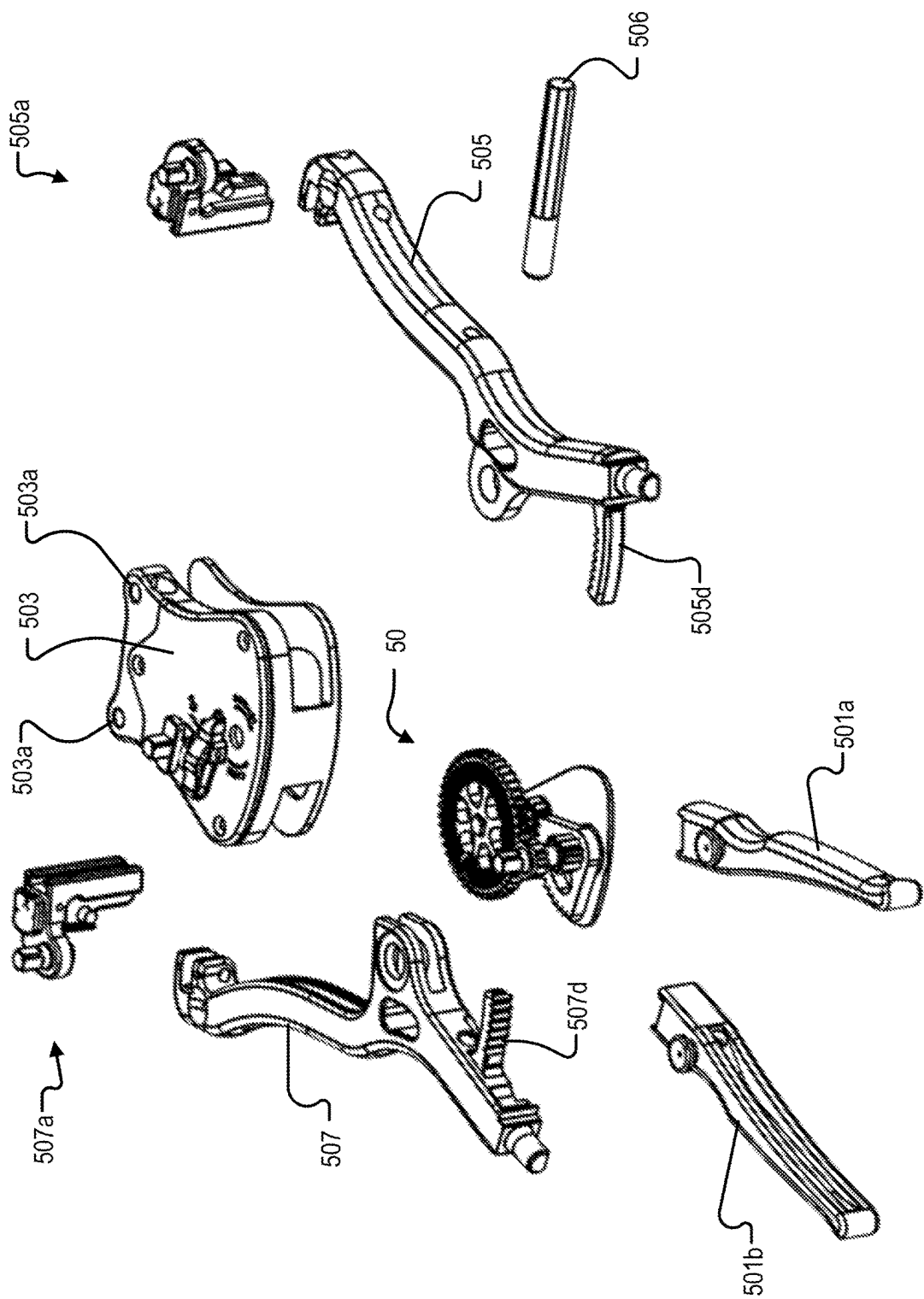
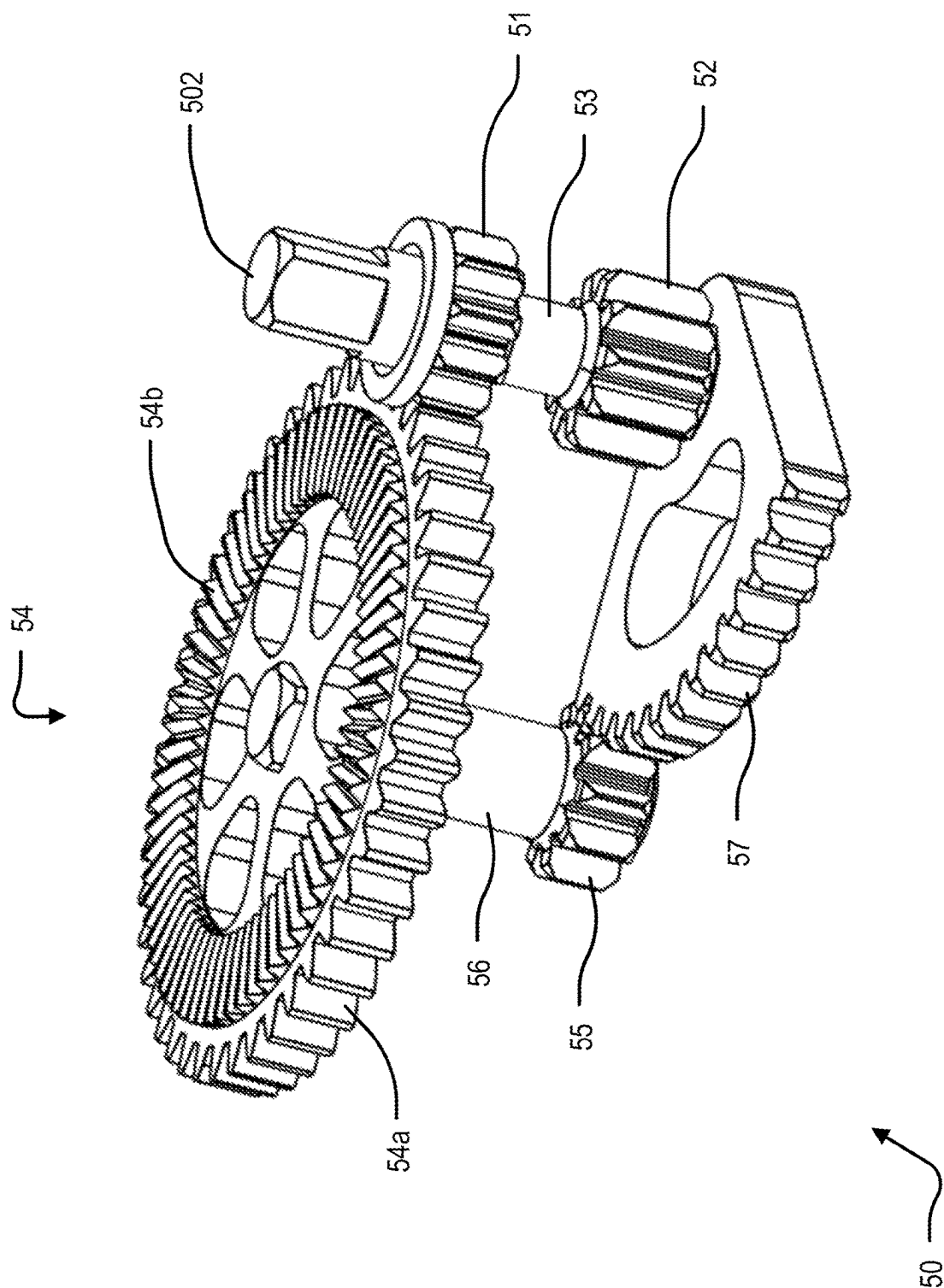


FIG. 25



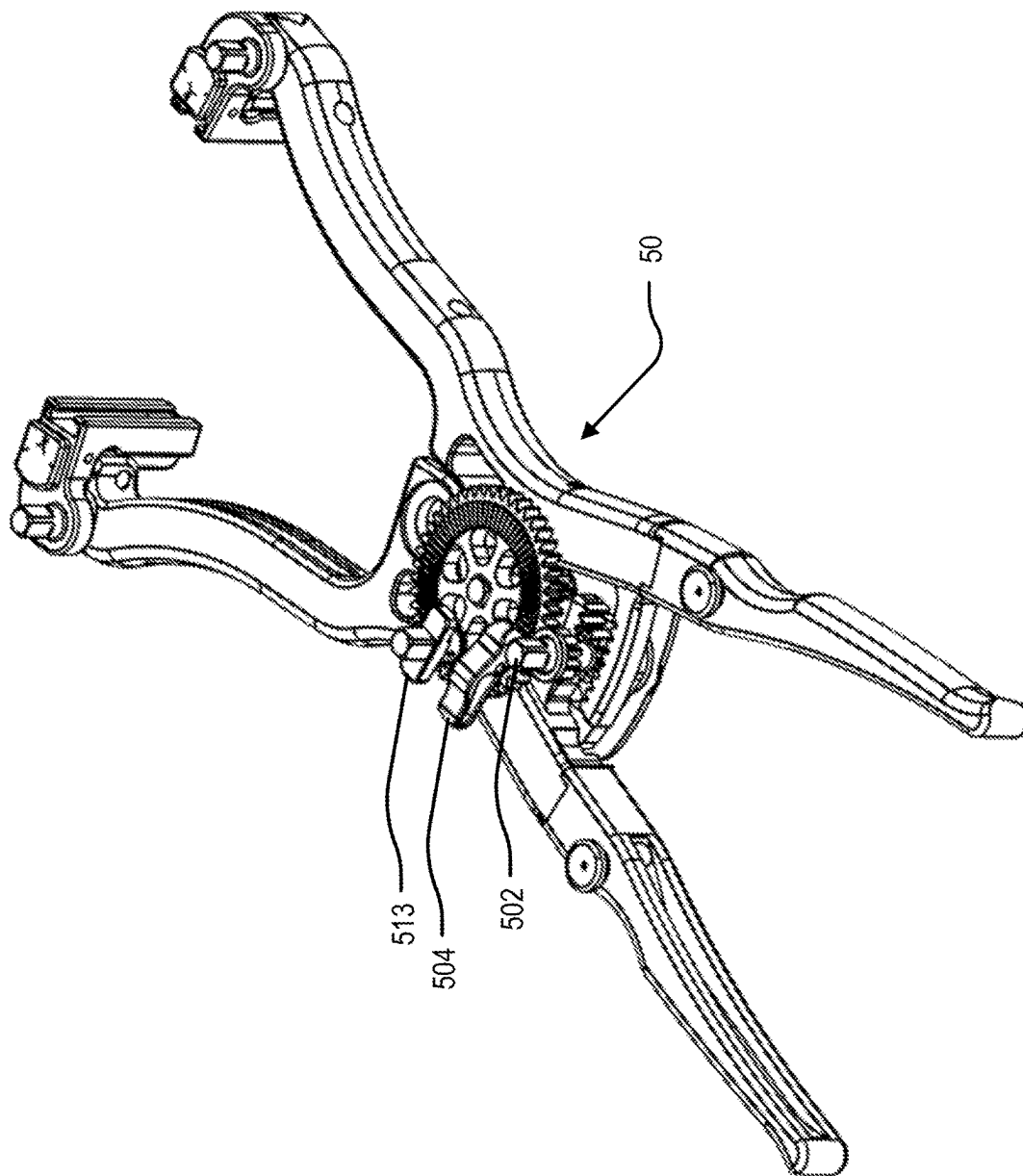


FIG. 26B

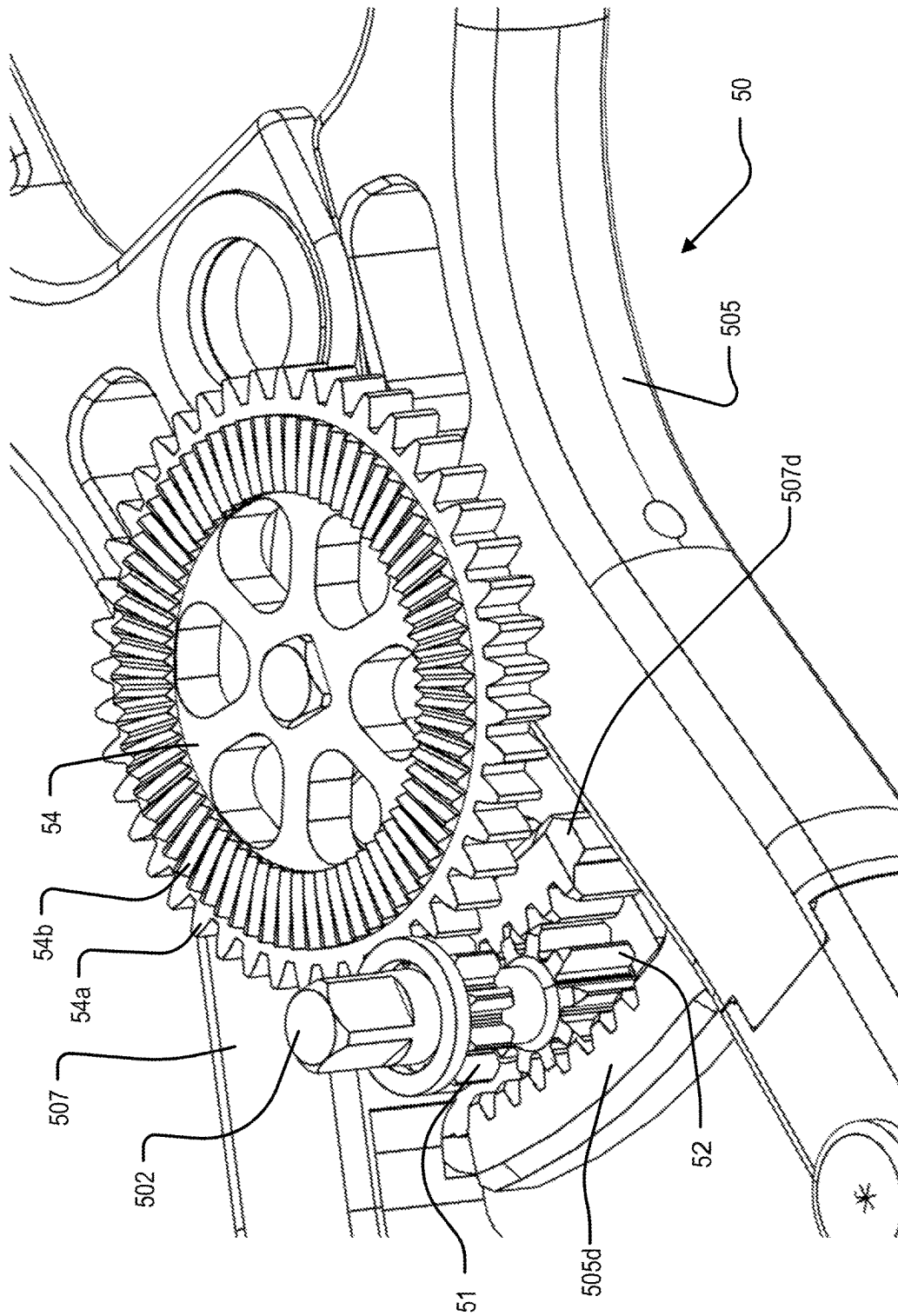


FIG. 26C

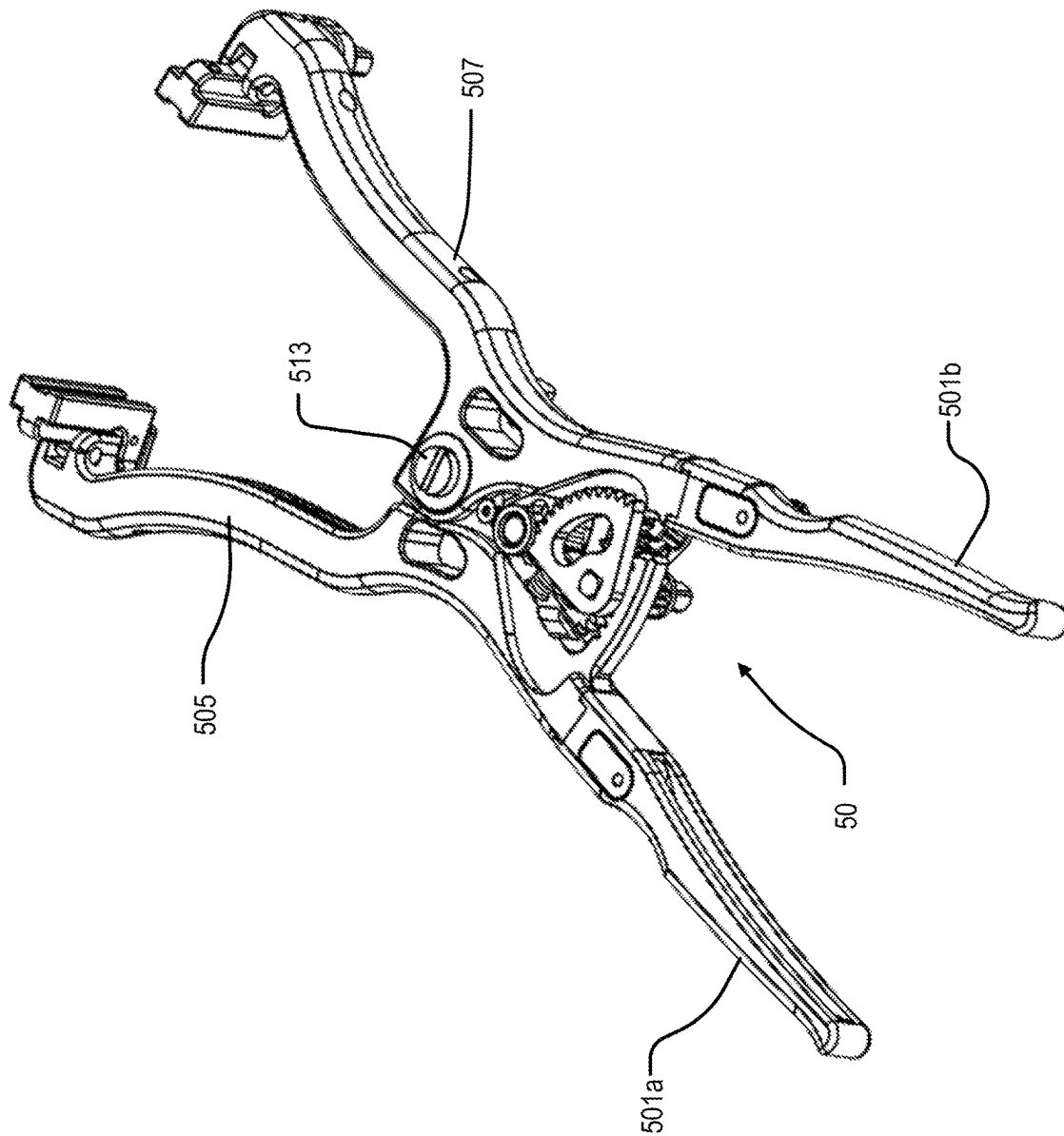


FIG. 26D

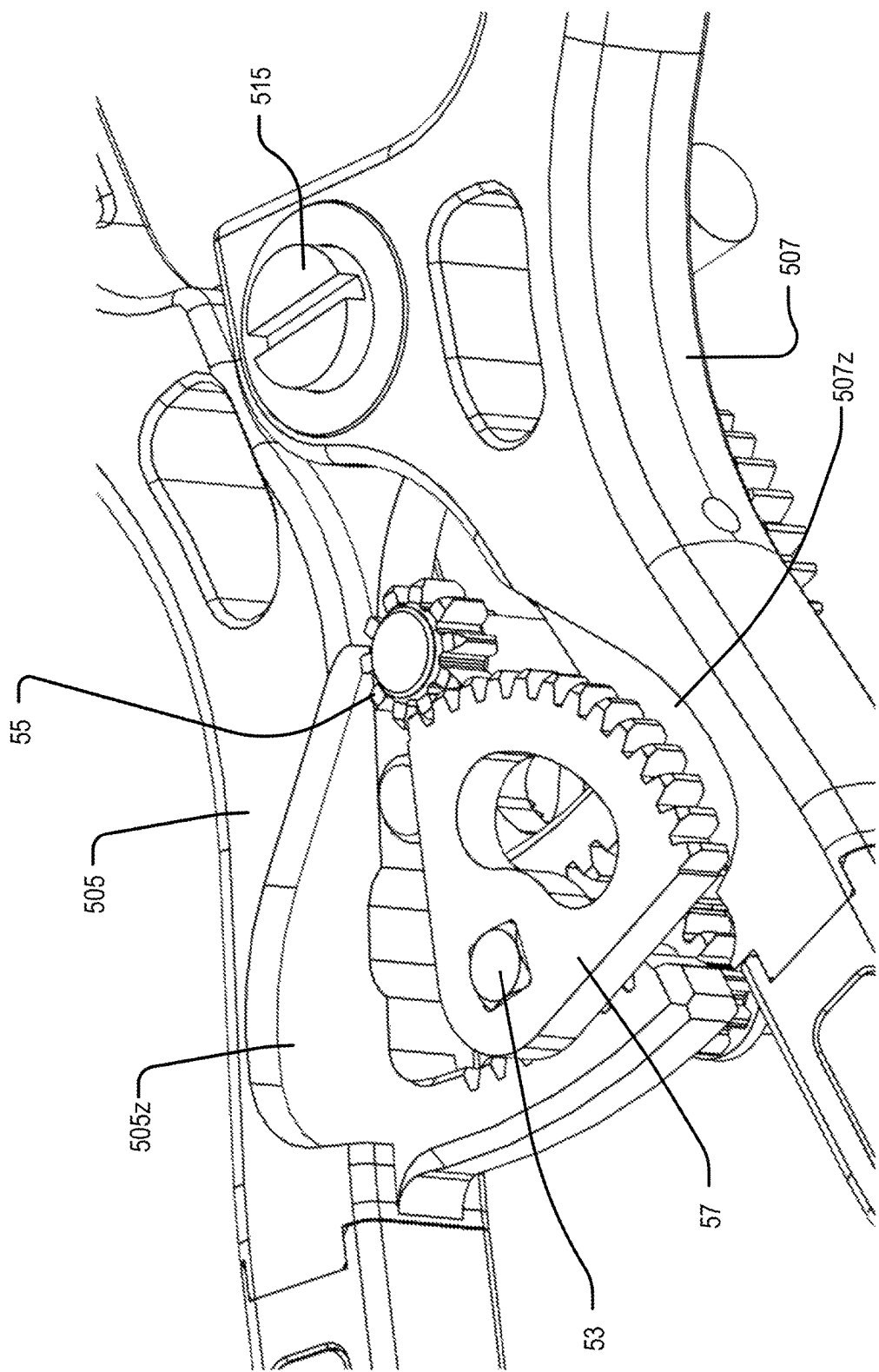


FIG. 26E

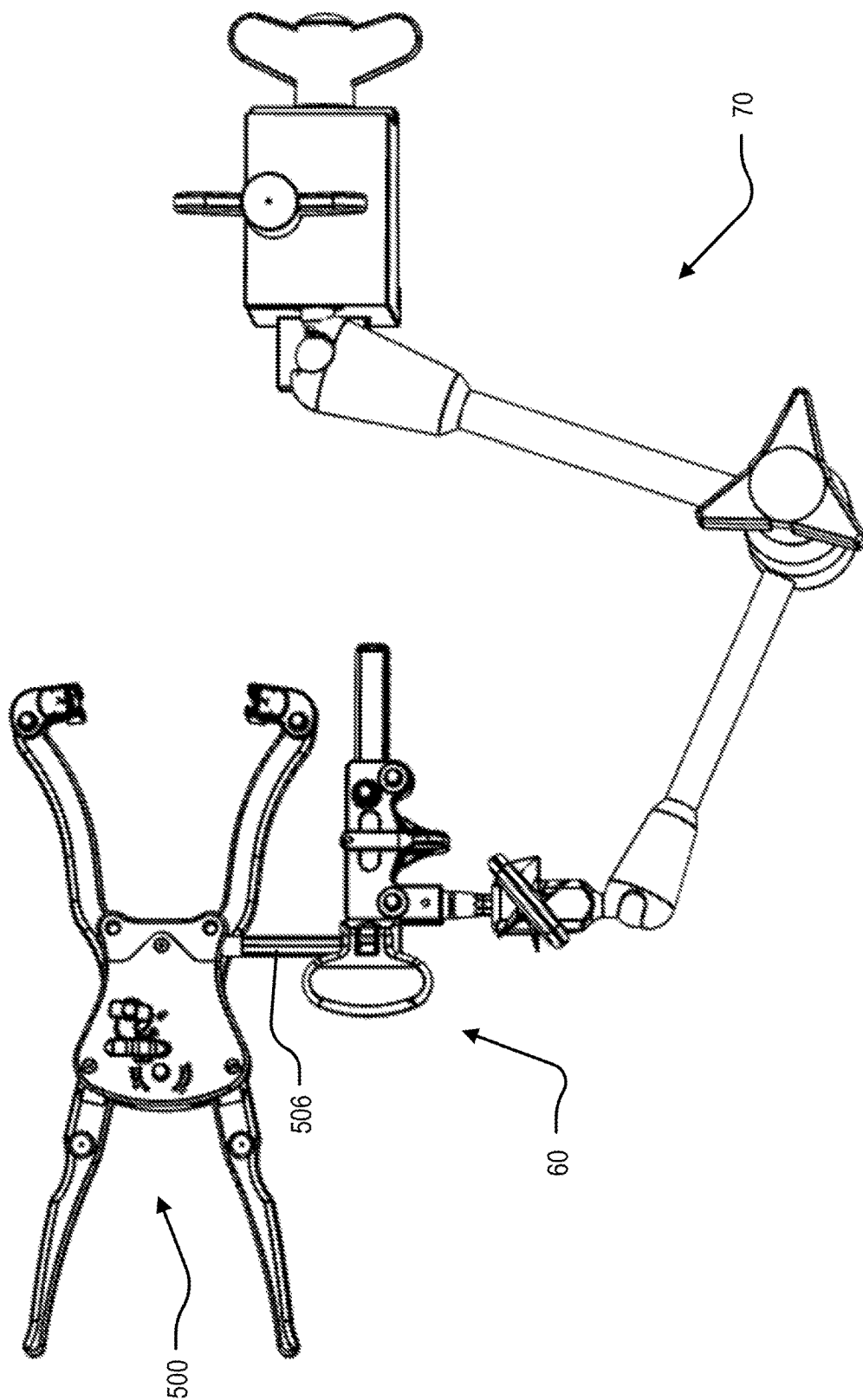


FIG. 27A

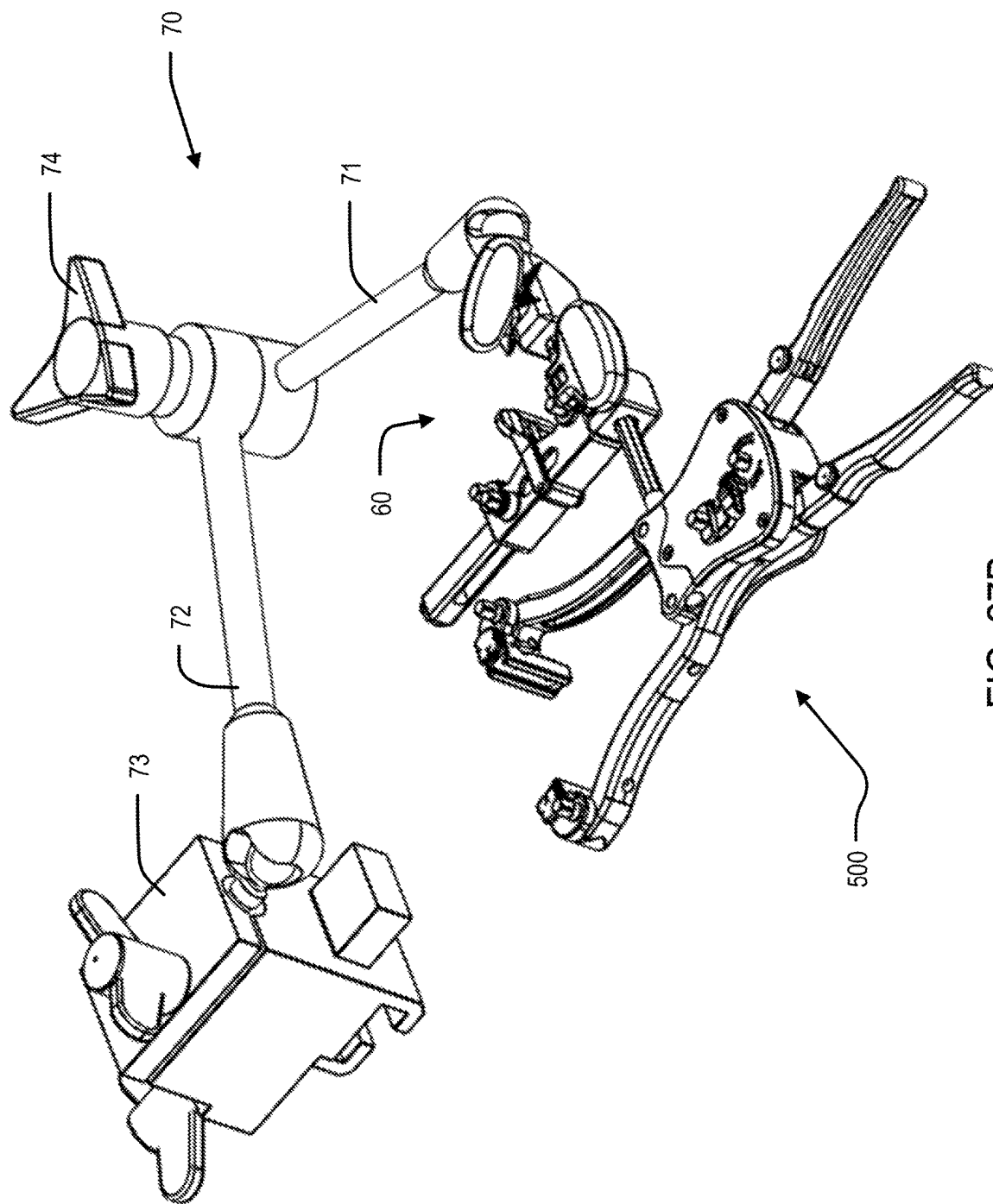


FIG. 27B

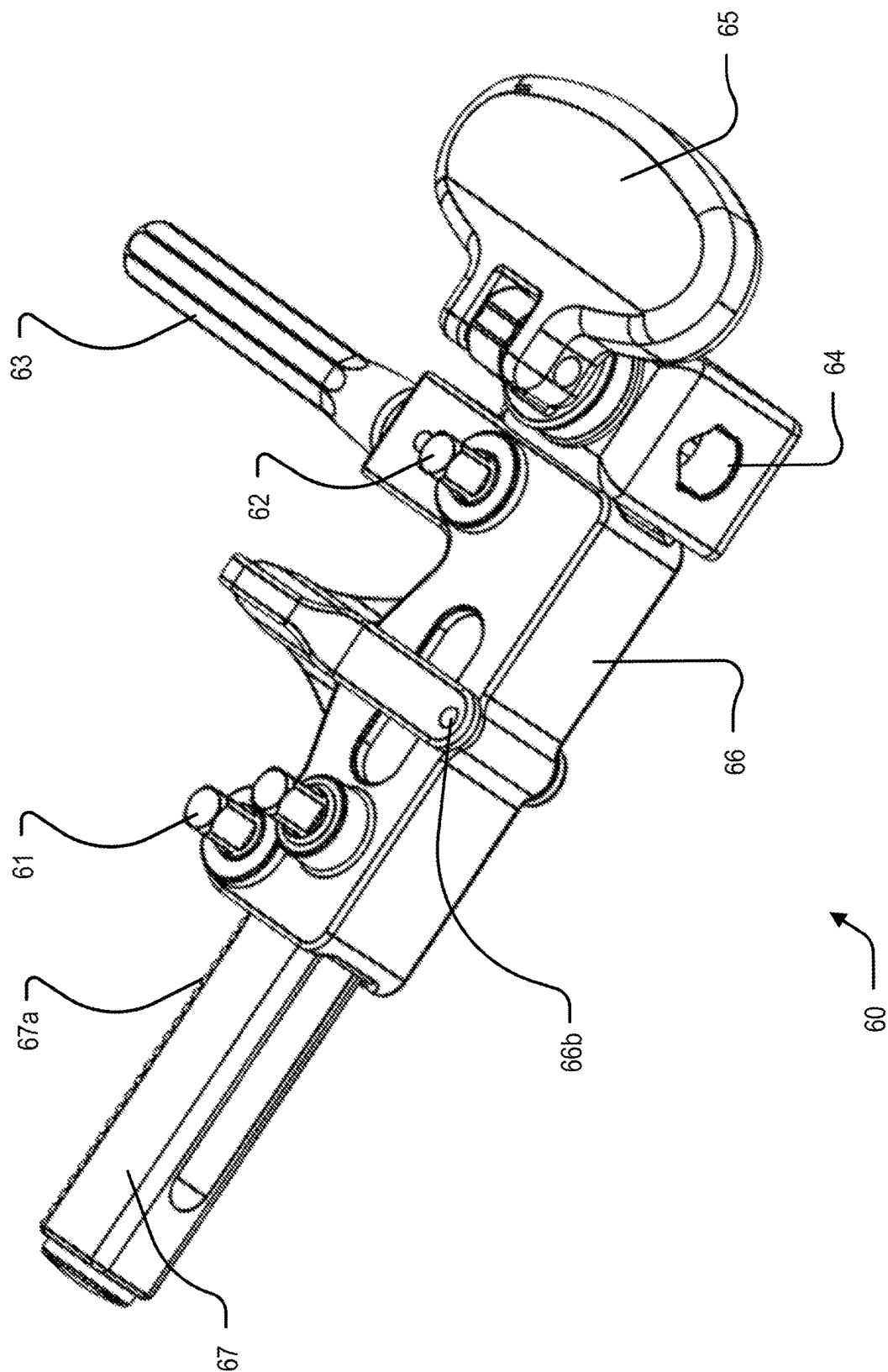


FIG. 28A

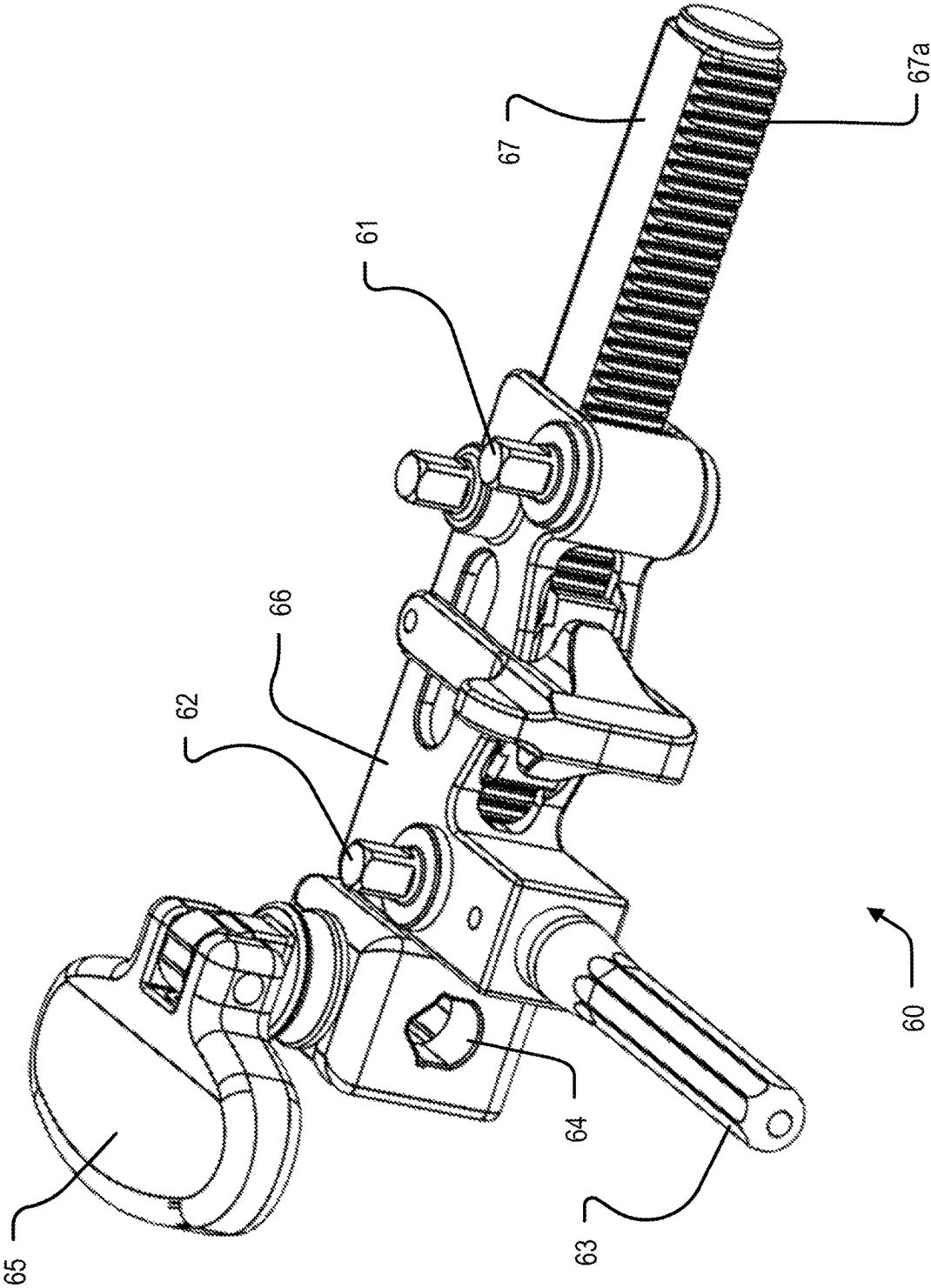


FIG. 28B

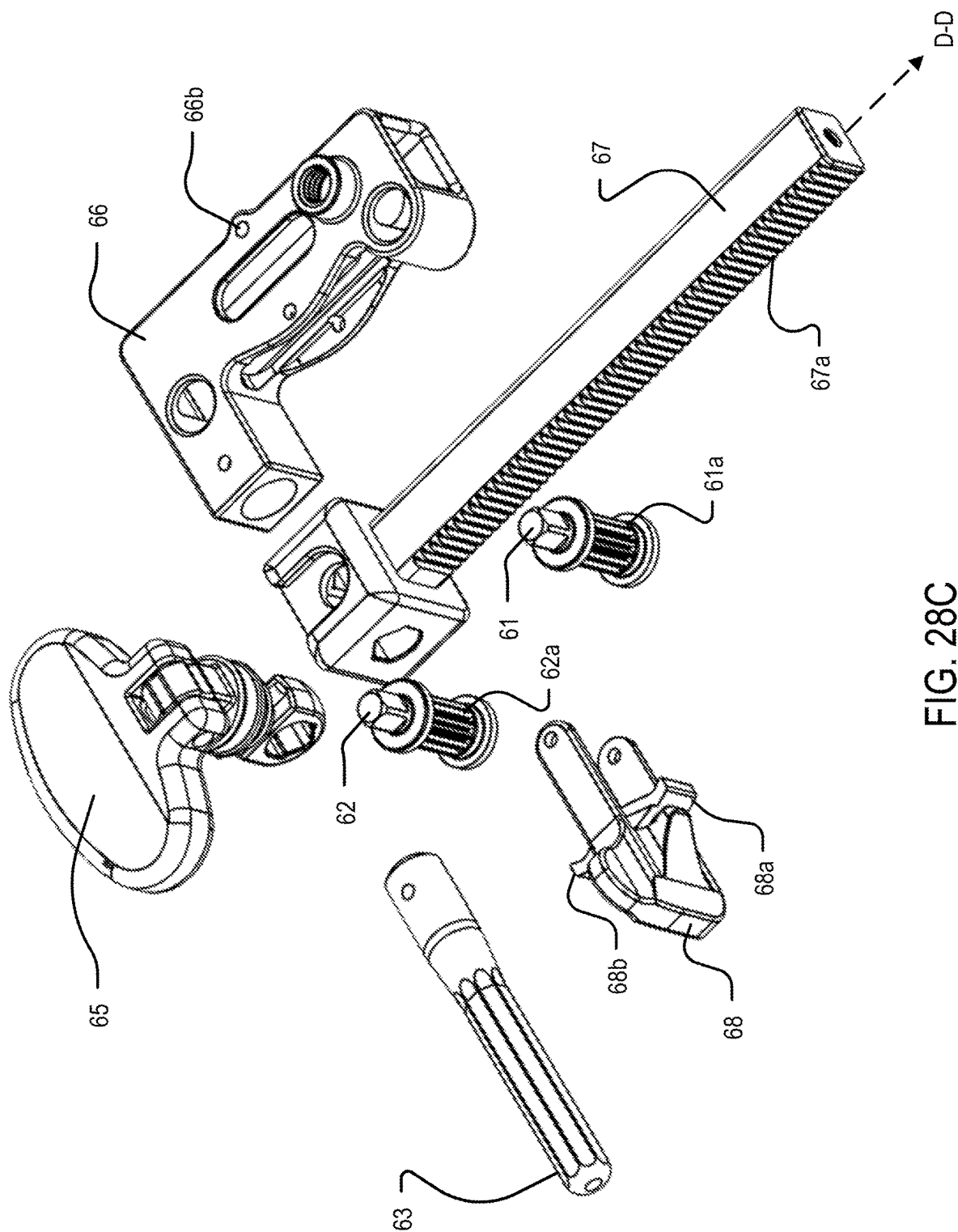


FIG. 28C

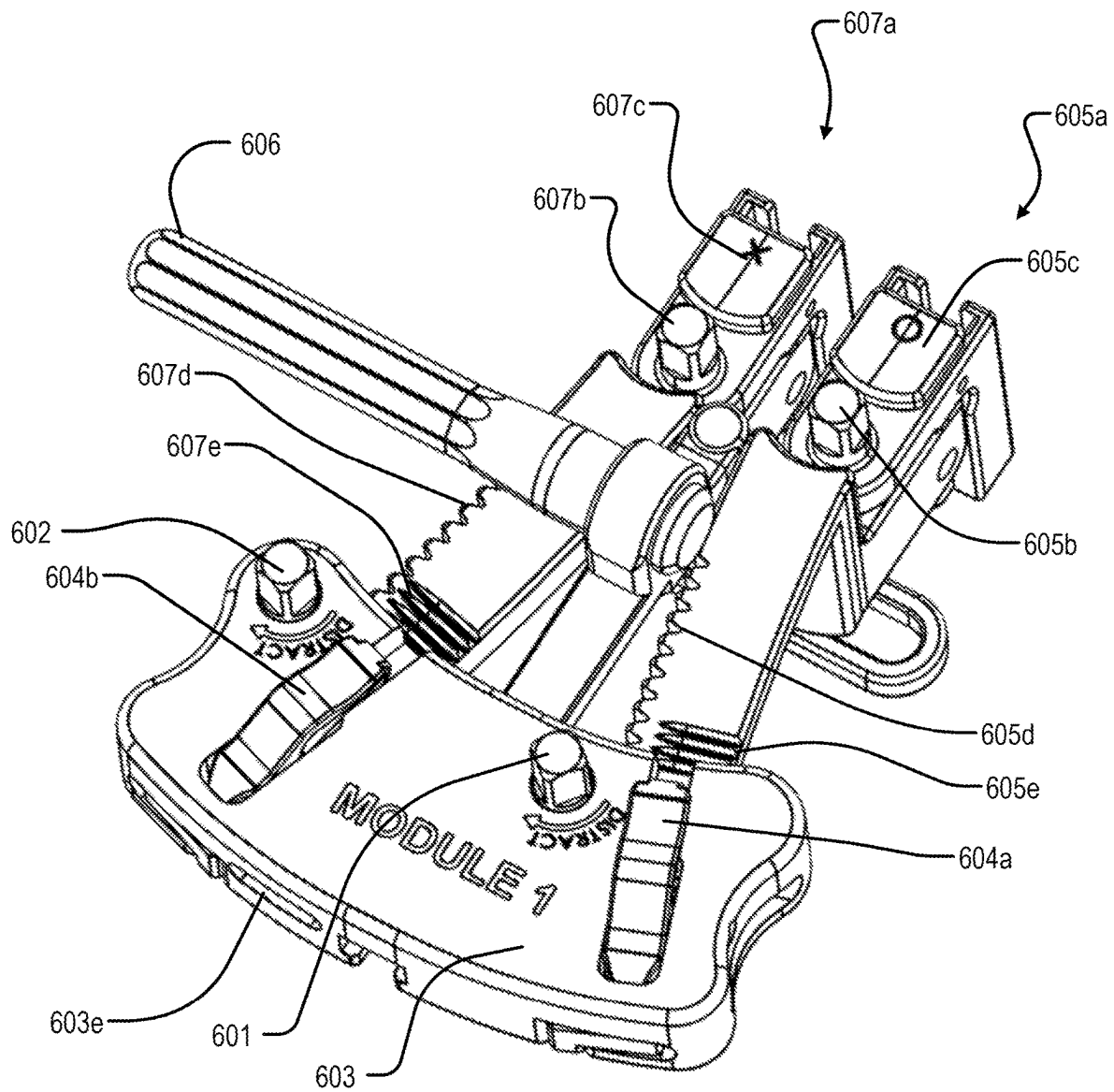


FIG. 29

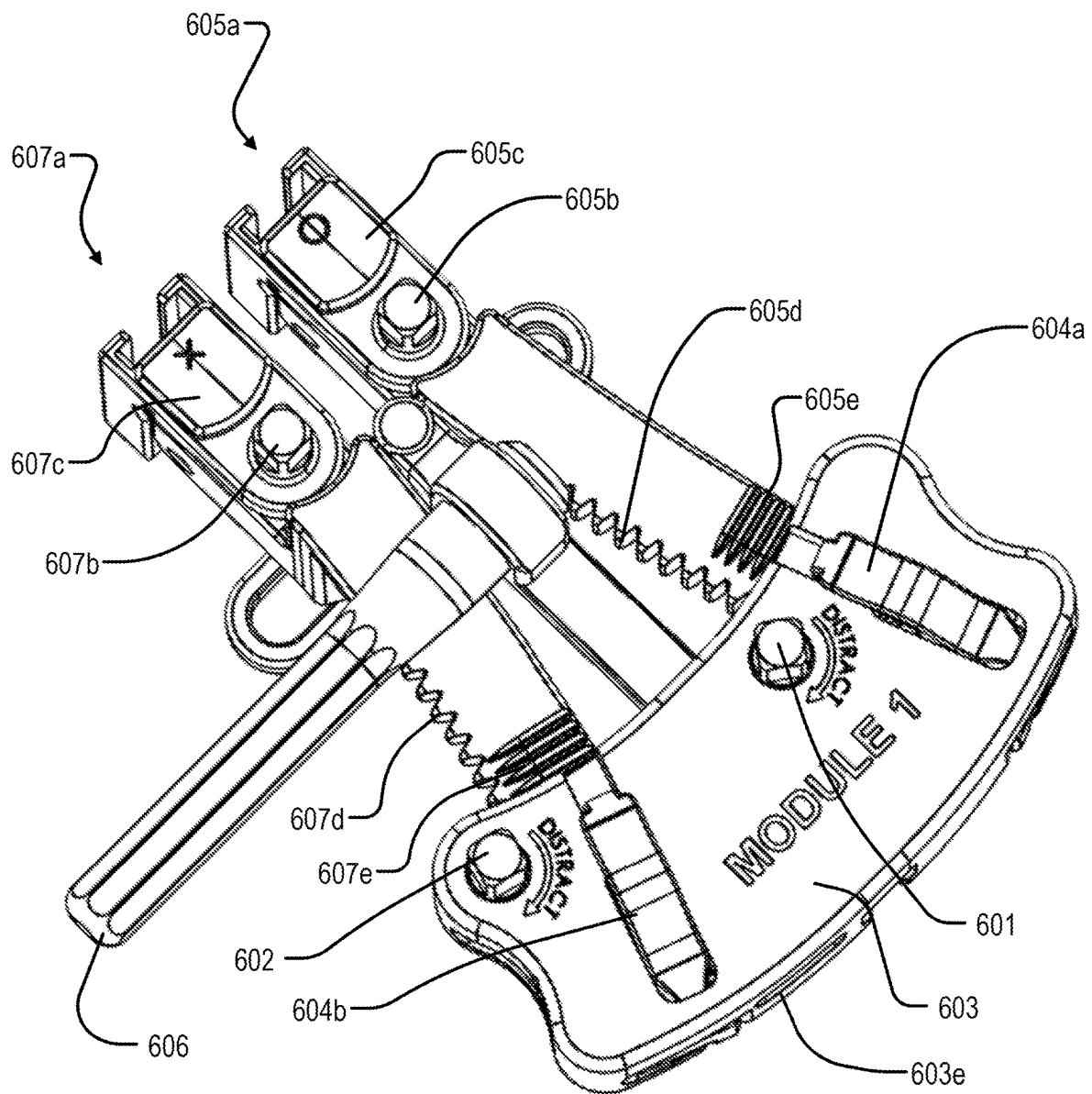


FIG. 30

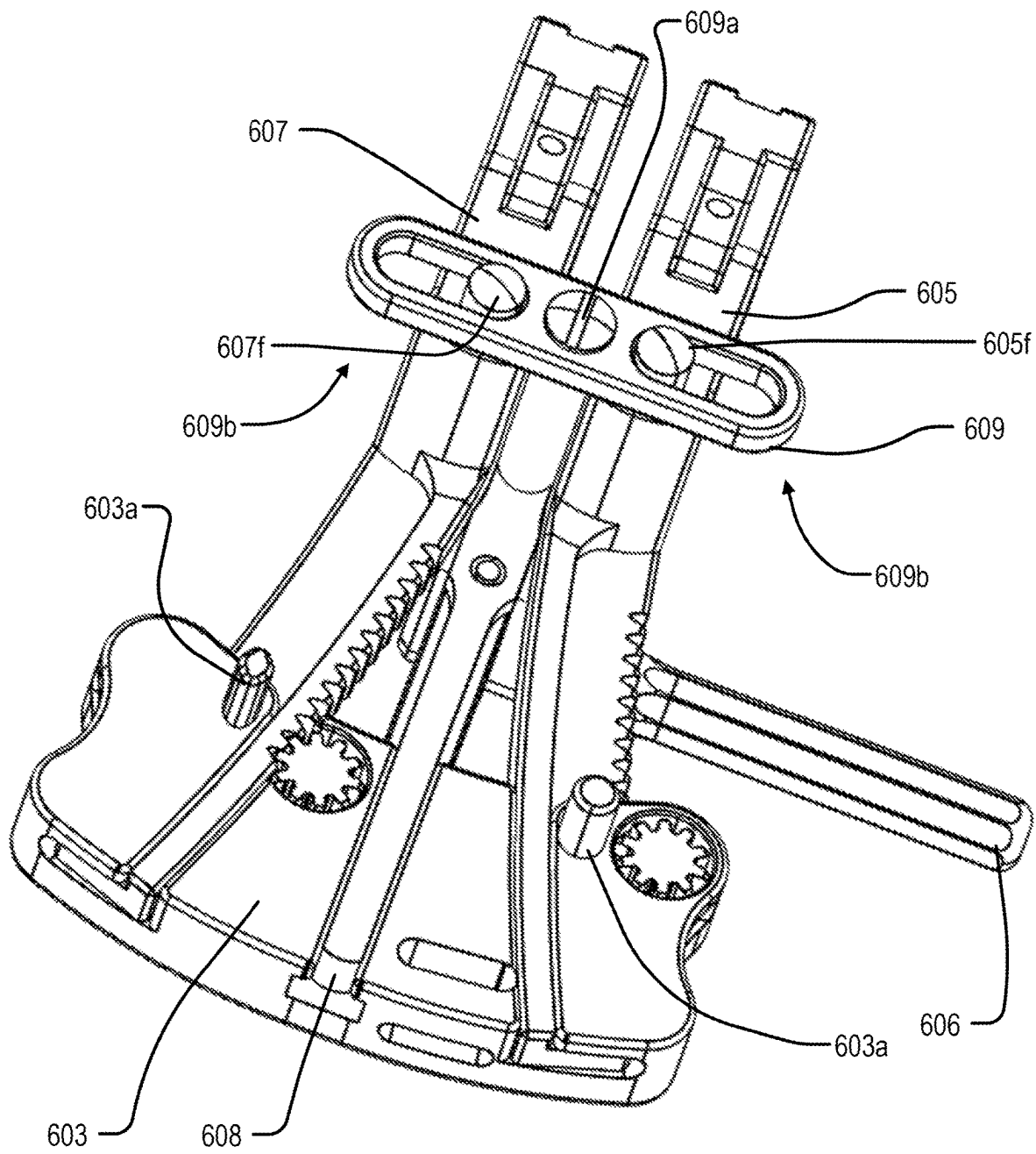


FIG. 31A

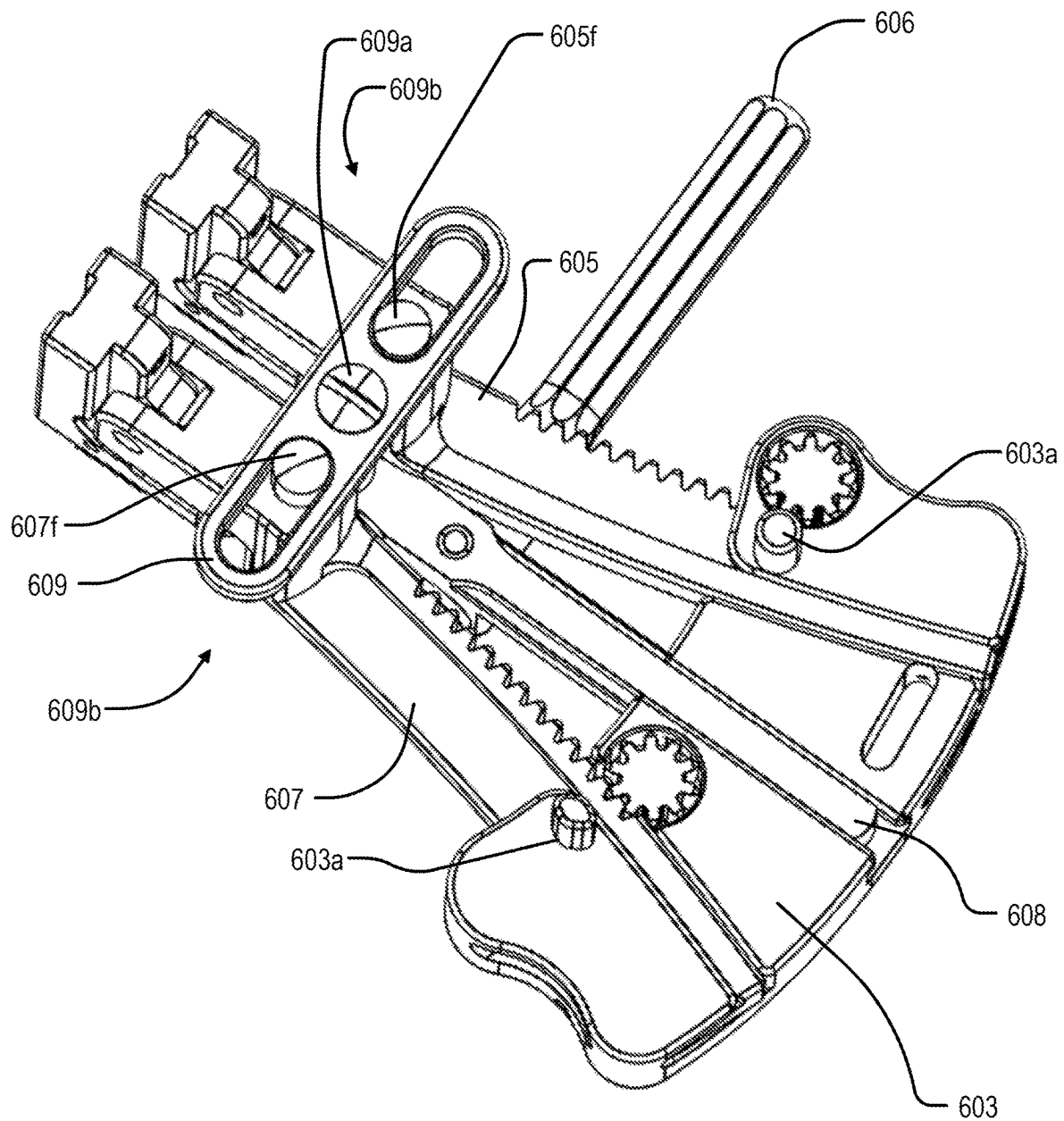


FIG. 31B

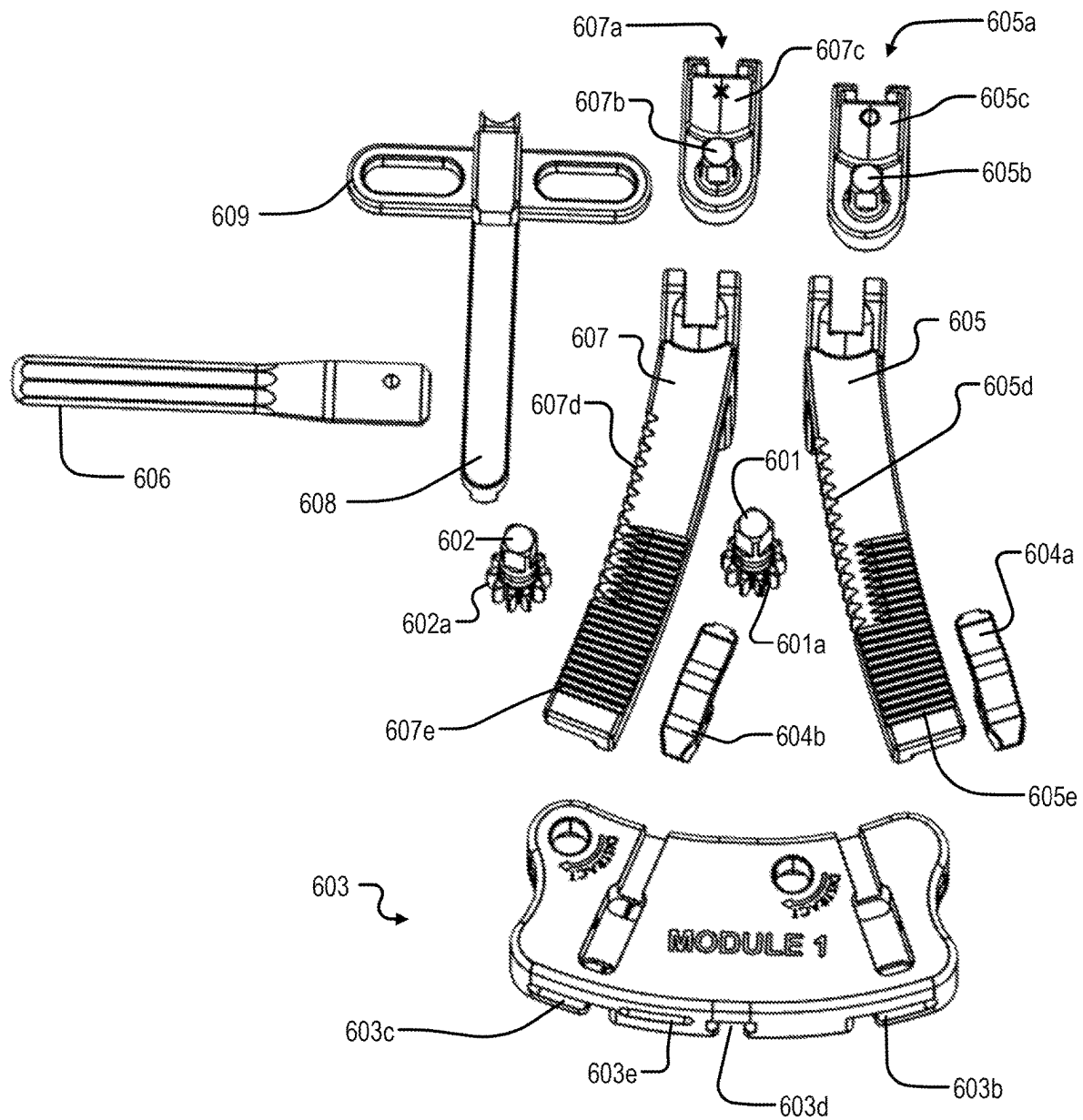


FIG. 32

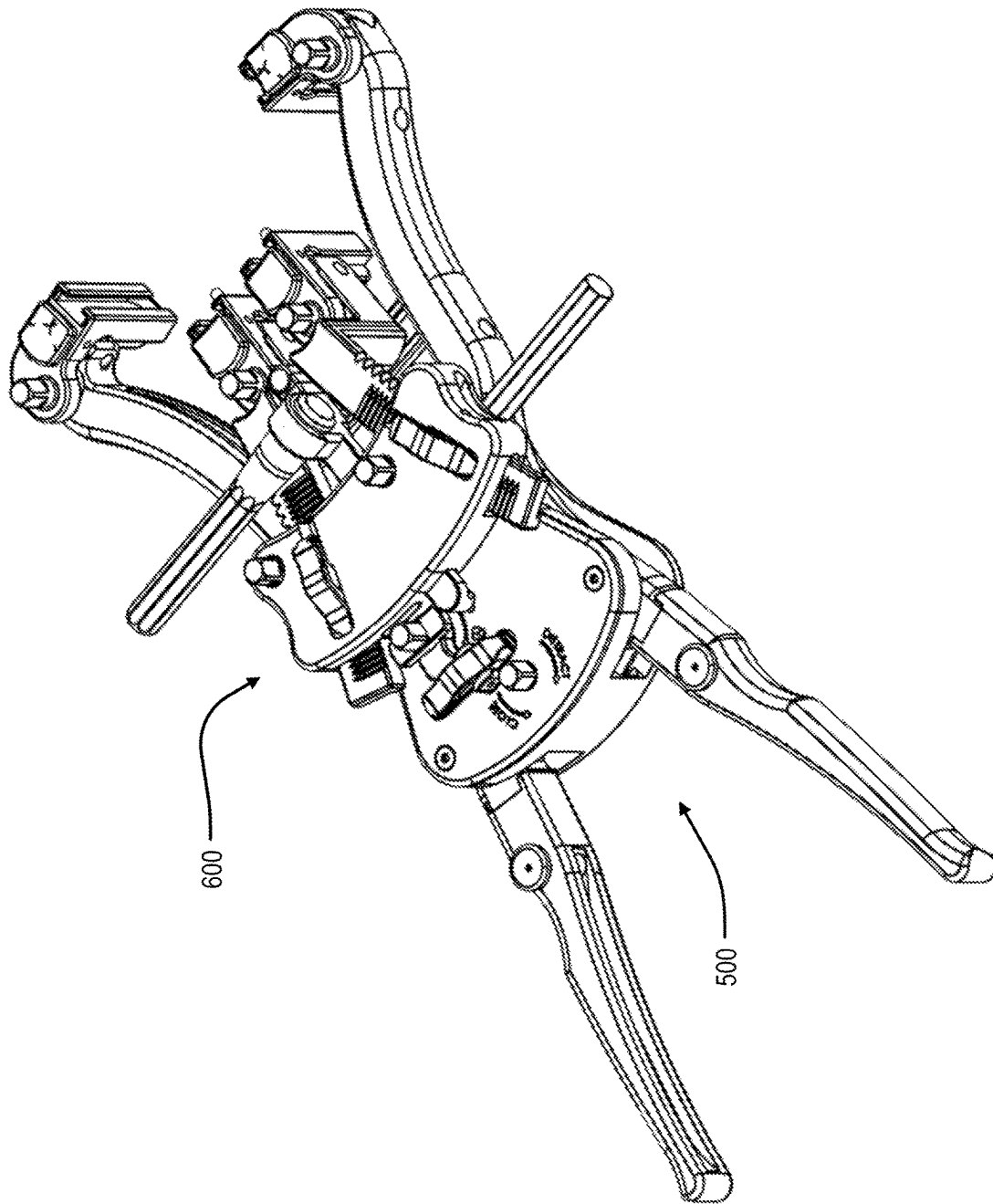


FIG. 33A

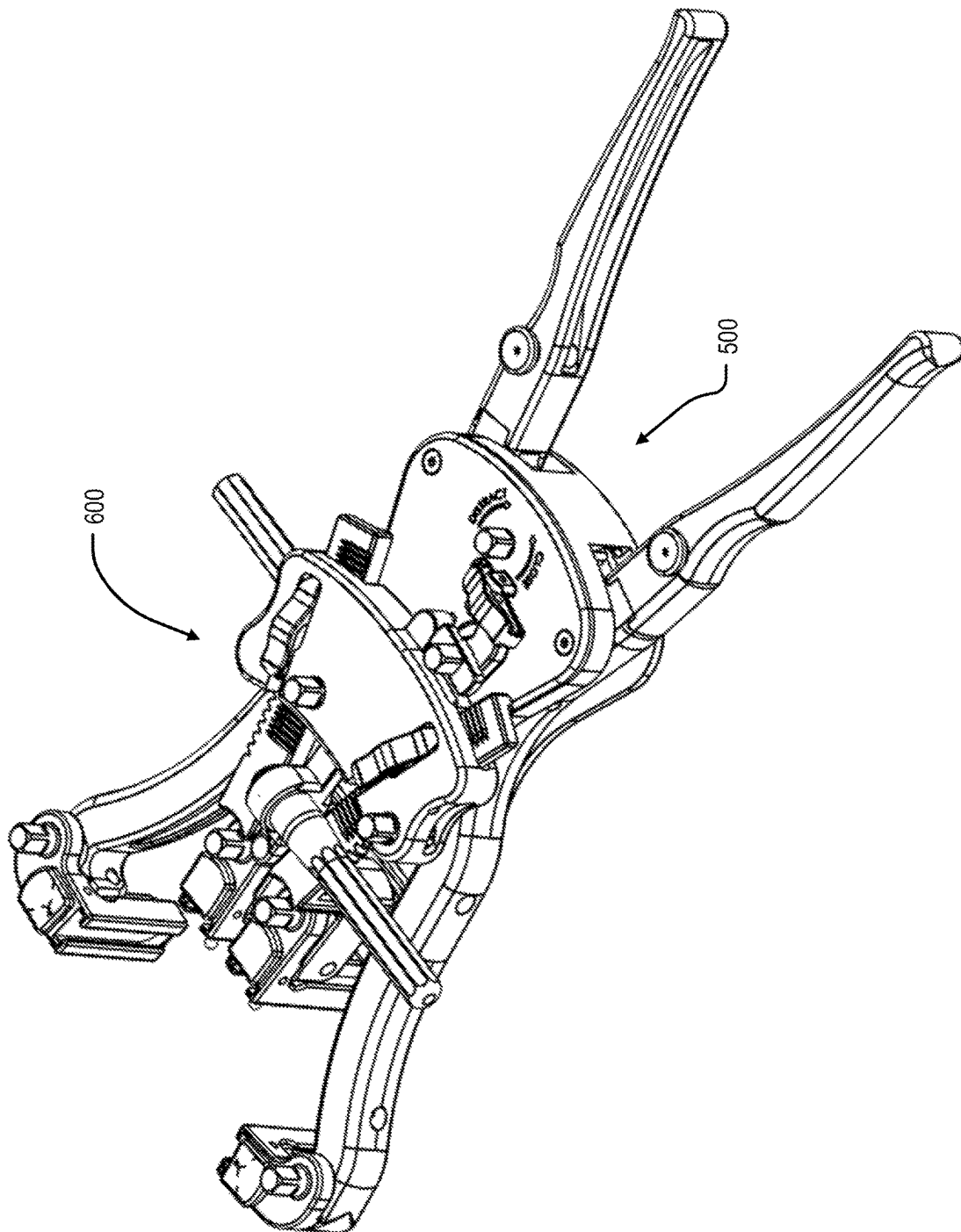


FIG. 33B

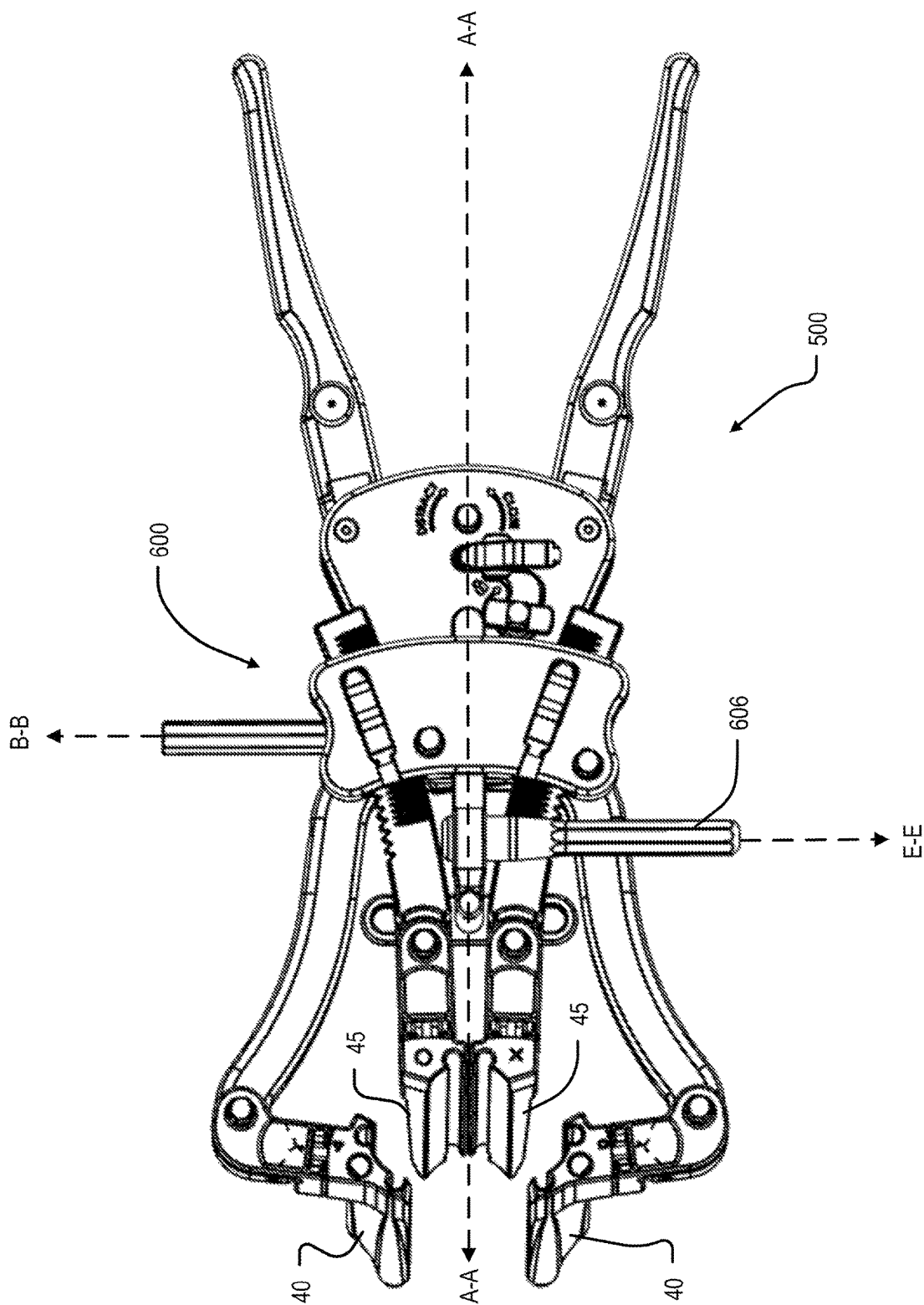


FIG. 34

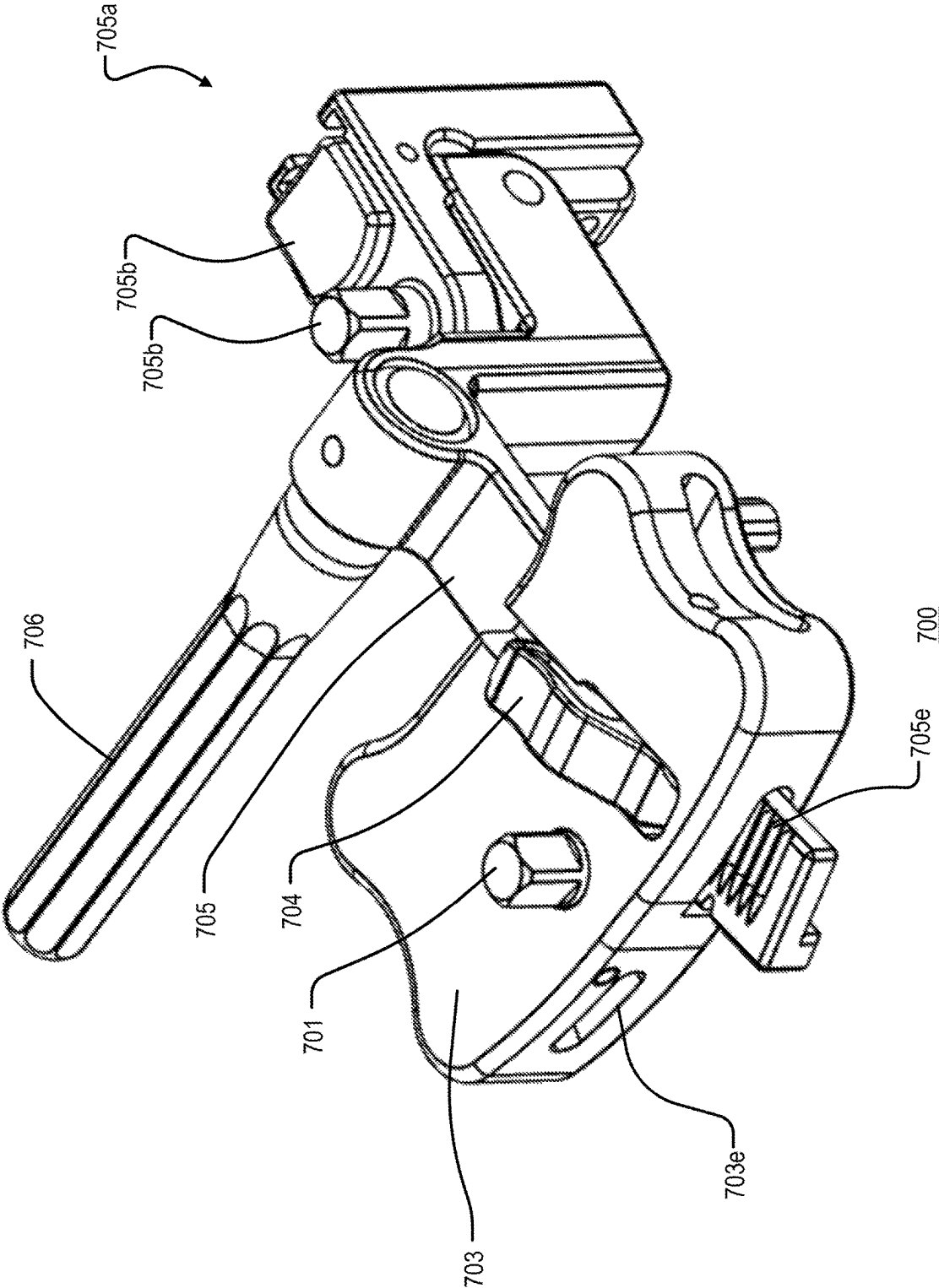
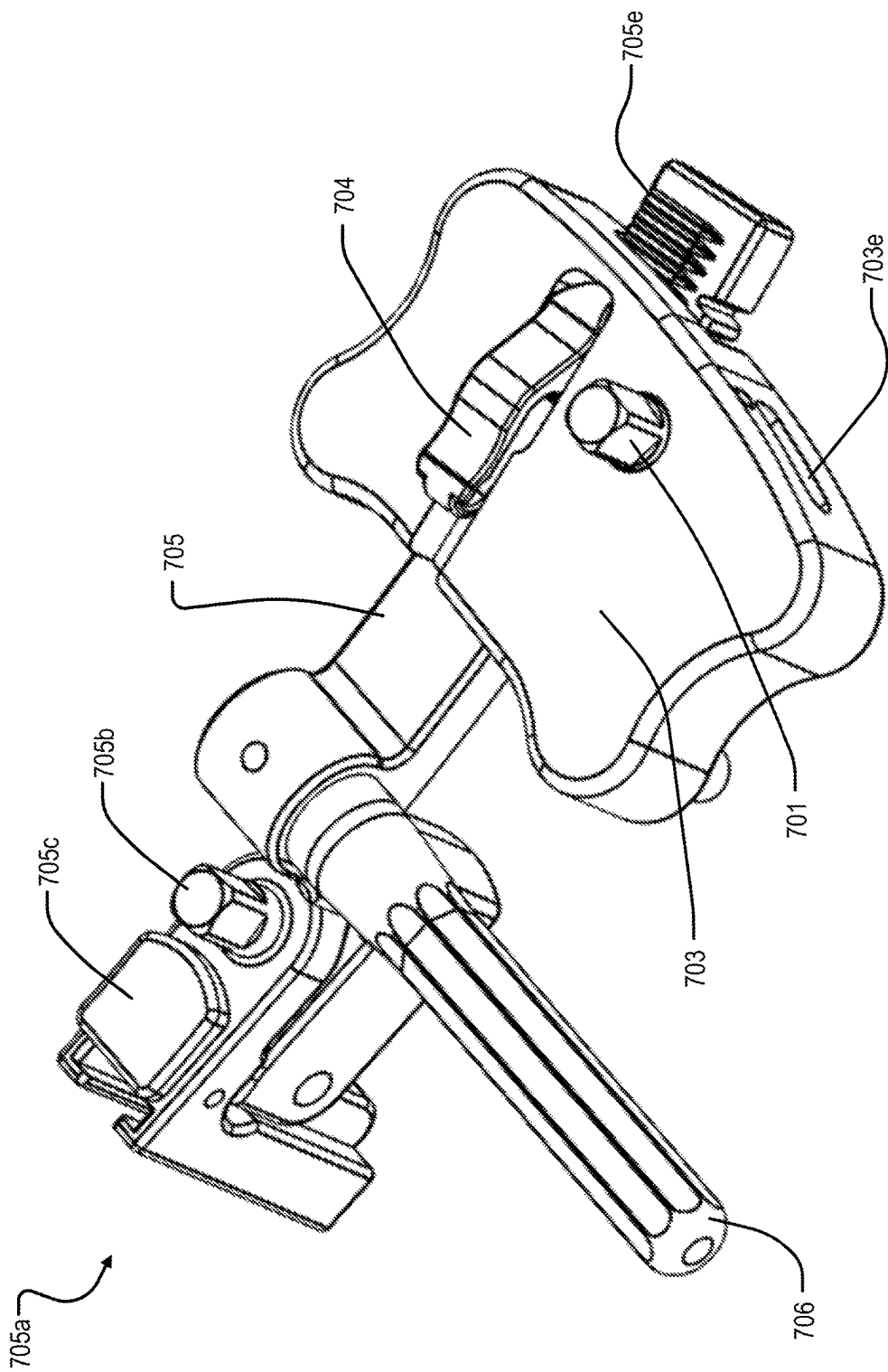
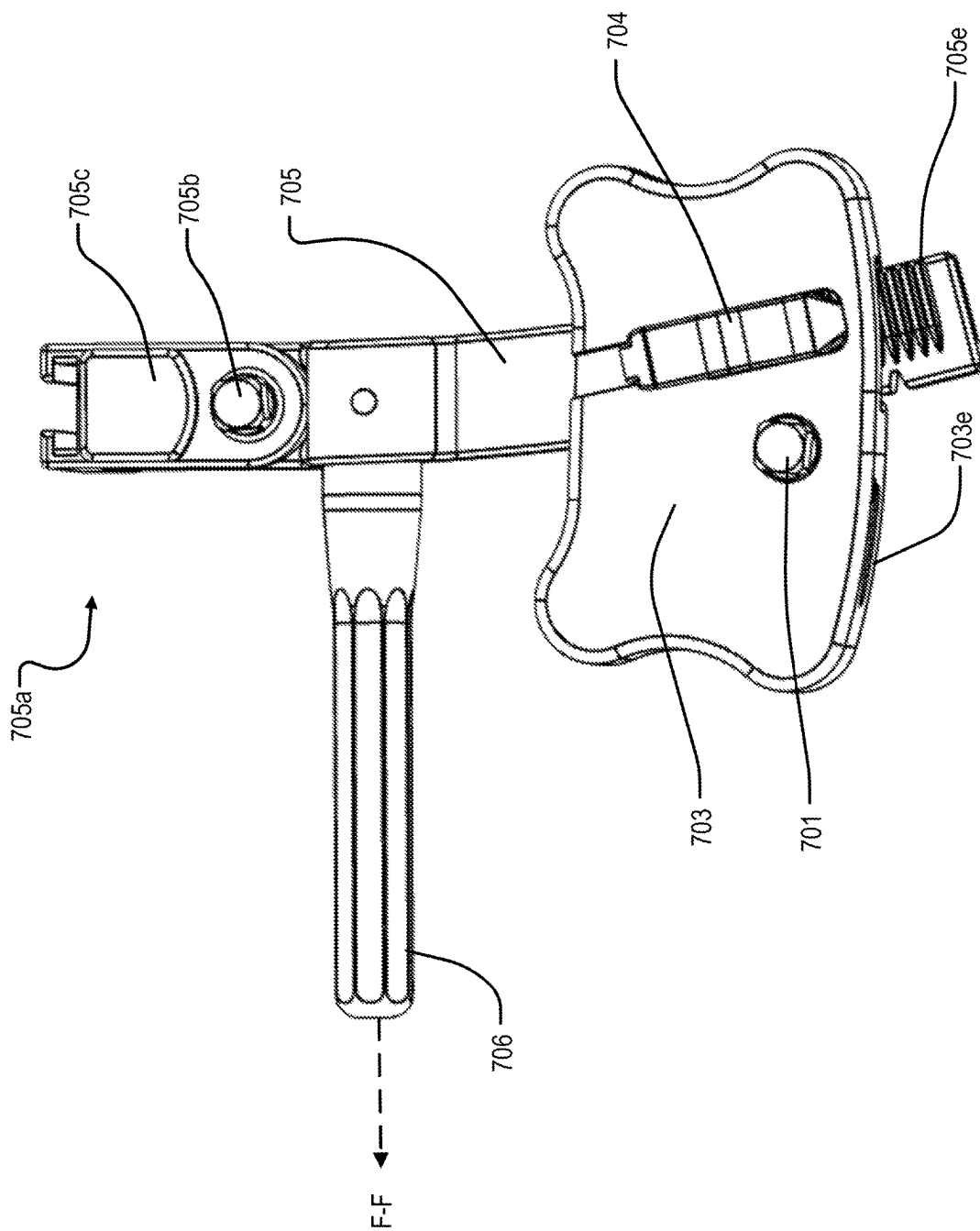


FIG. 35



700
FIG. 36



700
FIG. 37

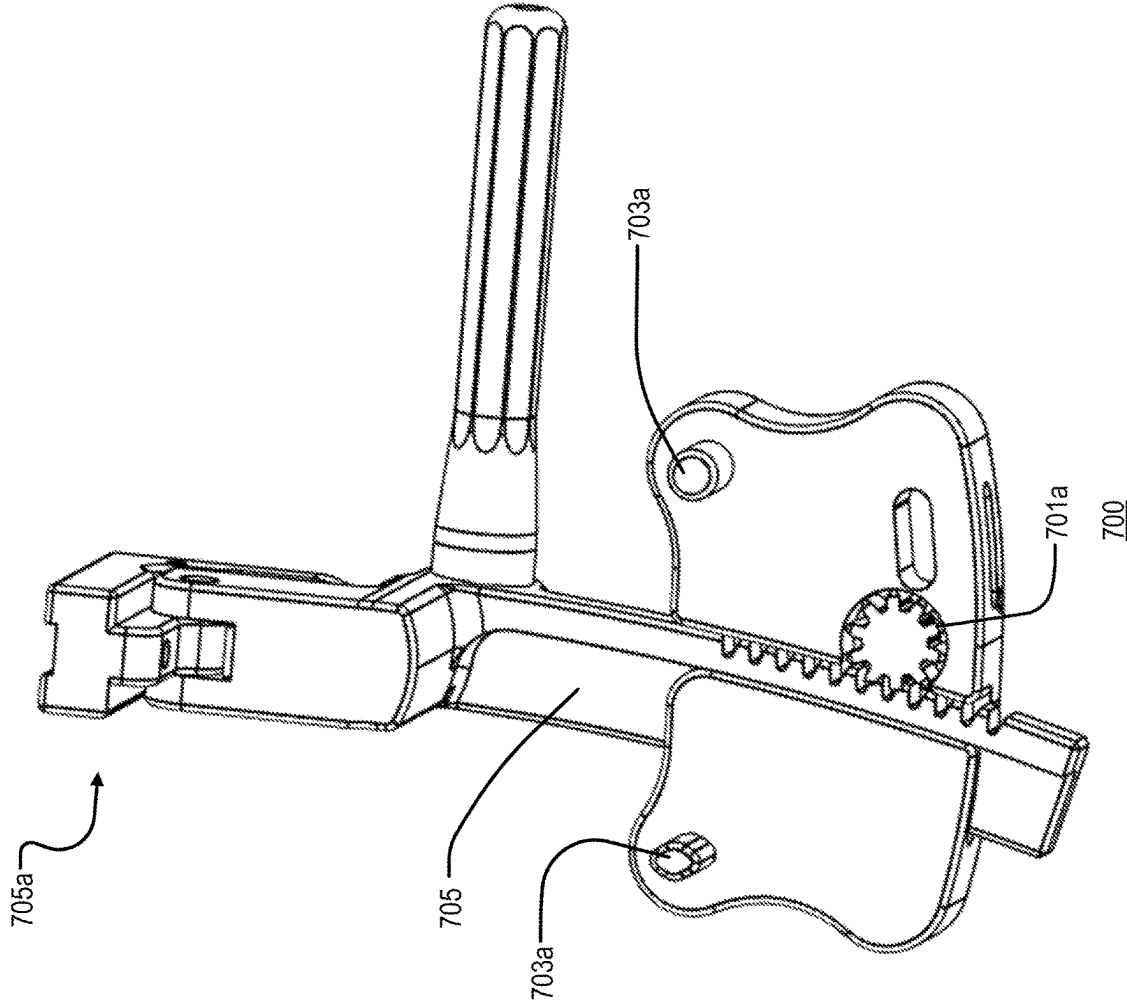


FIG. 38

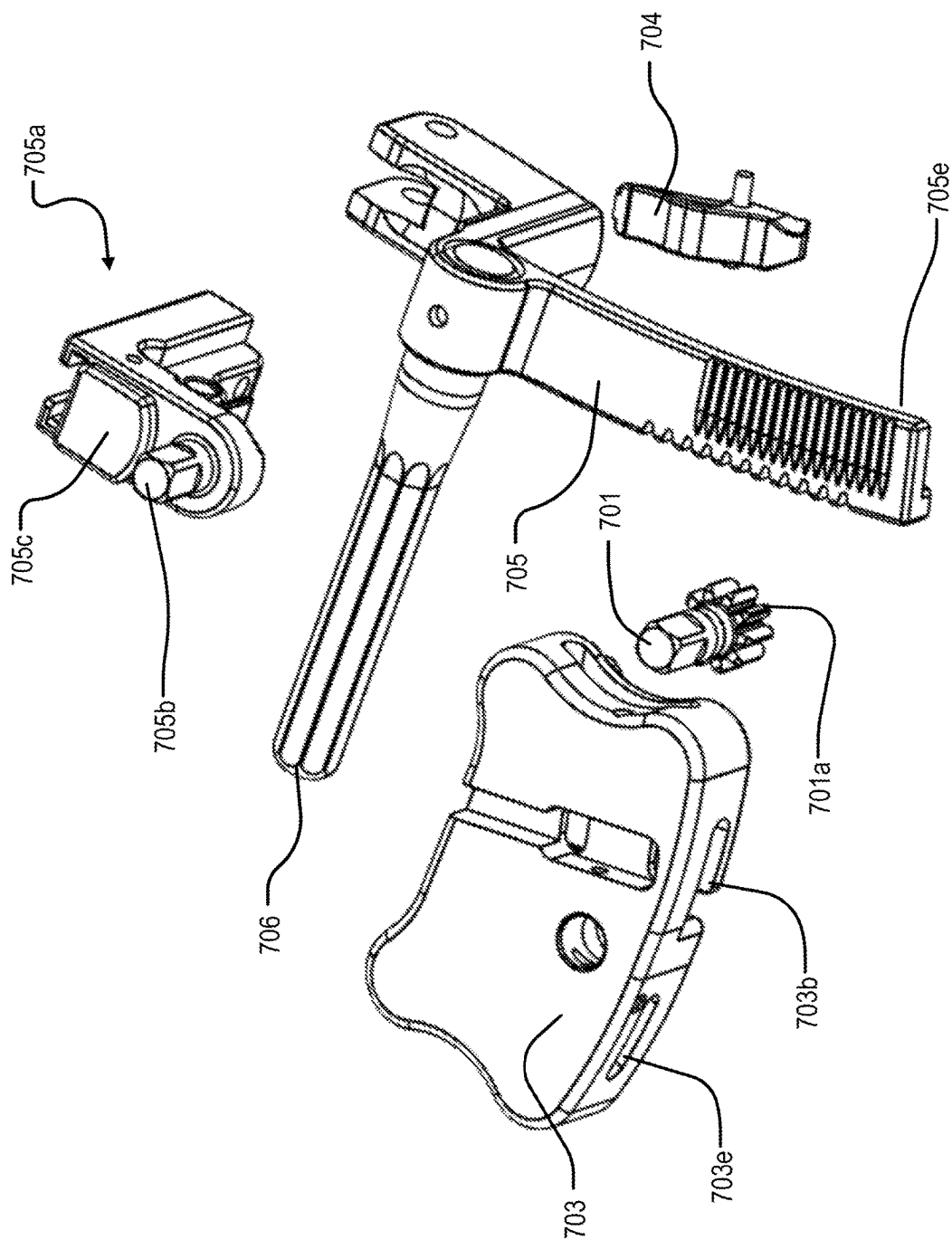


FIG. 39

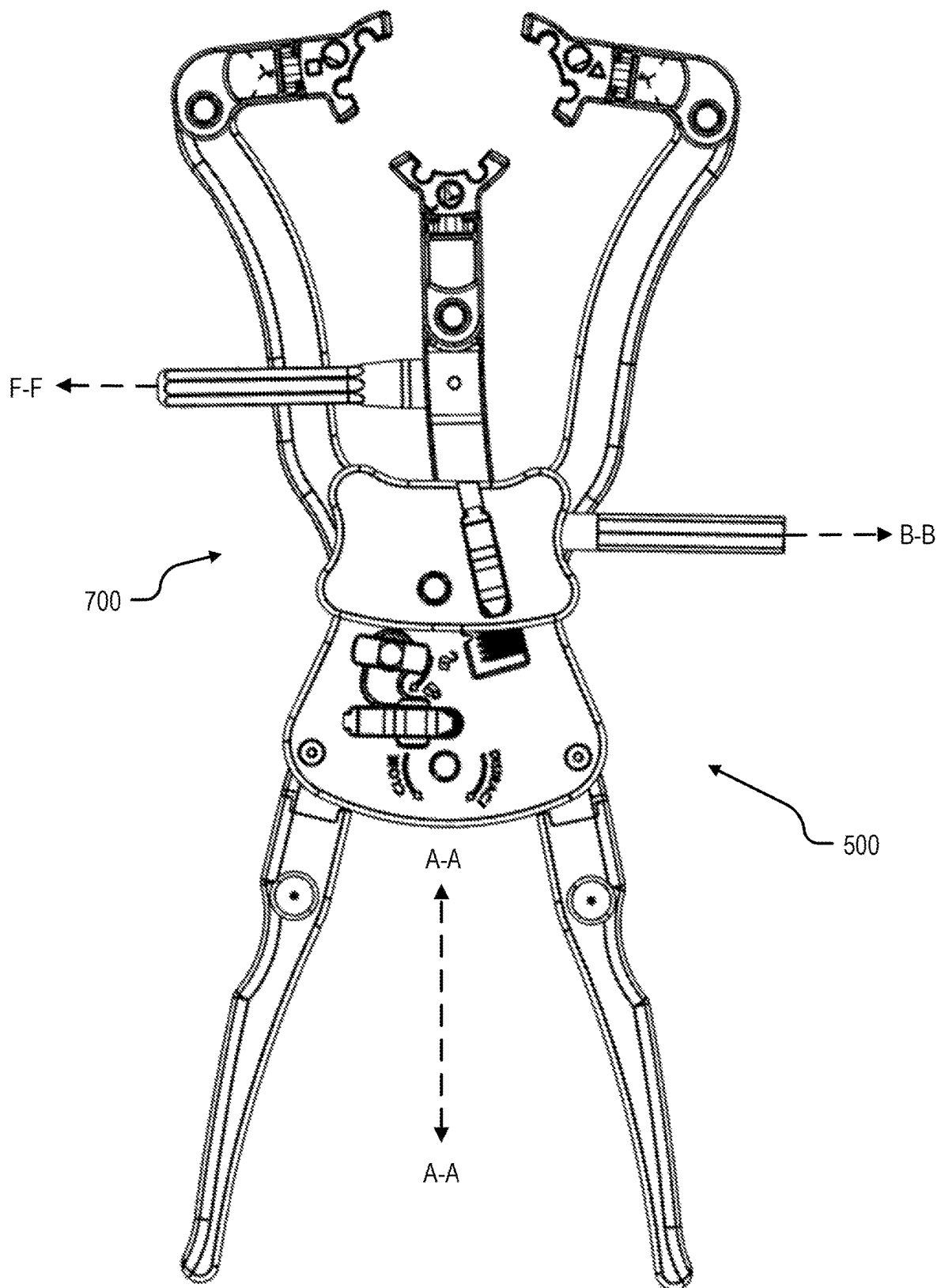


FIG. 40

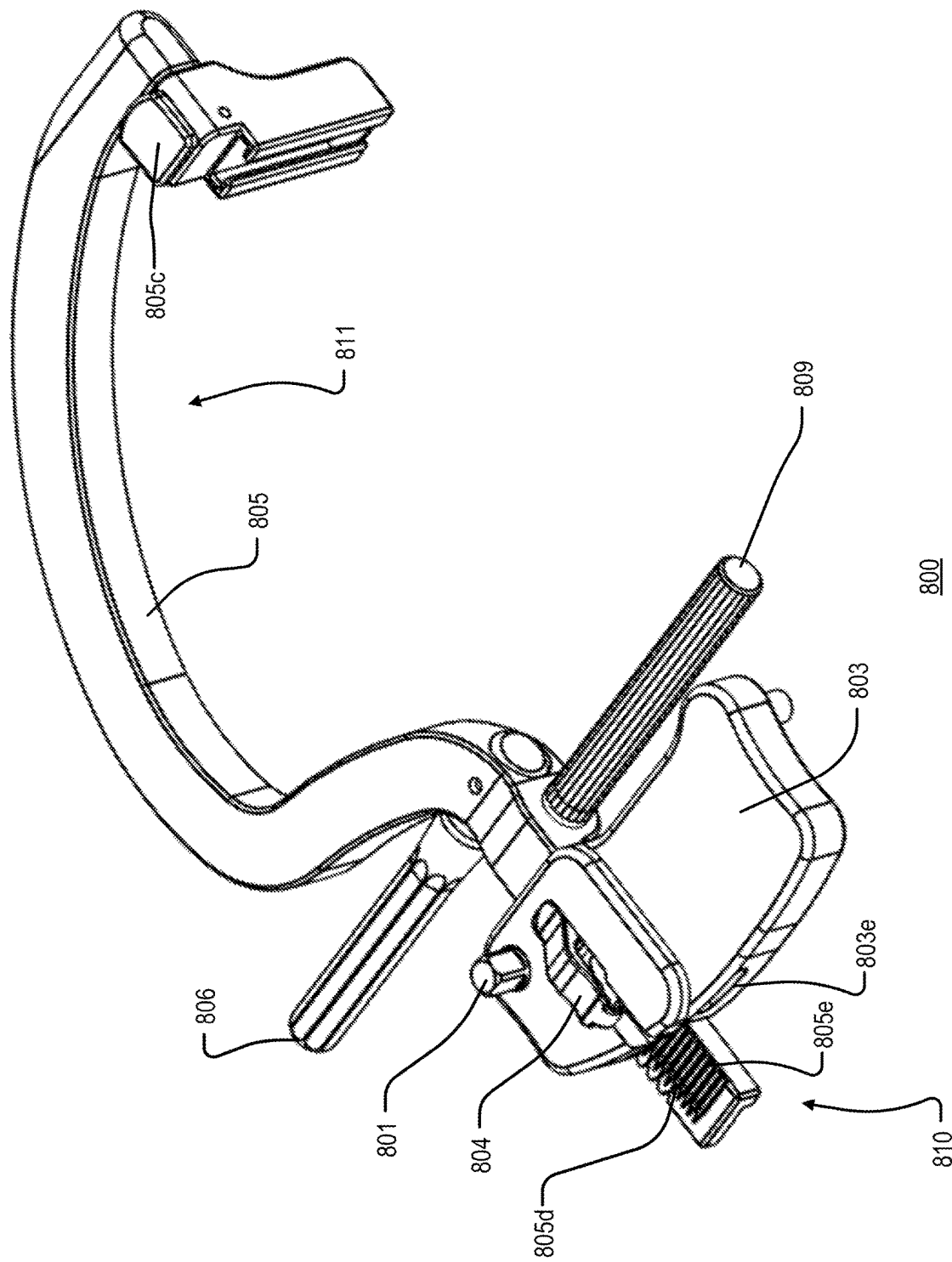
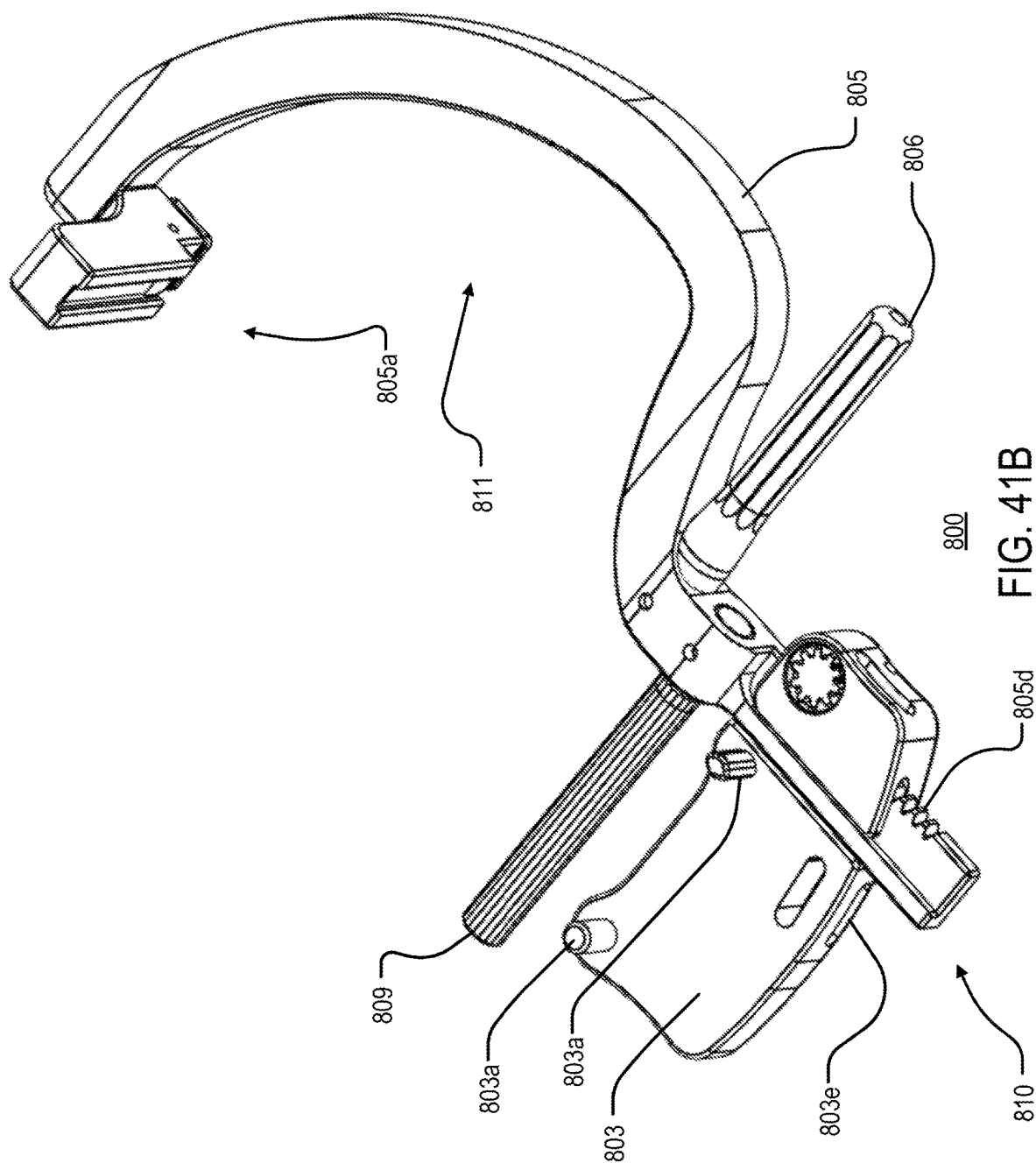
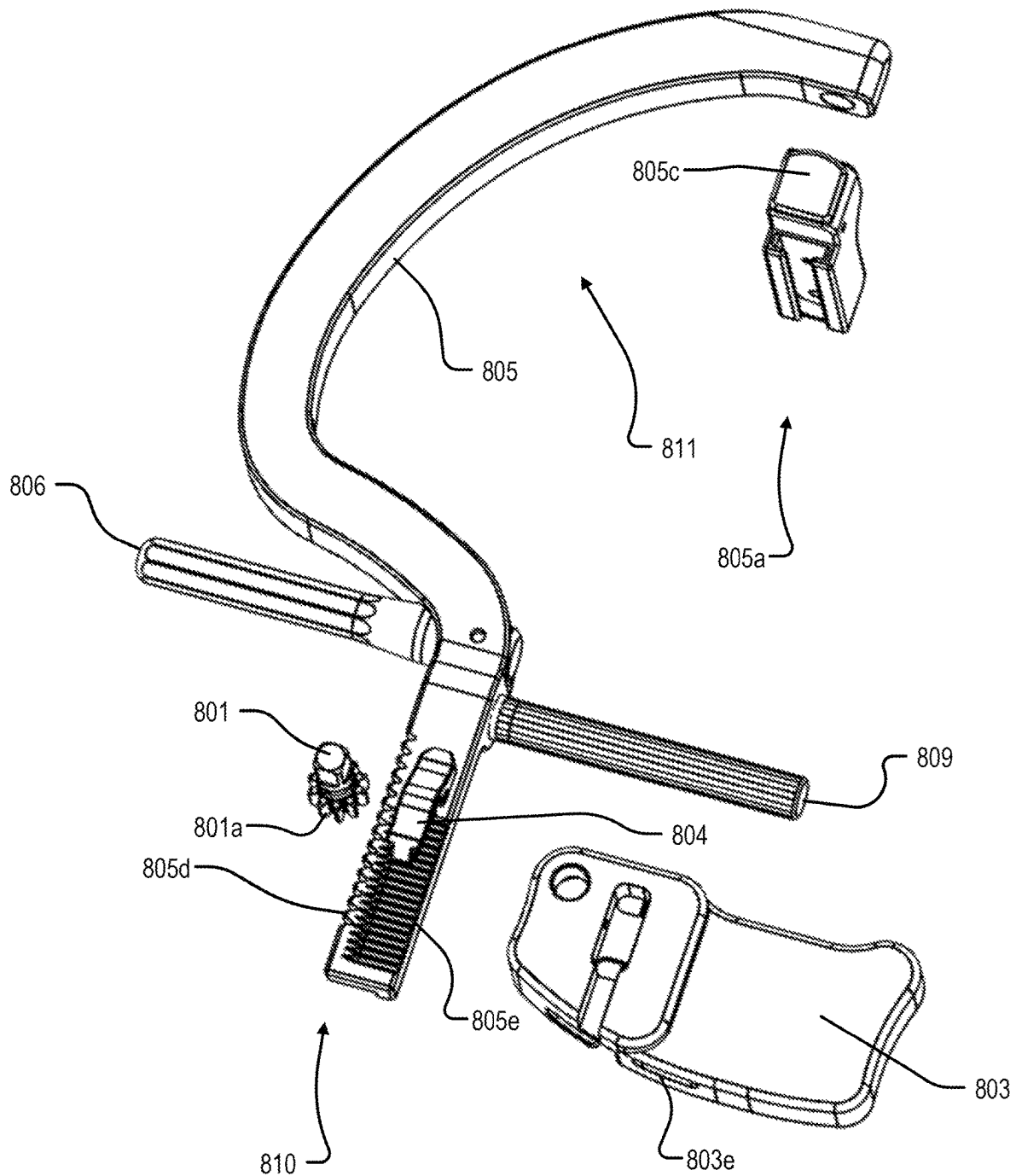


FIG. 41A





800

FIG. 42

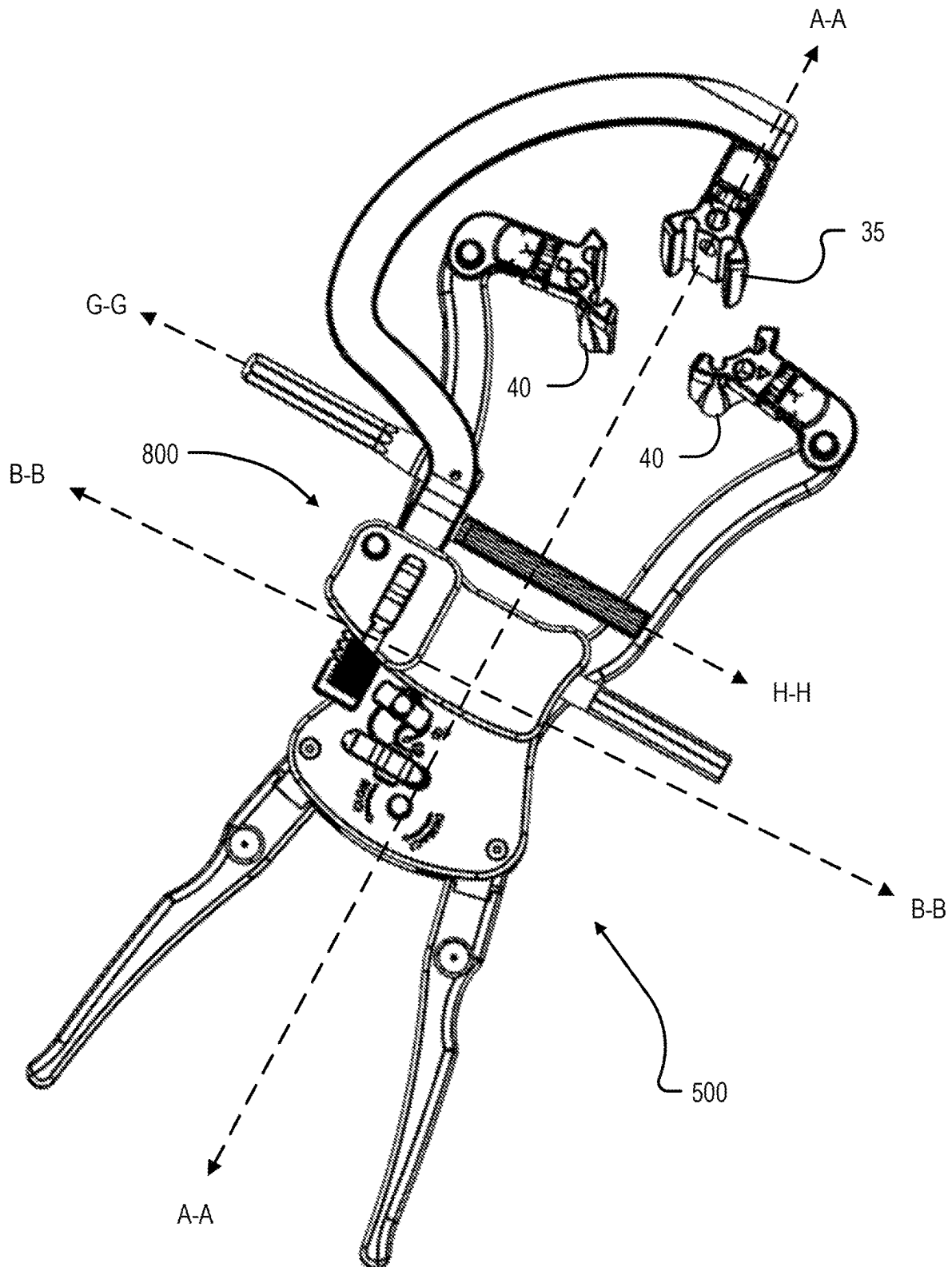


FIG. 43

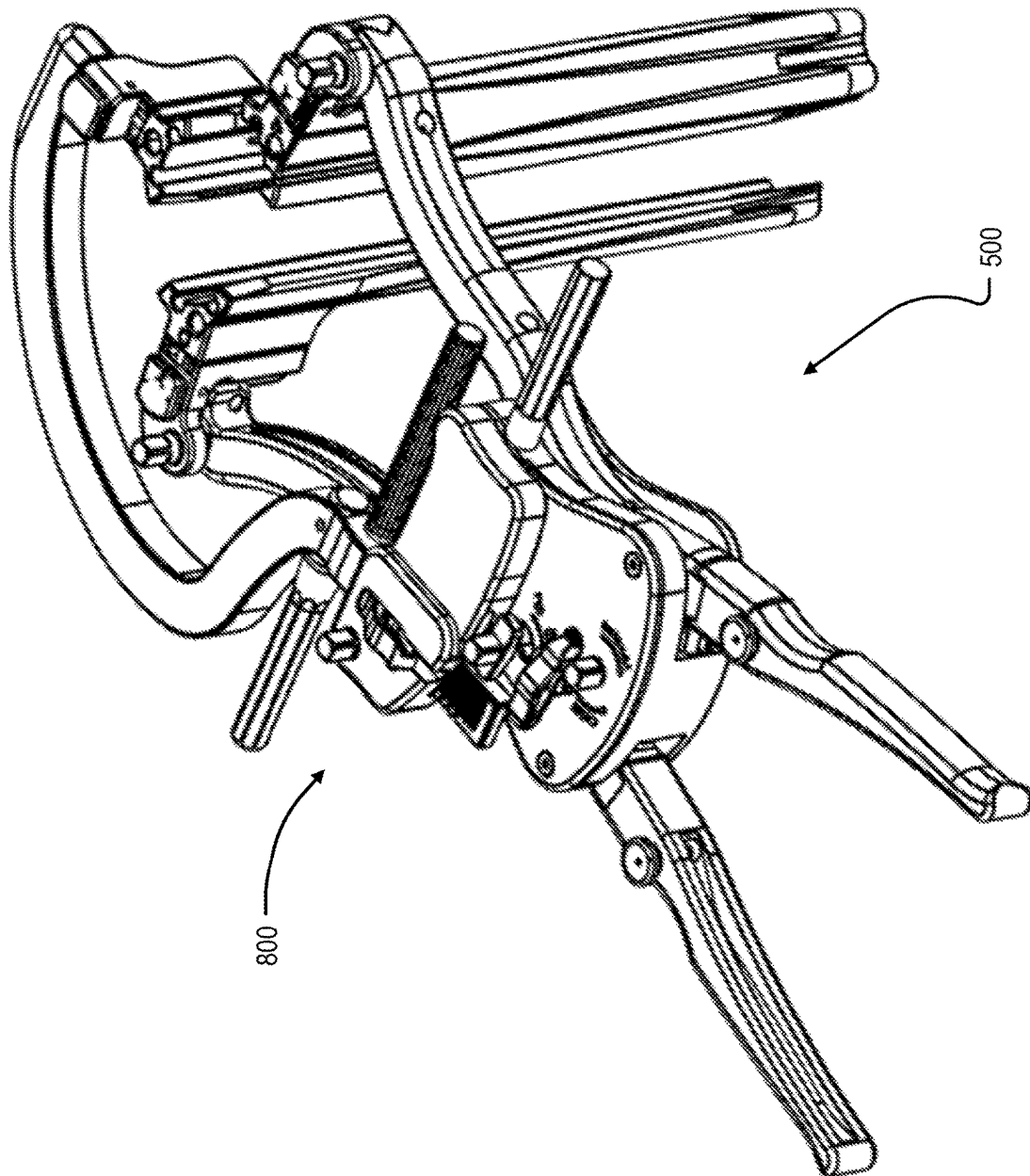
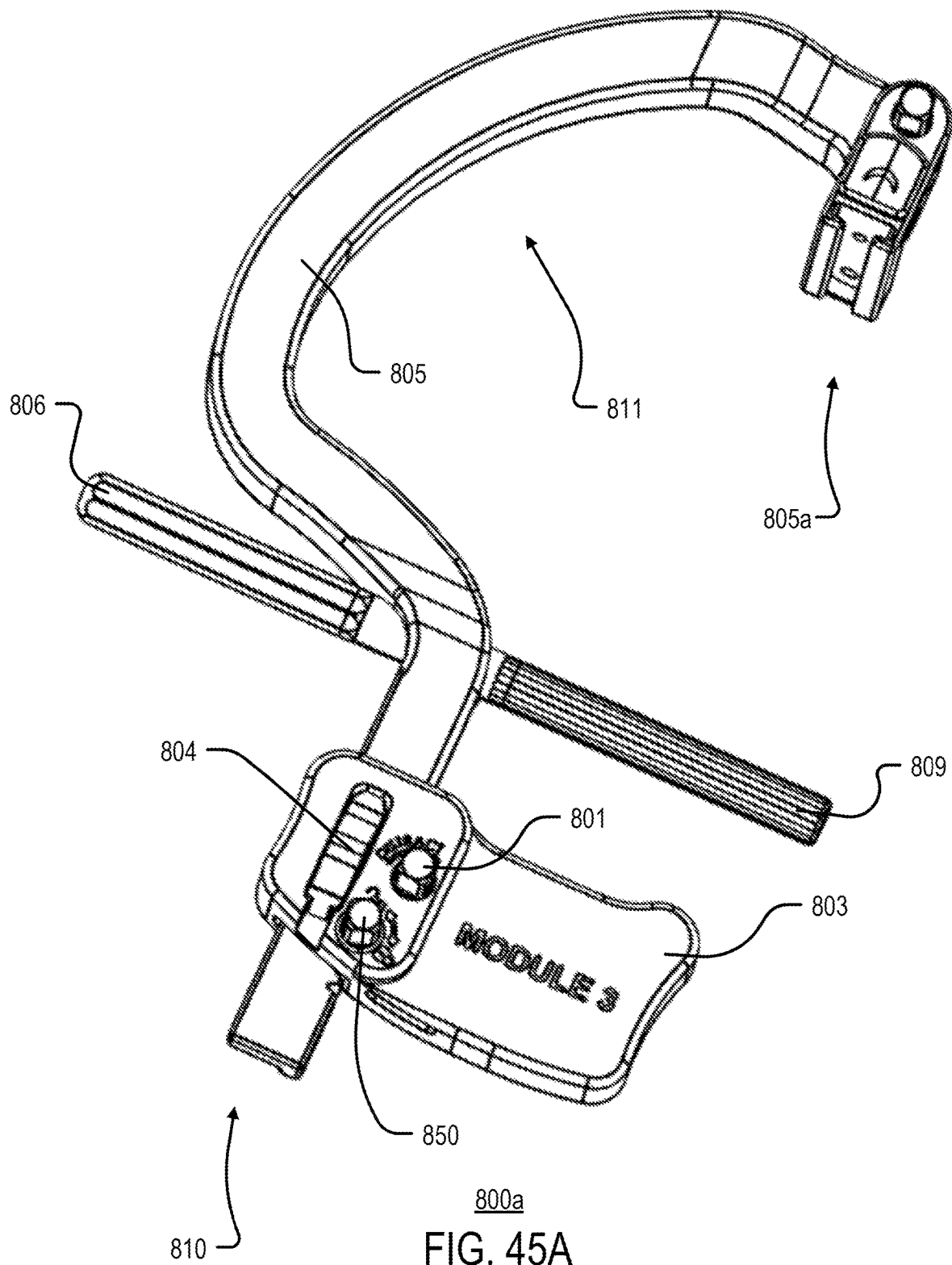
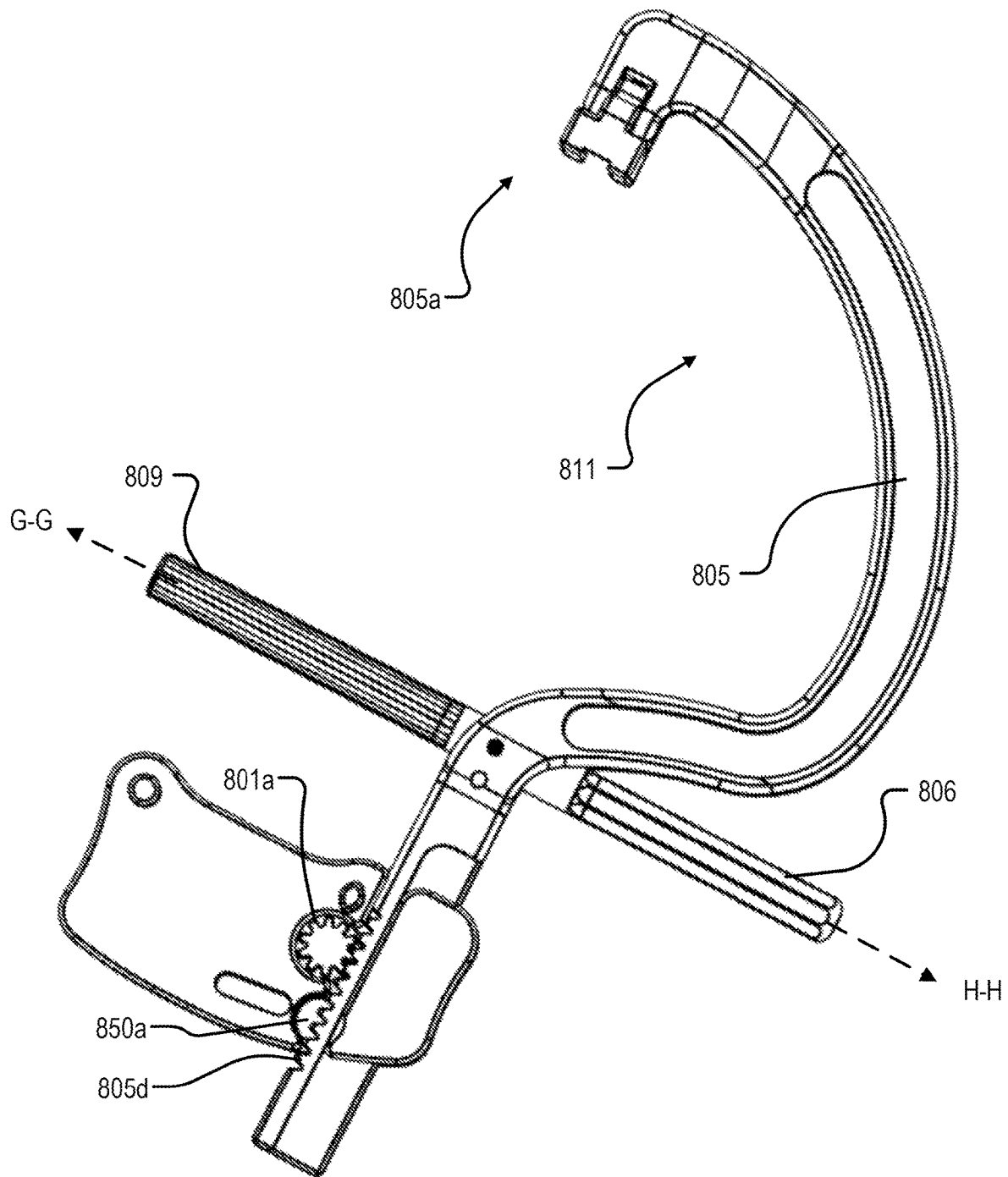


FIG. 44





800a
FIG. 45B

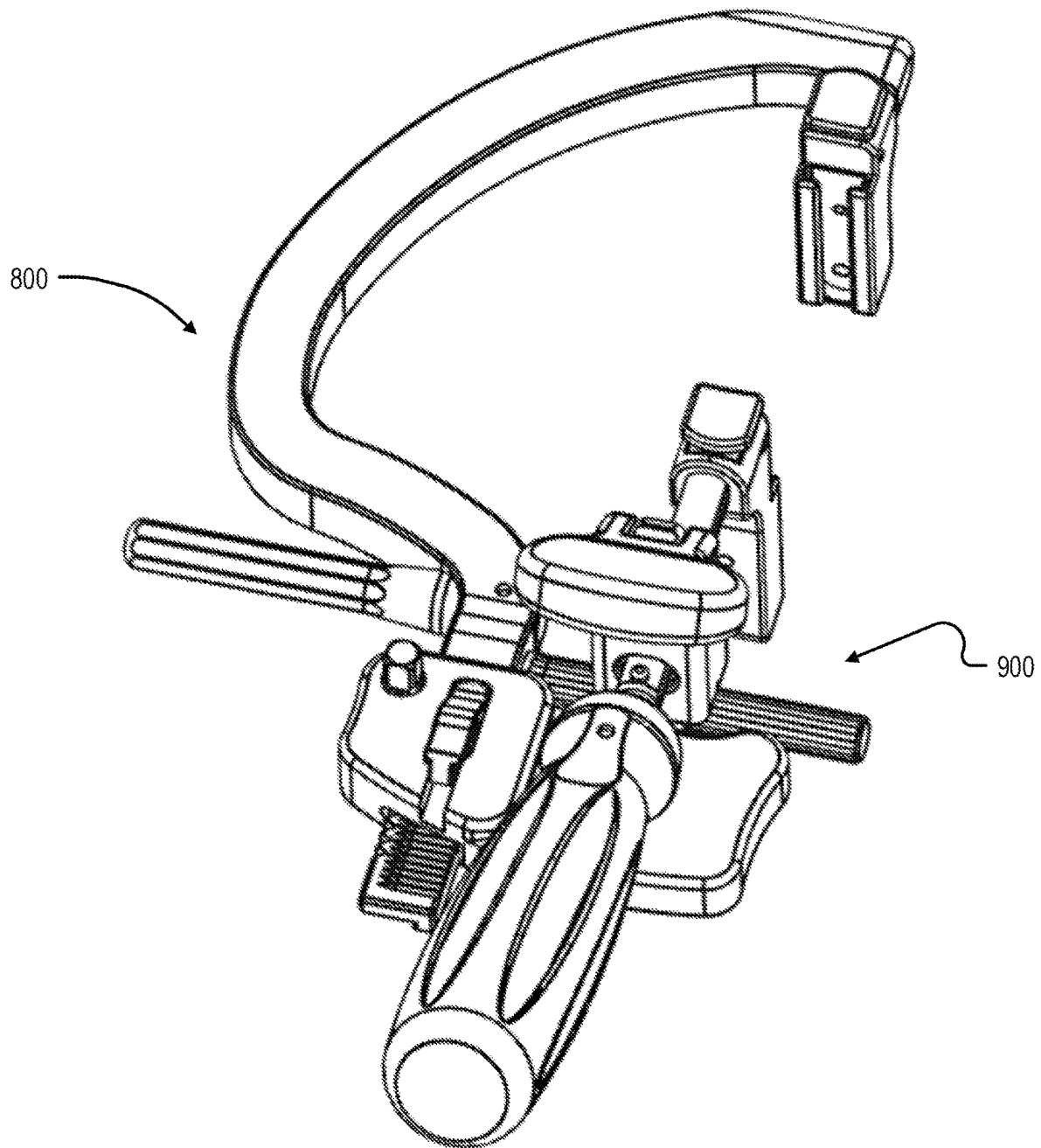
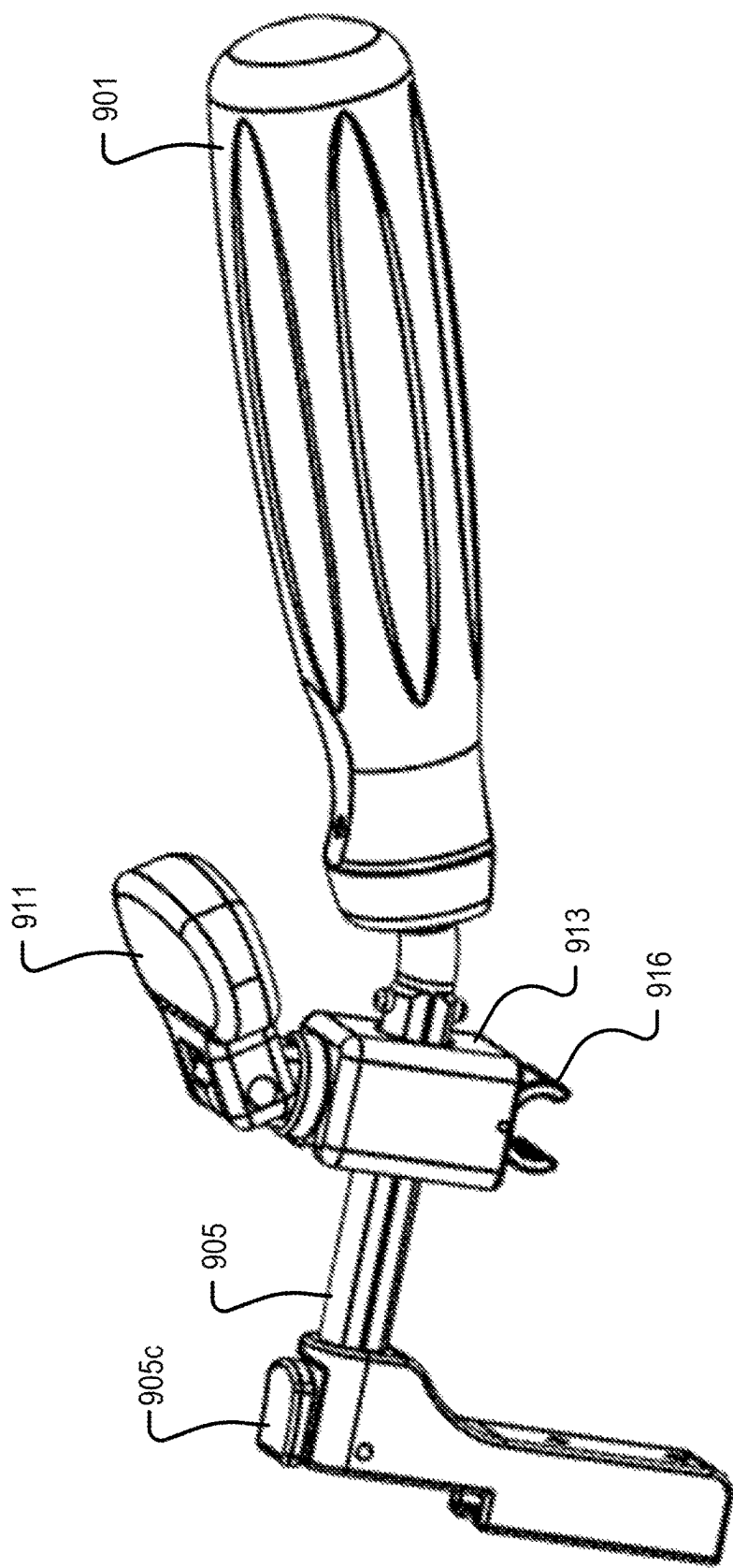
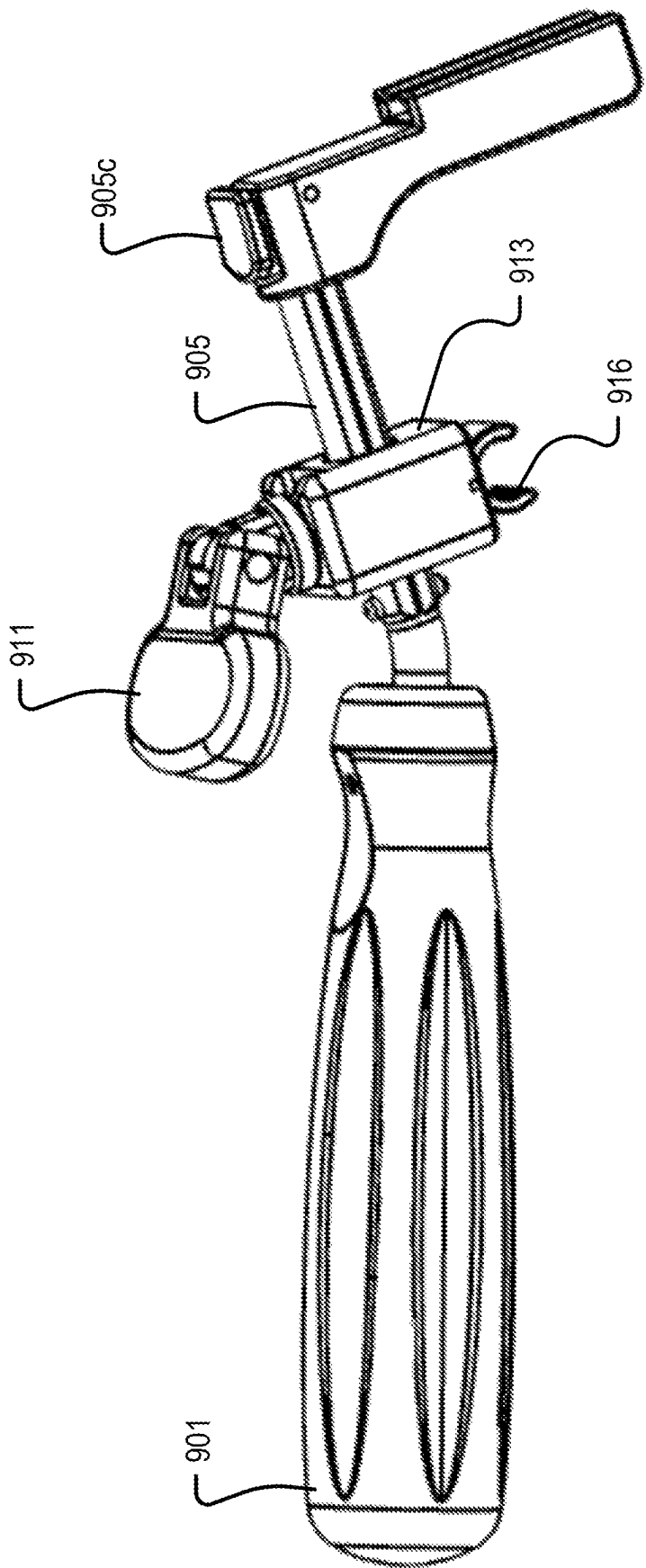


FIG. 46



900

FIG. 47A



900

FIG. 47B

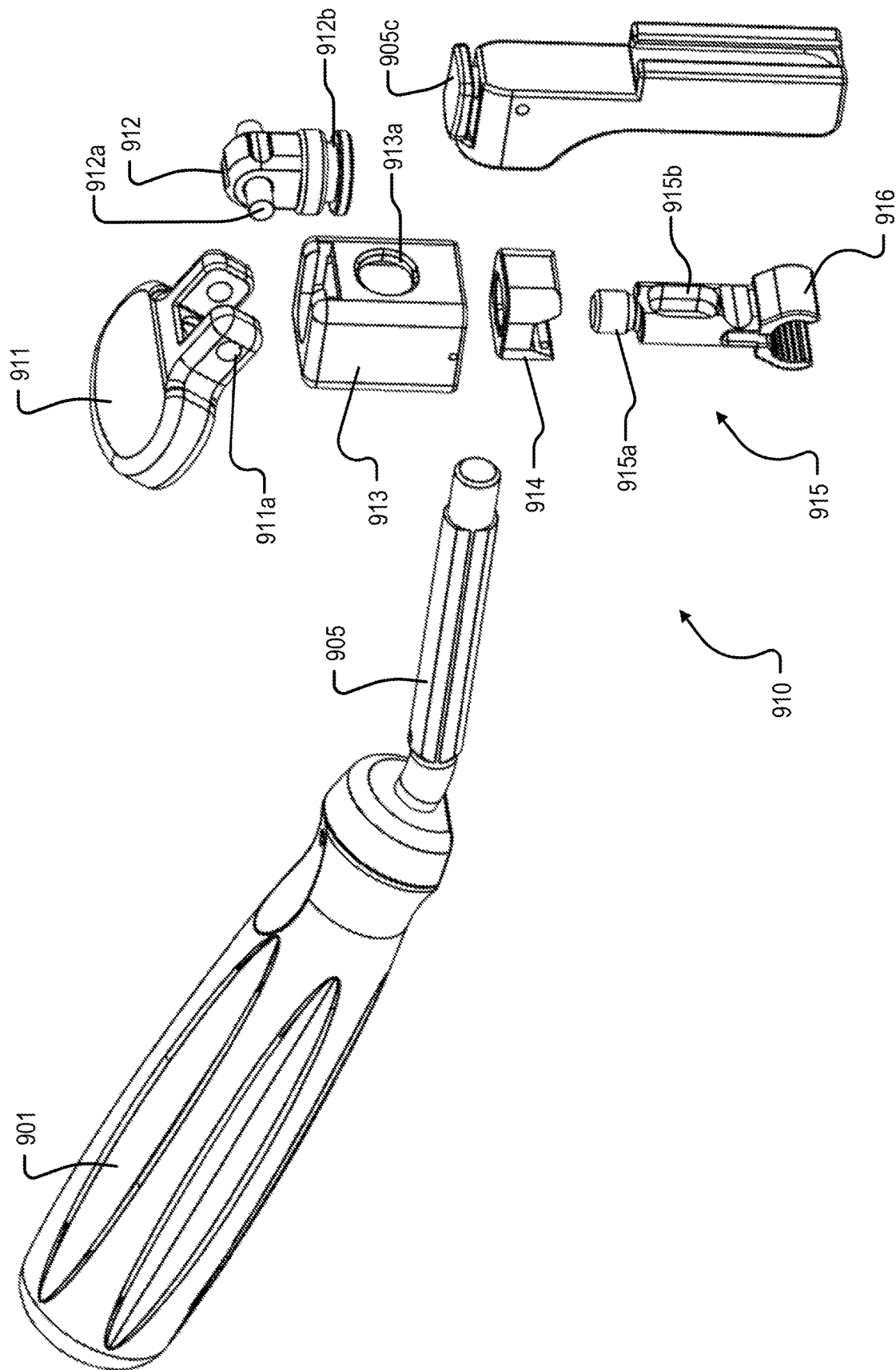


FIG. 48A

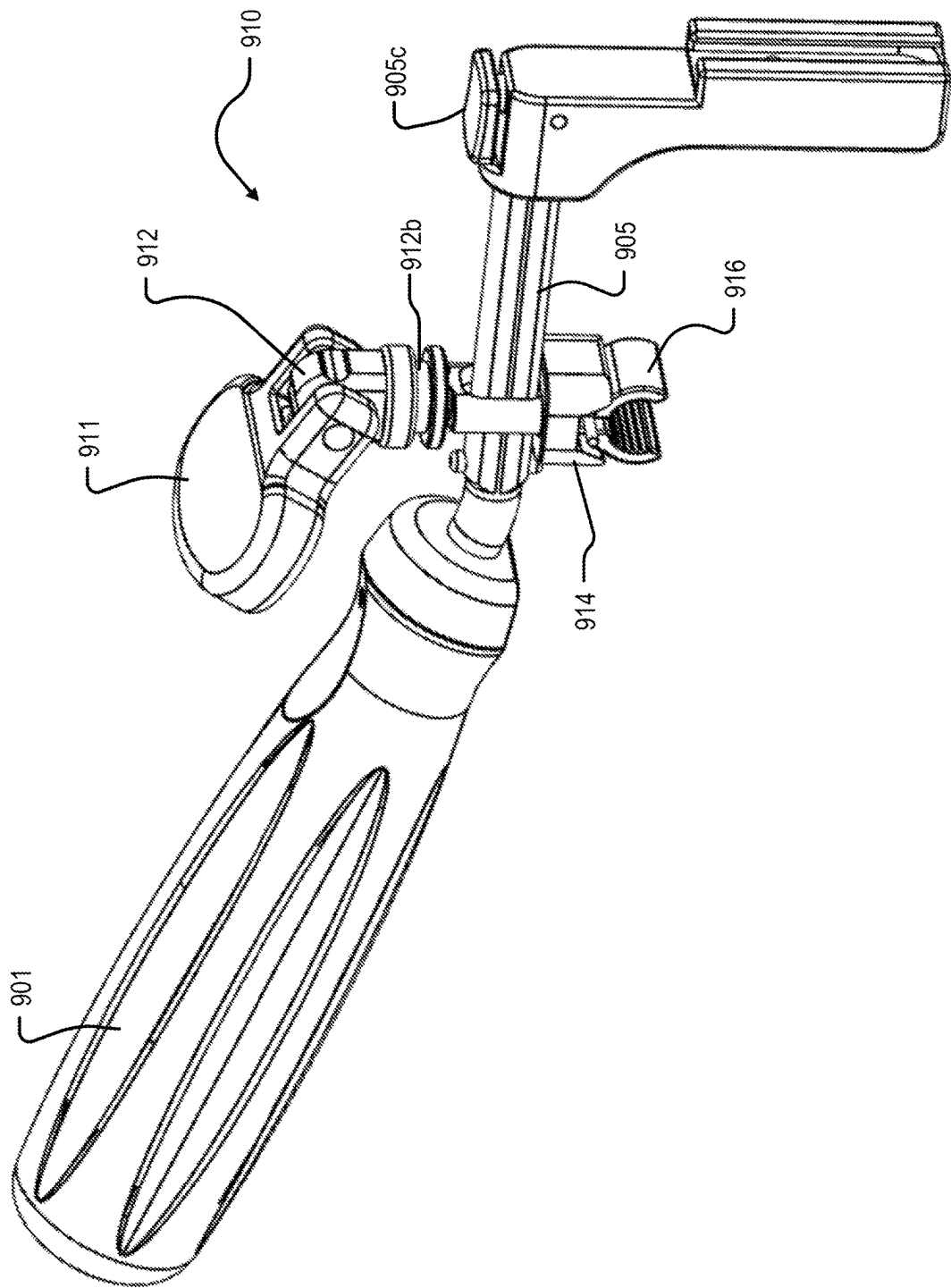
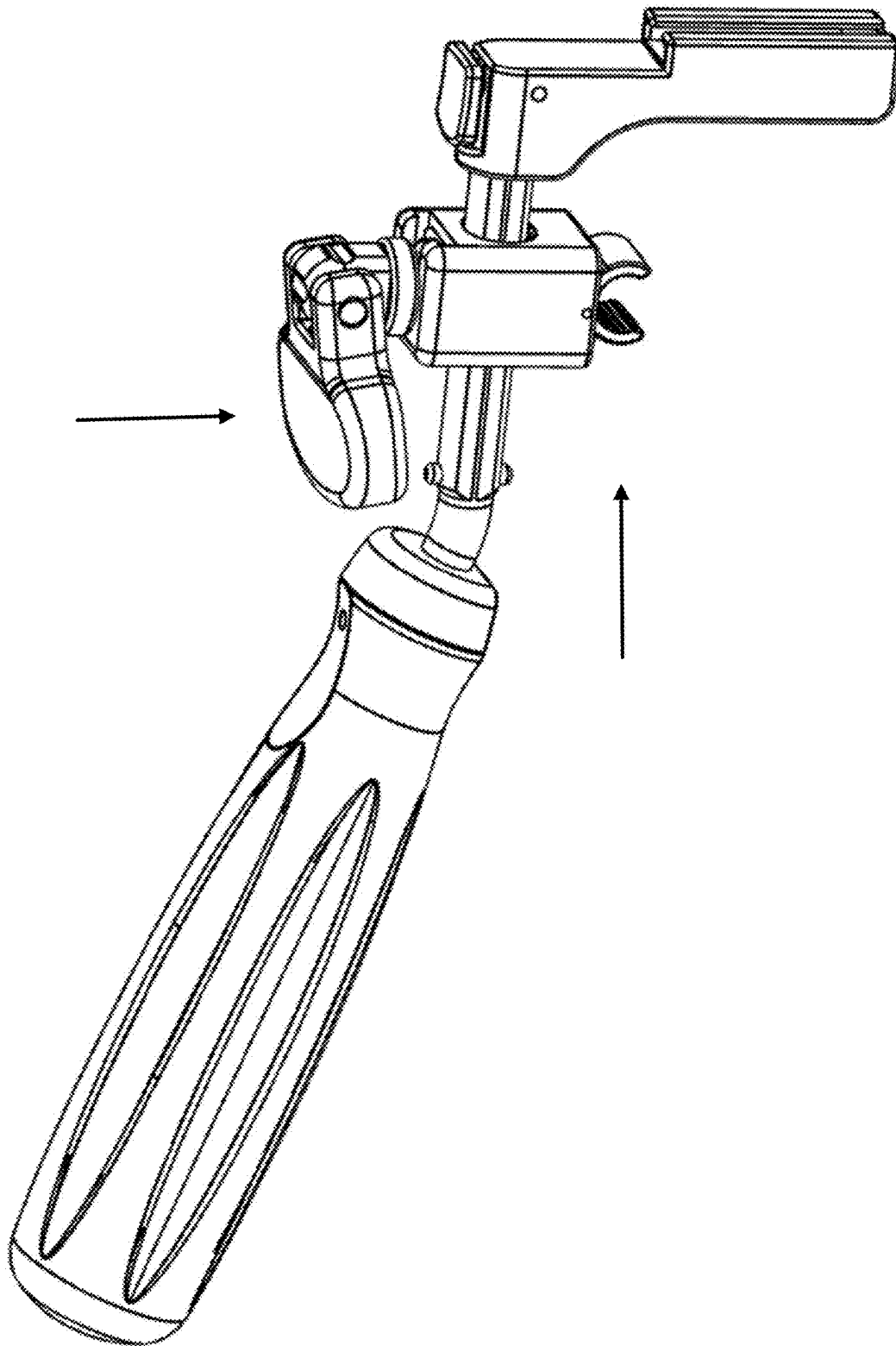
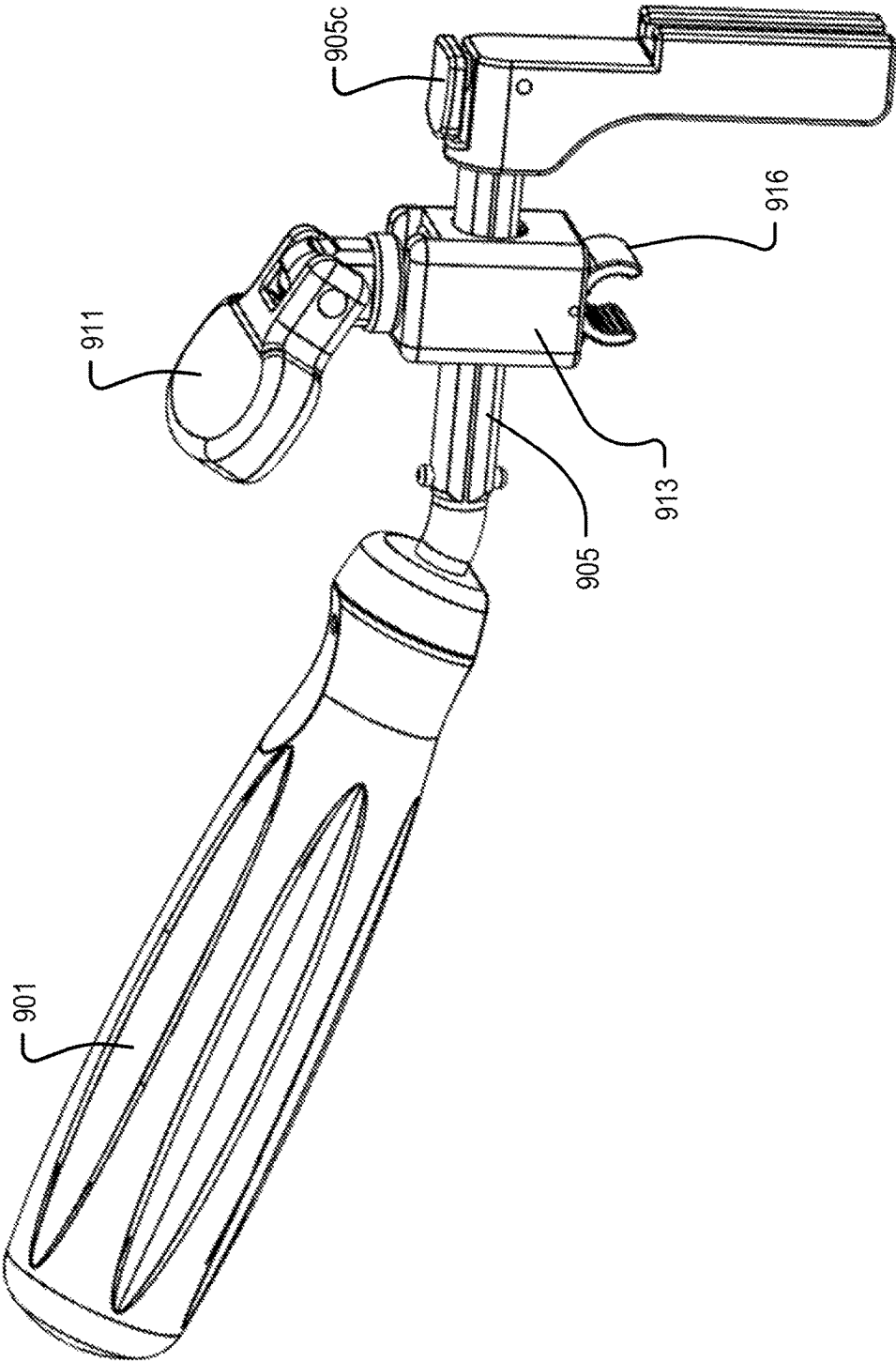


FIG. 48B

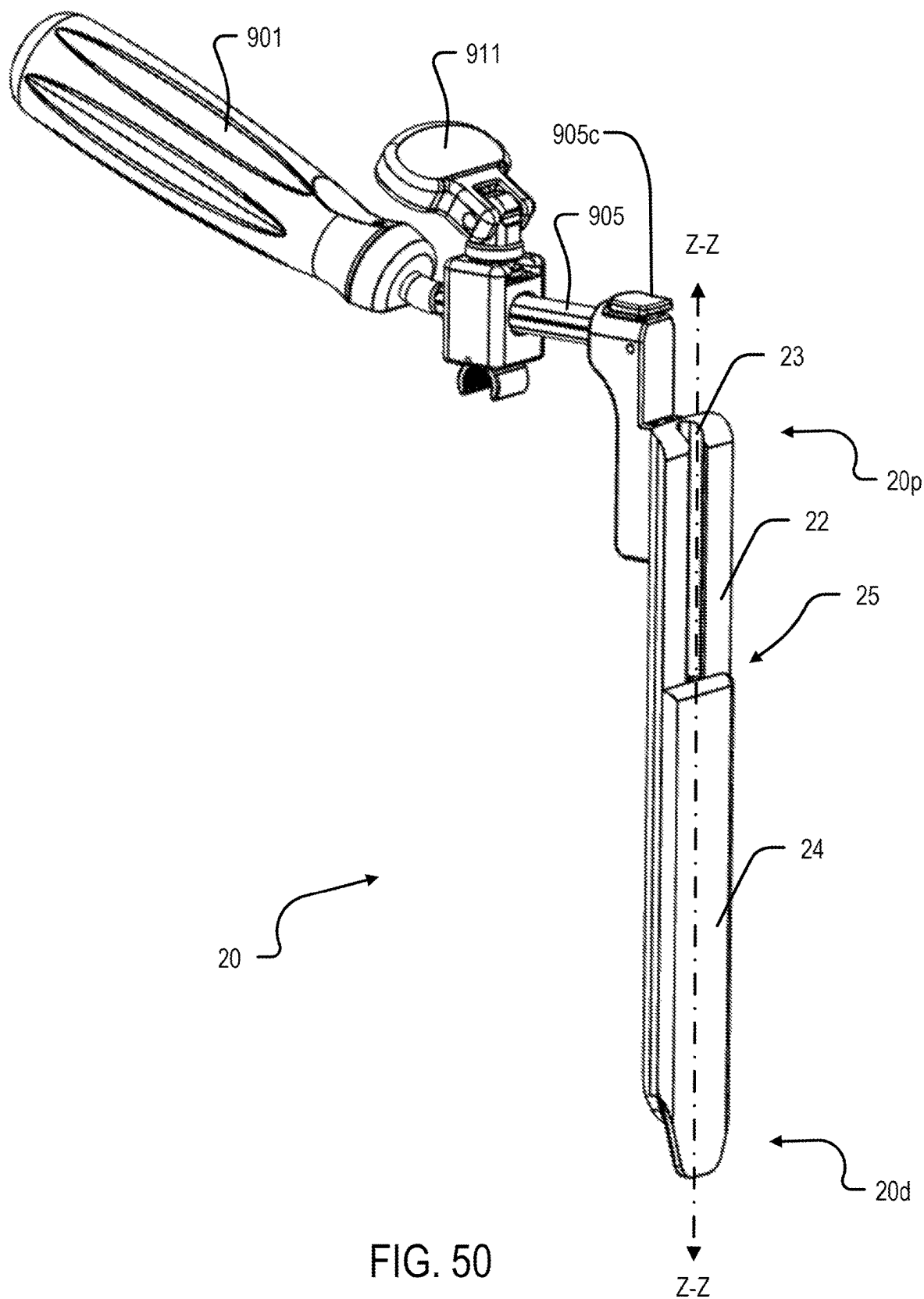


900

FIG. 49A



900
FIG. 49B



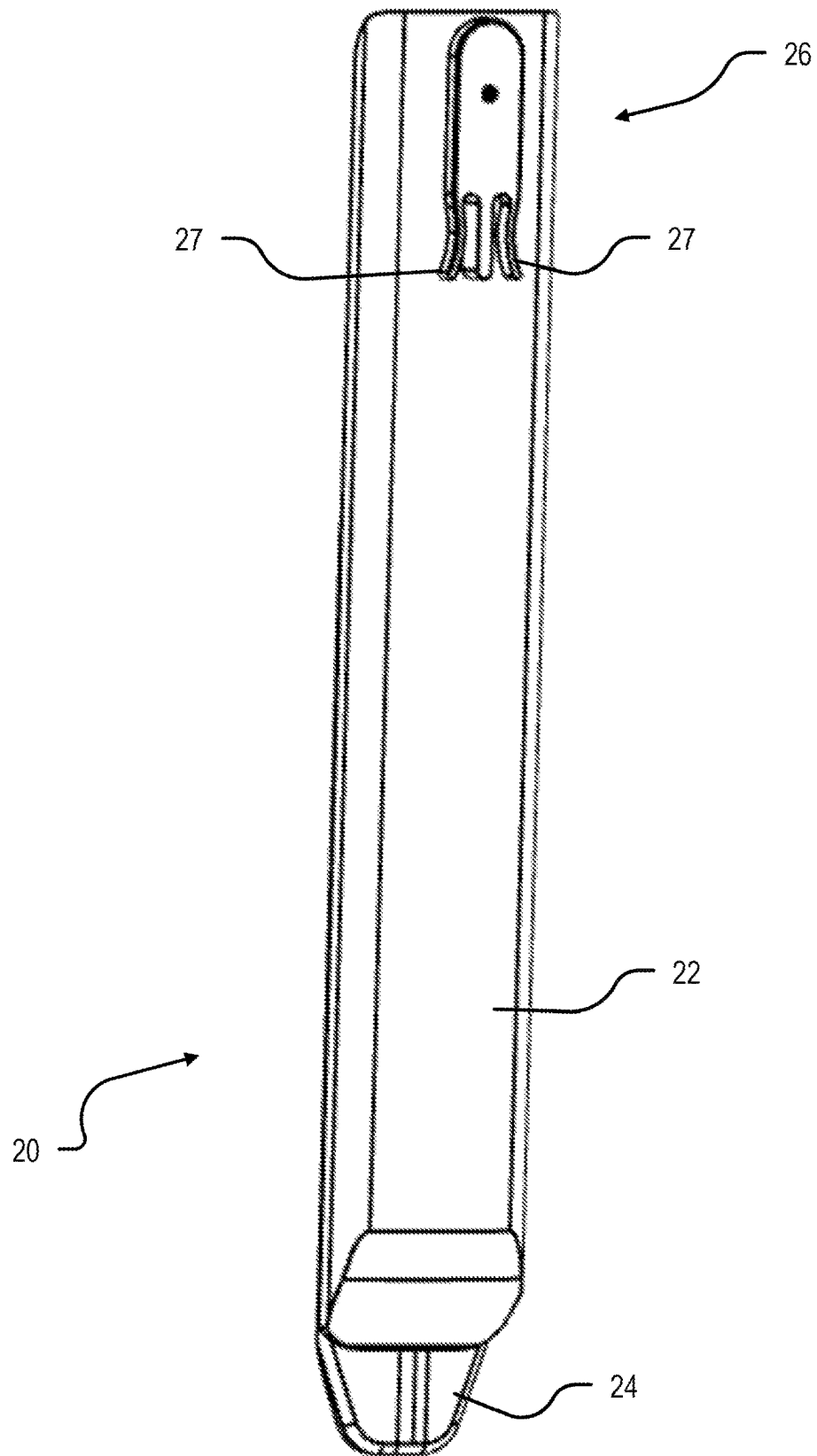


FIG. 51

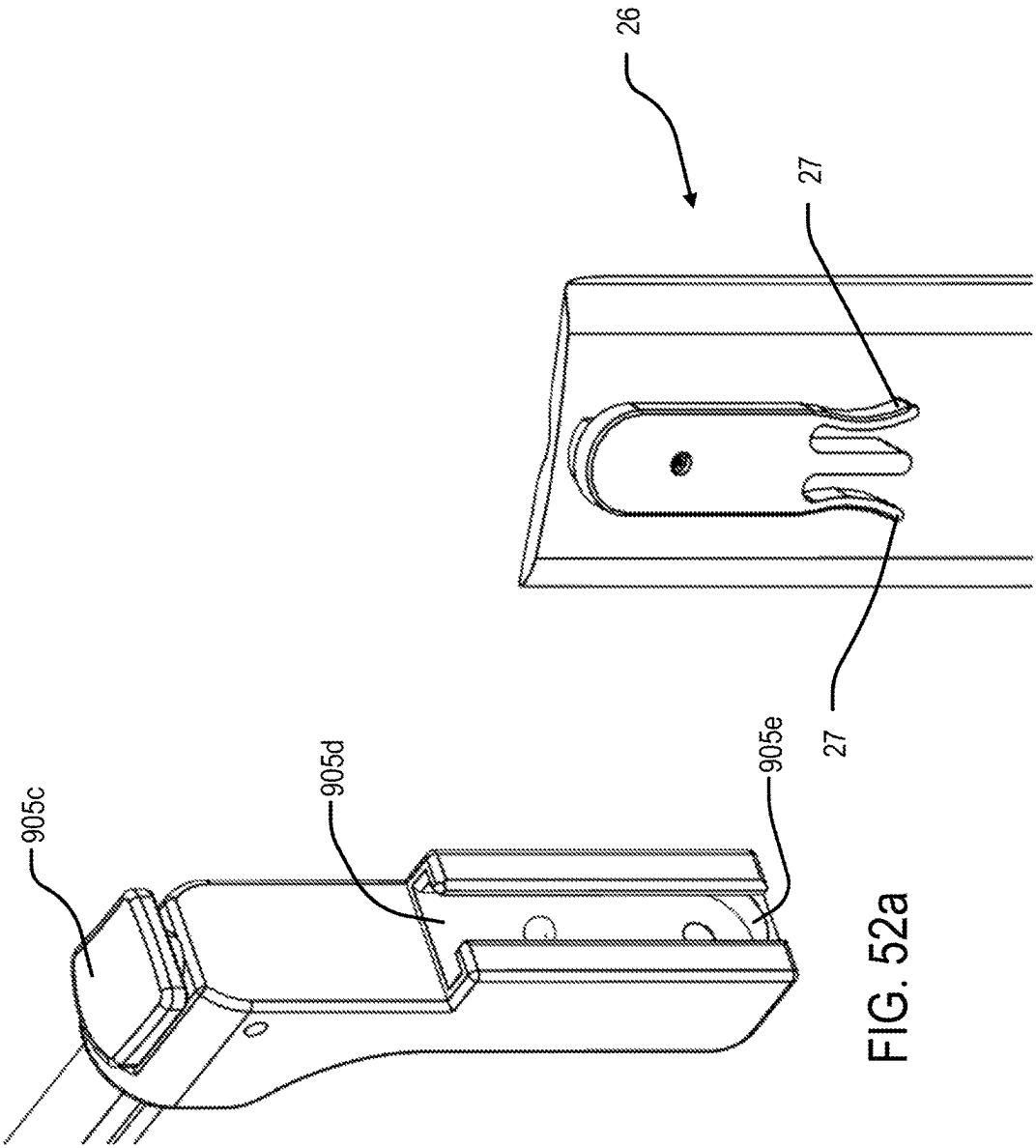


FIG. 52b

FIG. 52a

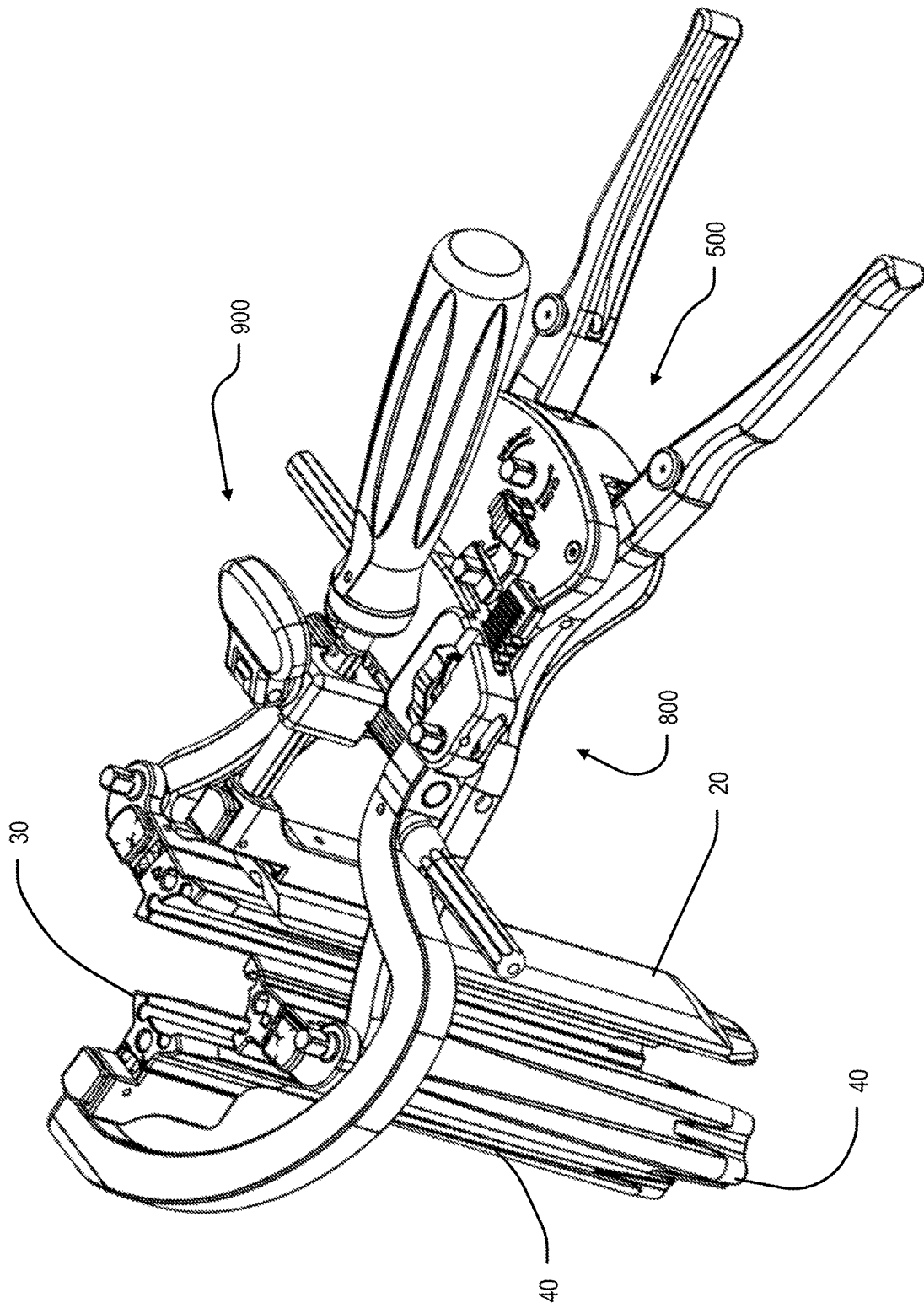


FIG. 53

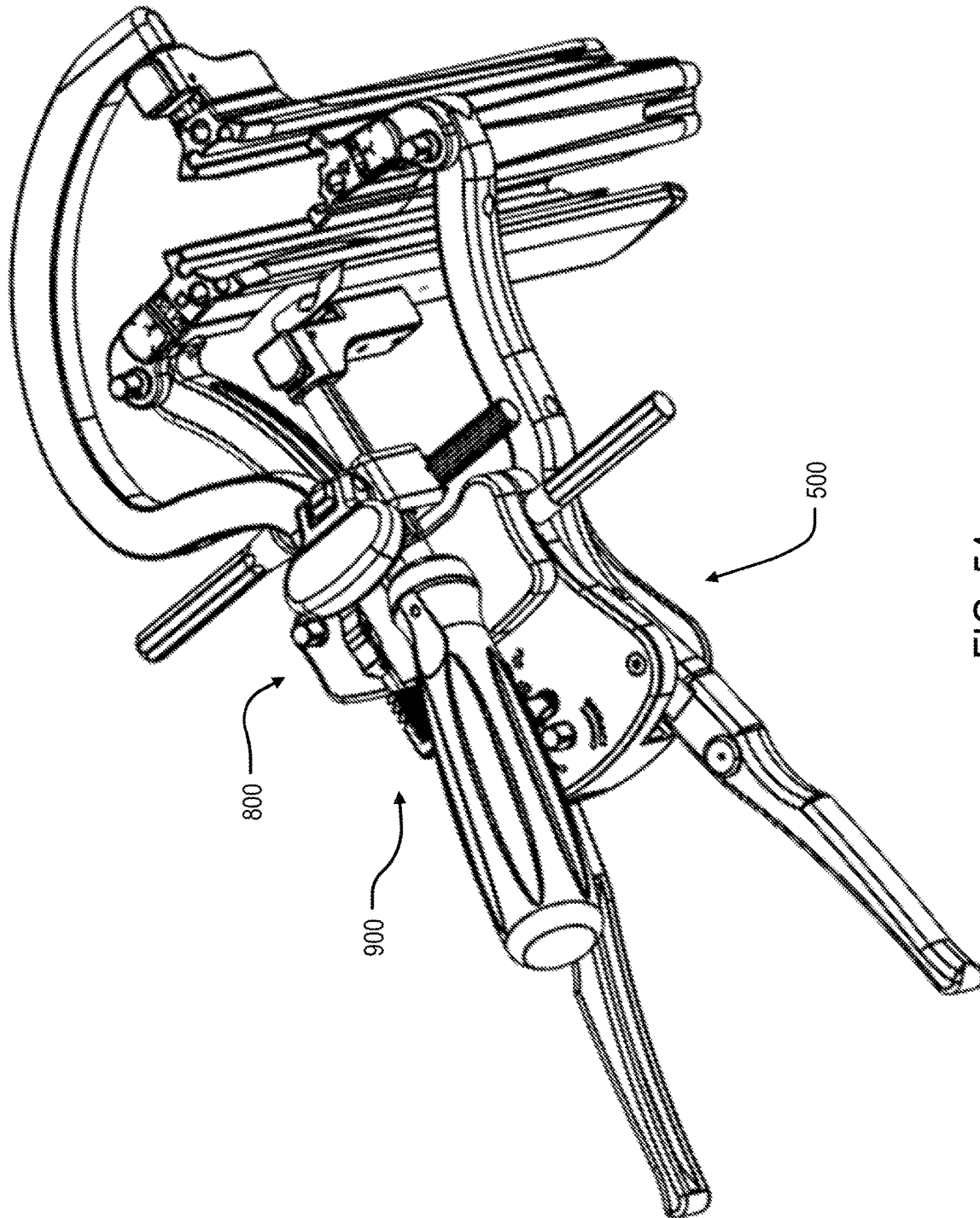
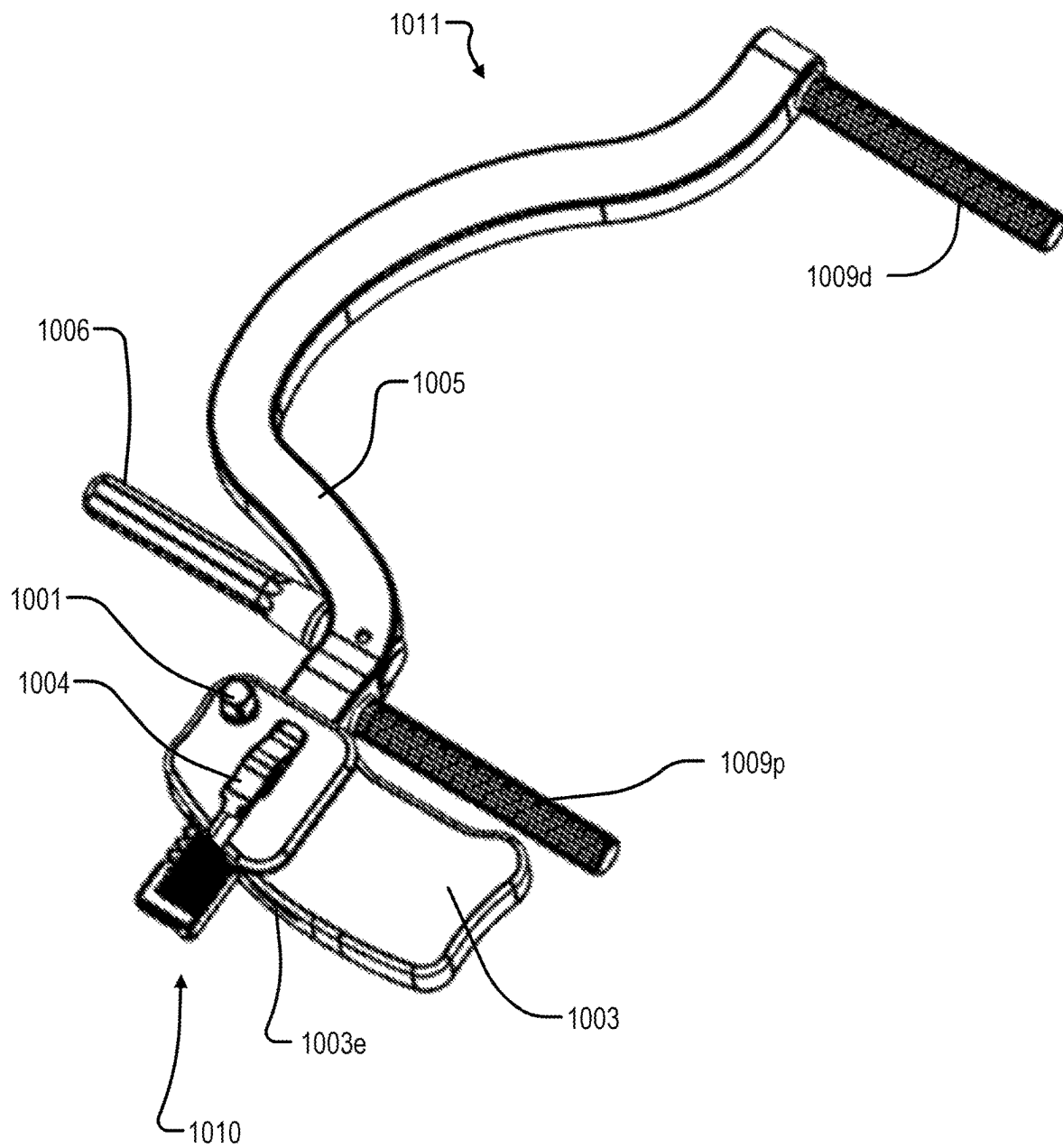
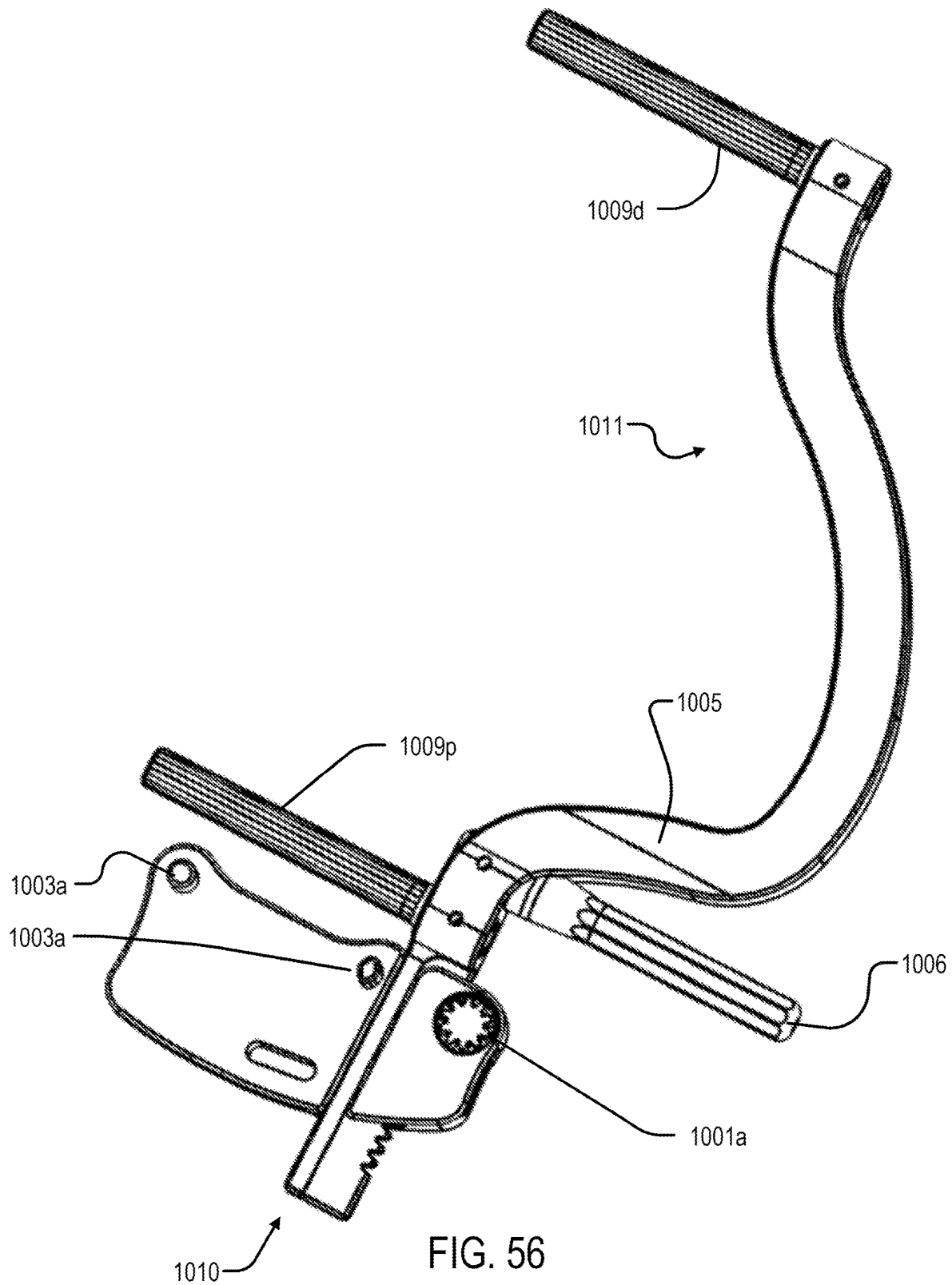


FIG. 54



1000

FIG. 55



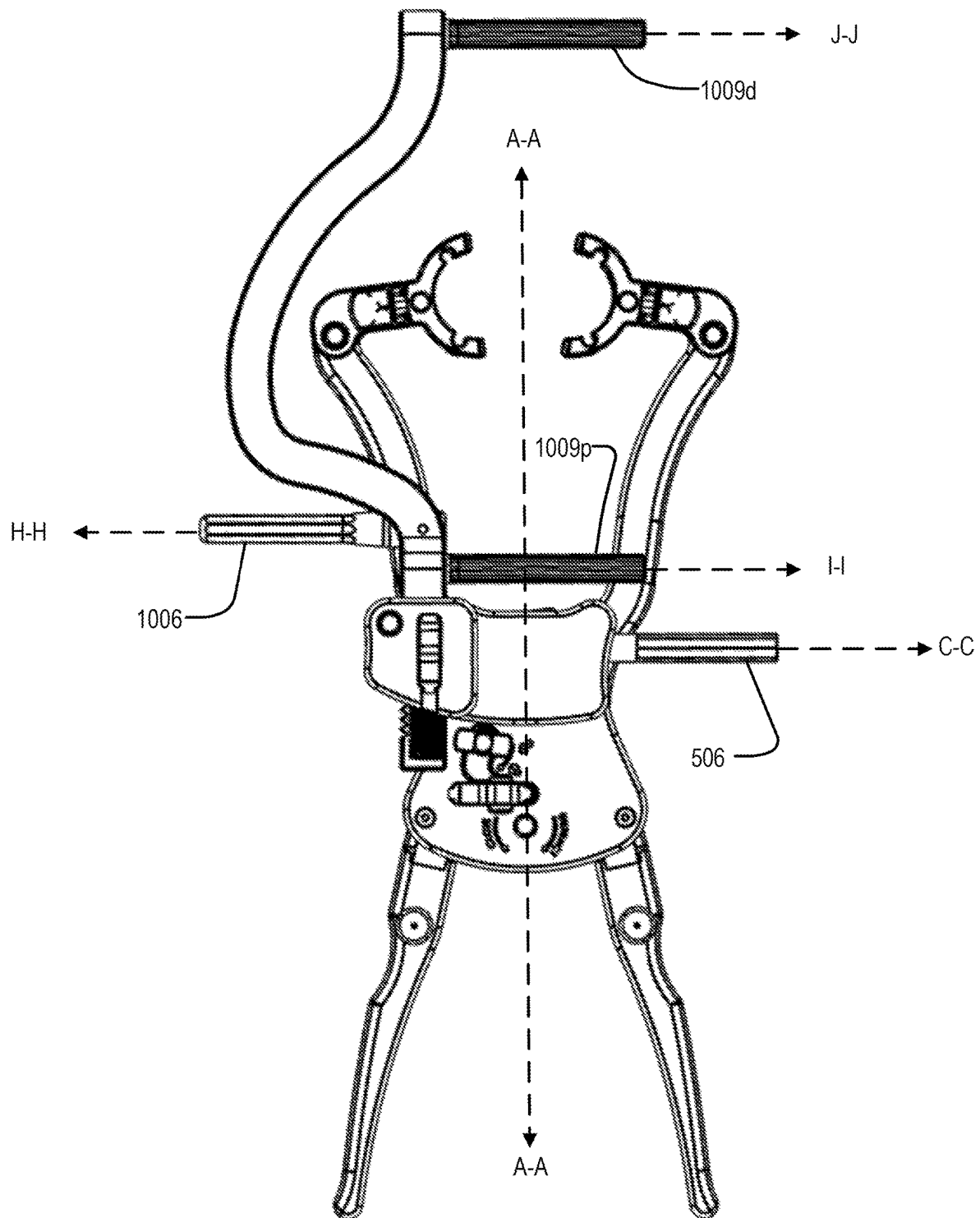


FIG. 58

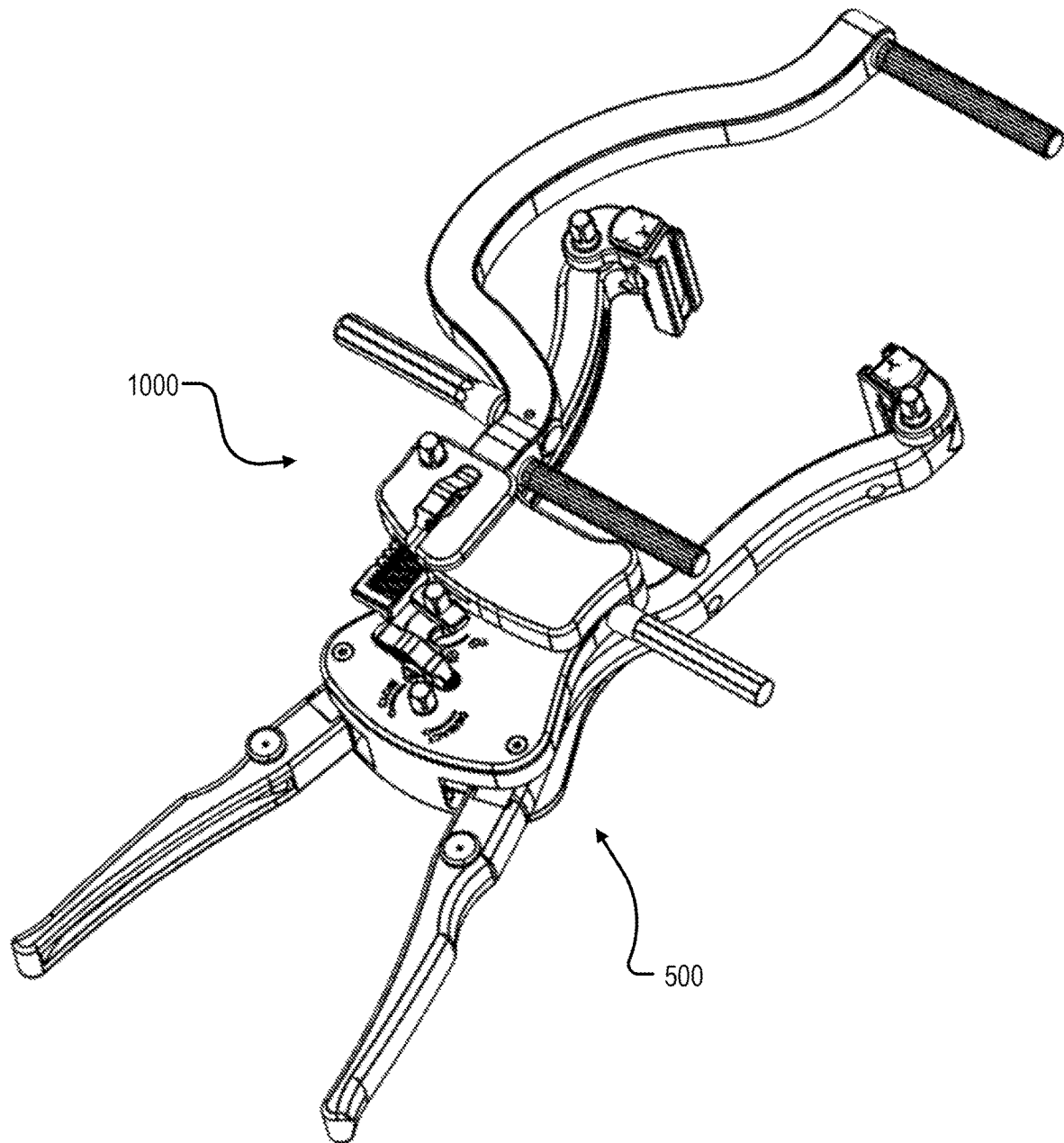


FIG. 59

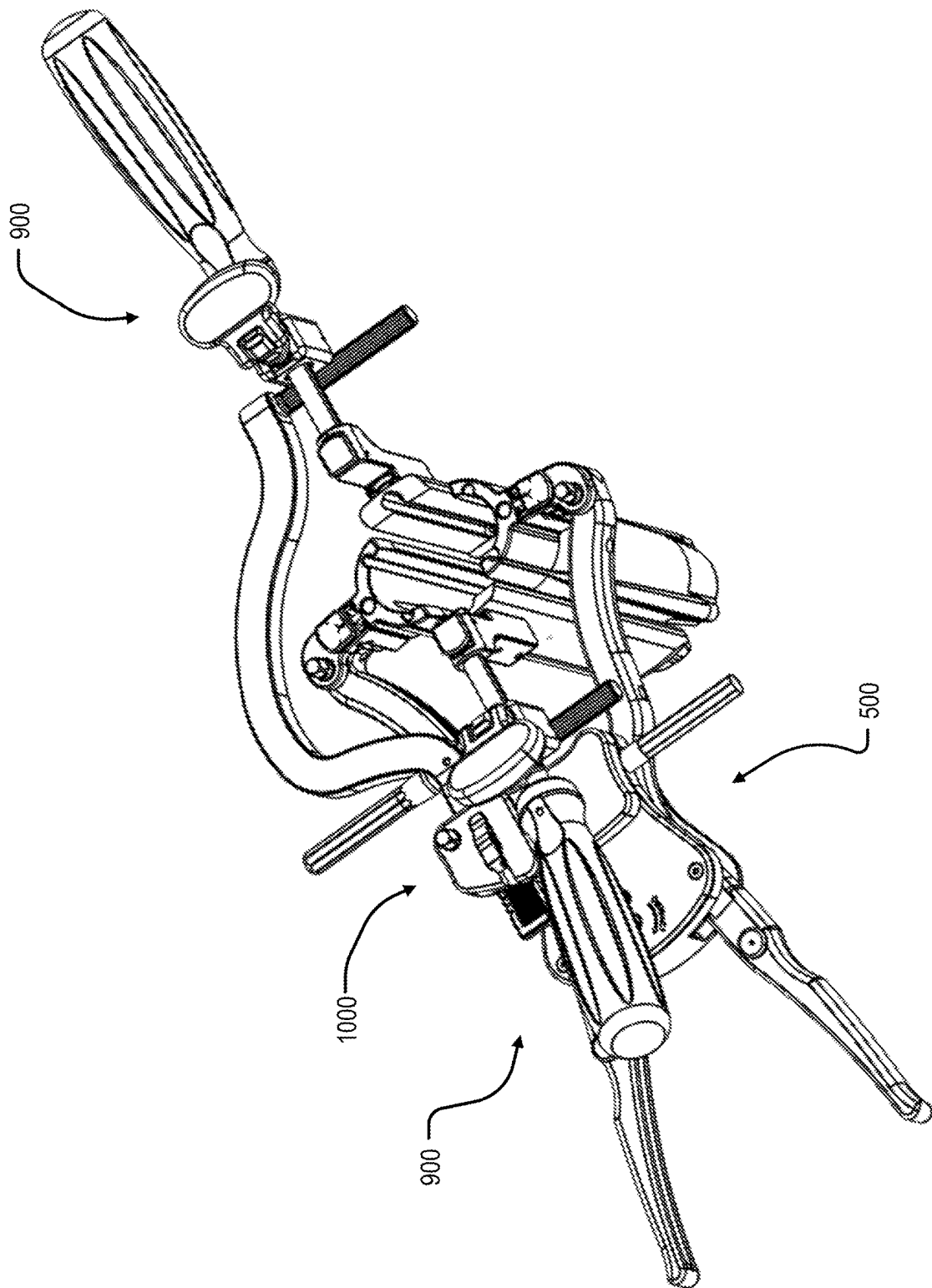


FIG. 60

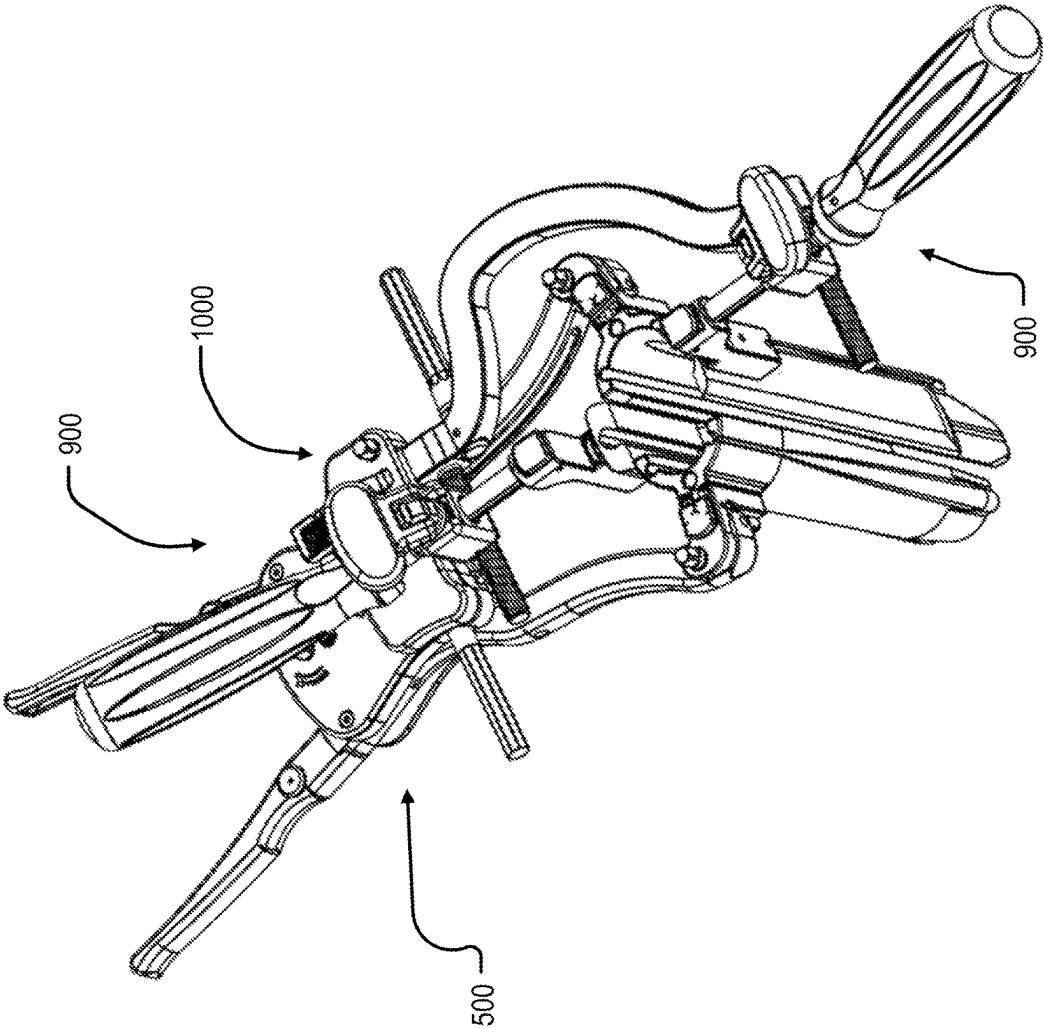
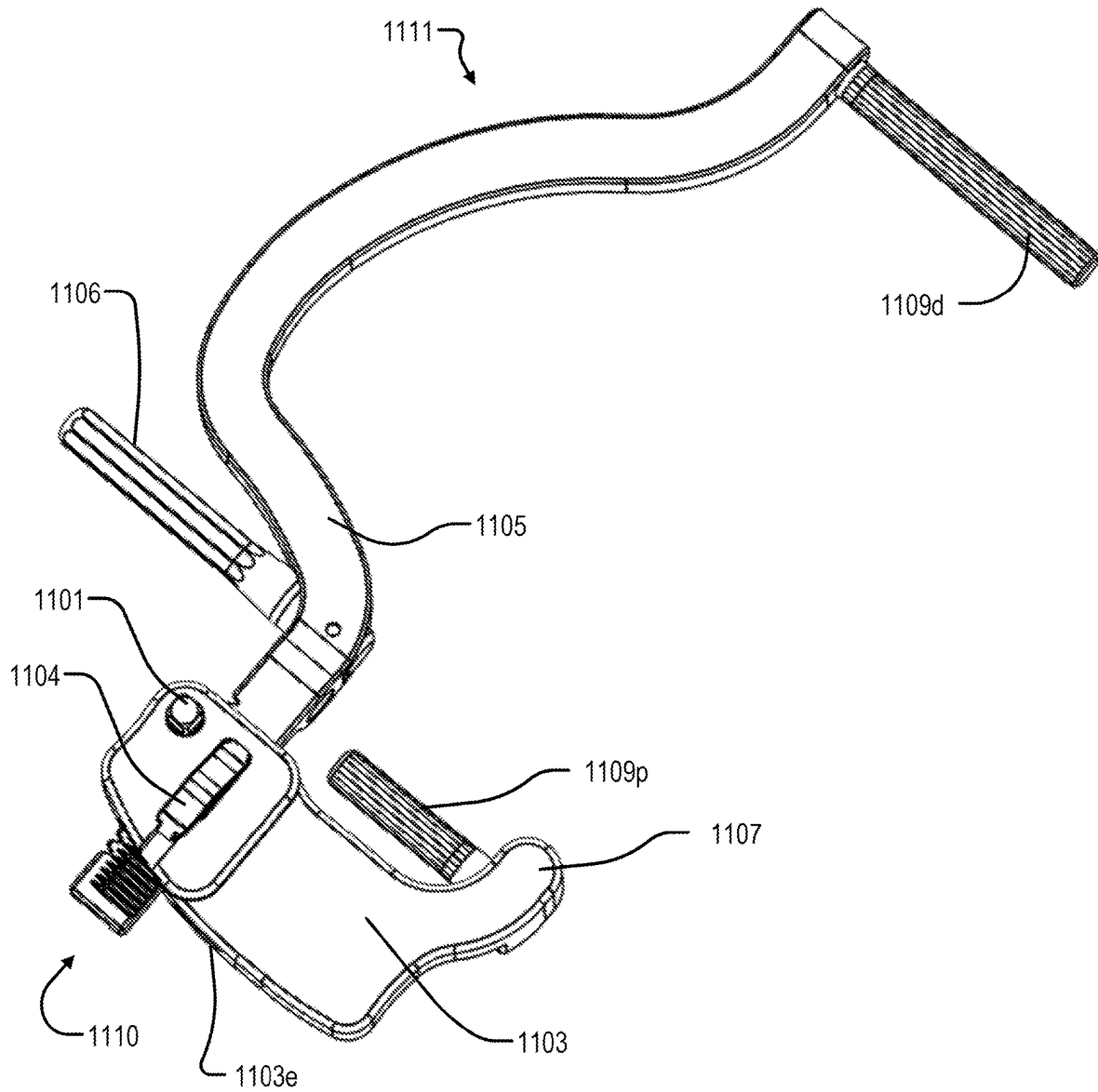
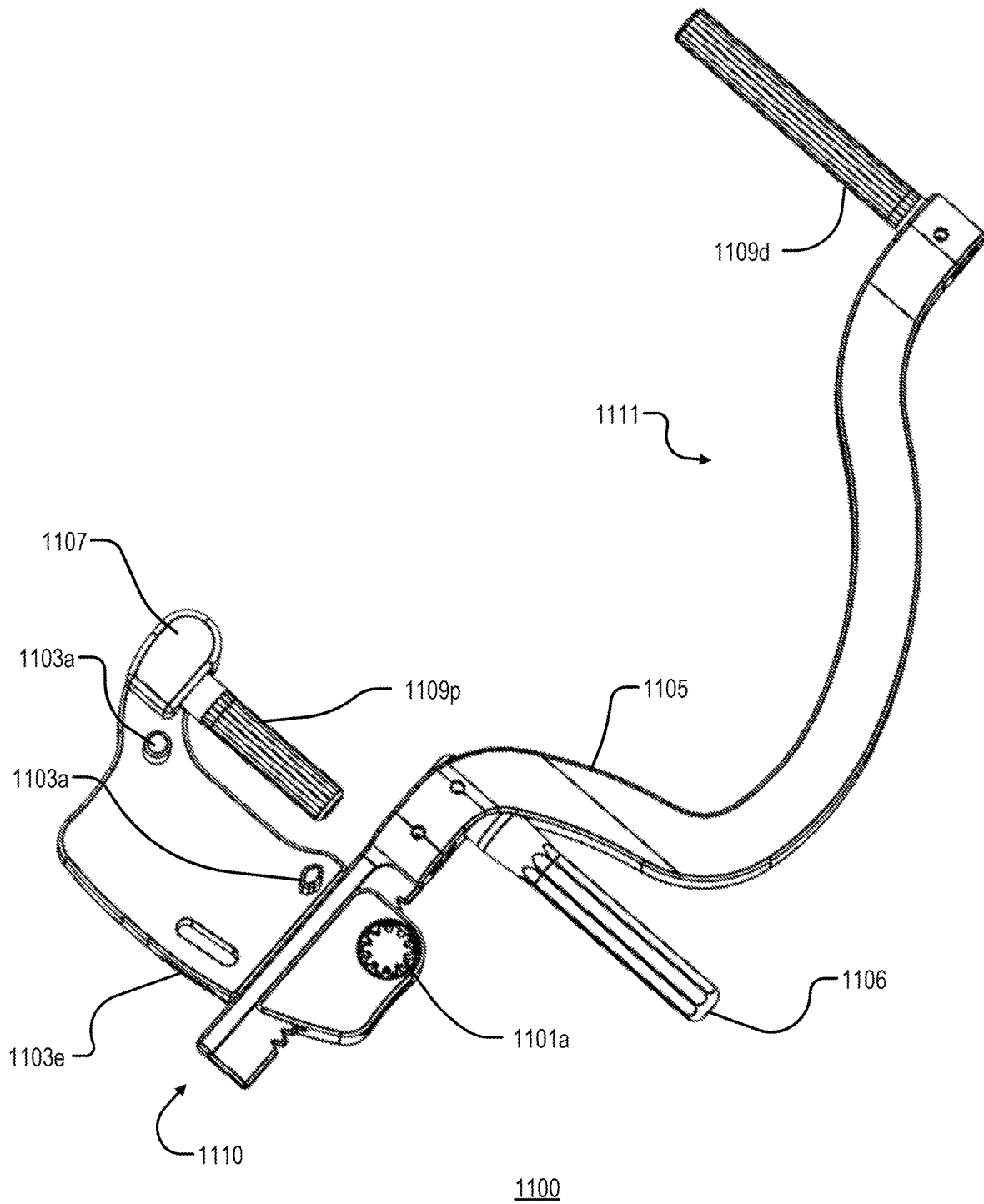


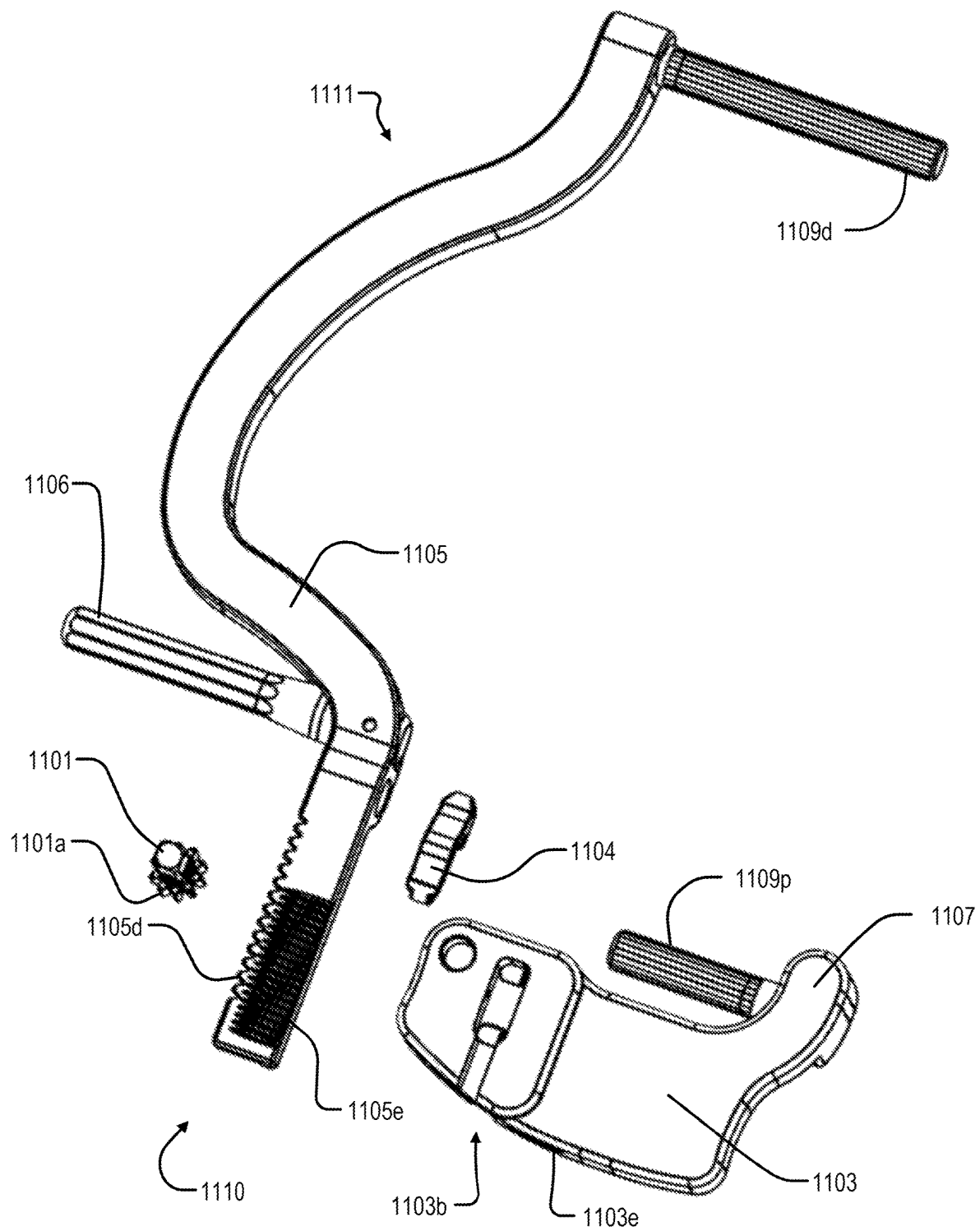
FIG. 61



1100
FIG. 62



1100
FIG. 63



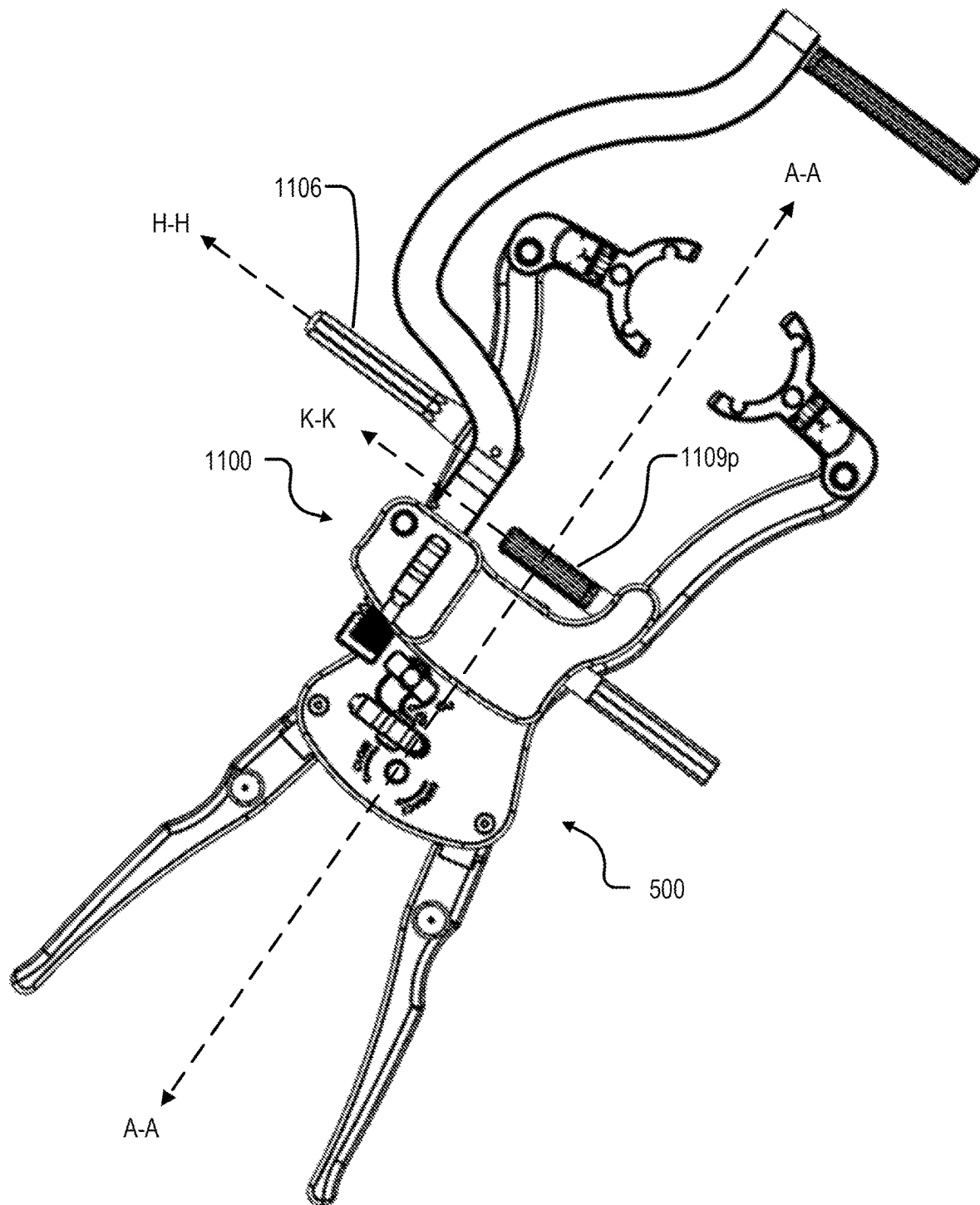


FIG. 65

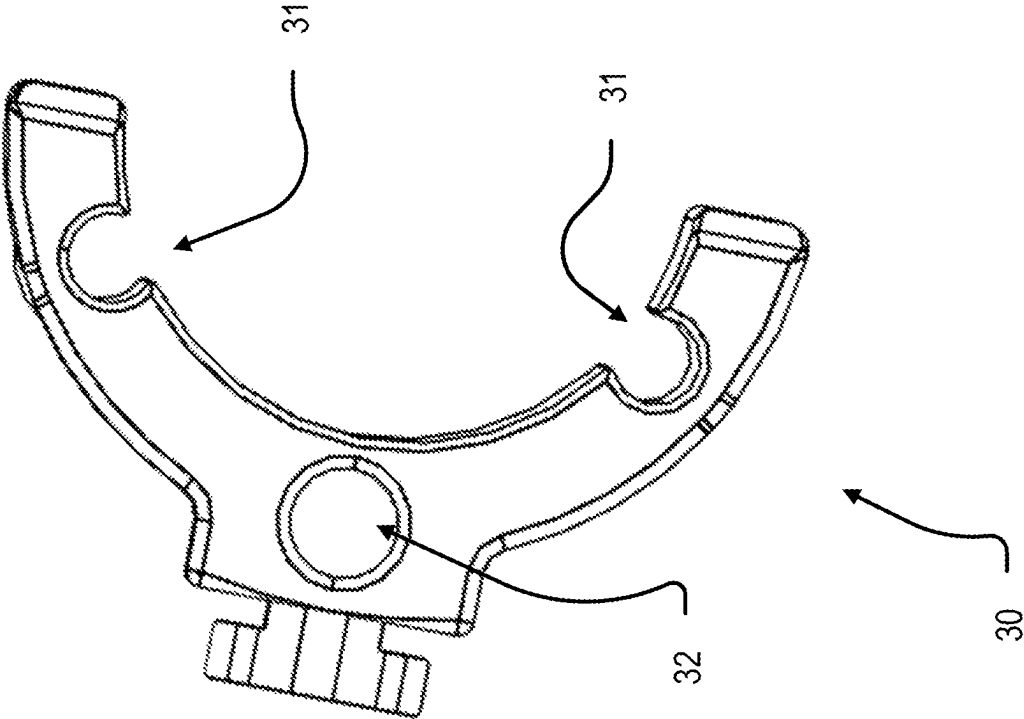
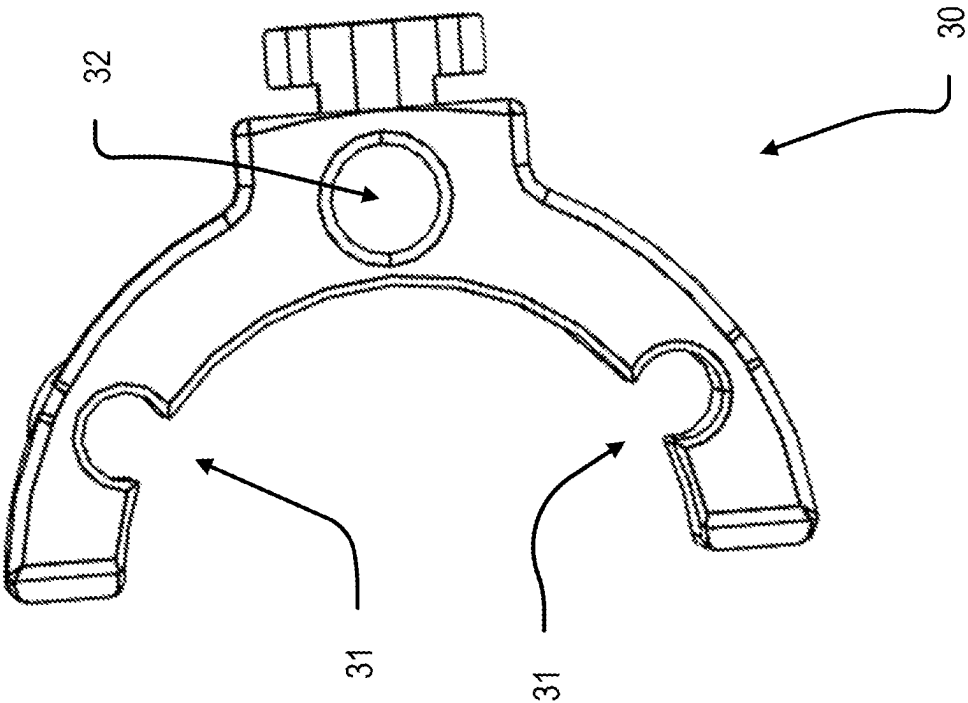


FIG. 66

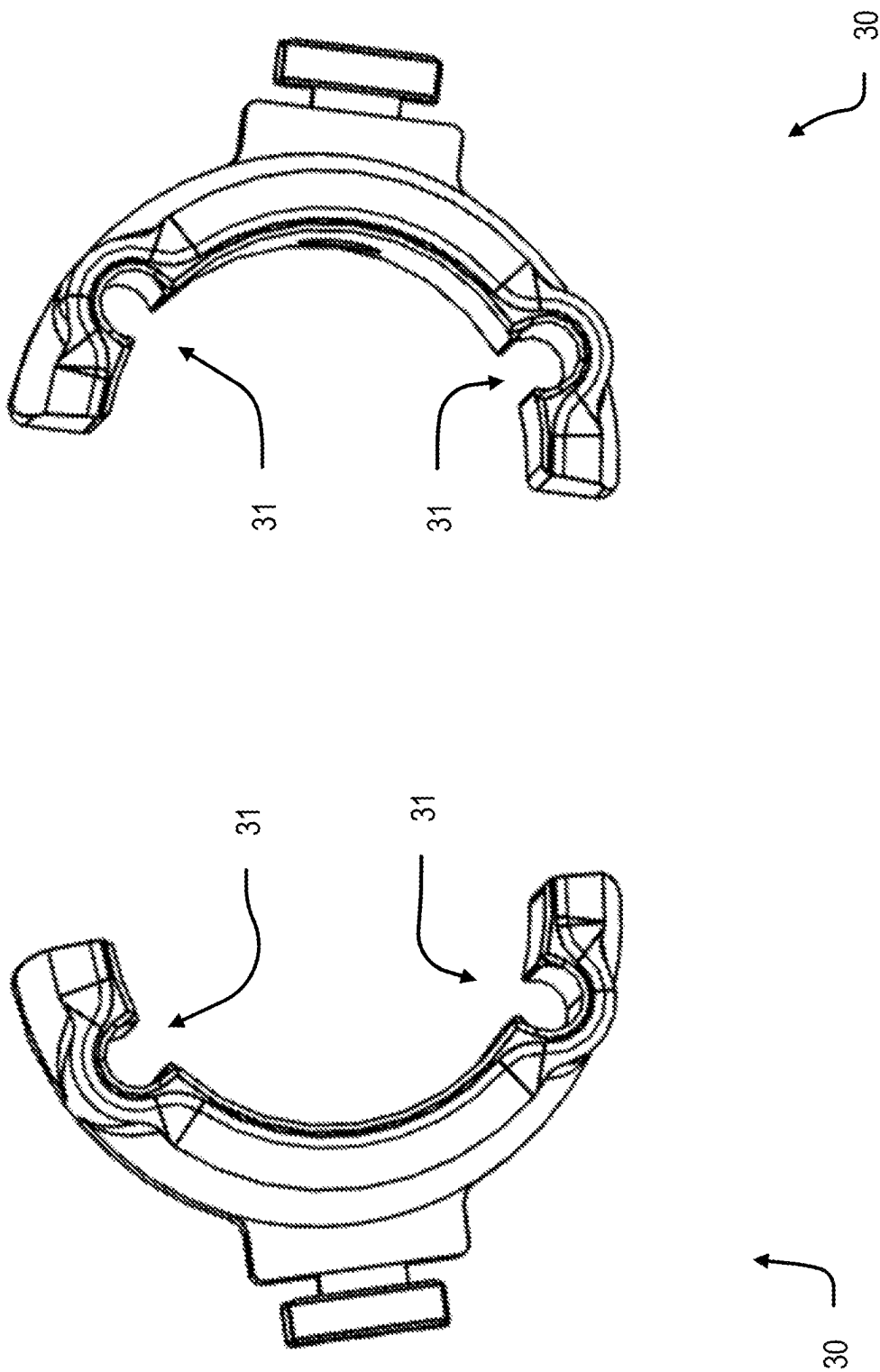


FIG. 67

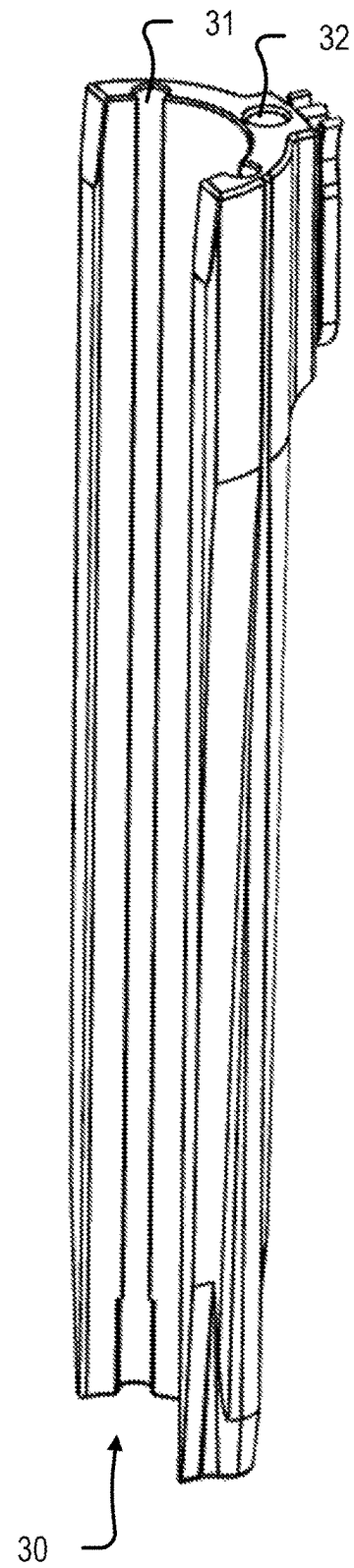
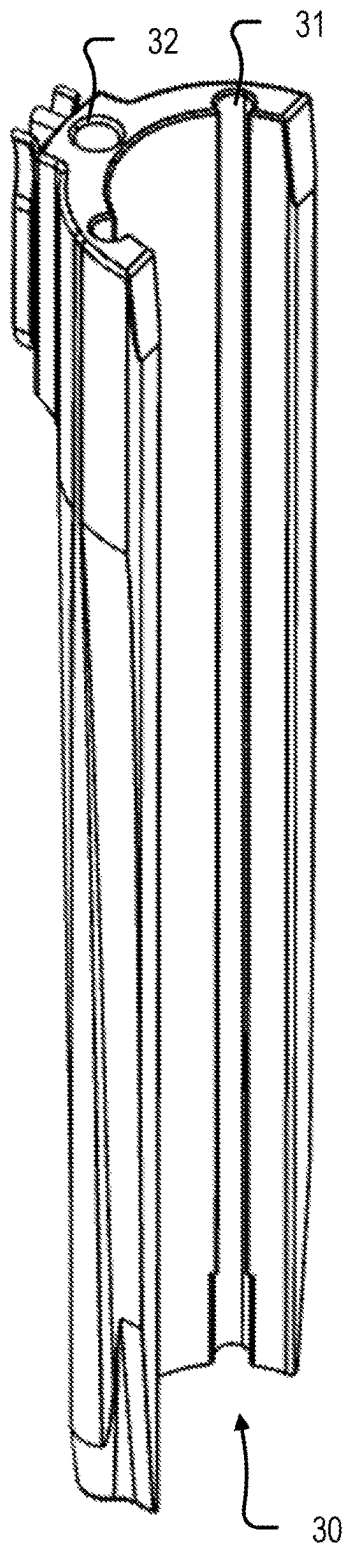


FIG. 68

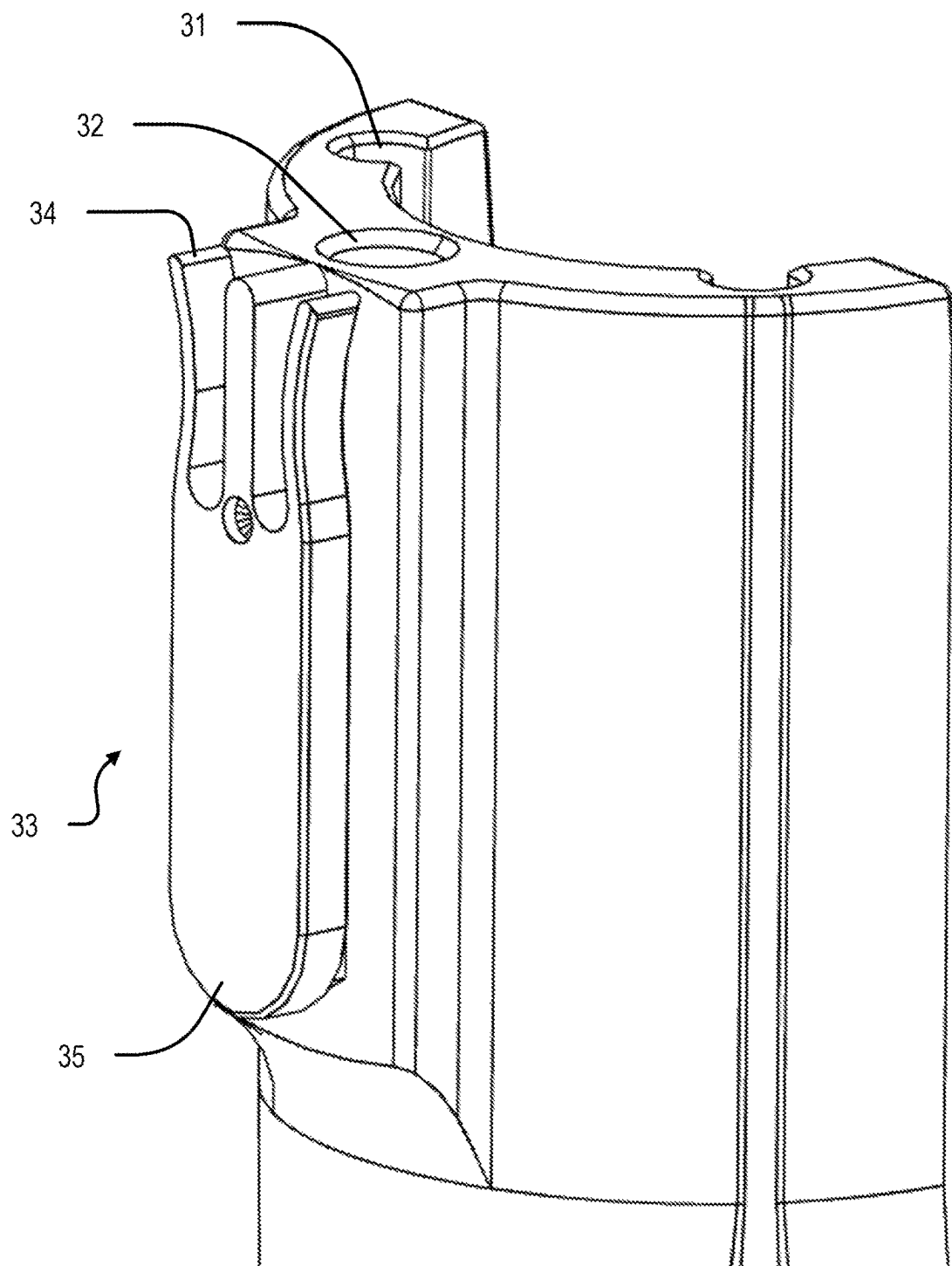


FIG. 69

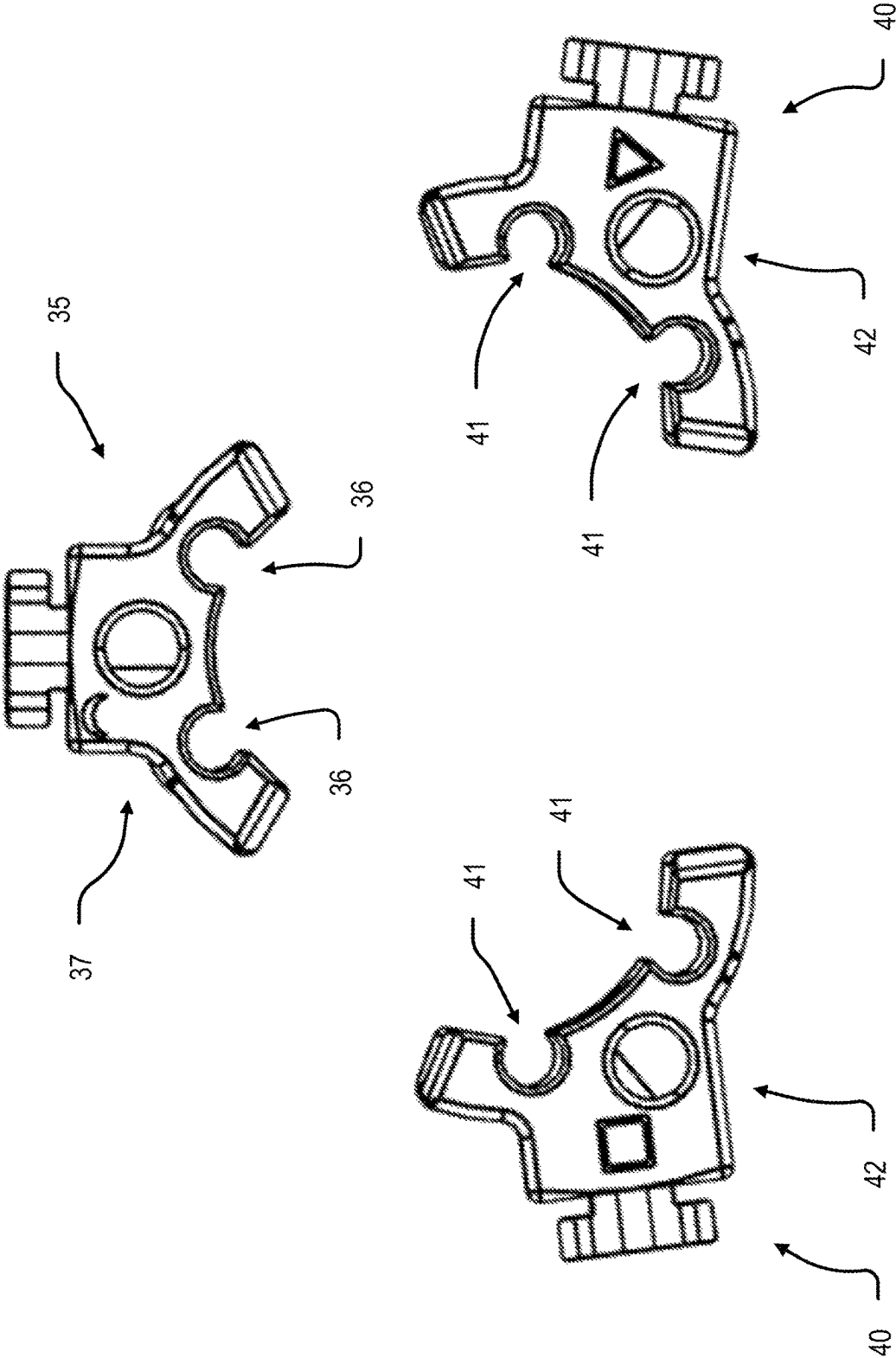


FIG. 70

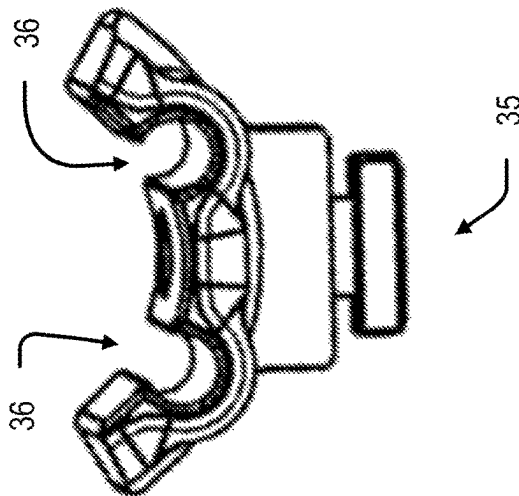
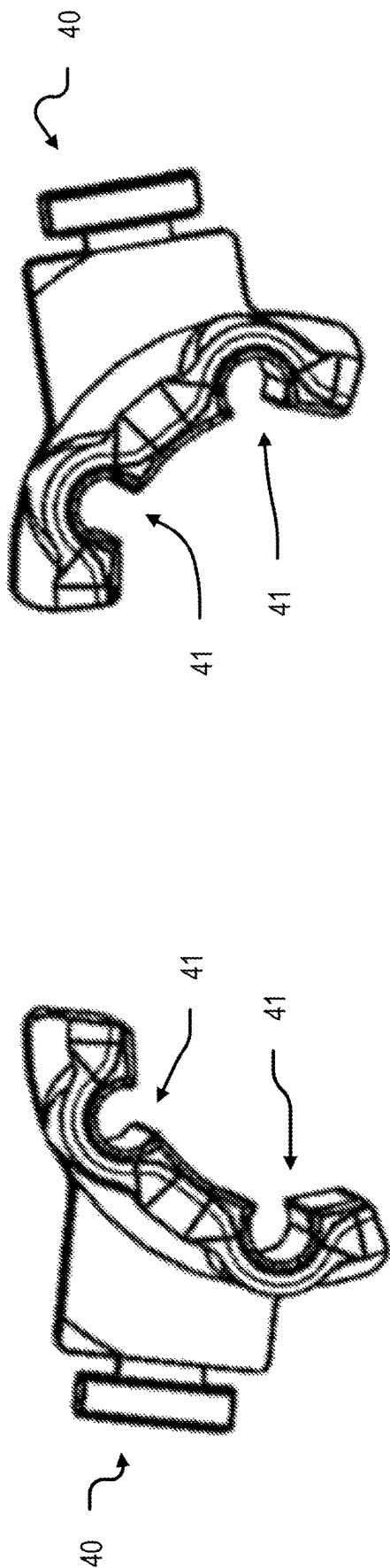


FIG. 71

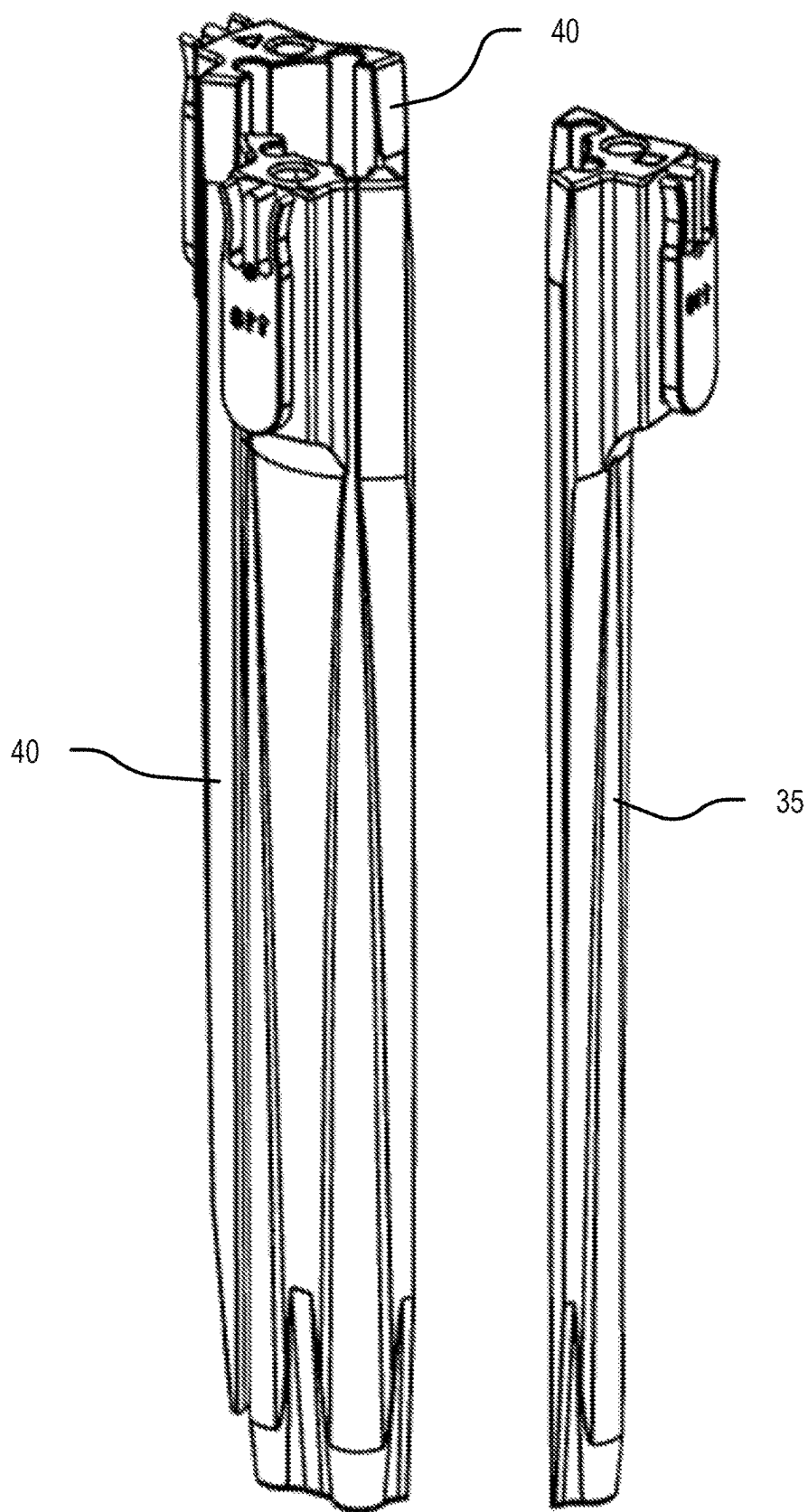


FIG. 72

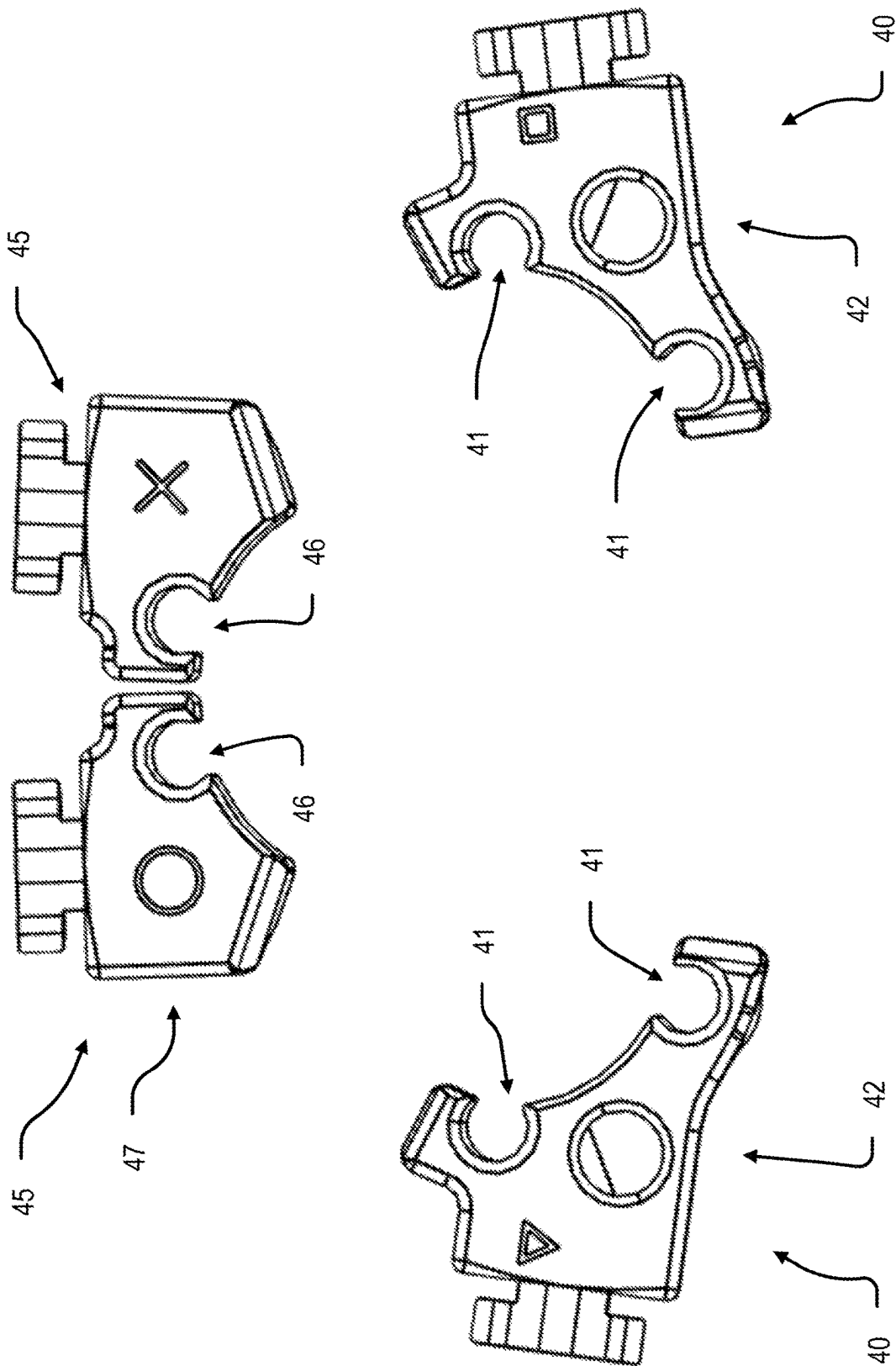


FIG. 73

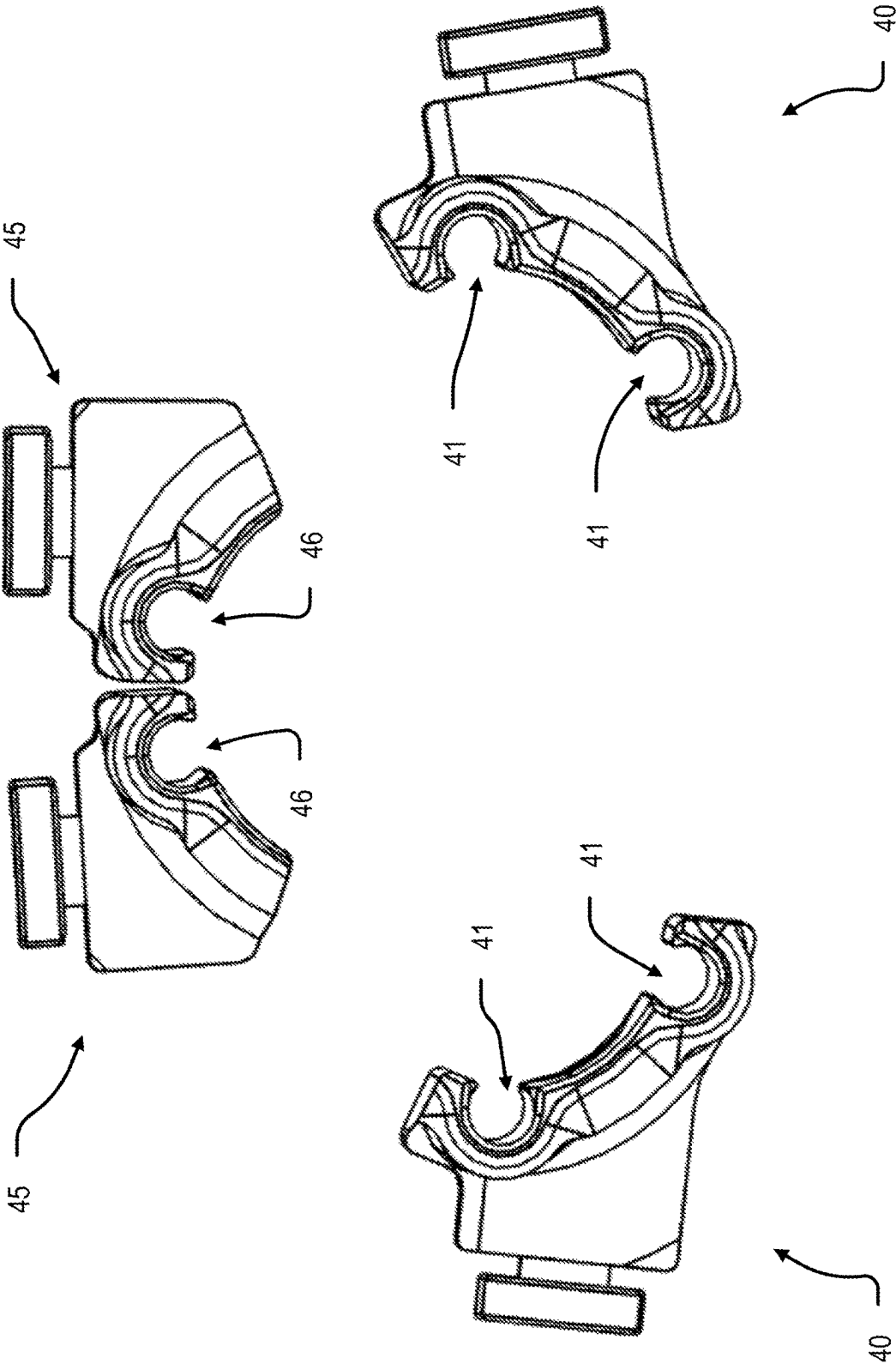


FIG. 74

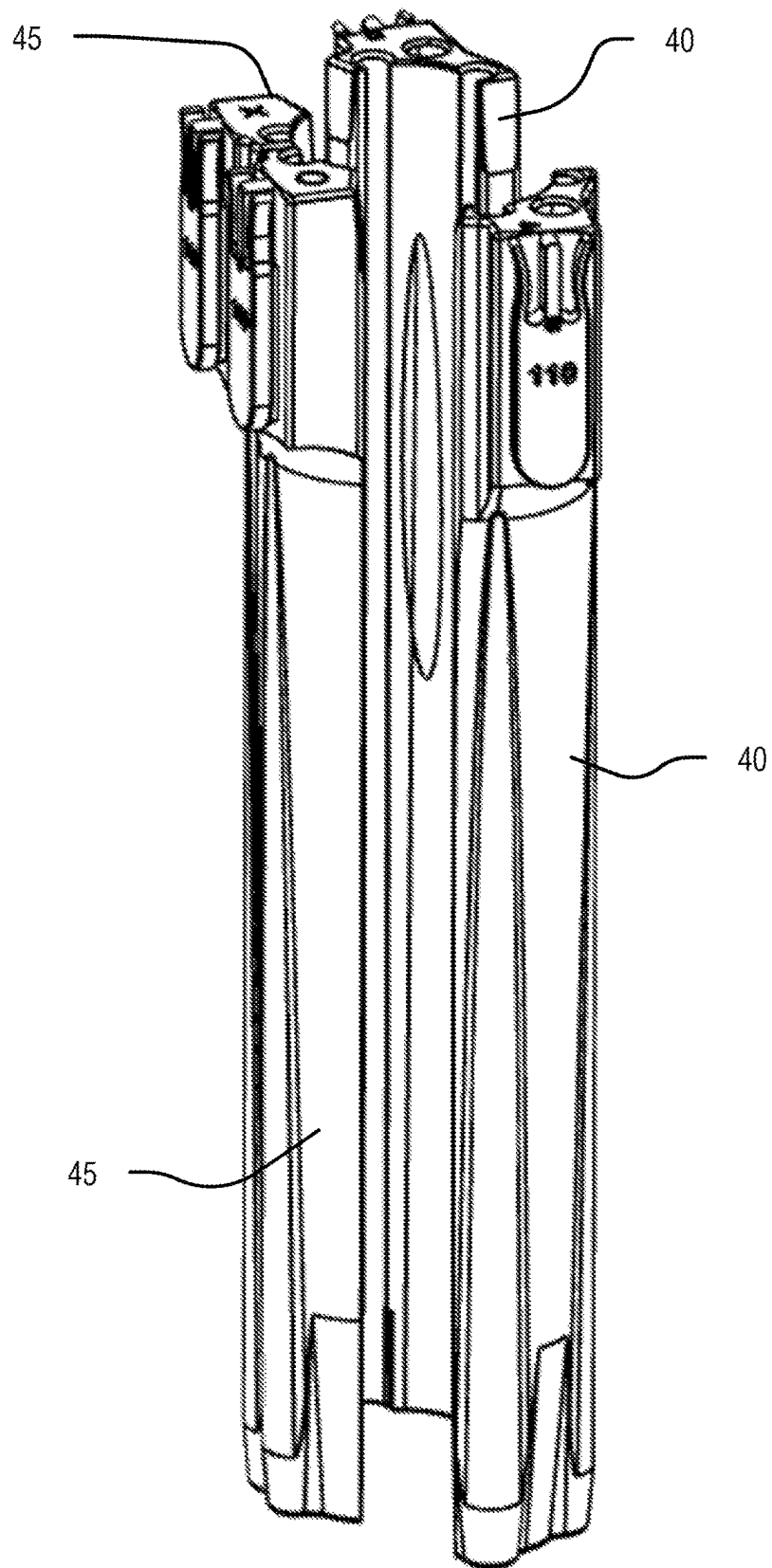


FIG. 75

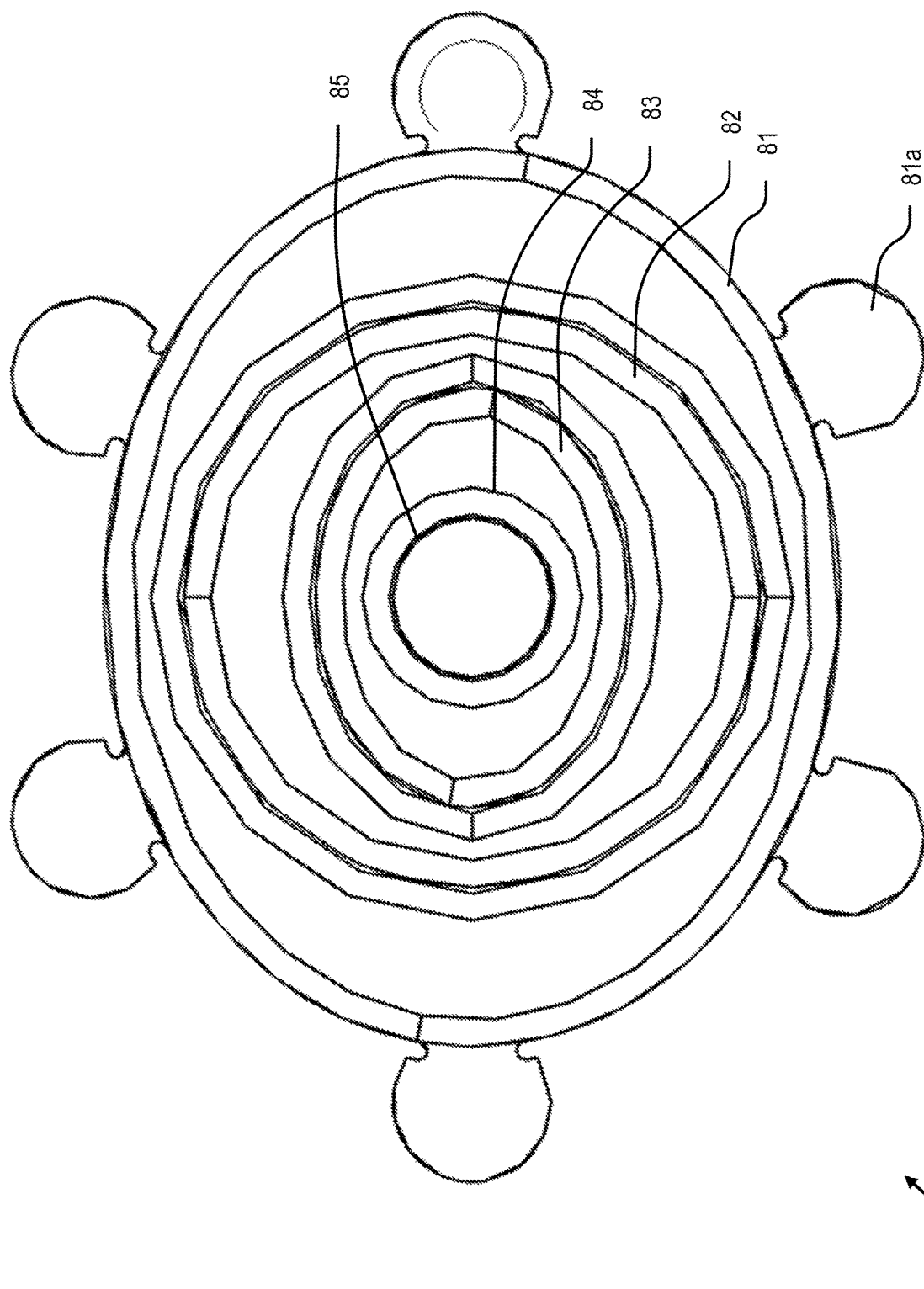


FIG. 76

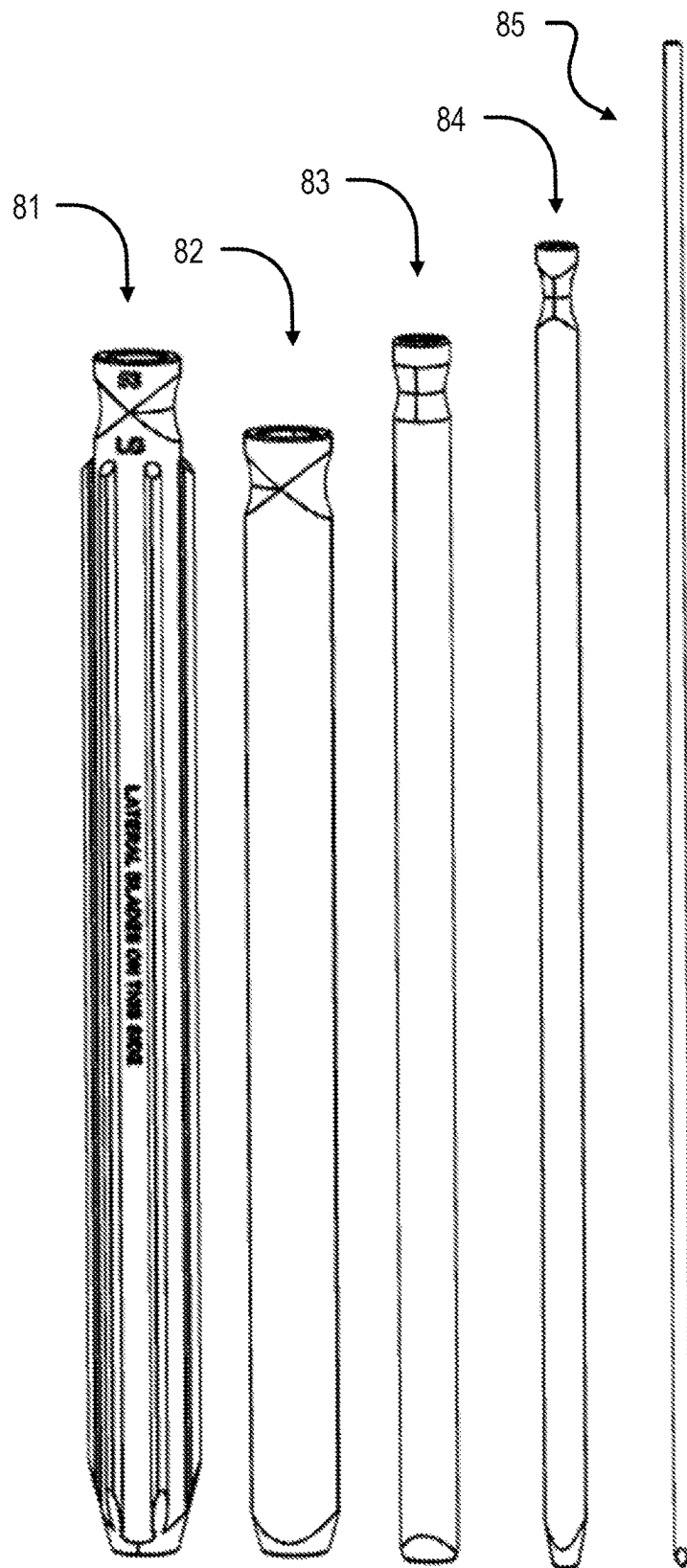


FIG. 77A

80

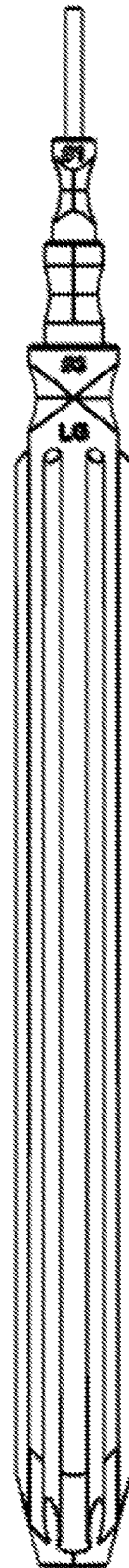


FIG. 77B

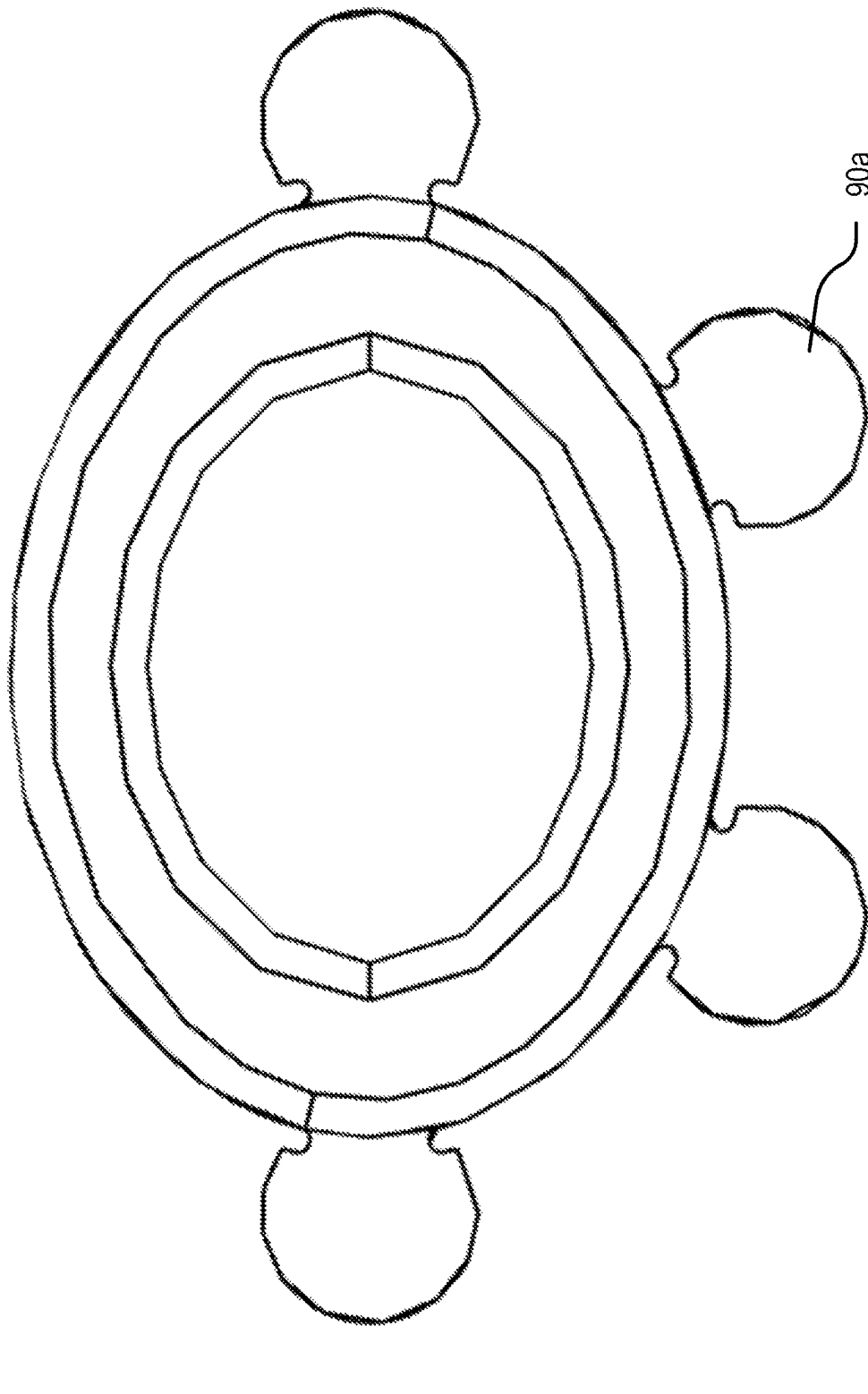


FIG. 78

90

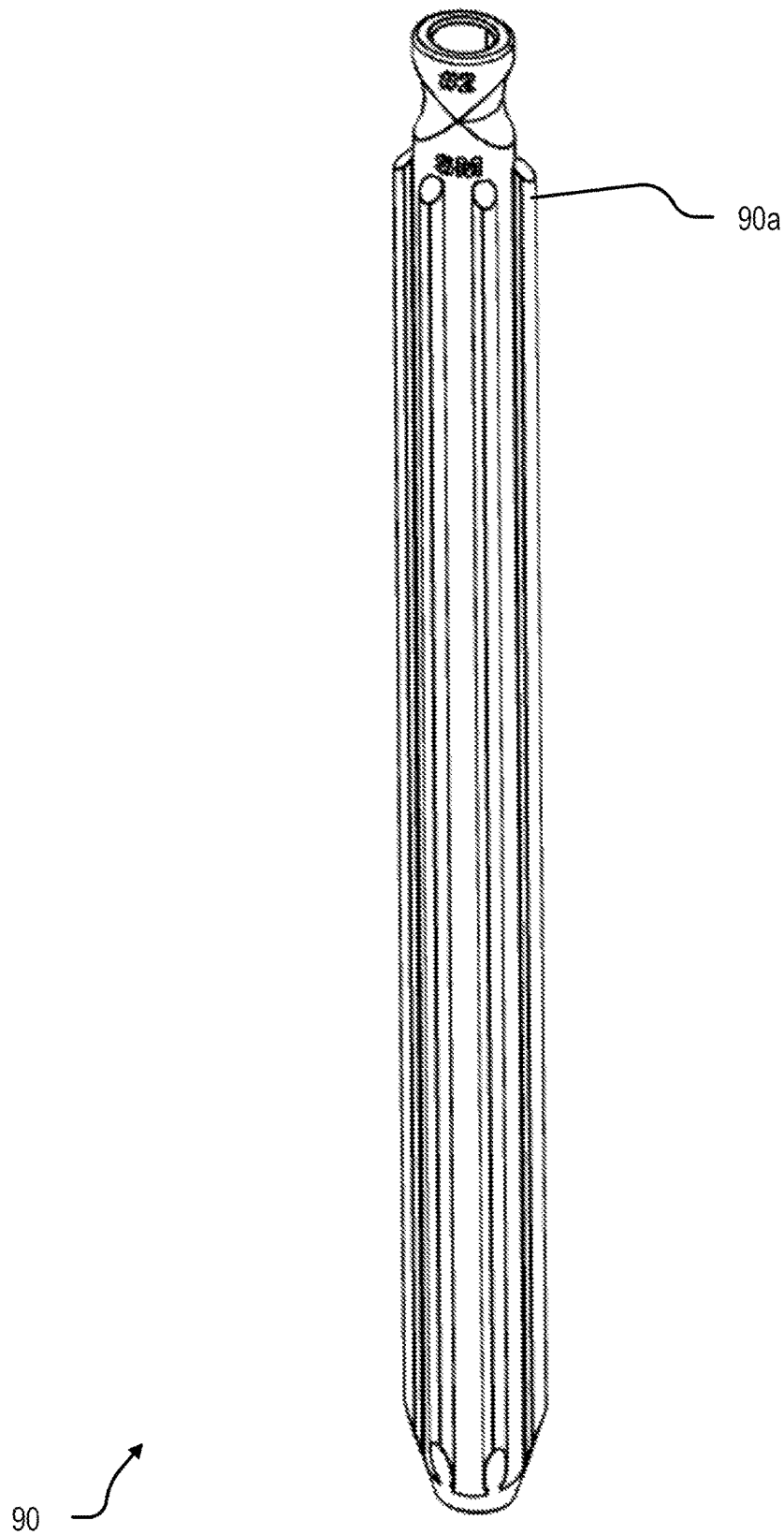


FIG. 79

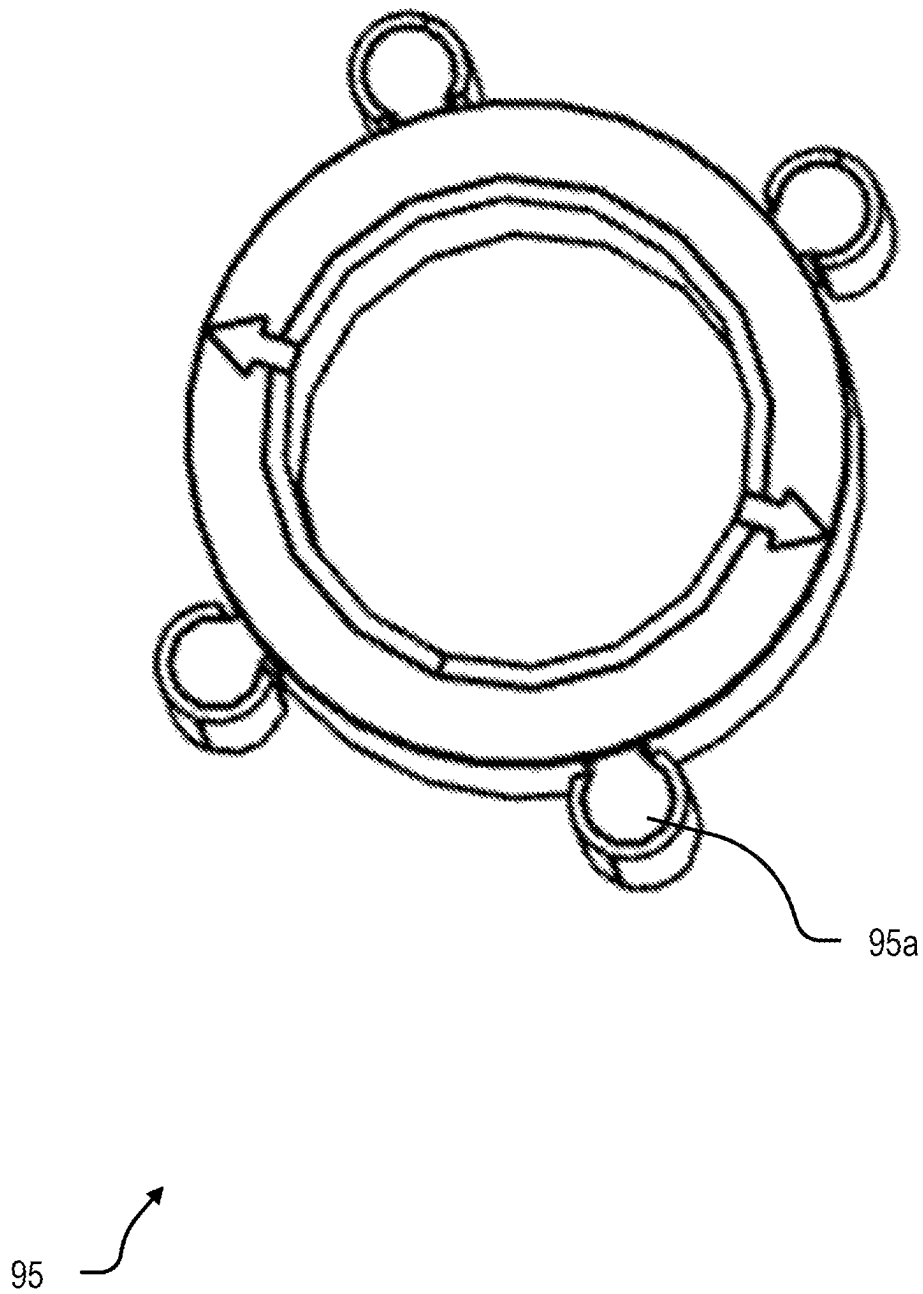


FIG. 80A

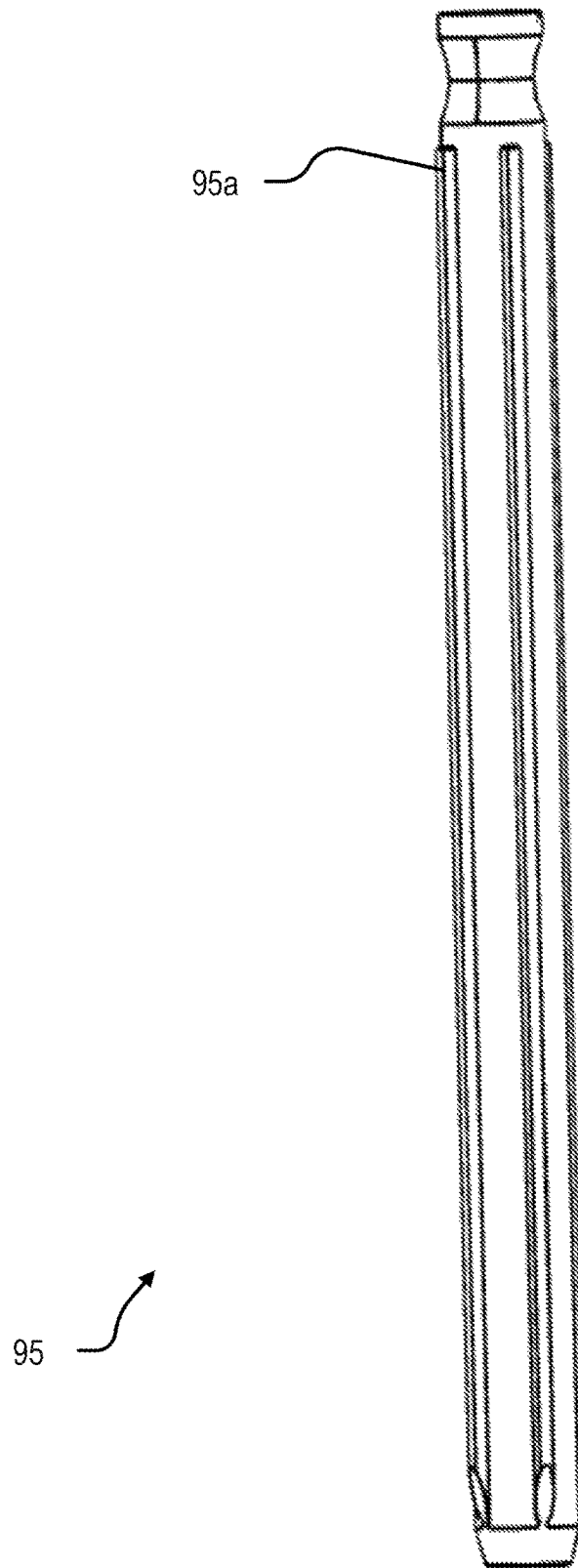


FIG. 80B

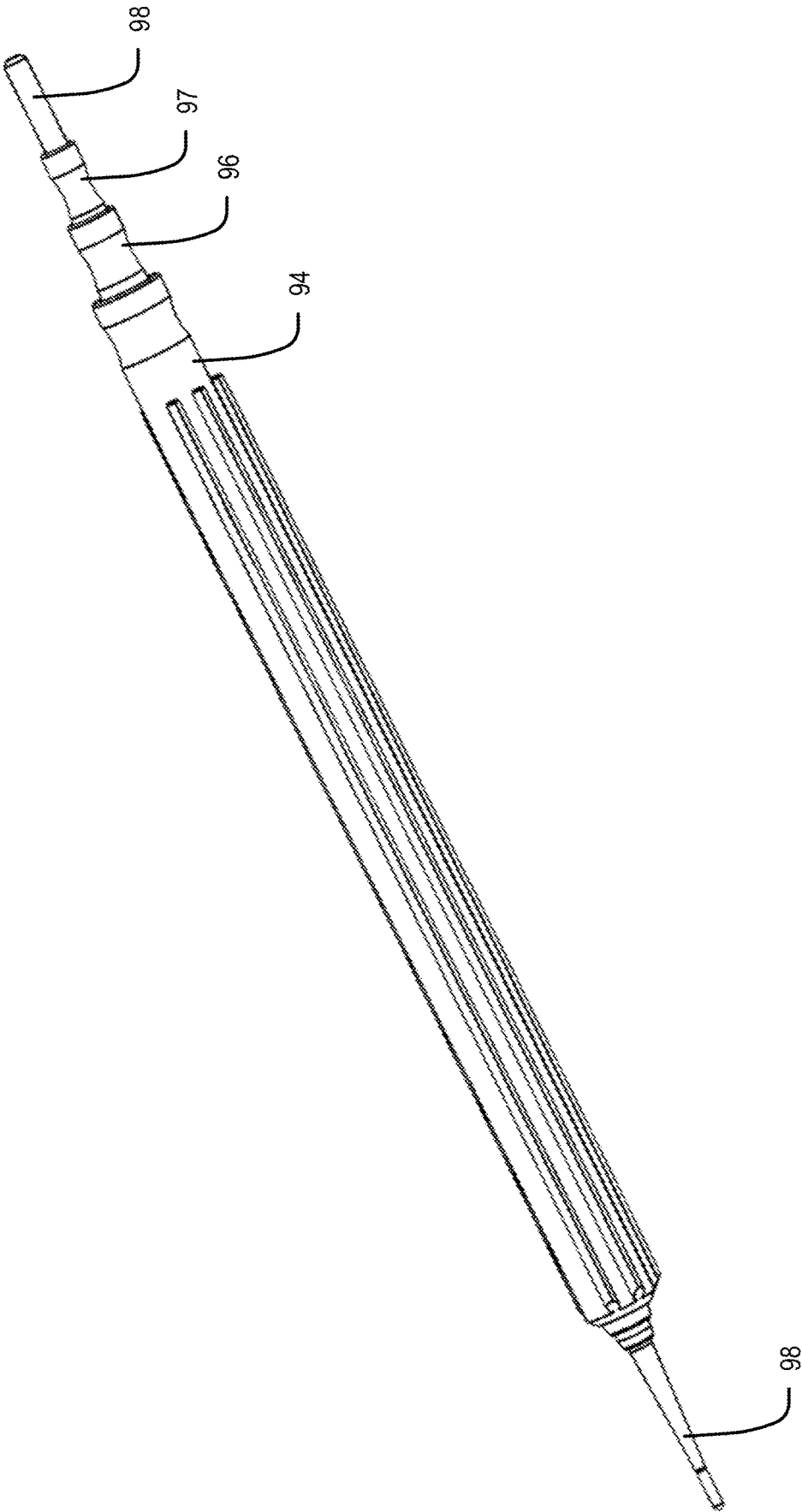


FIG. 80C

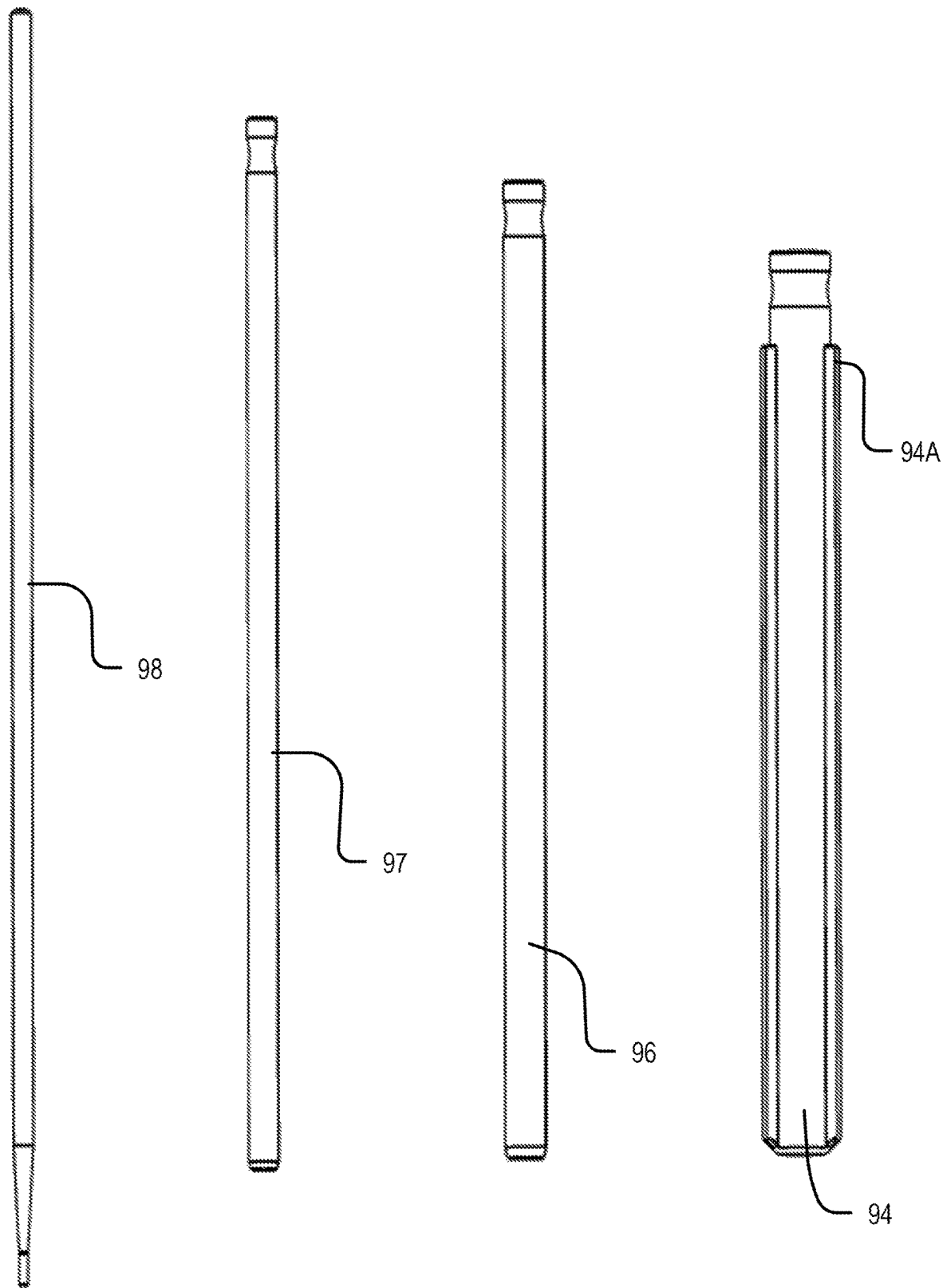


FIG. 80D

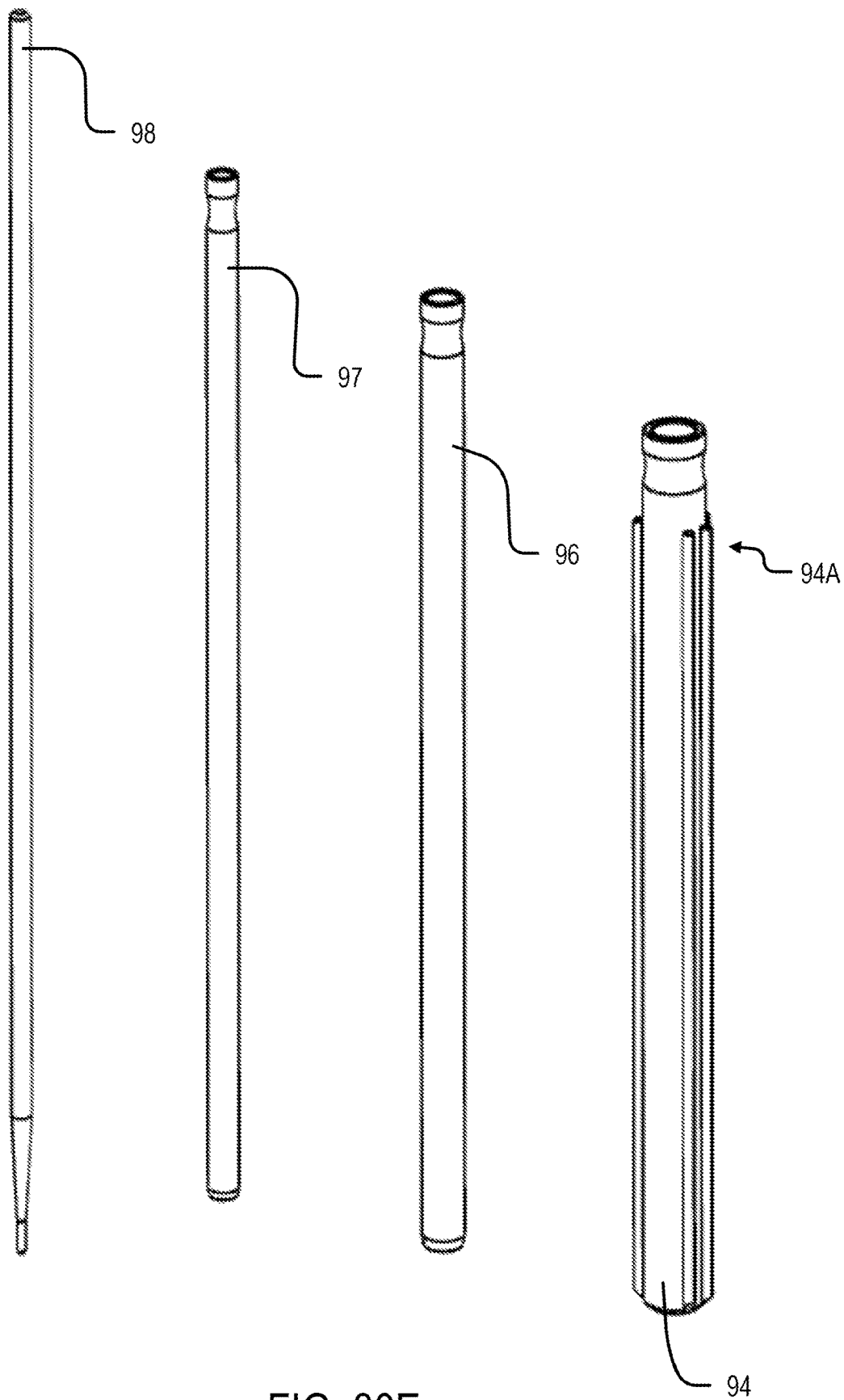
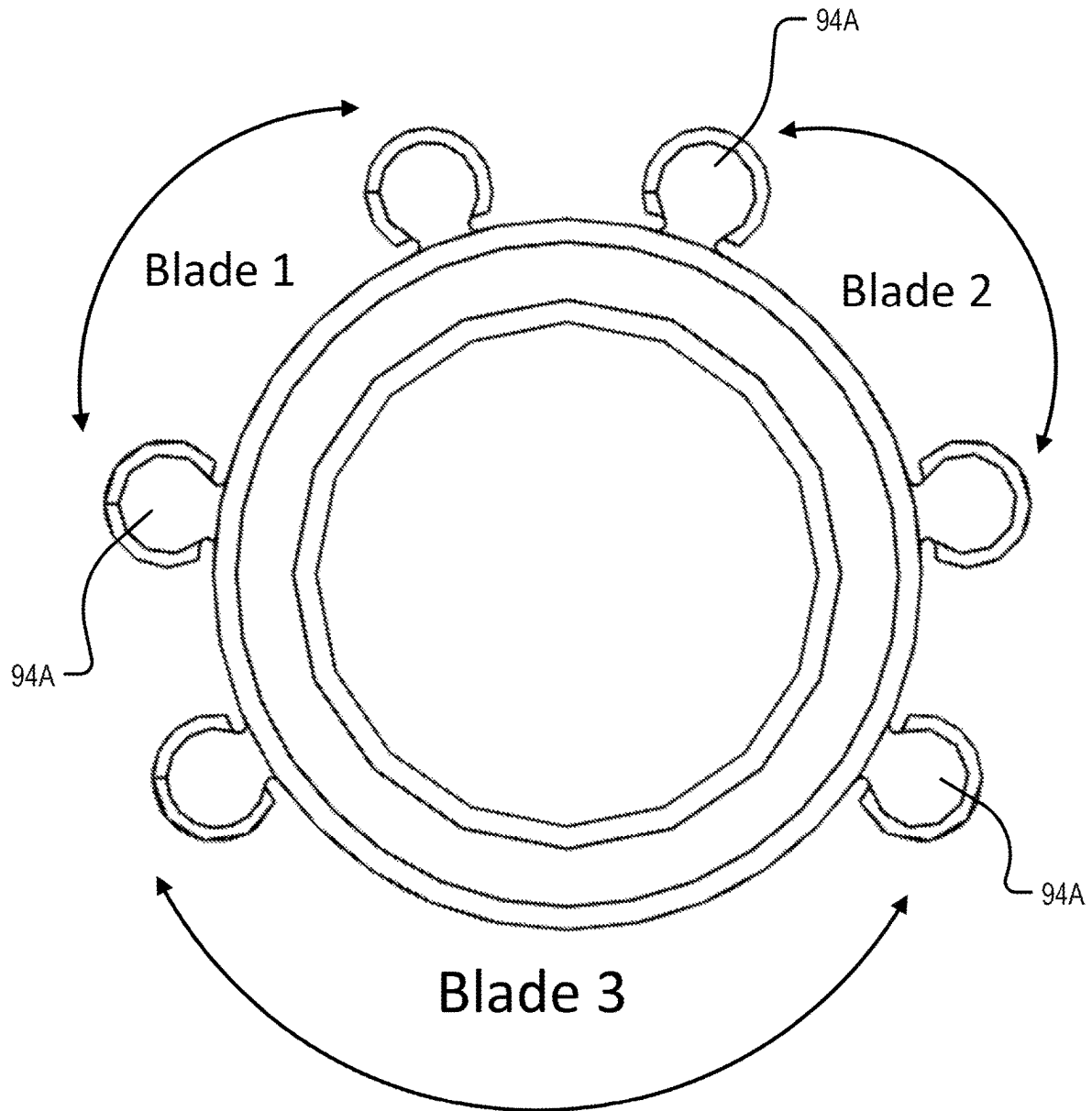


FIG. 80E



94
FIG. 81

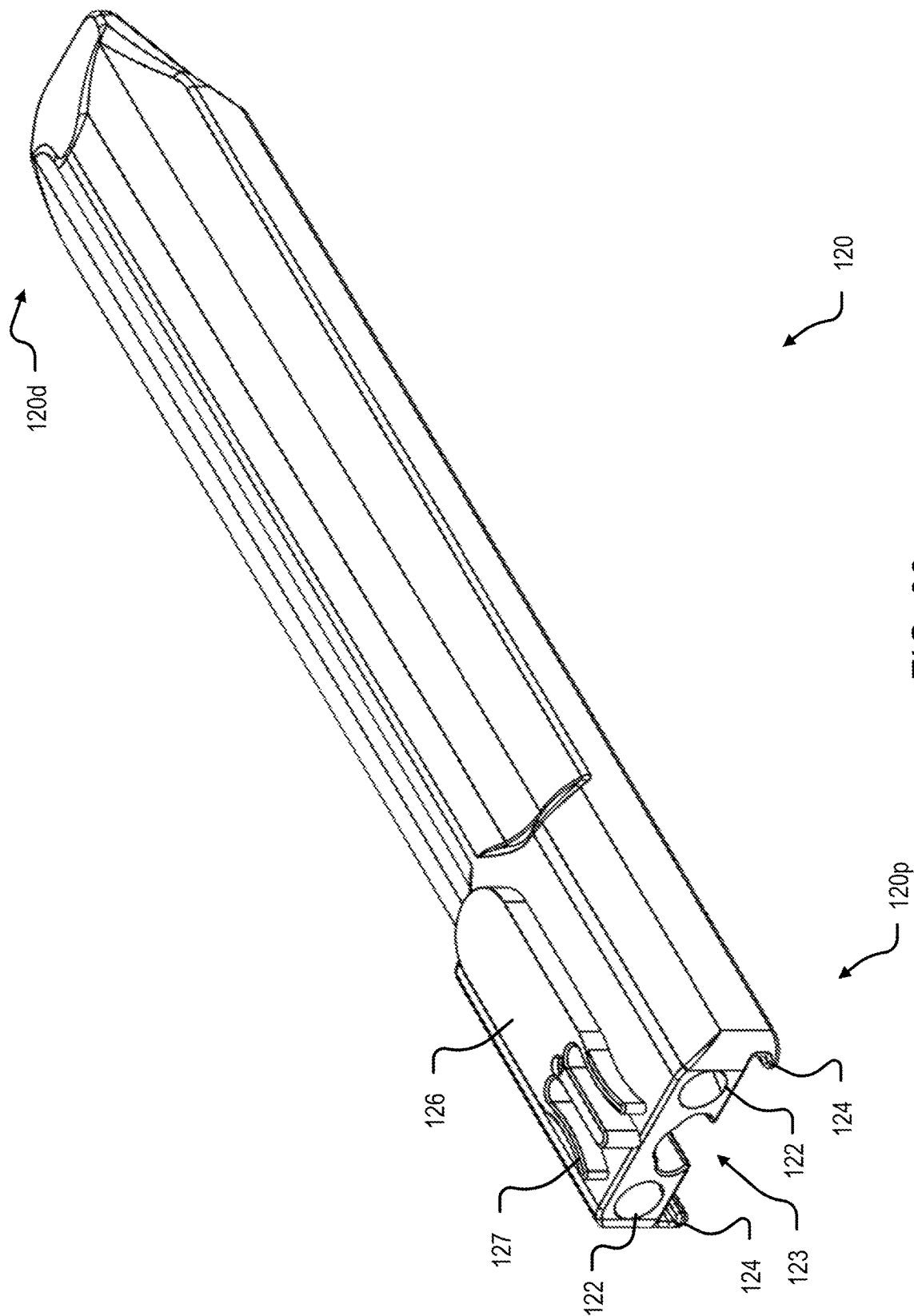


FIG. 82

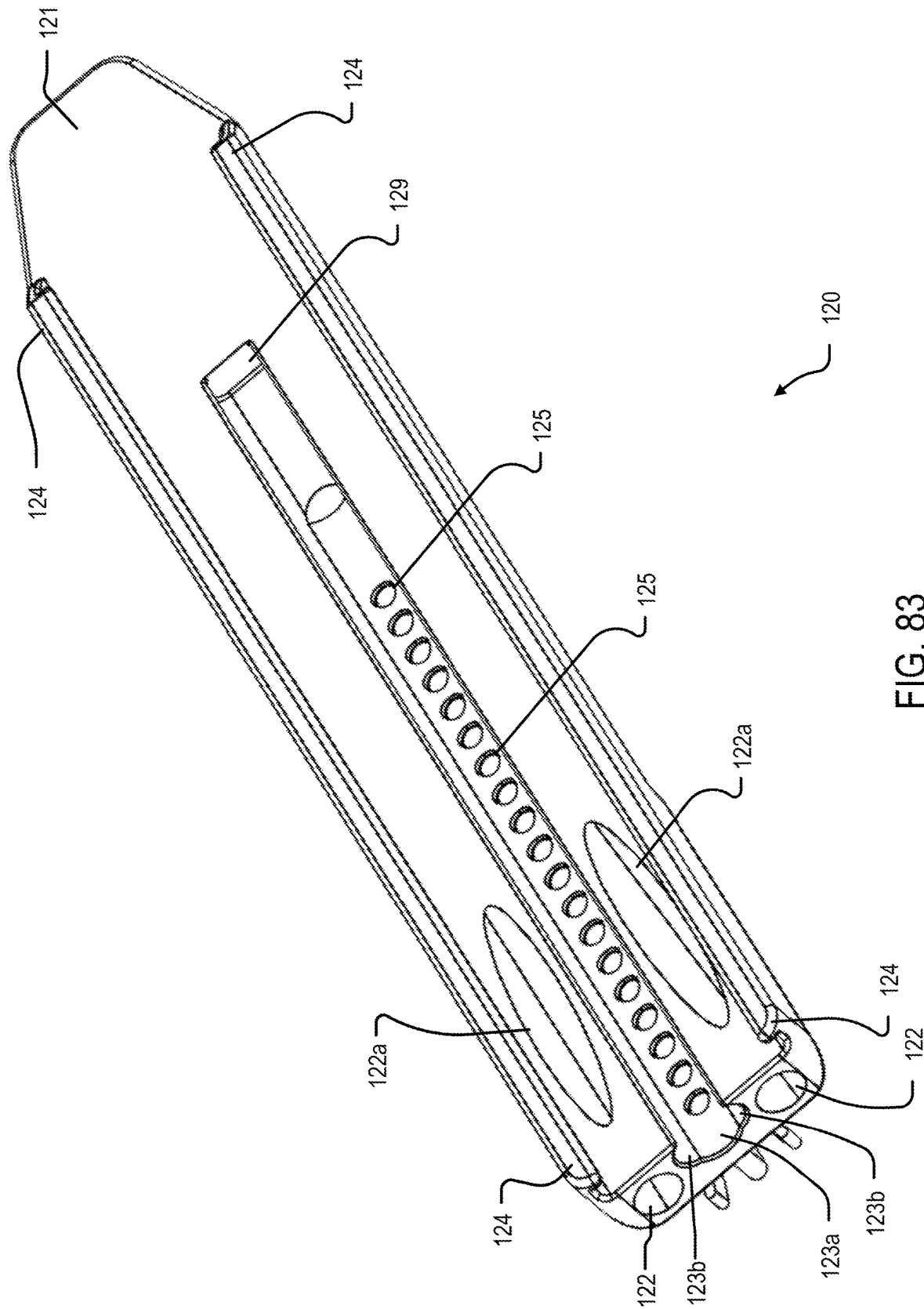


FIG. 83

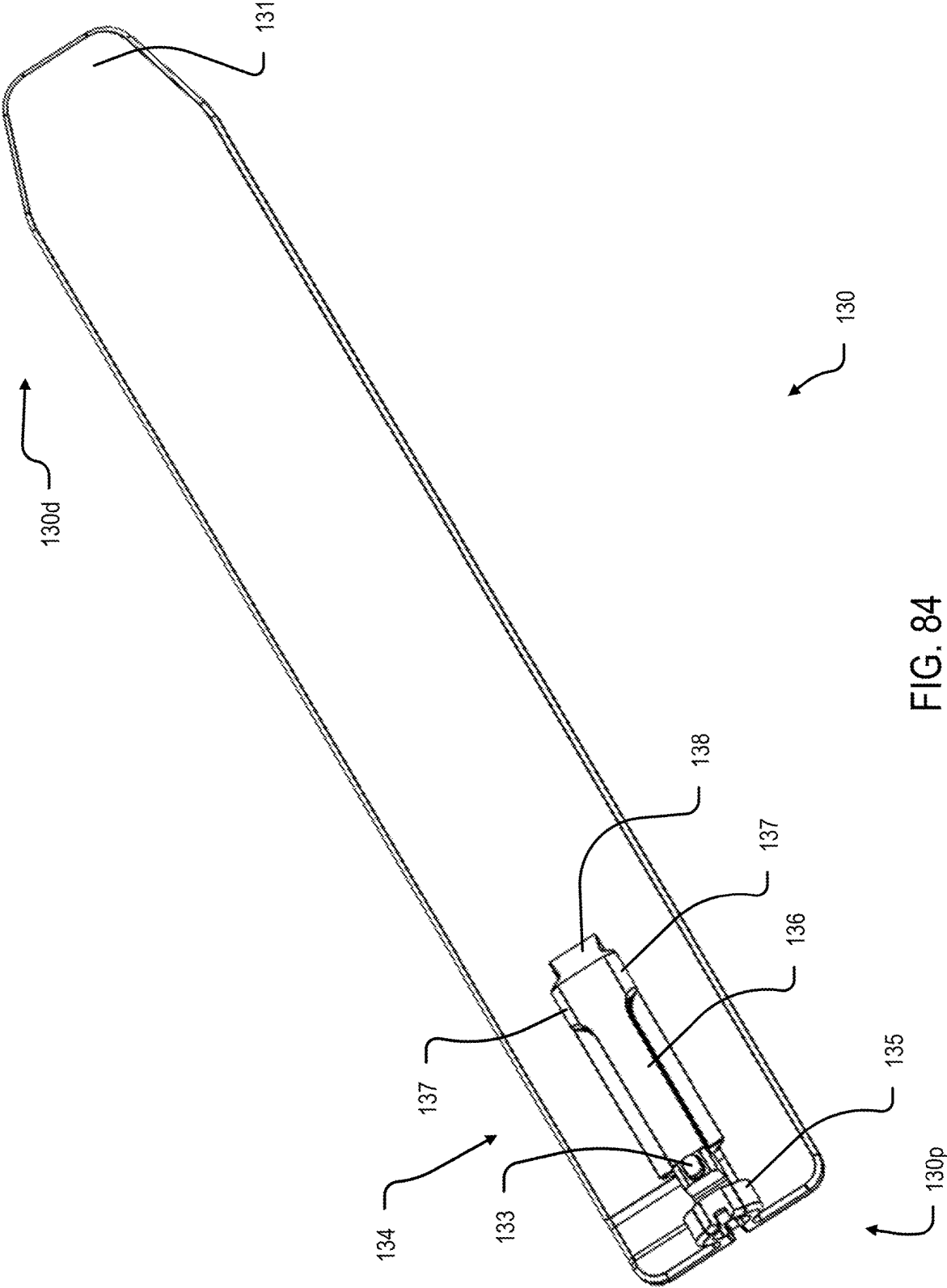


FIG. 84

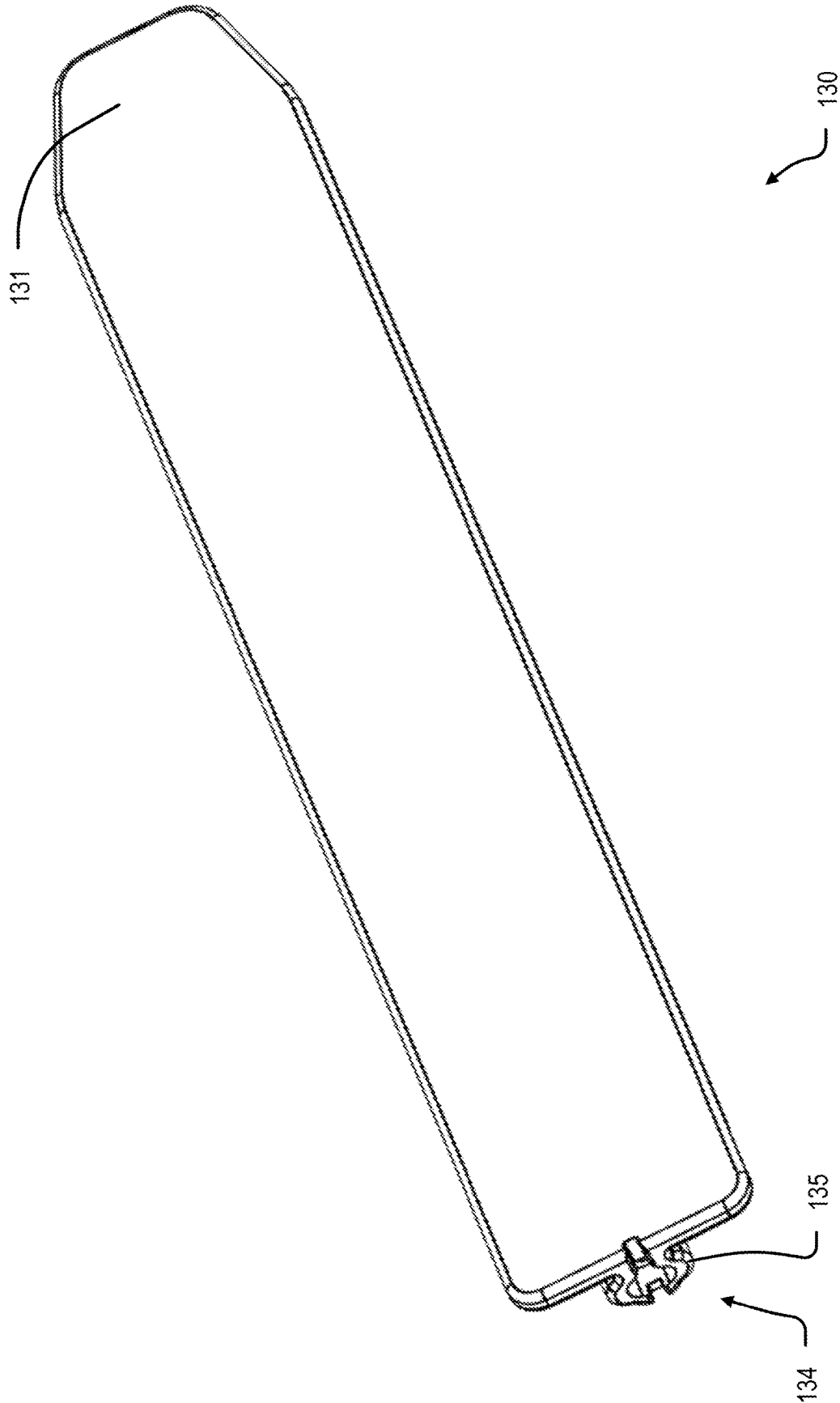


FIG. 85

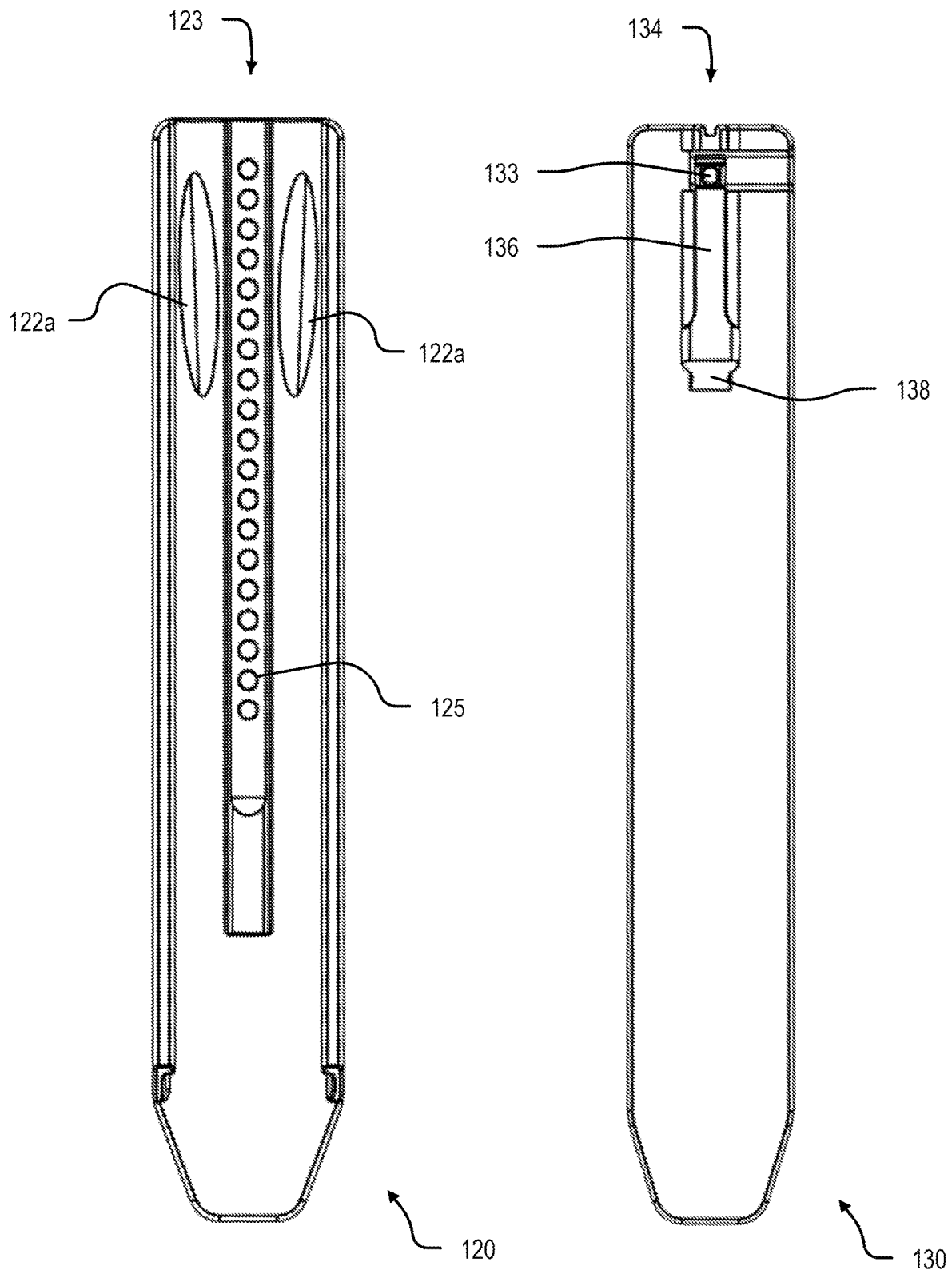


FIG. 86

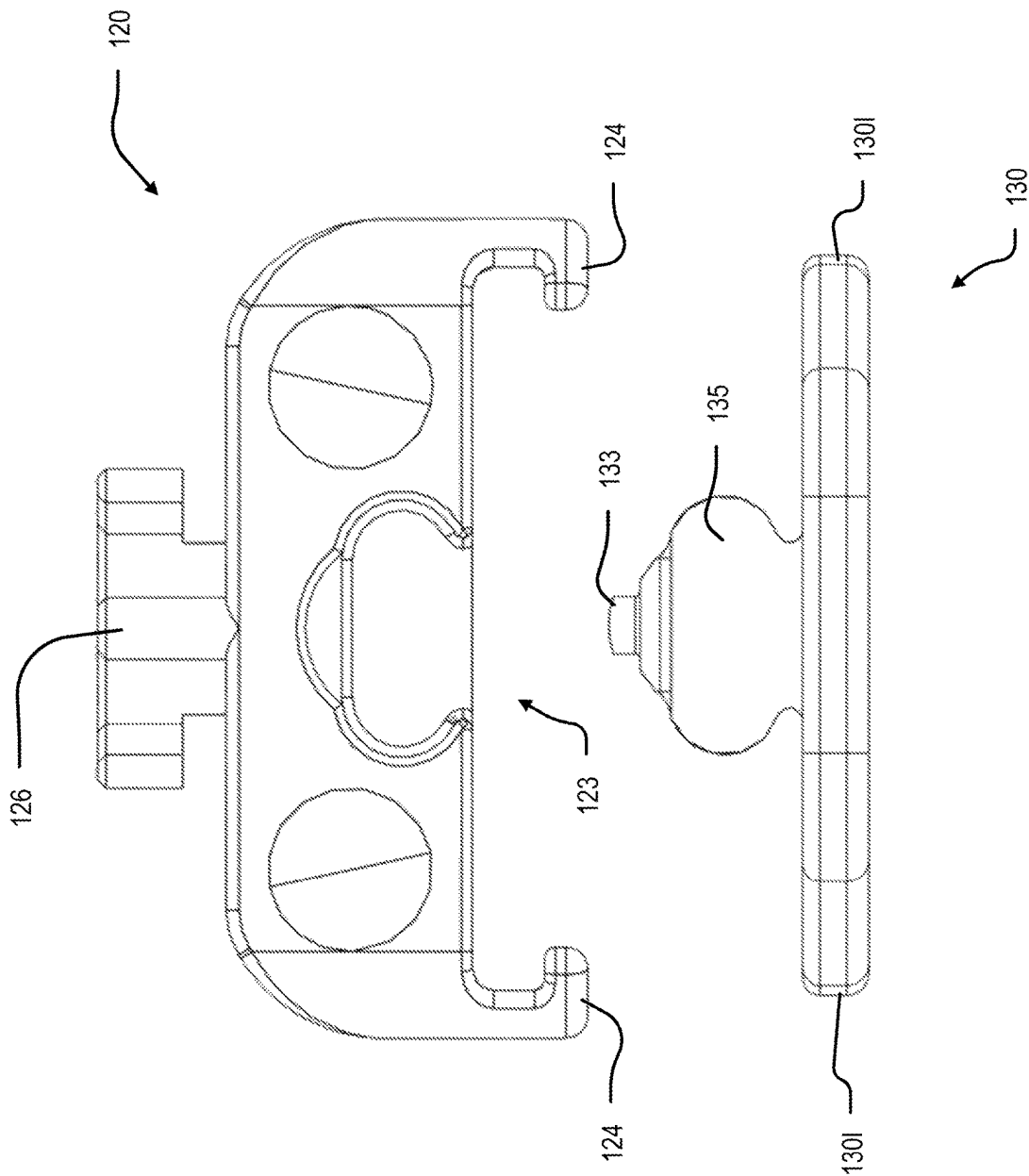


FIG. 87

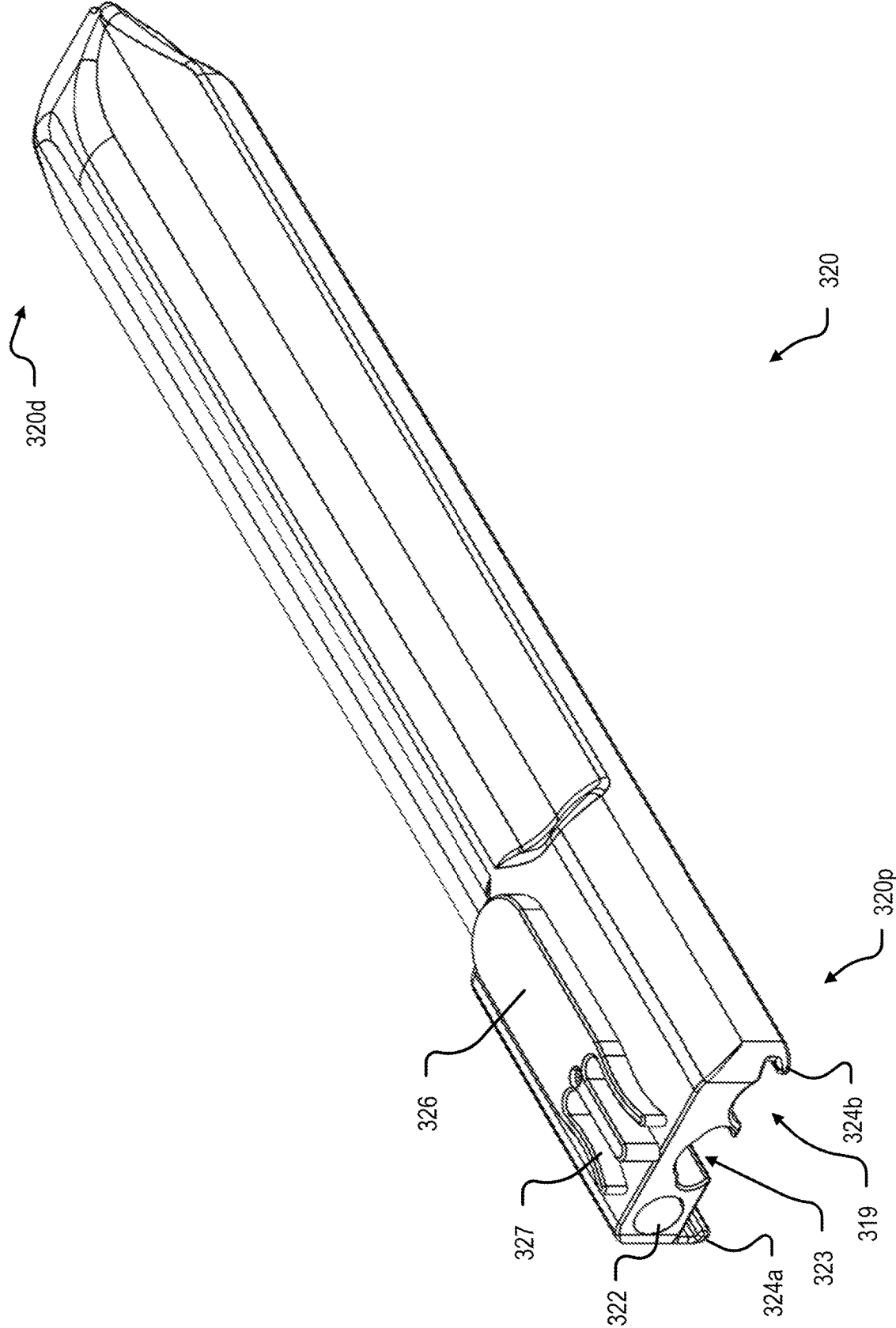


FIG. 88

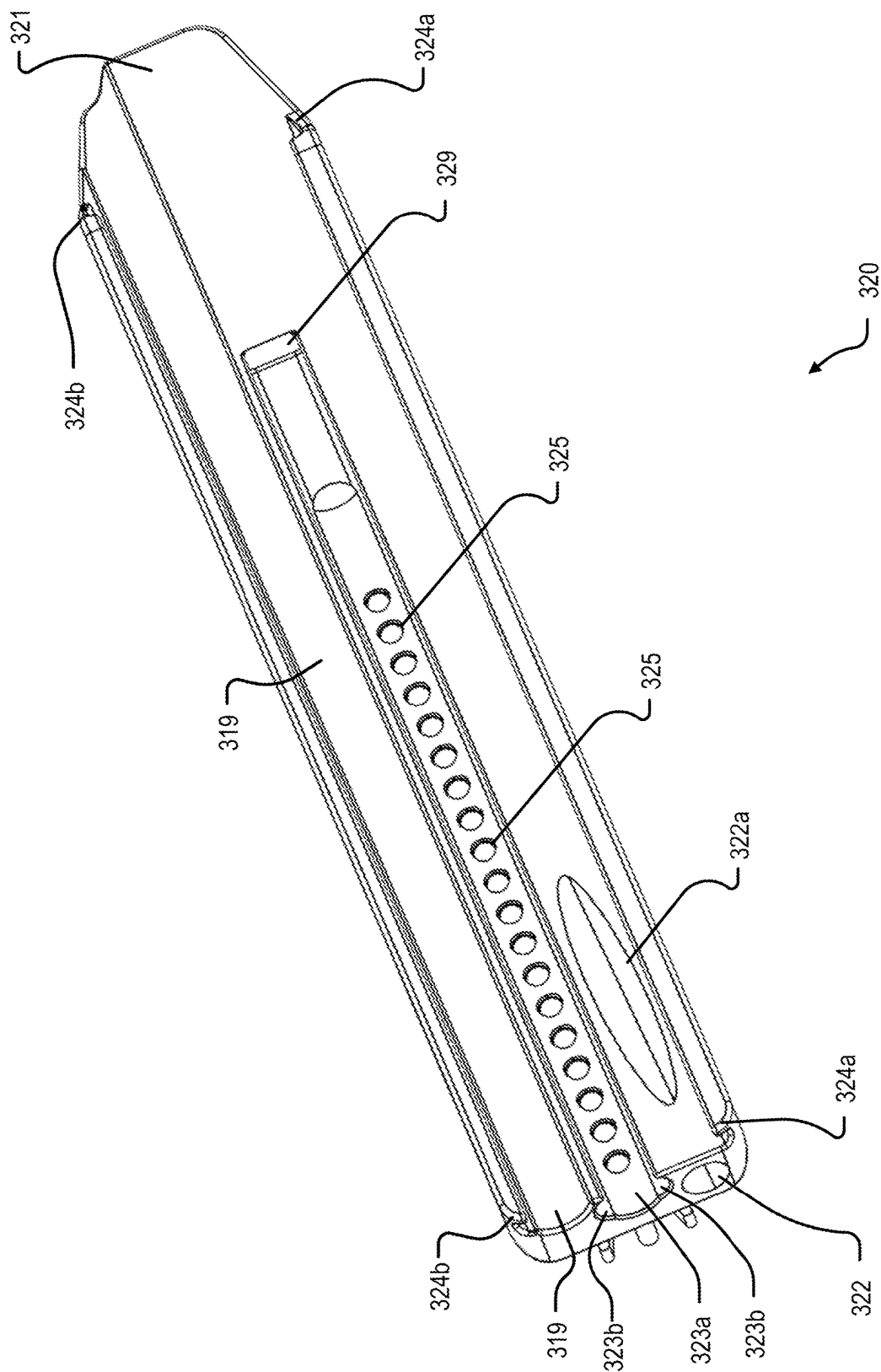
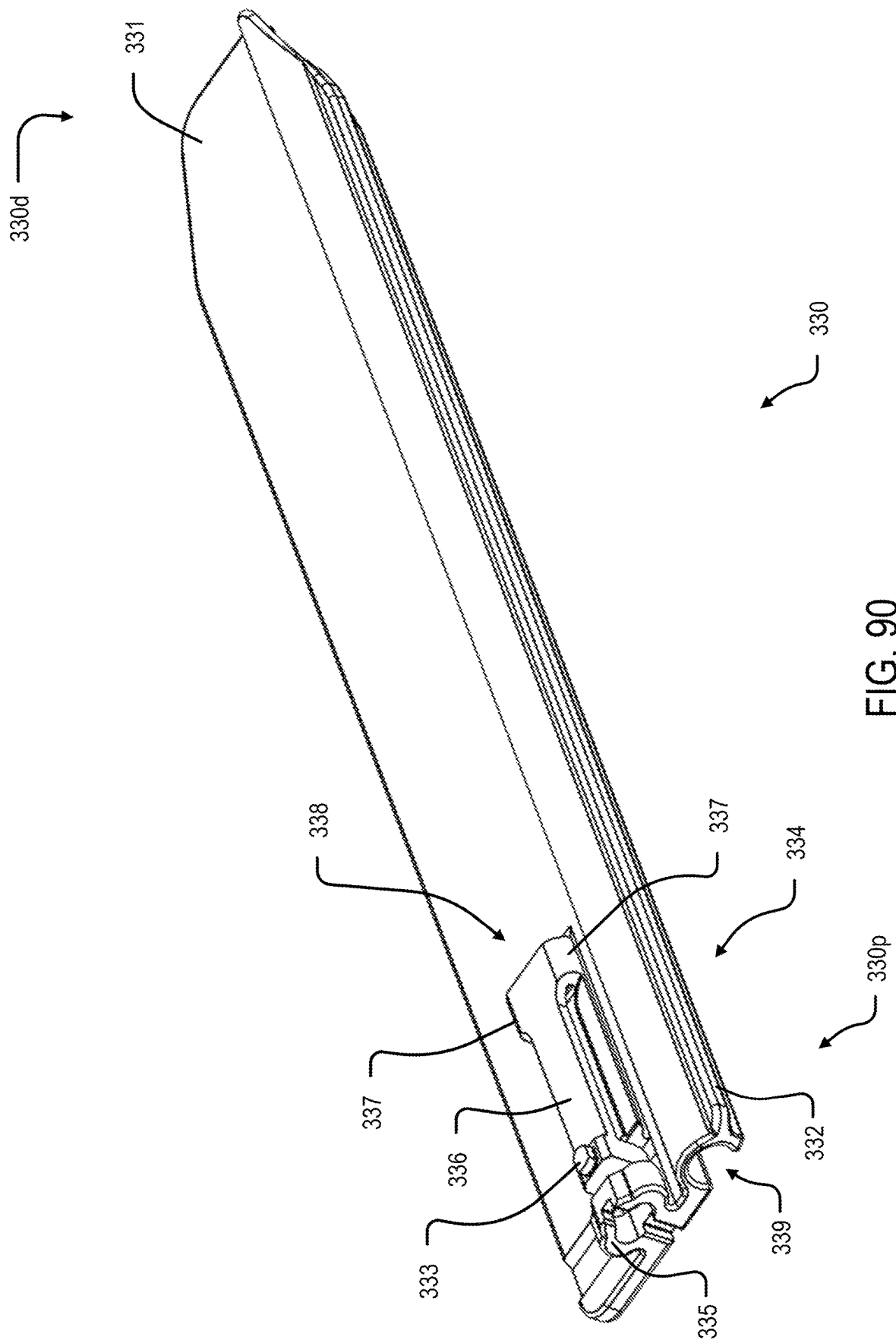


FIG. 89



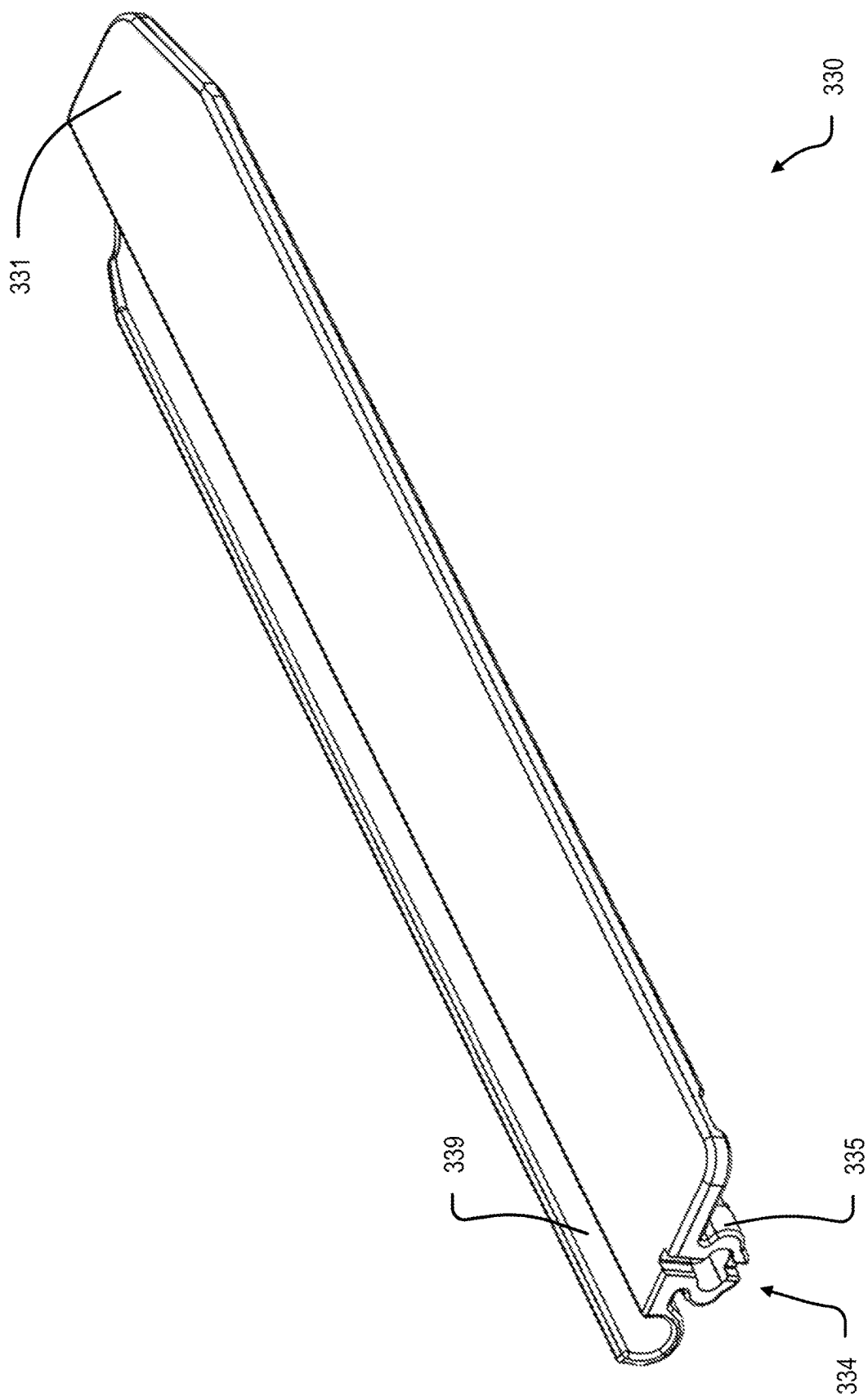


FIG. 91

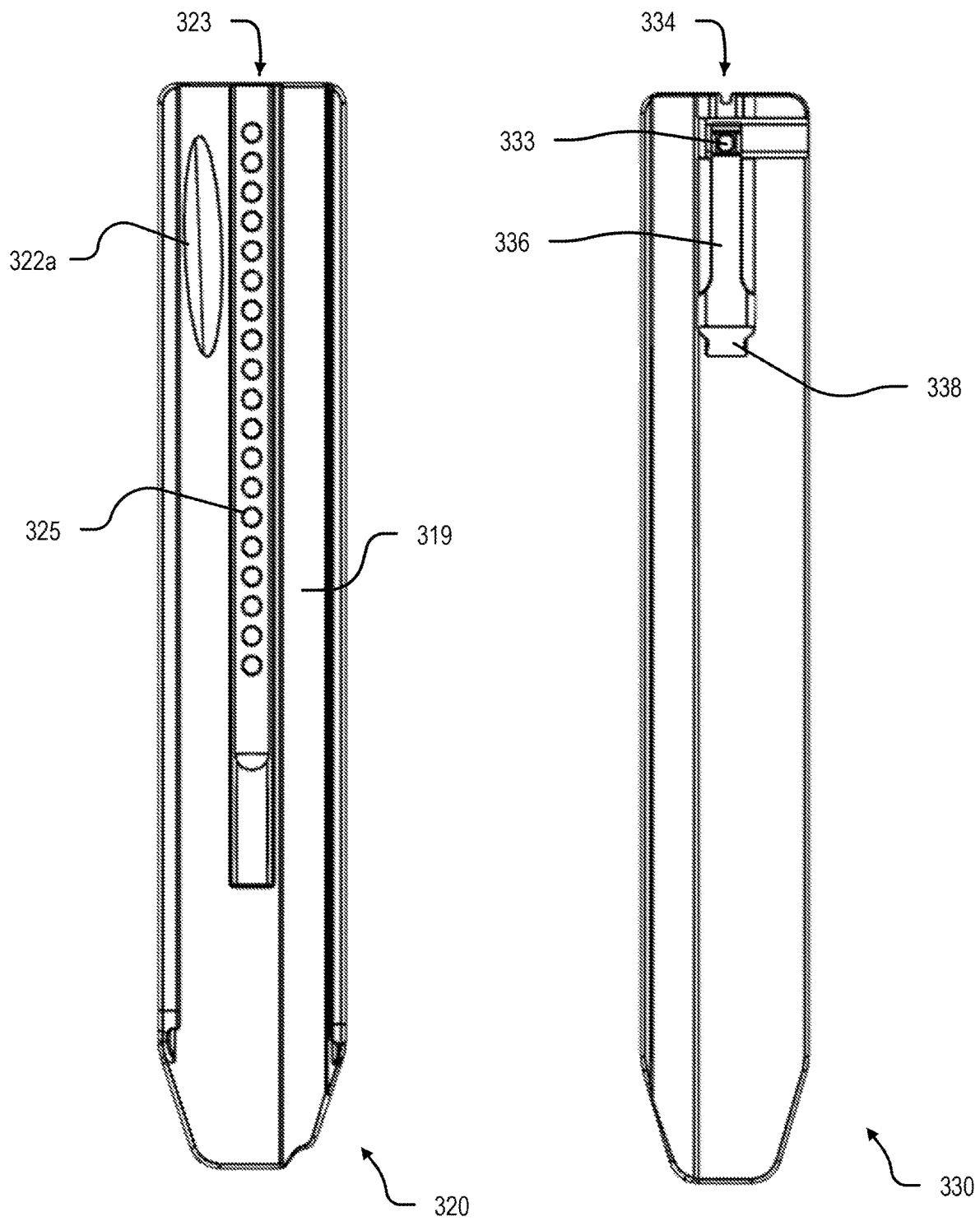


FIG. 92

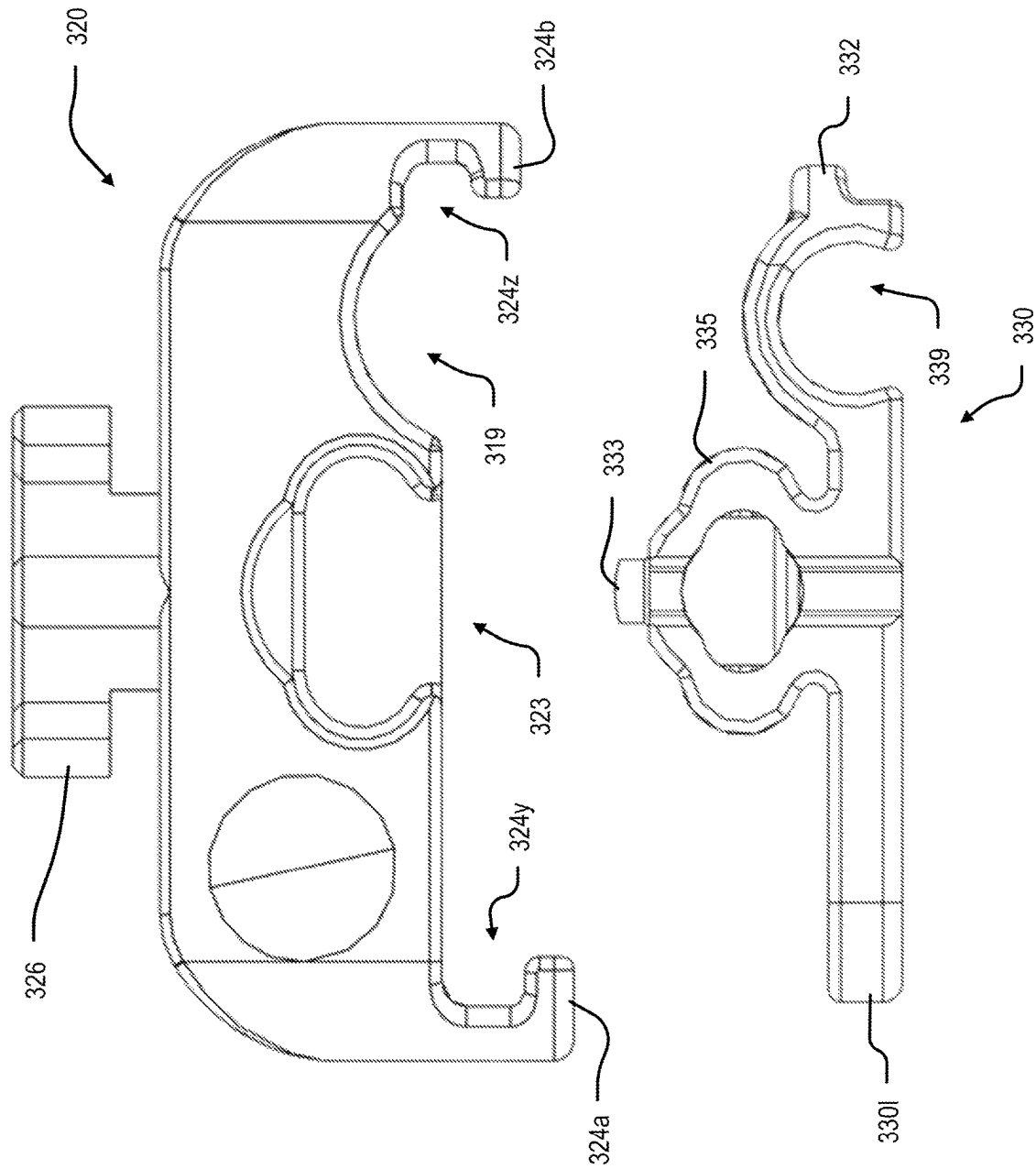
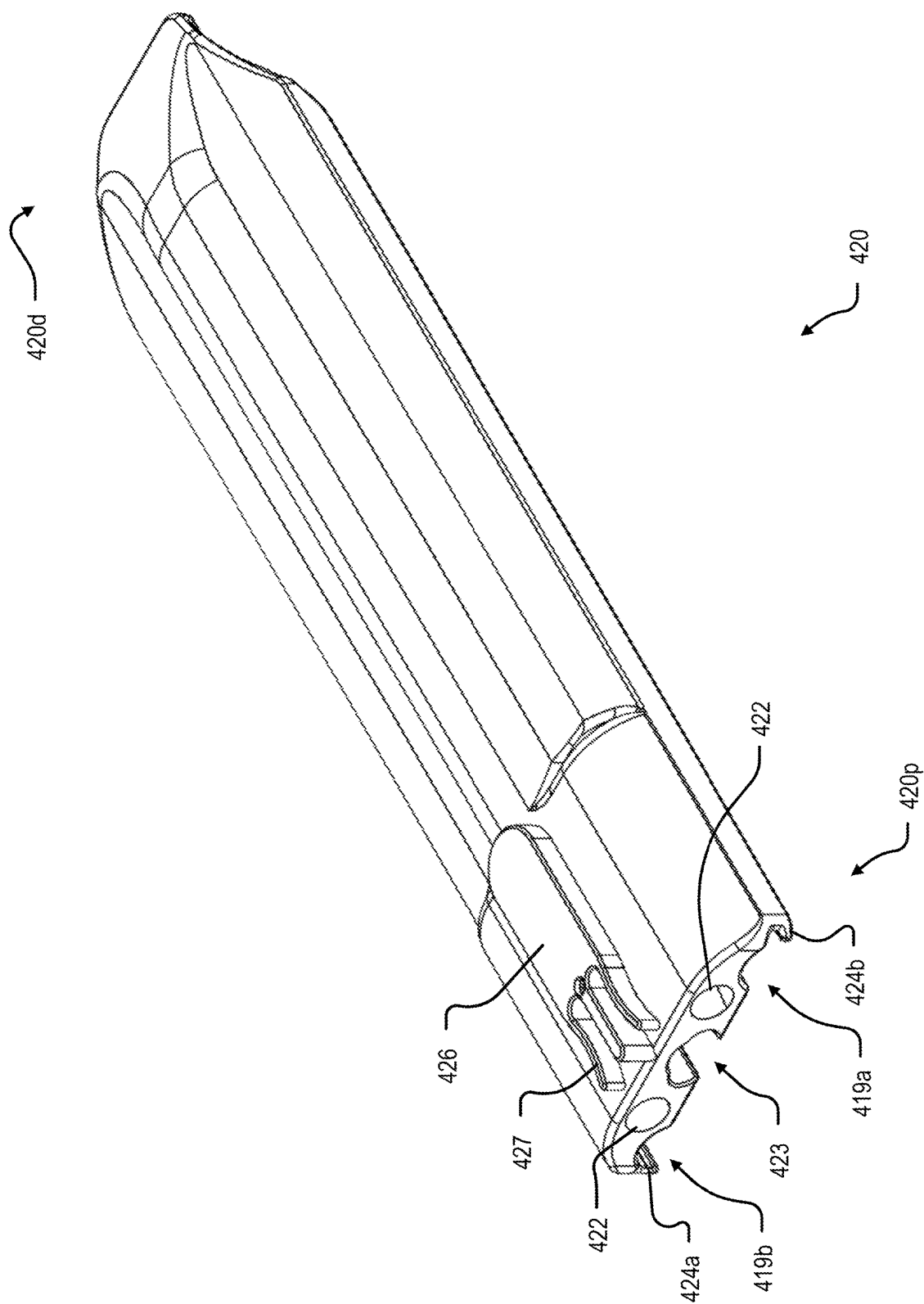


FIG. 93



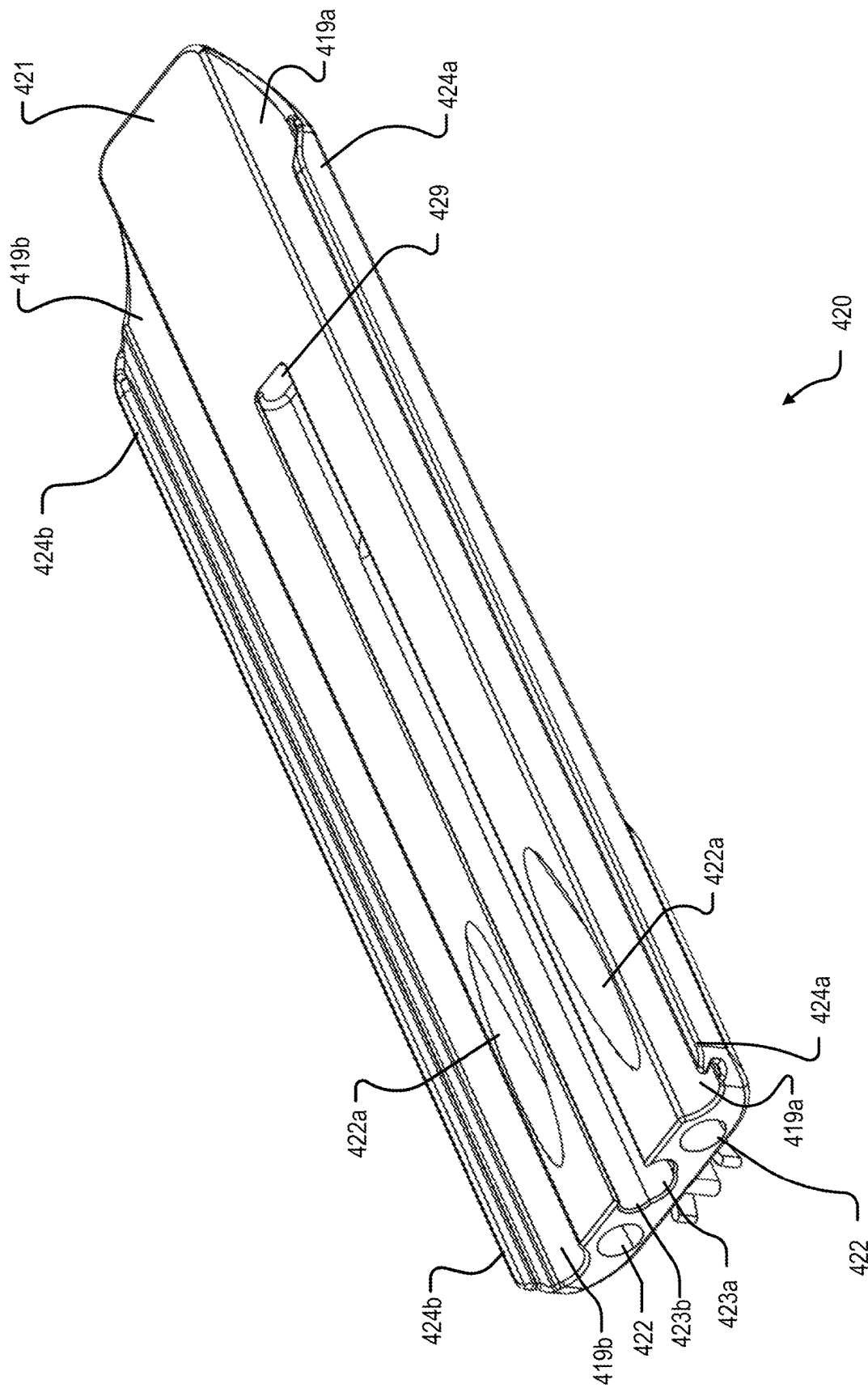
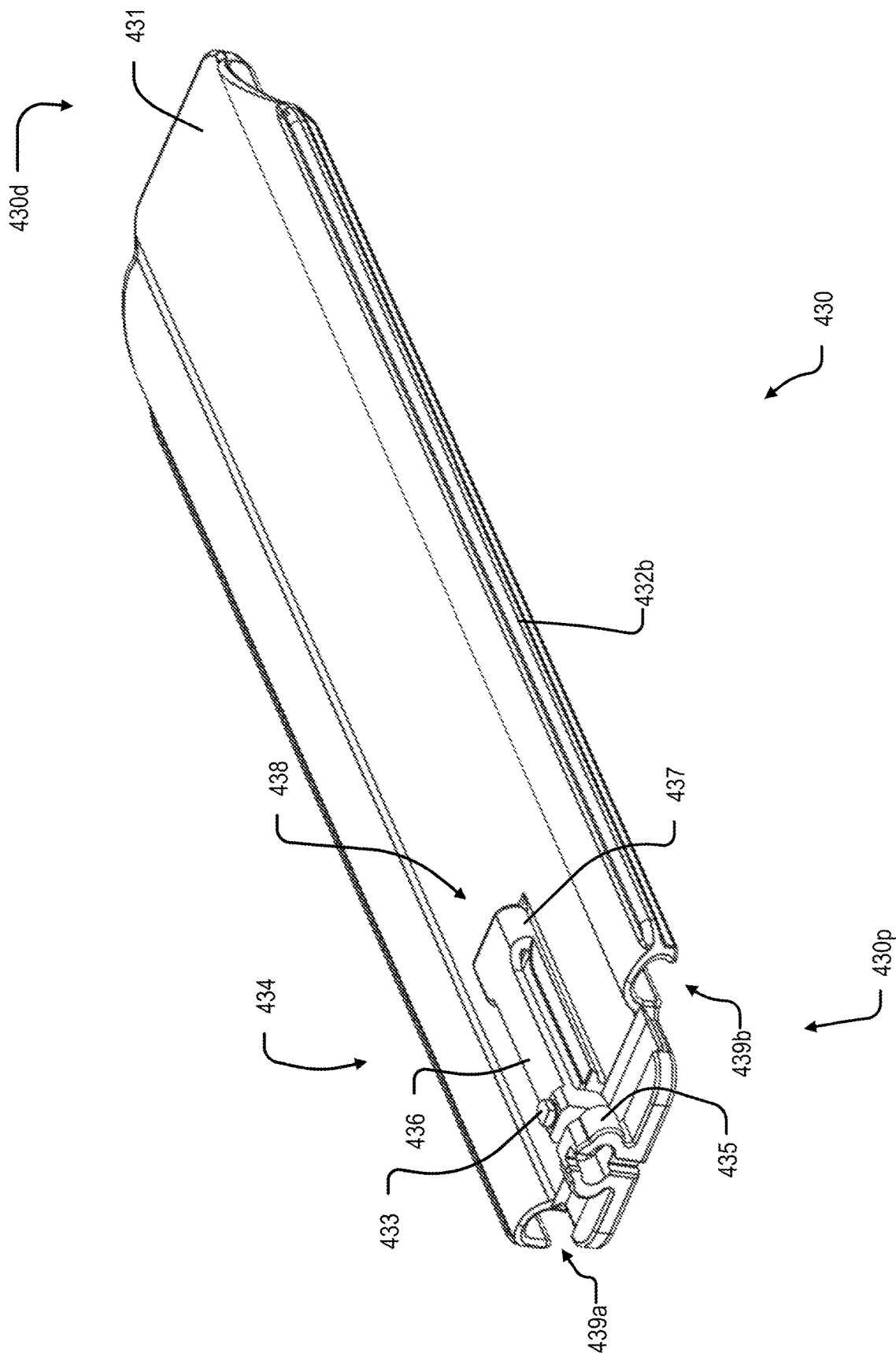


FIG. 95



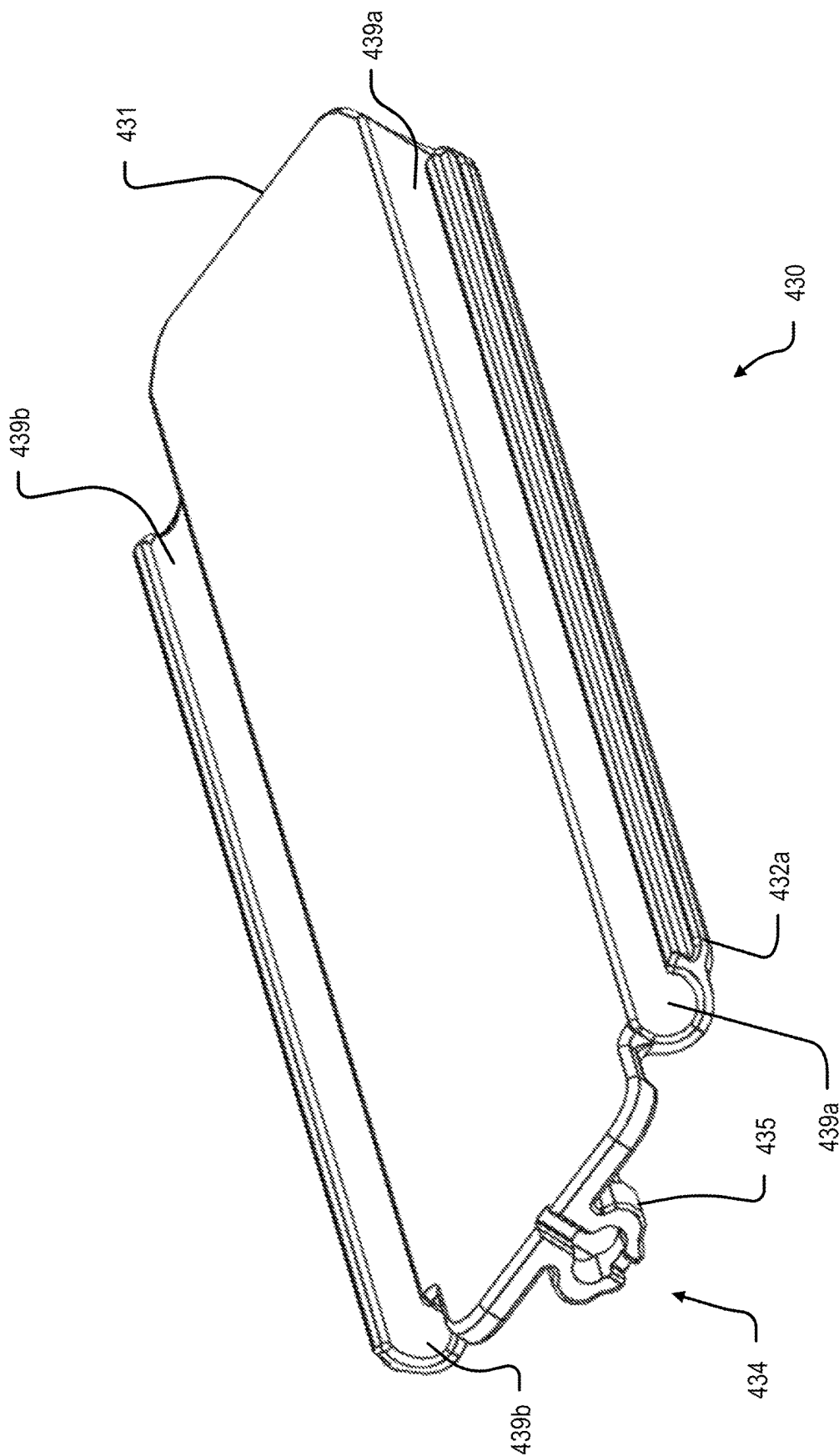


FIG. 97

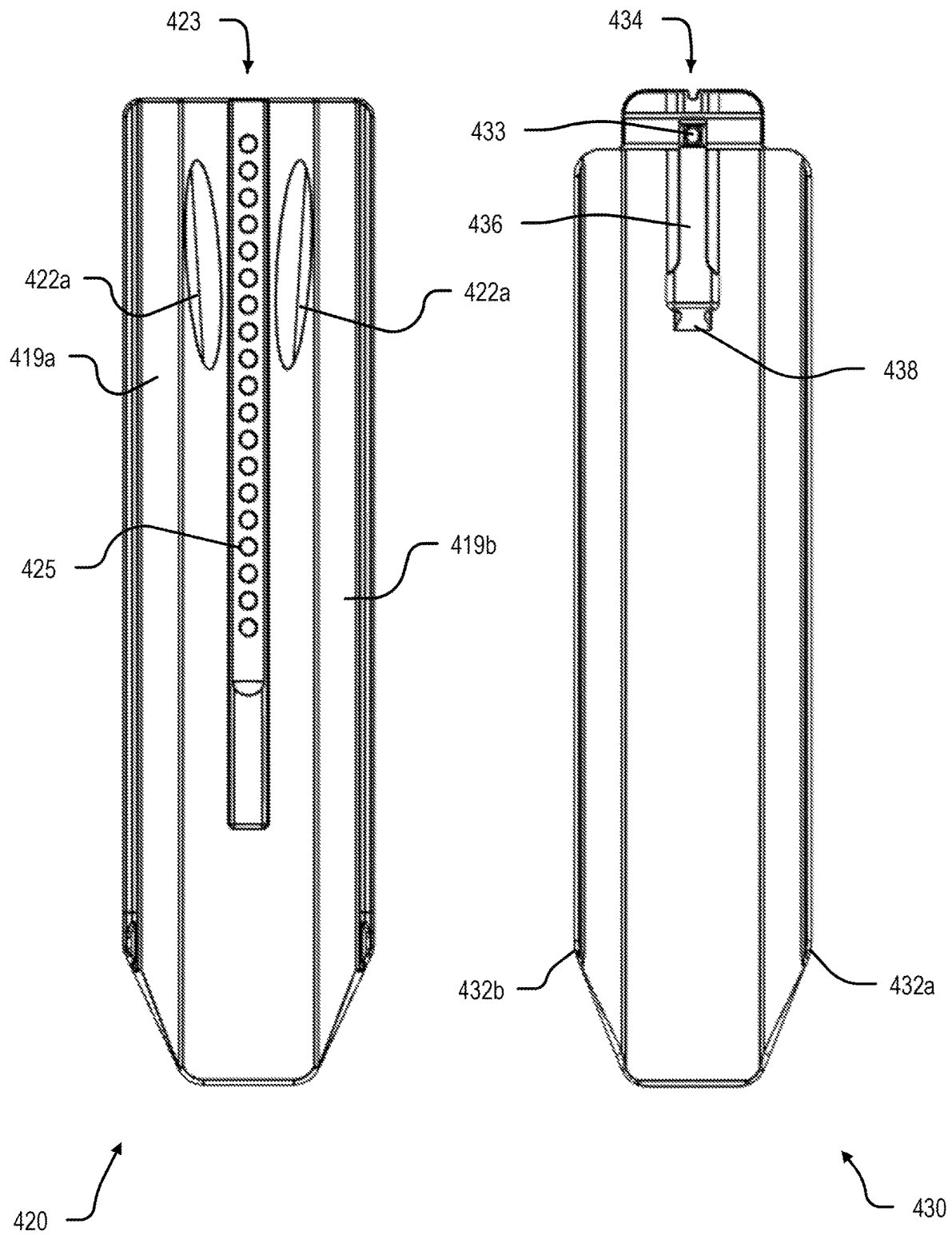
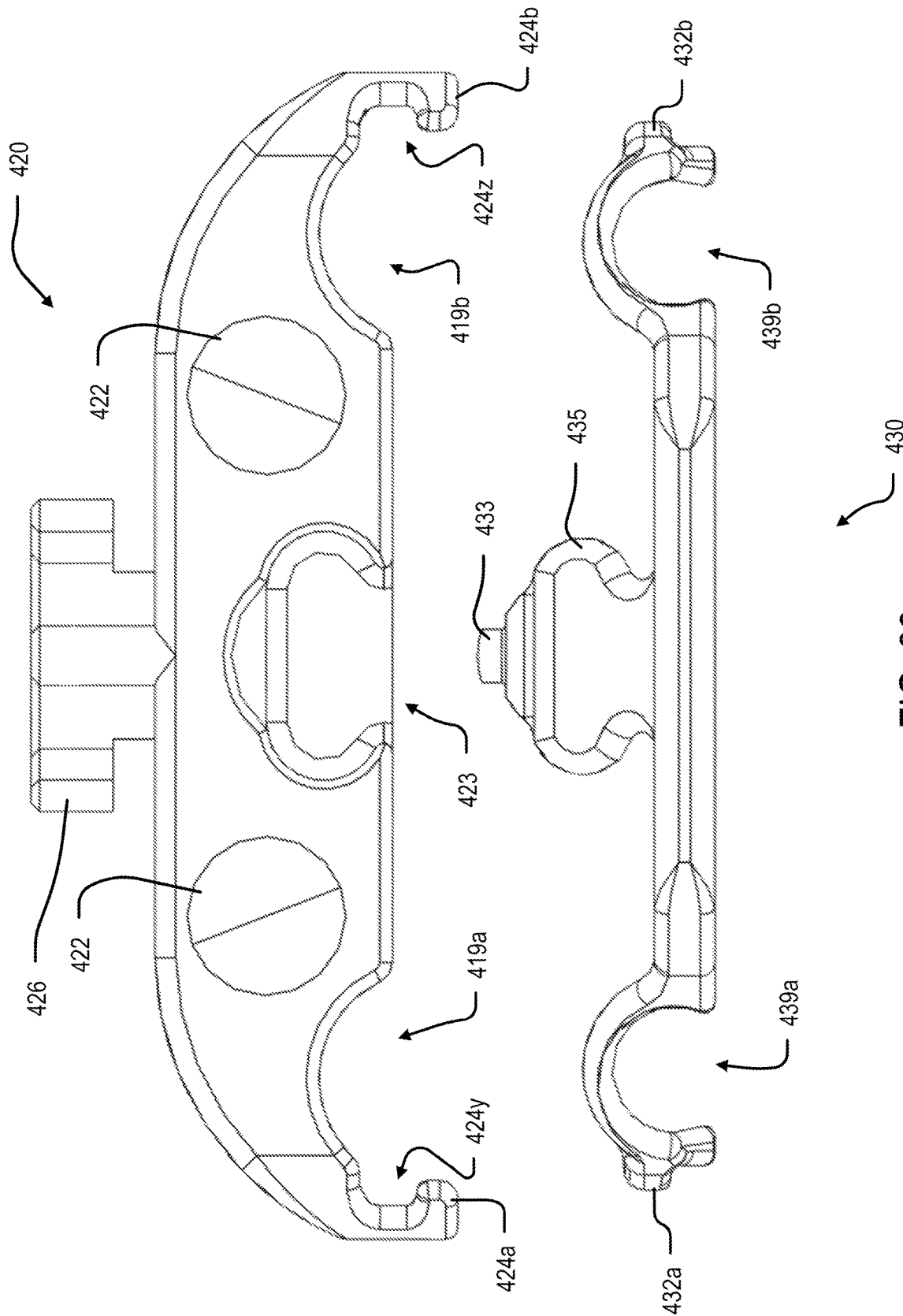


FIG. 98



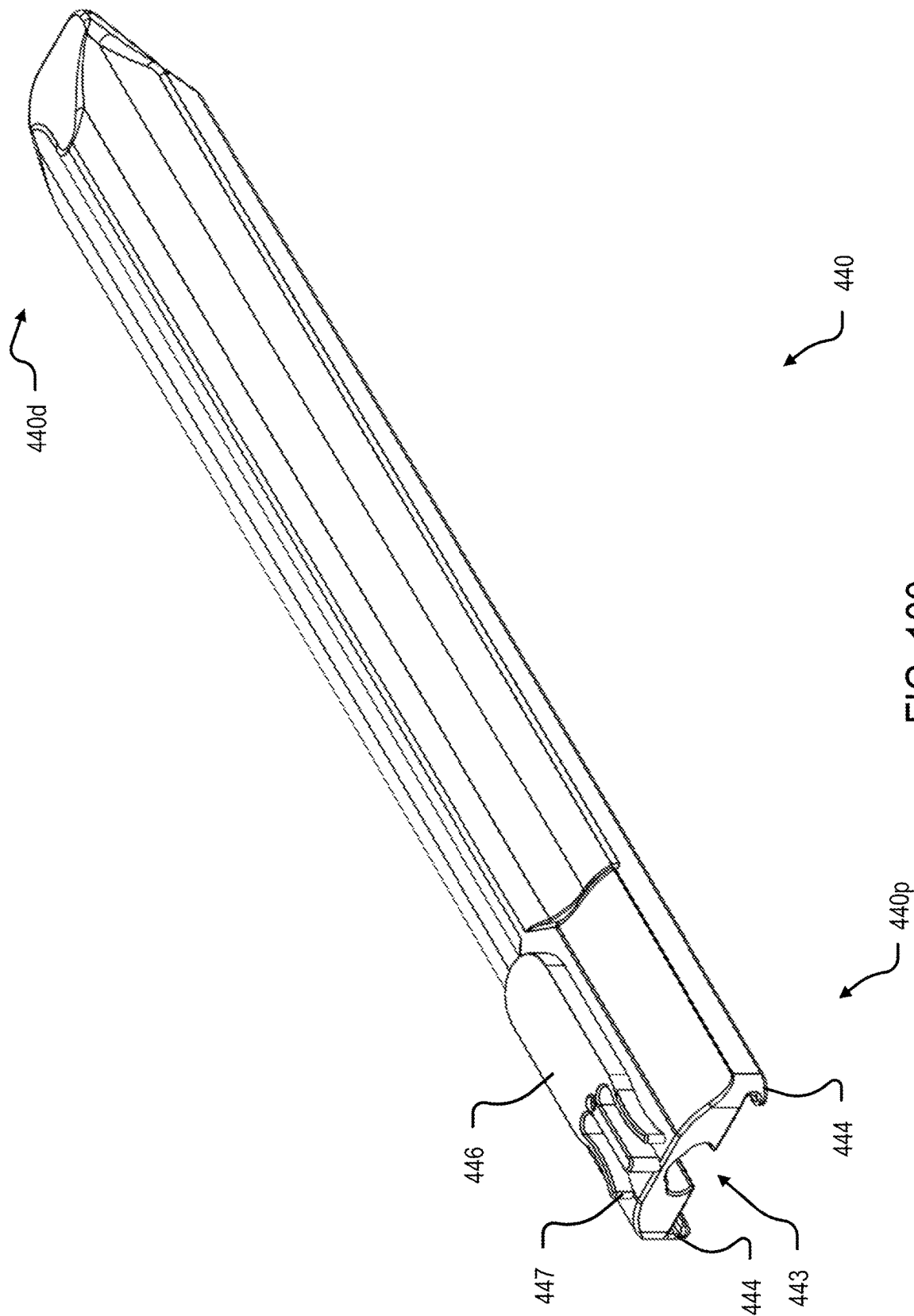


FIG. 100

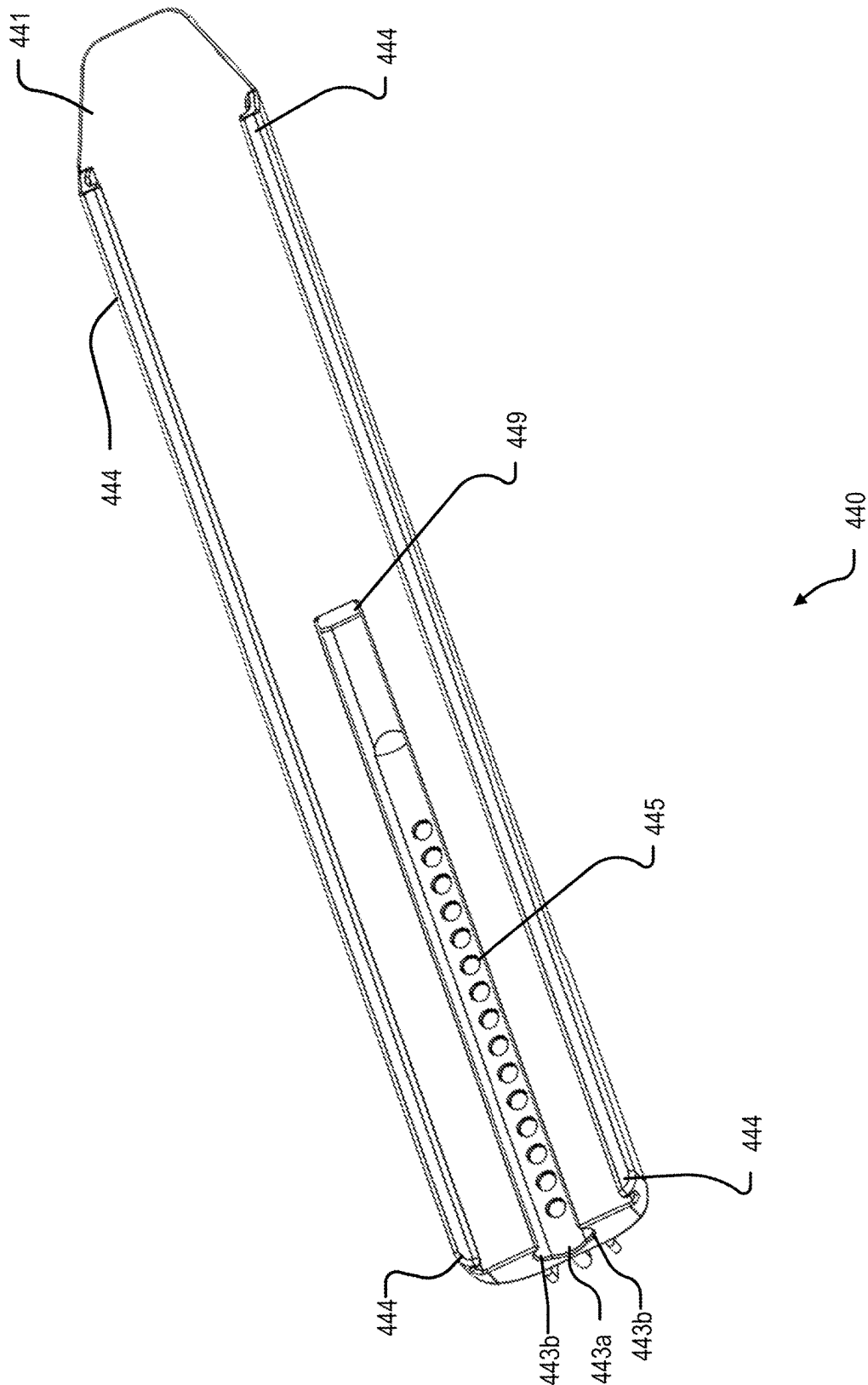


FIG. 101

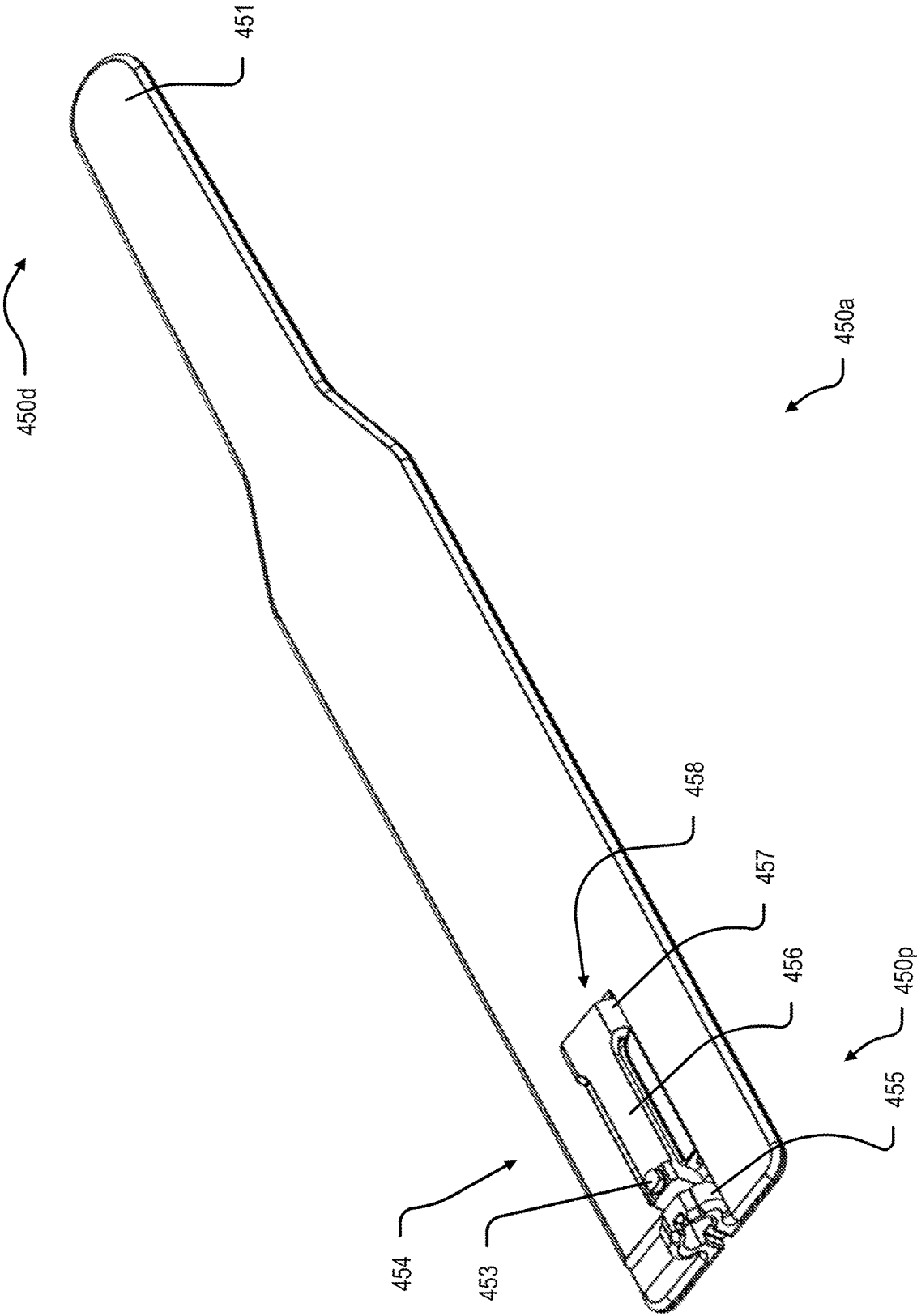
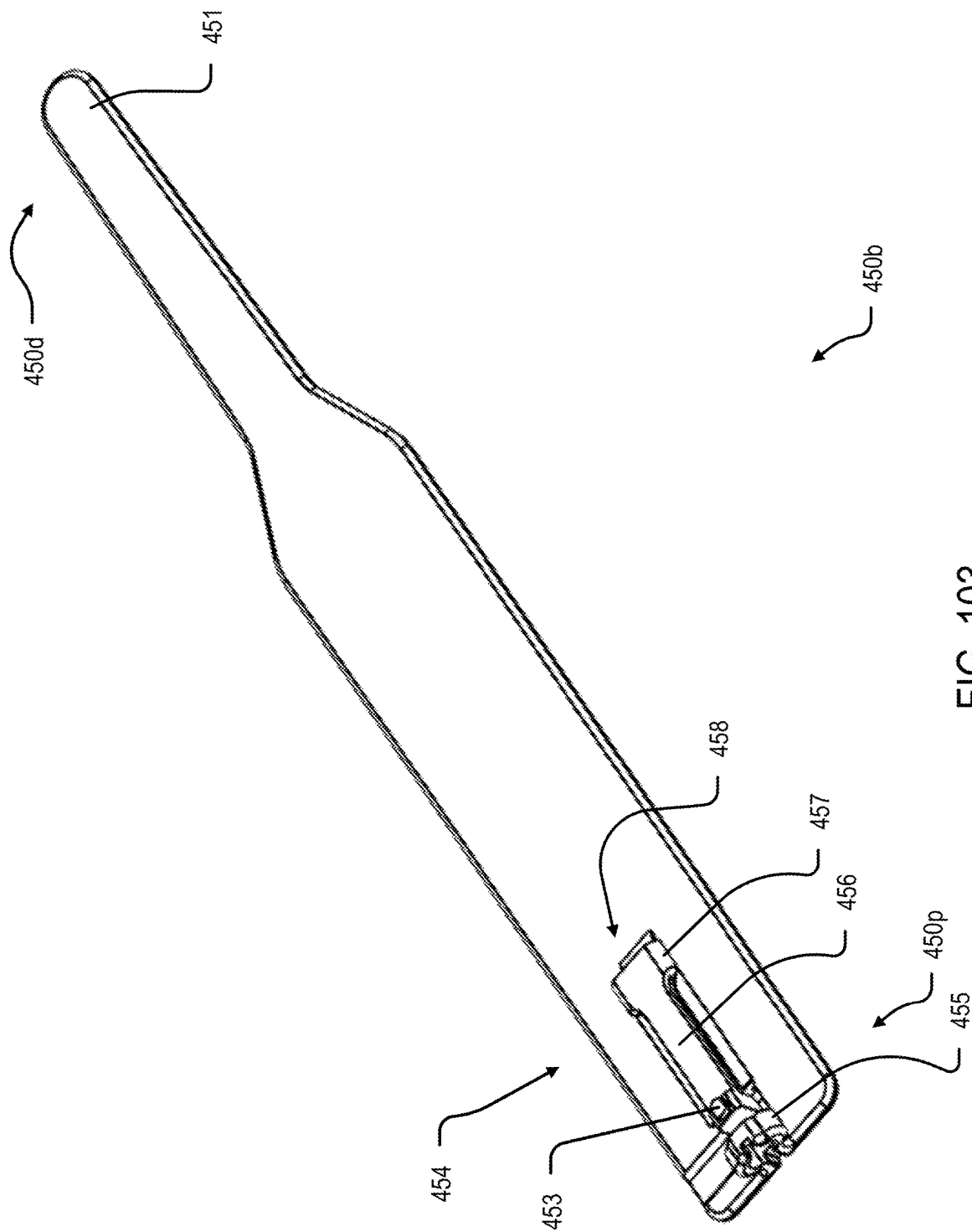


FIG. 102



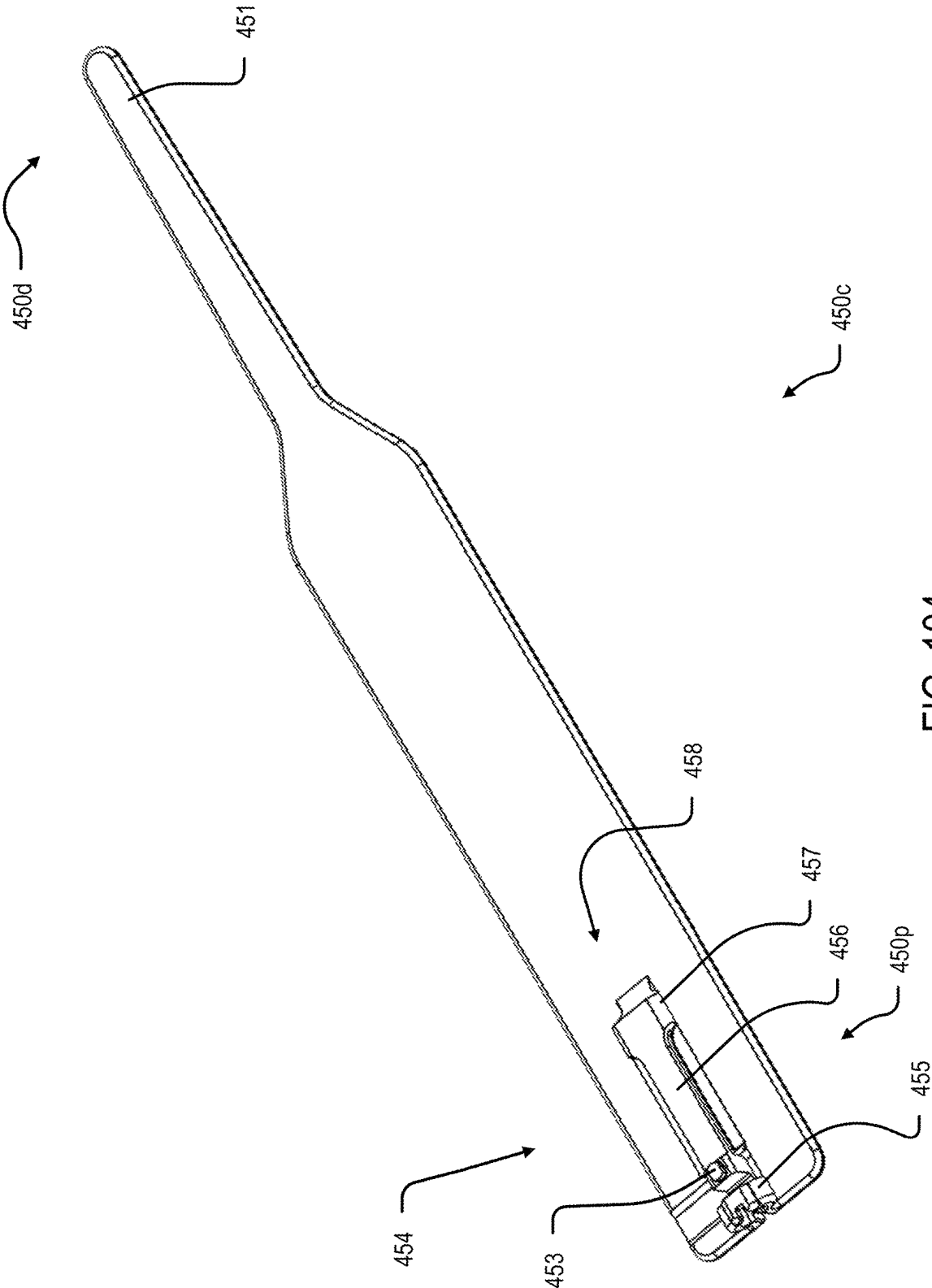


FIG. 104

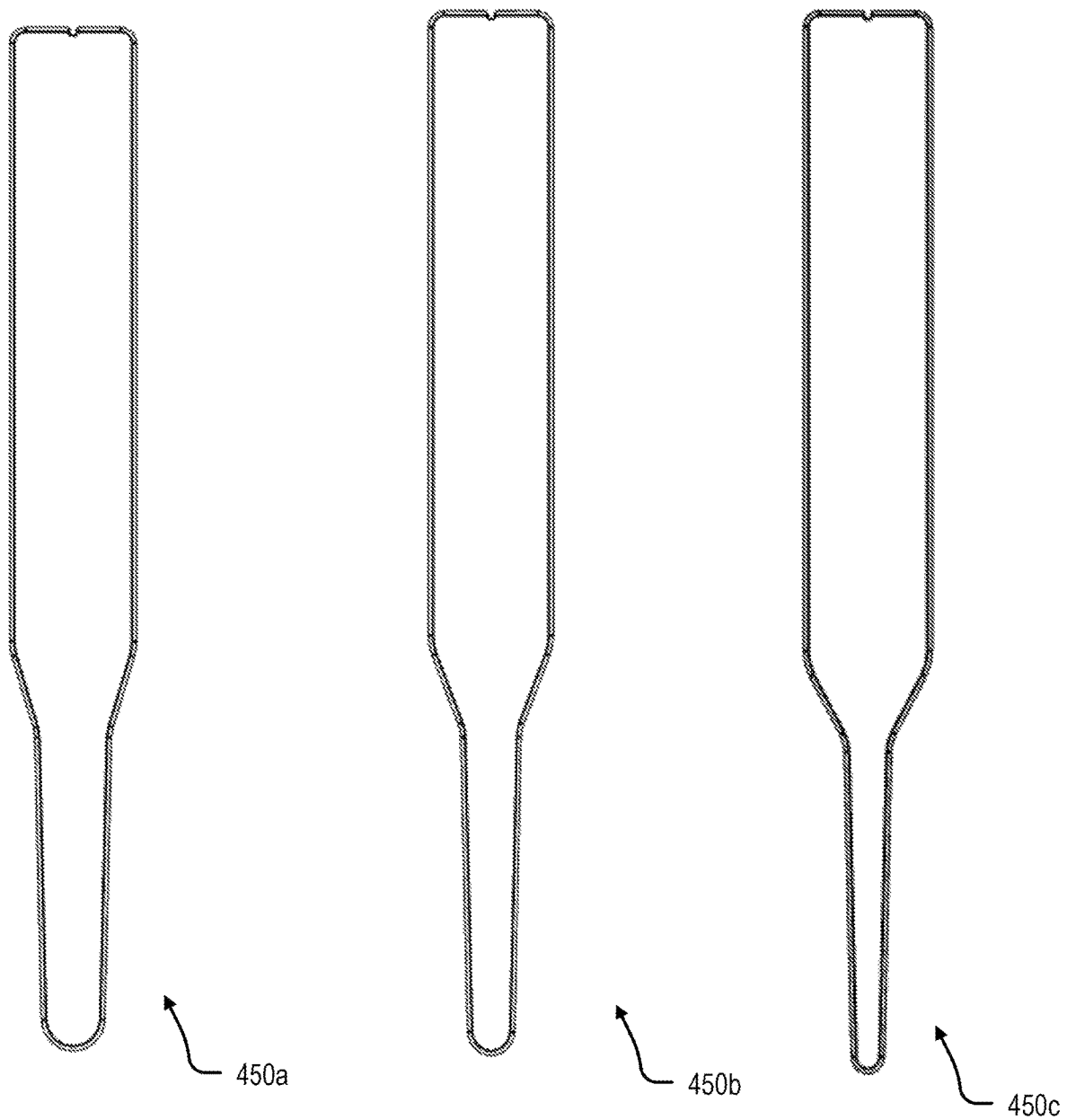


FIG. 105

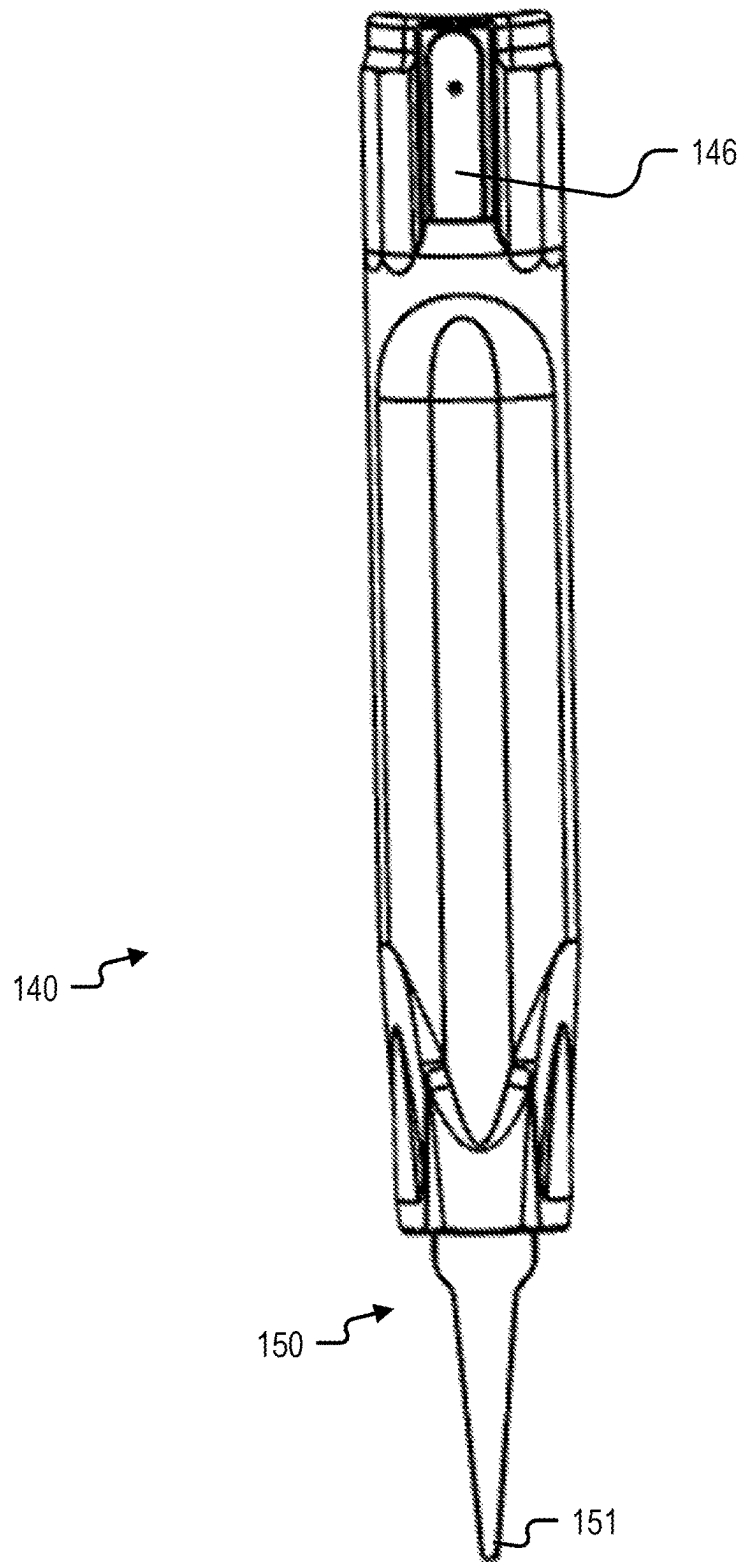


FIG. 106

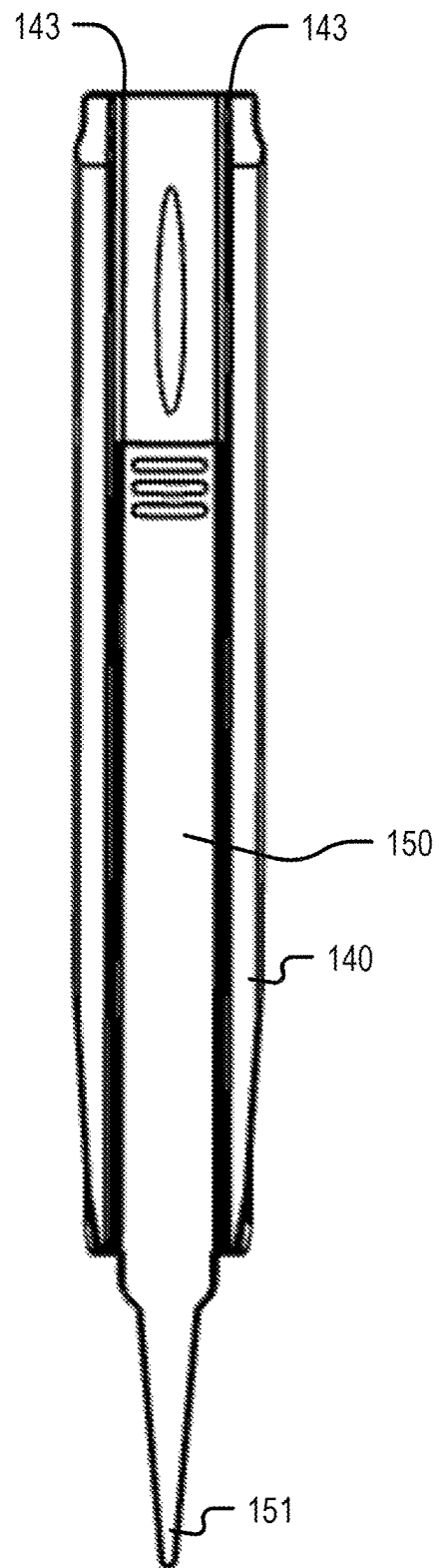


FIG. 107

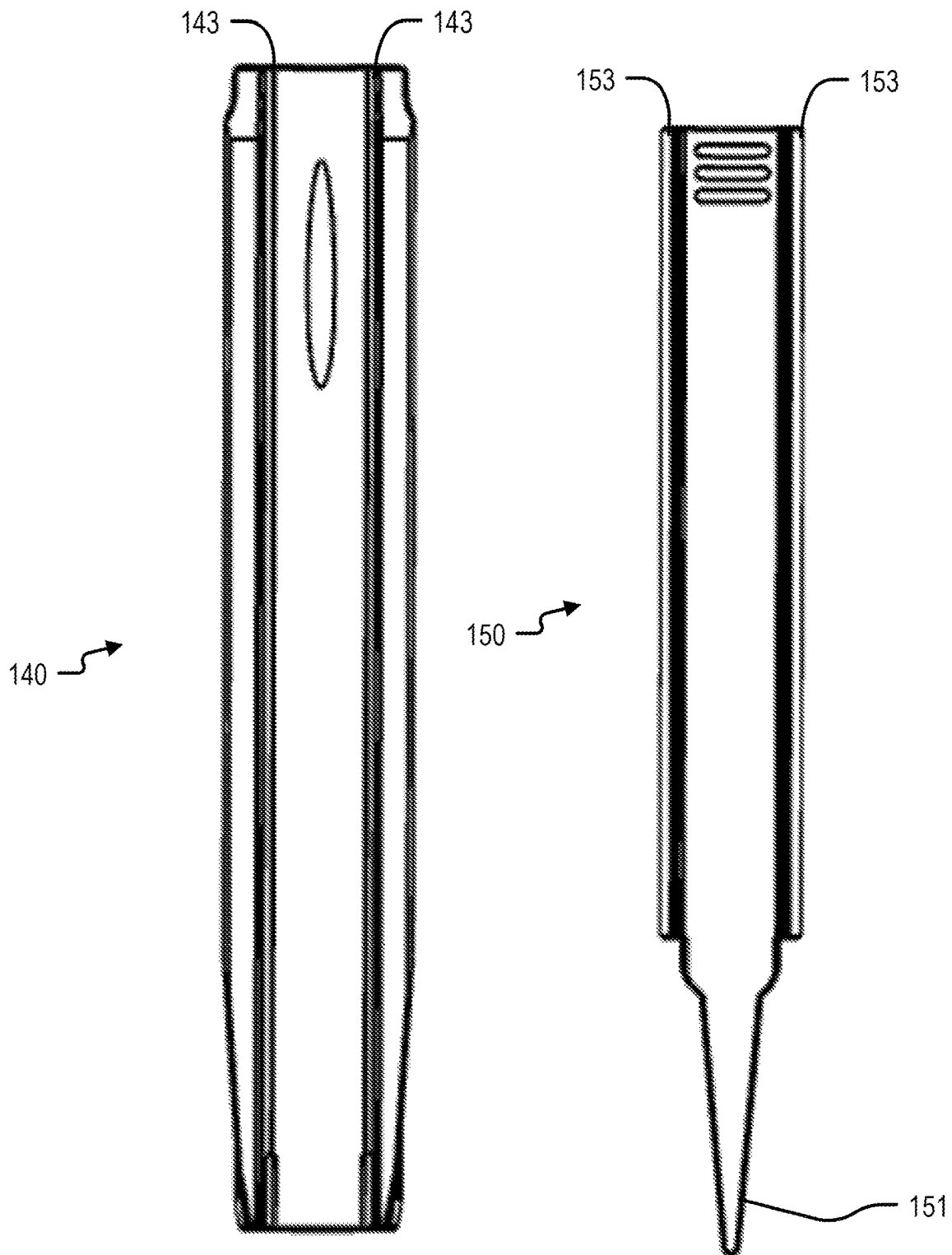
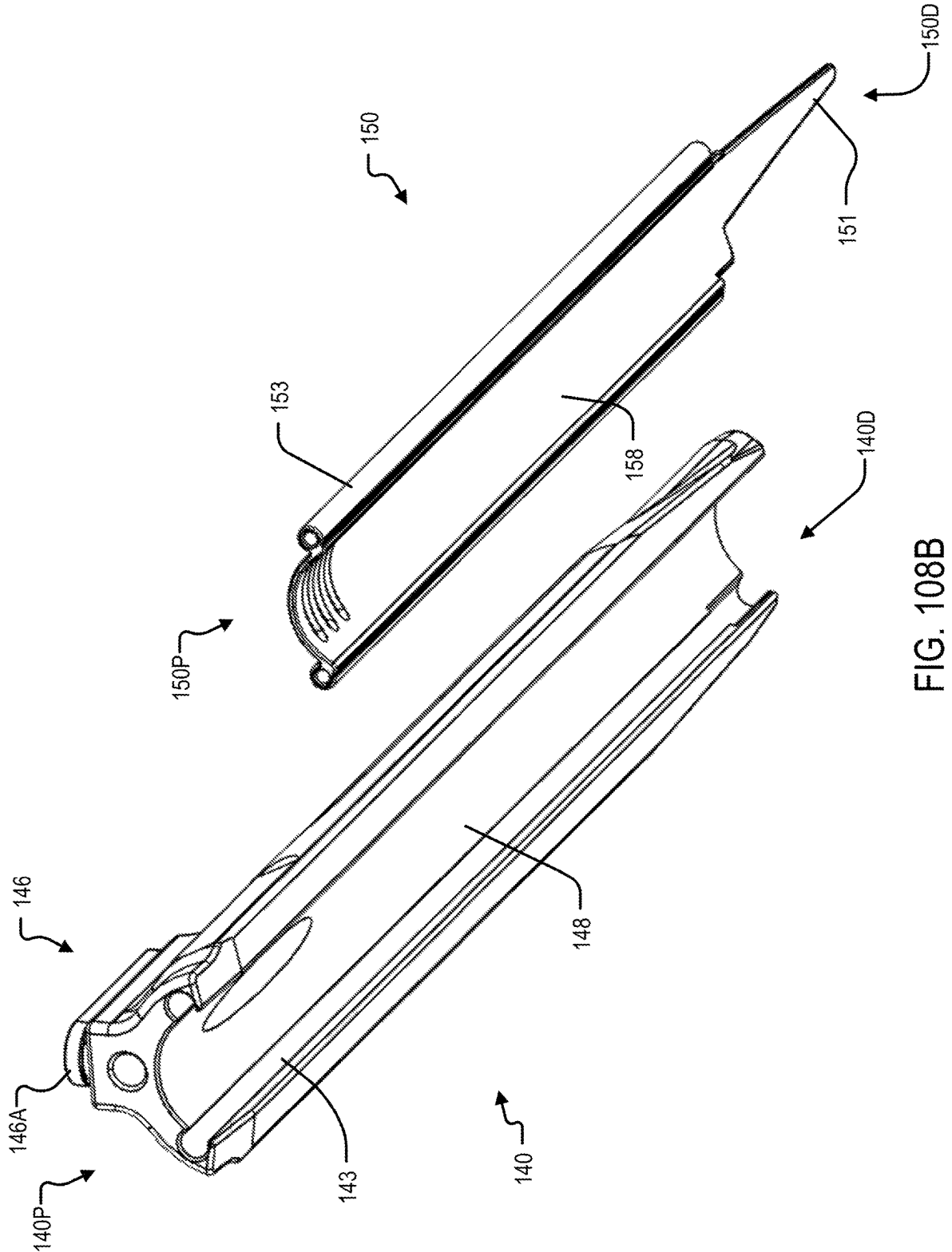


FIG. 108A



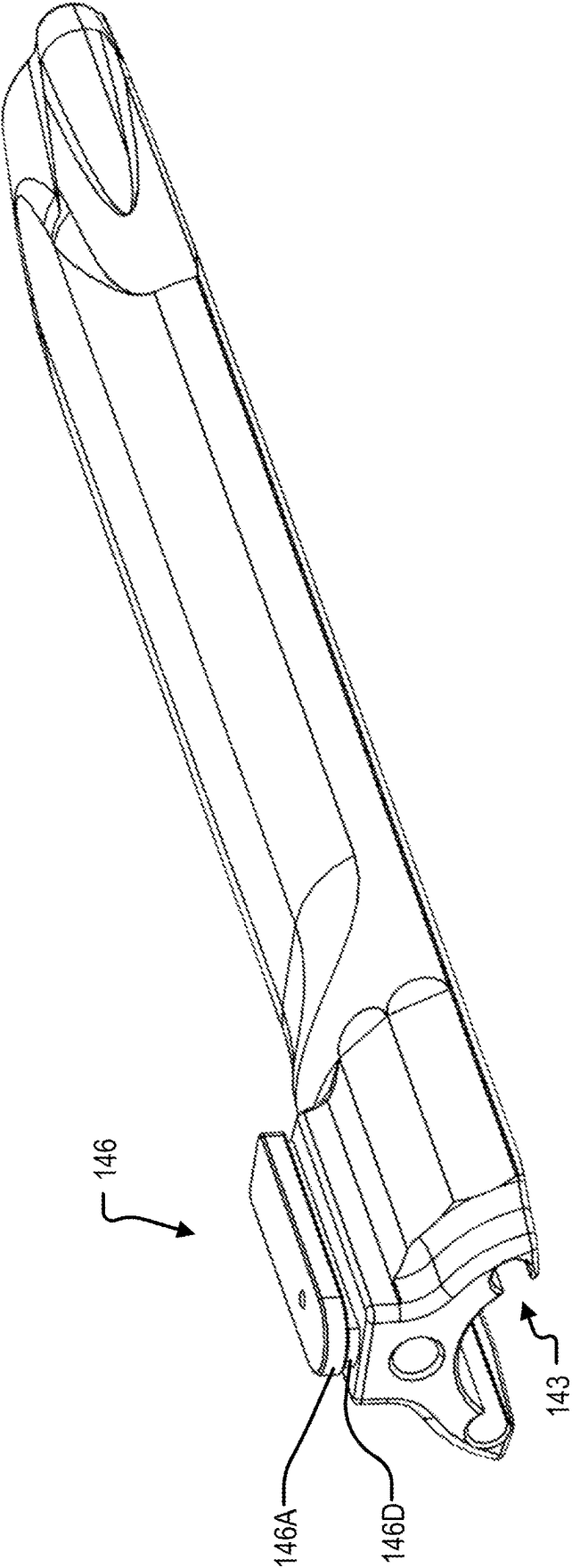
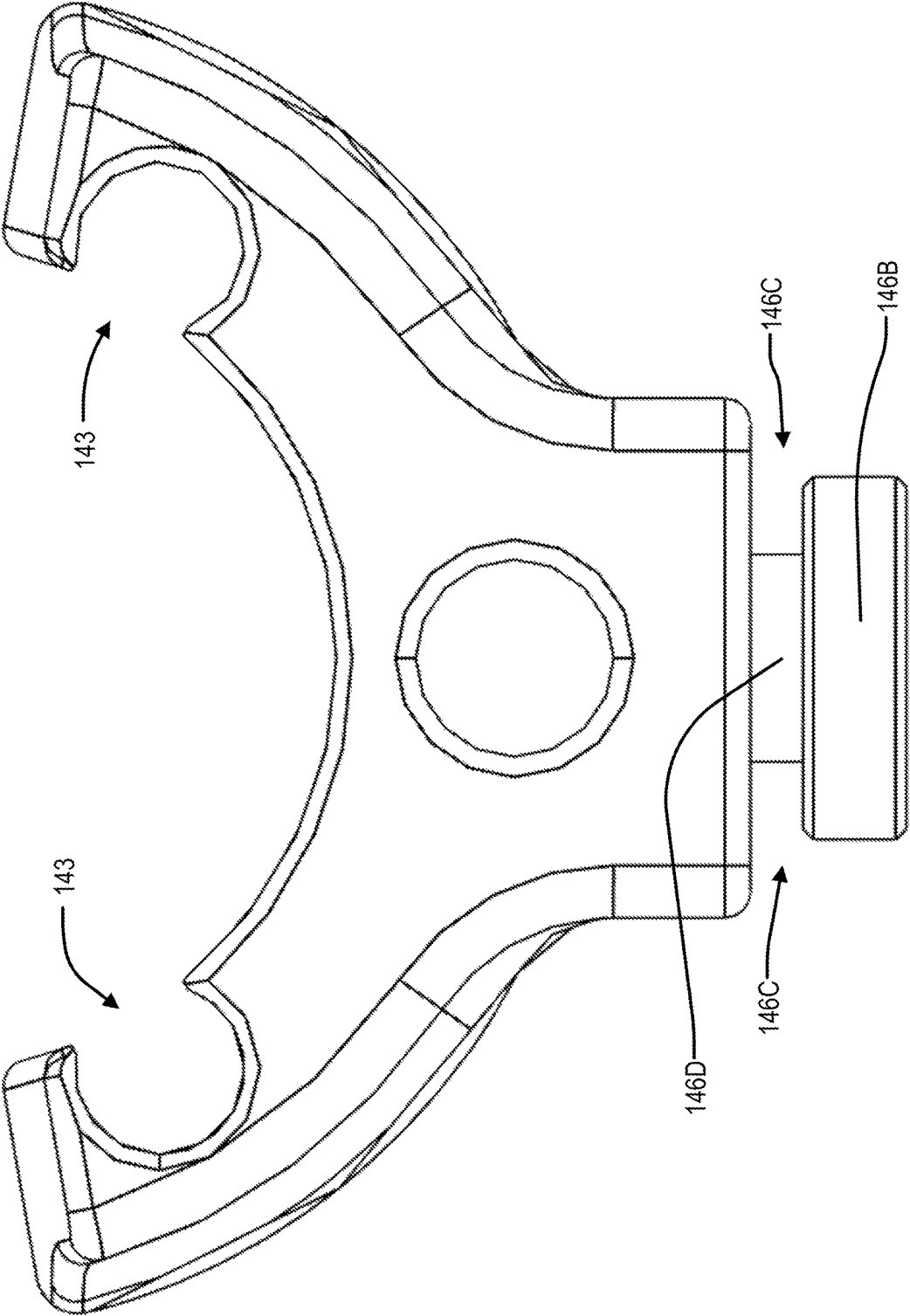


FIG. 108C



140
FIG. 108D

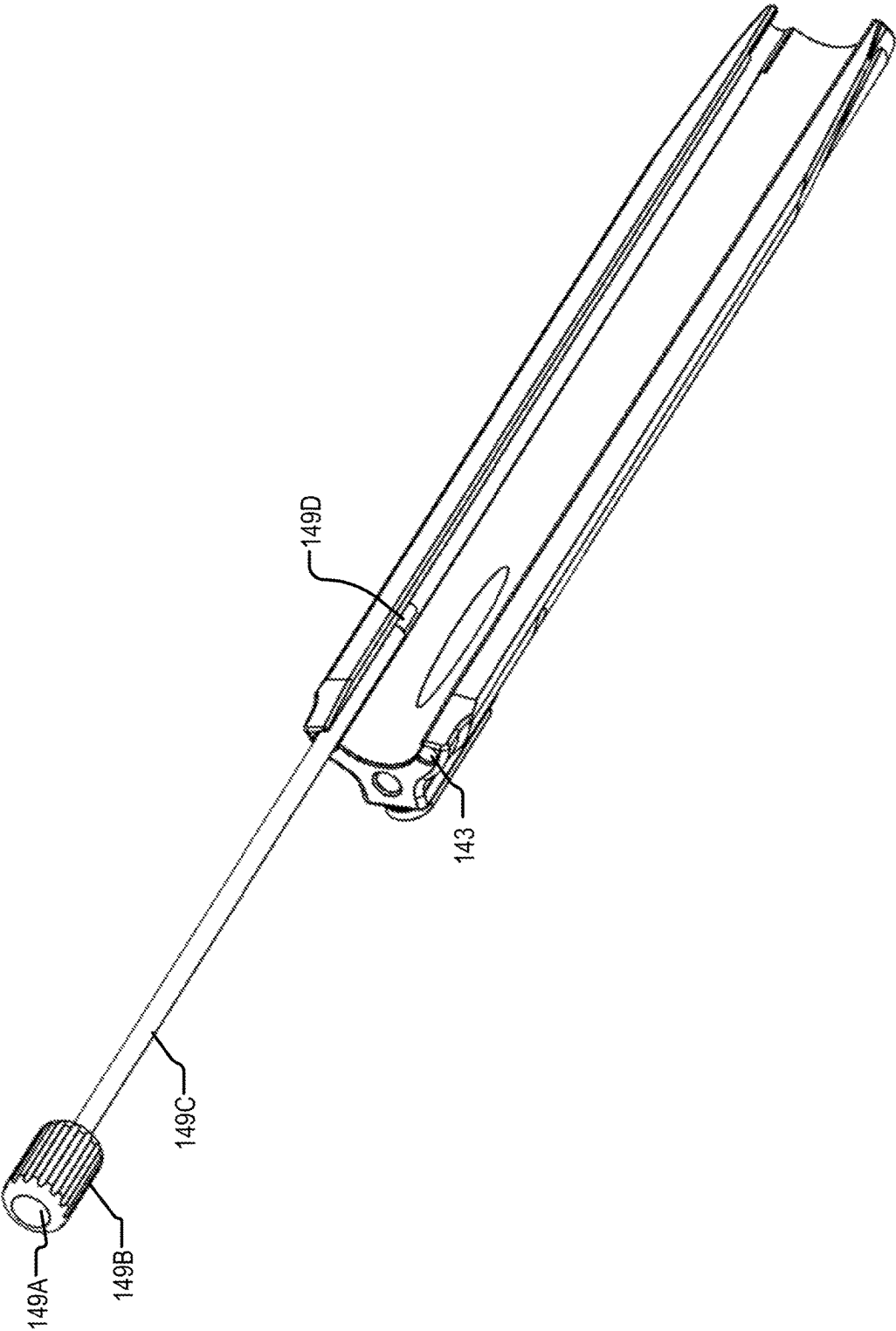
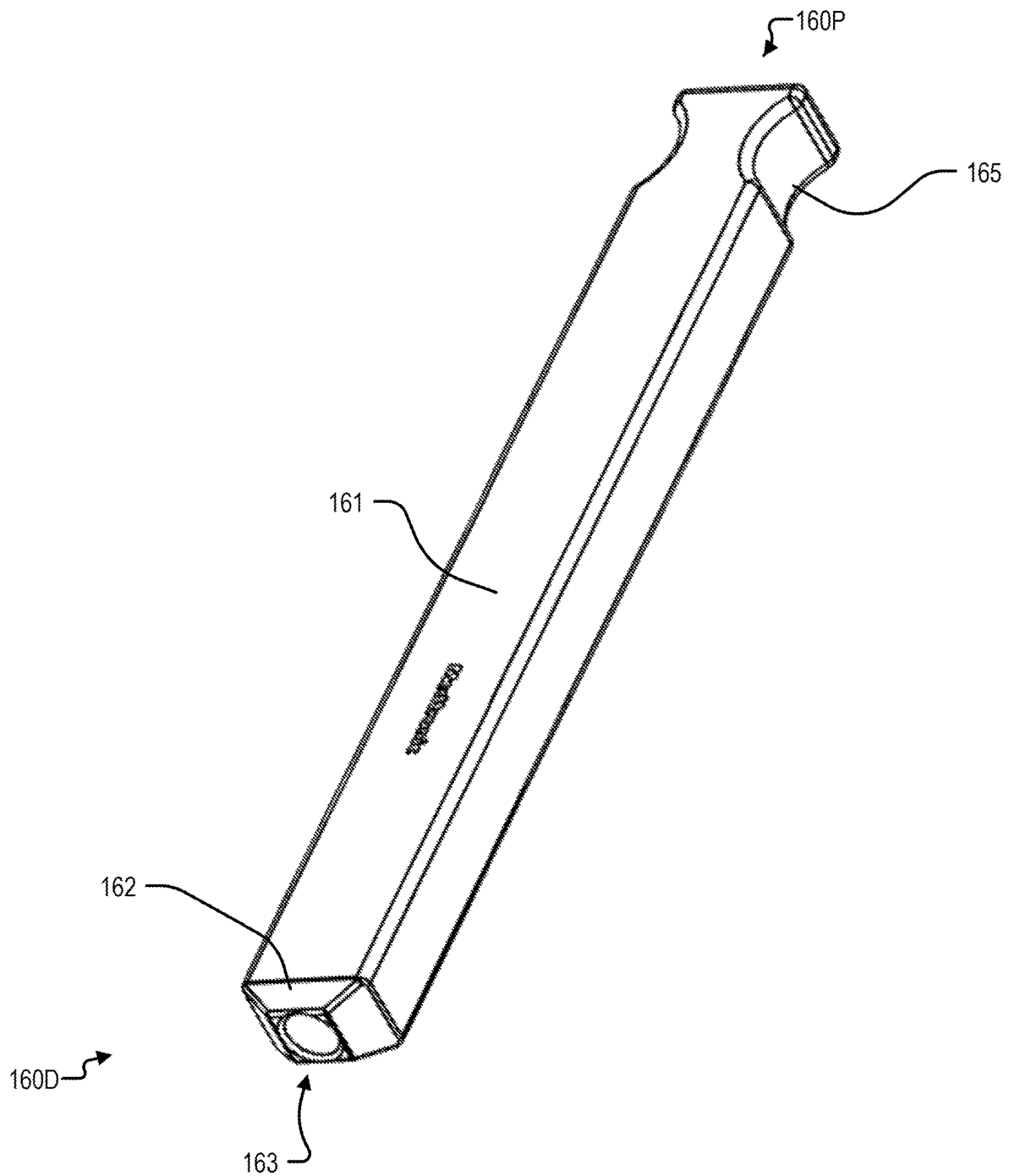


FIG. 108E



160
FIG. 109

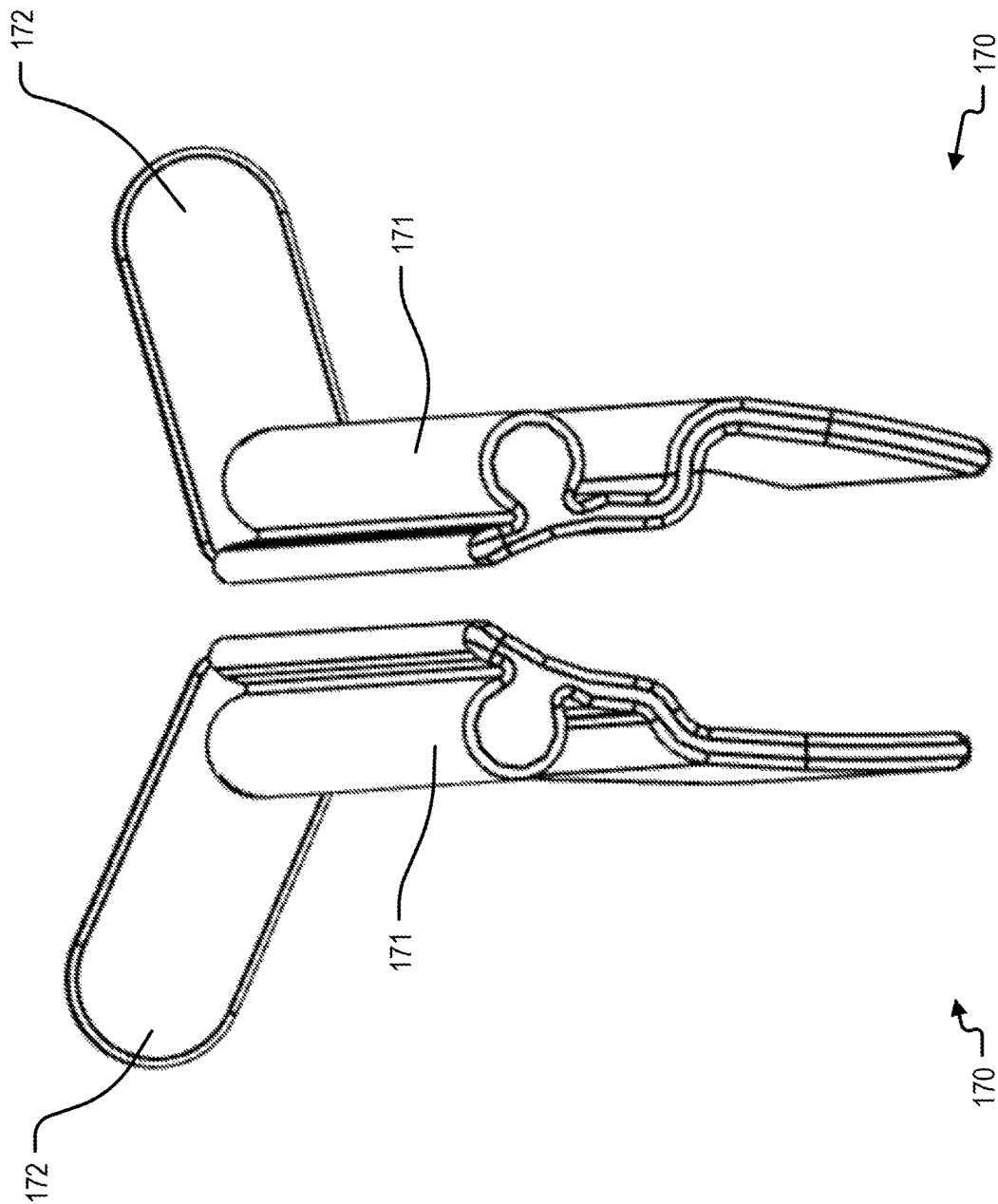


FIG. 110

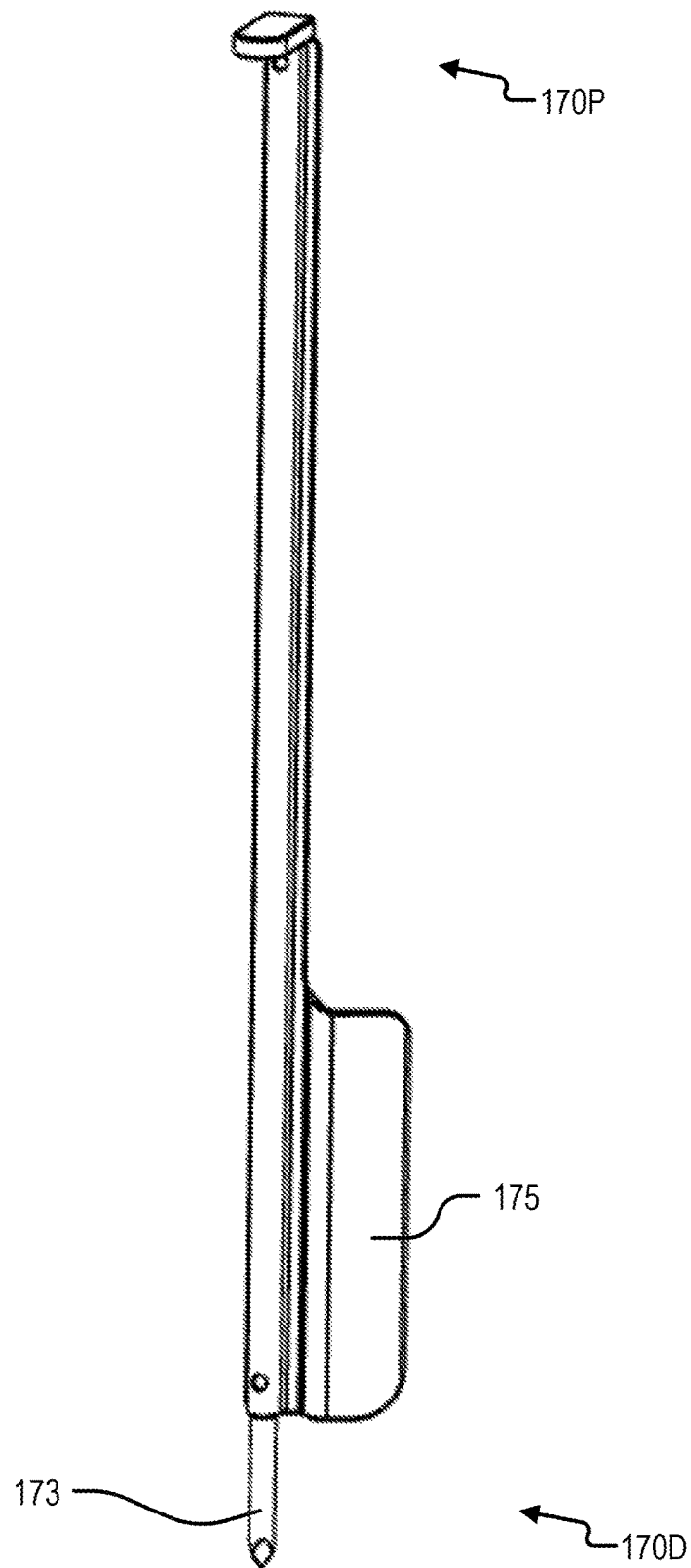


FIG. 111

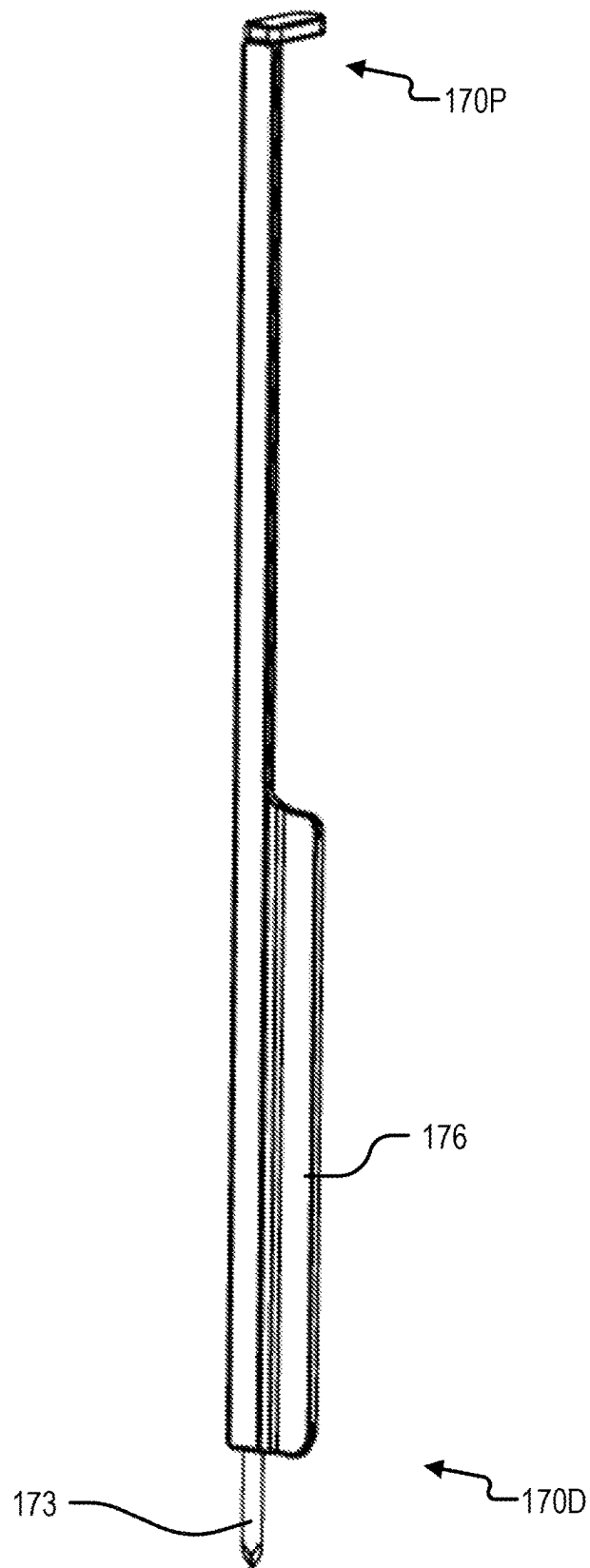


FIG. 112

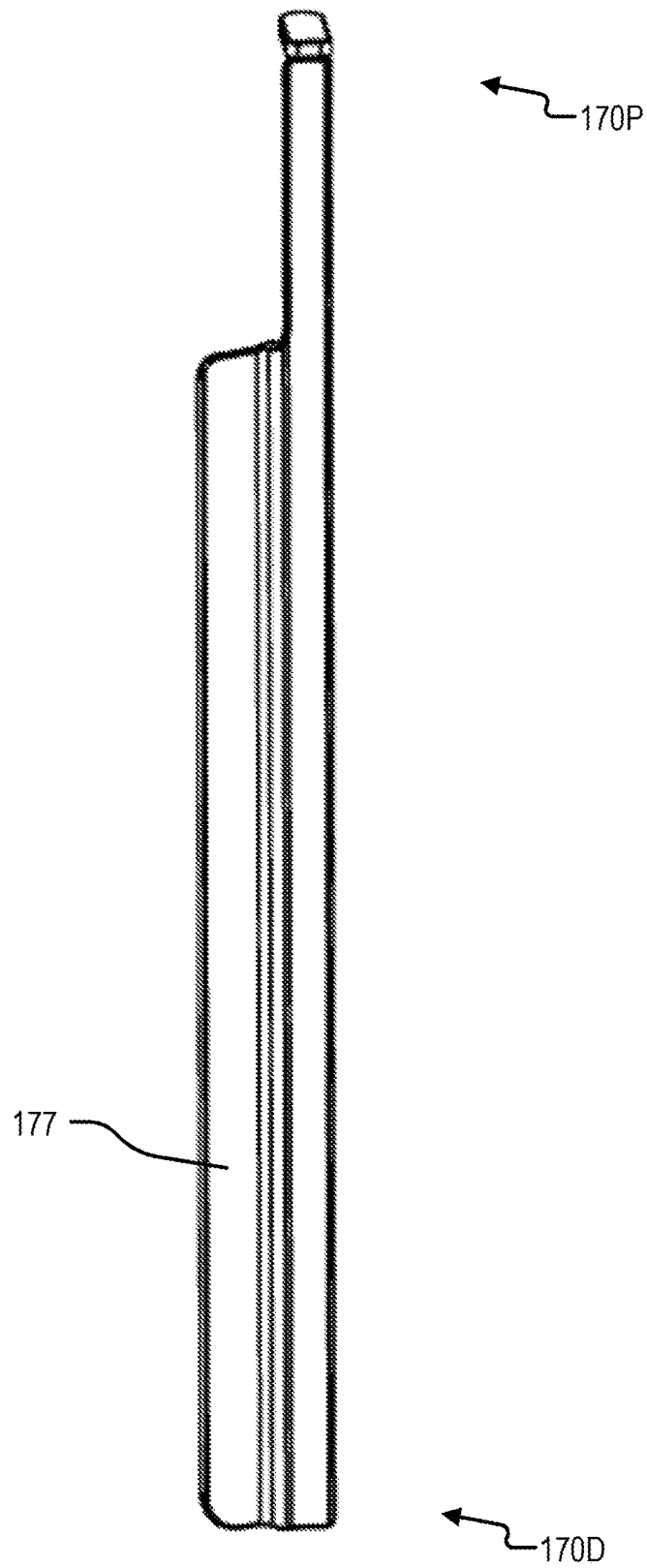
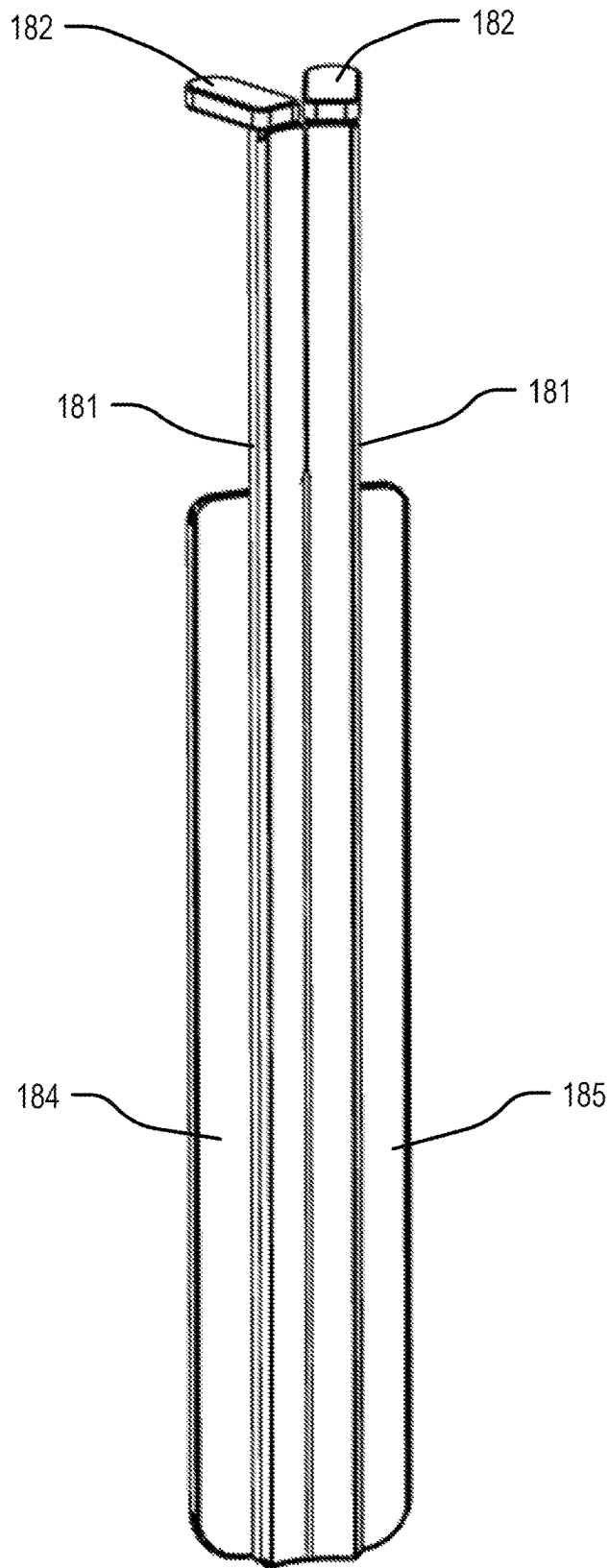
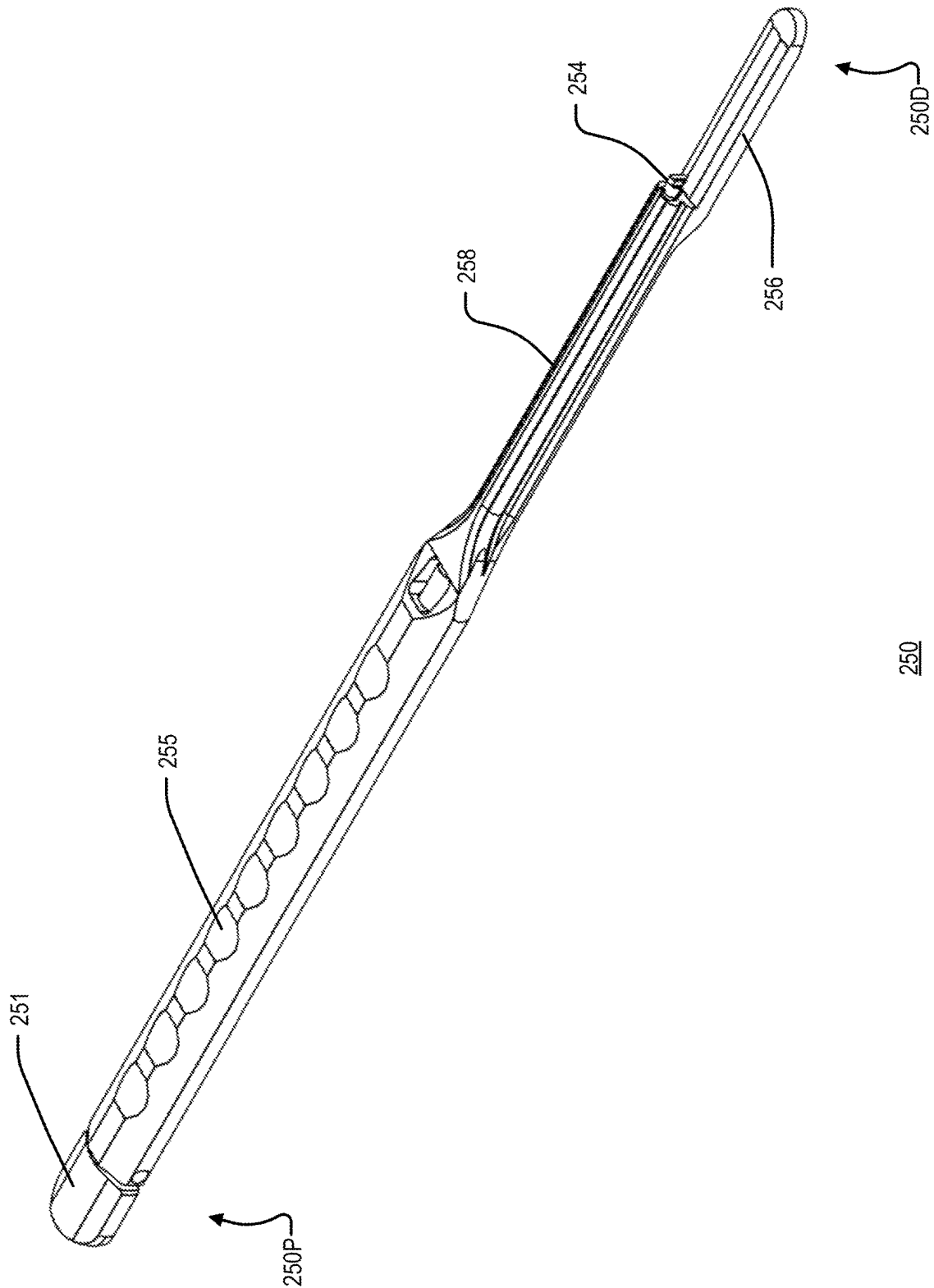


FIG. 113



180
FIG. 114



250
FIG. 115A

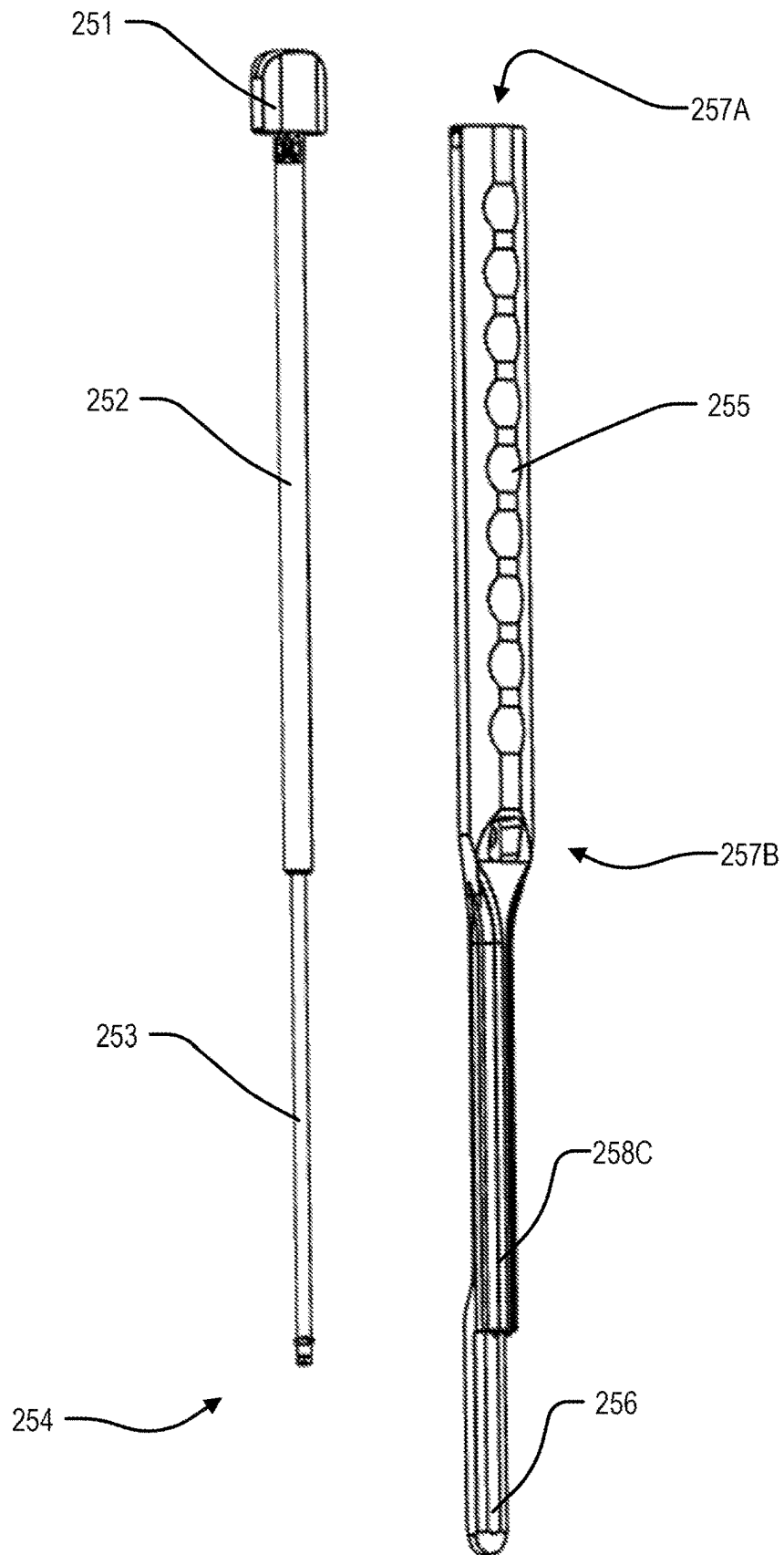


FIG. 115B

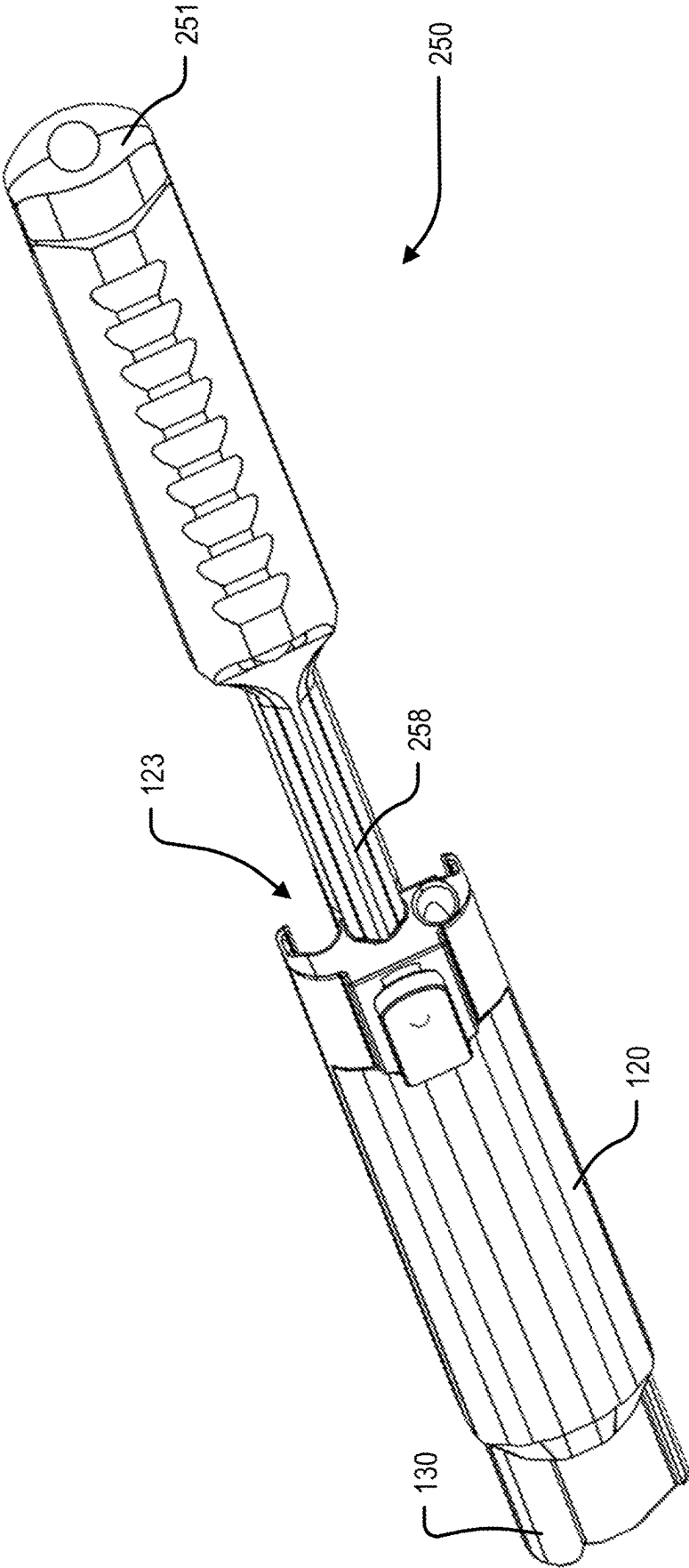


FIG. 116A

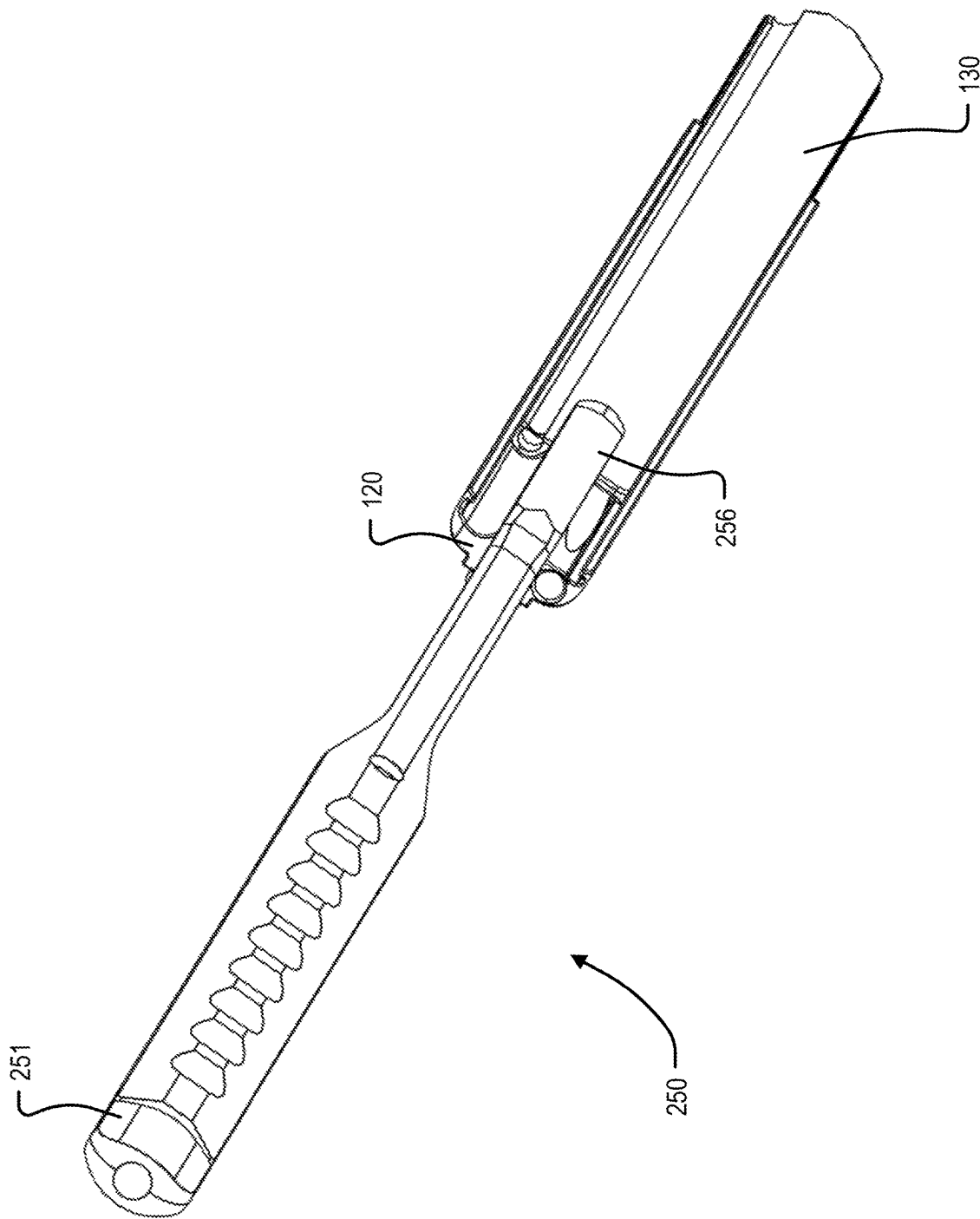


FIG. 116B

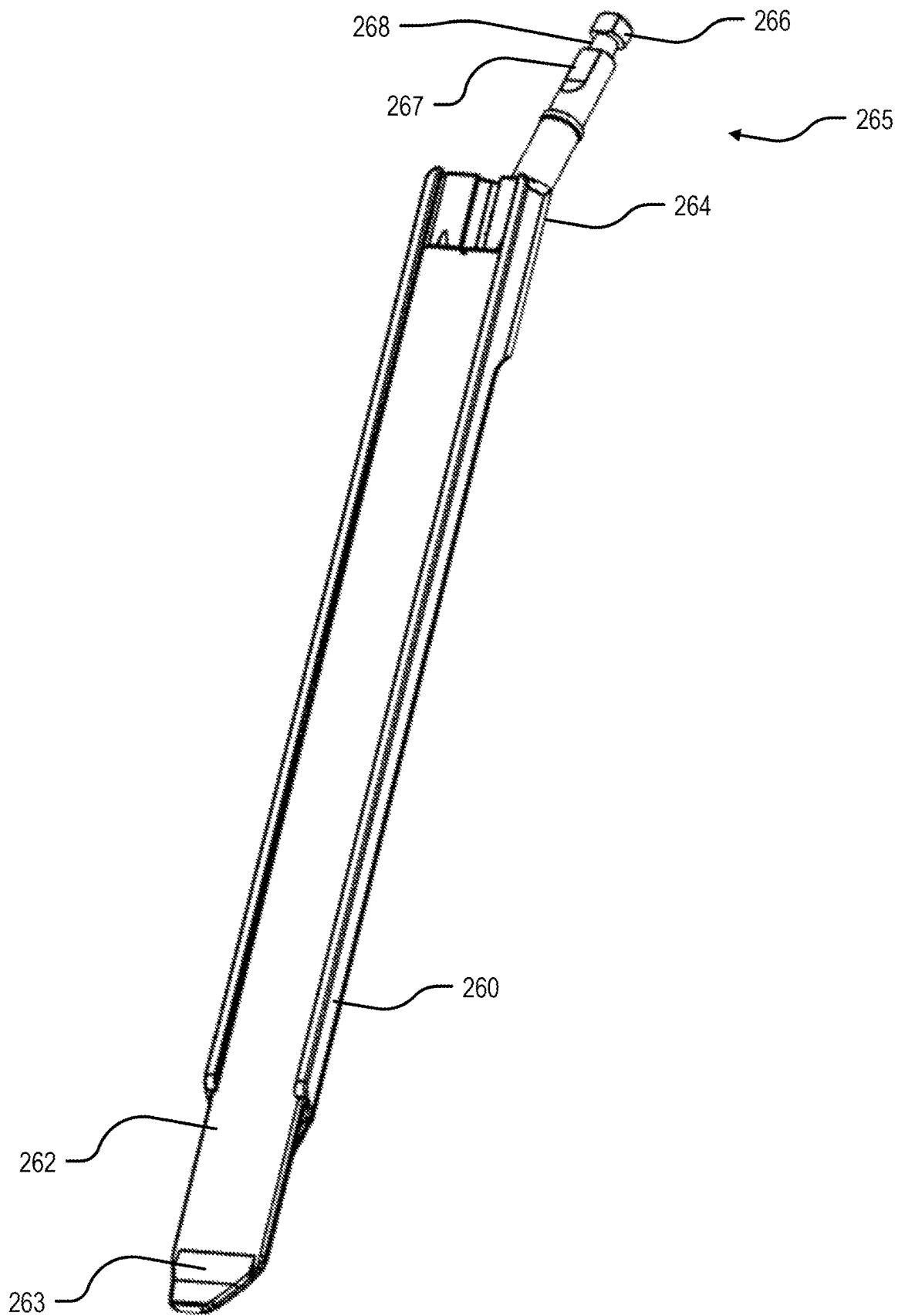


FIG. 117A

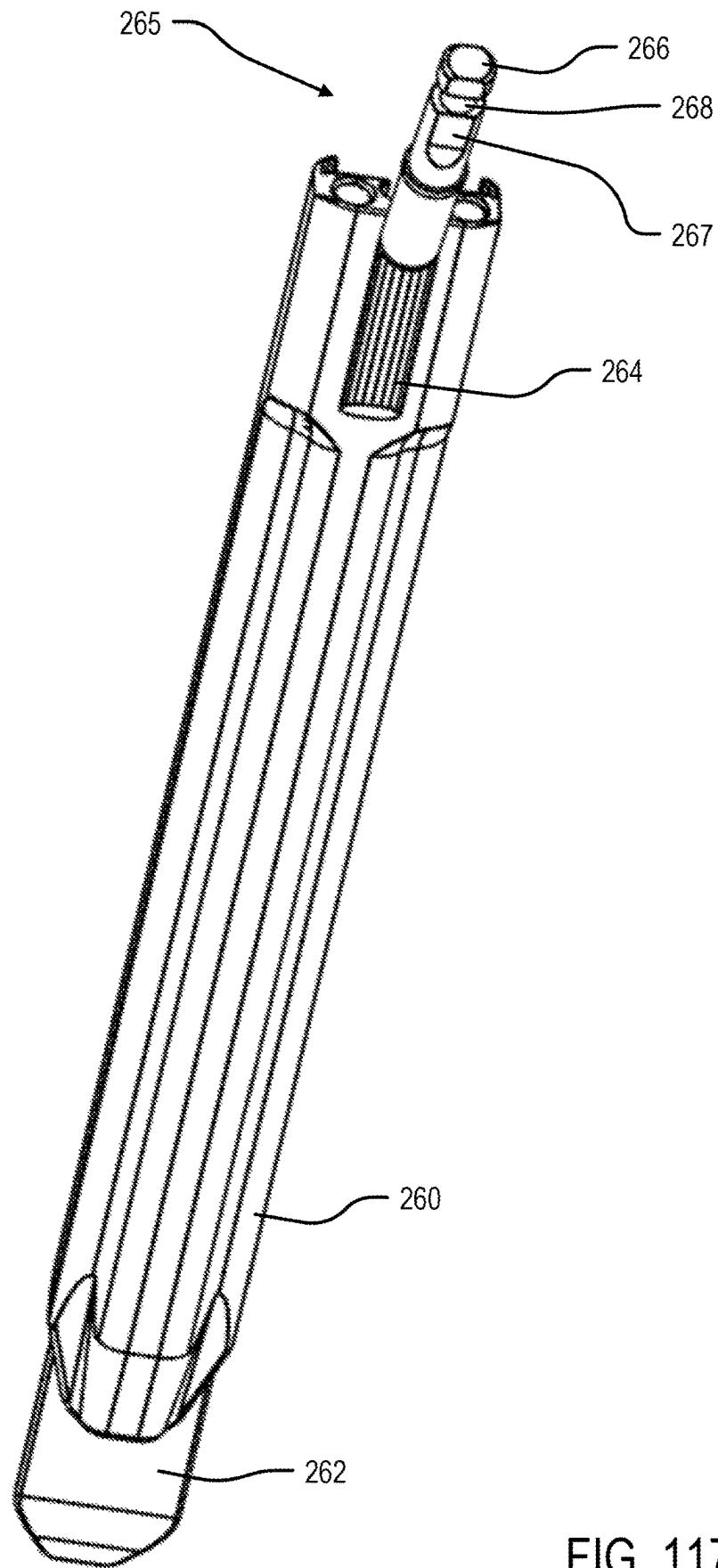


FIG. 117B

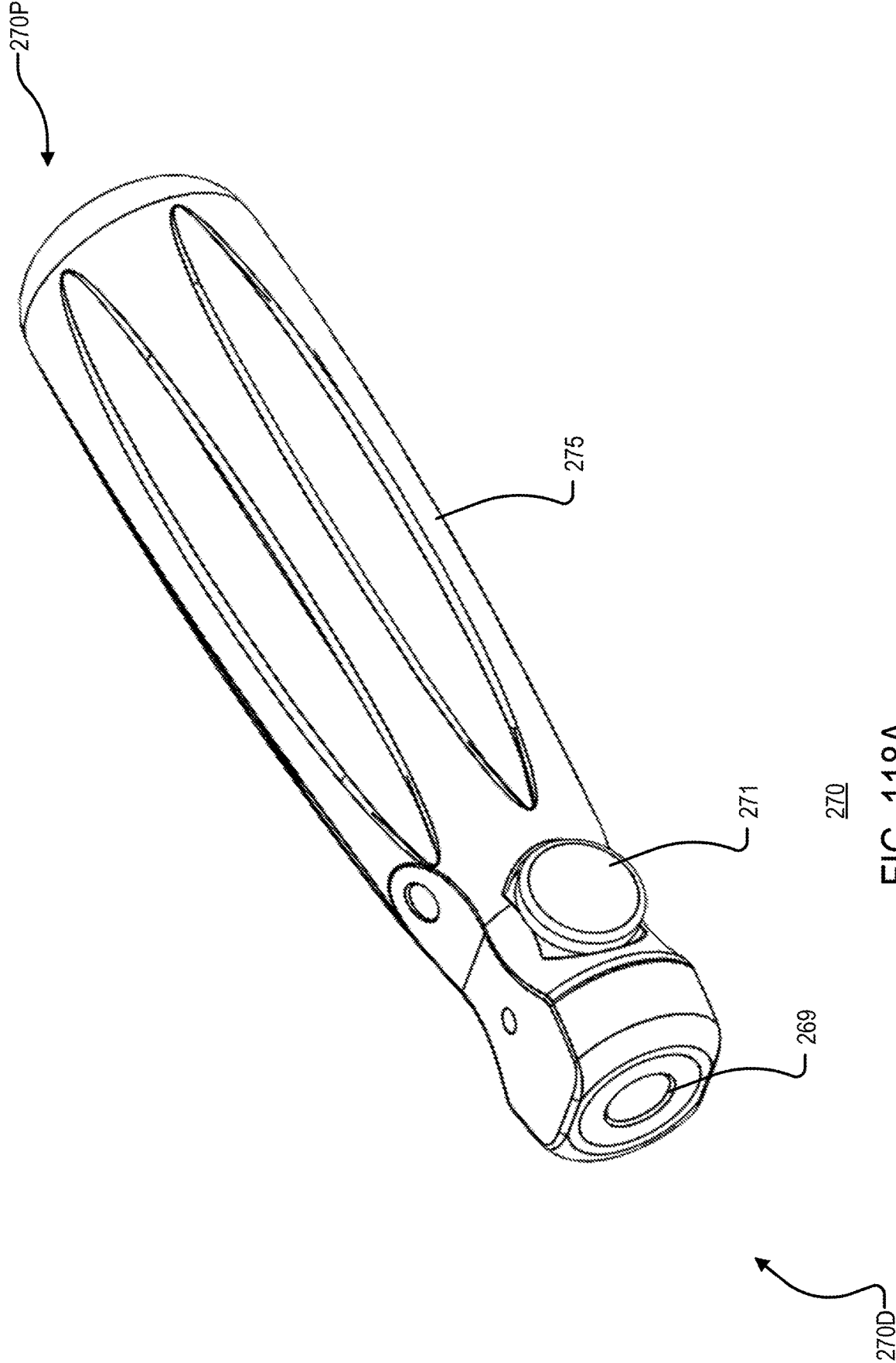


FIG. 118A

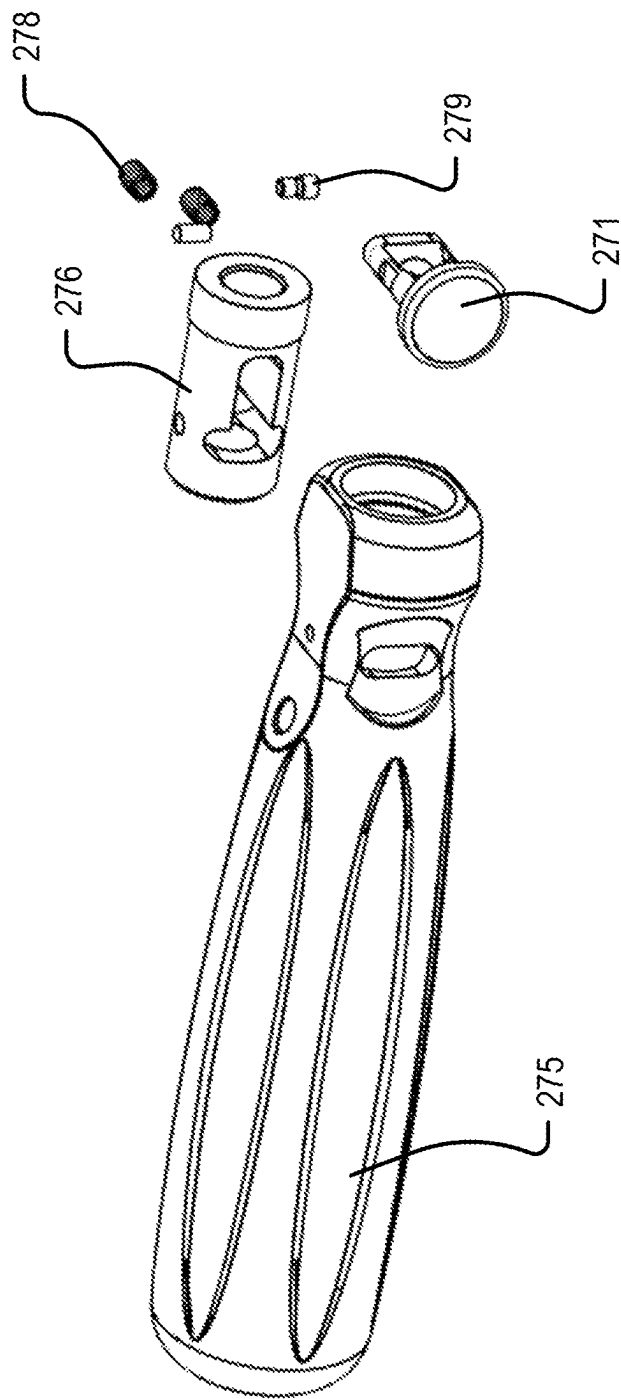
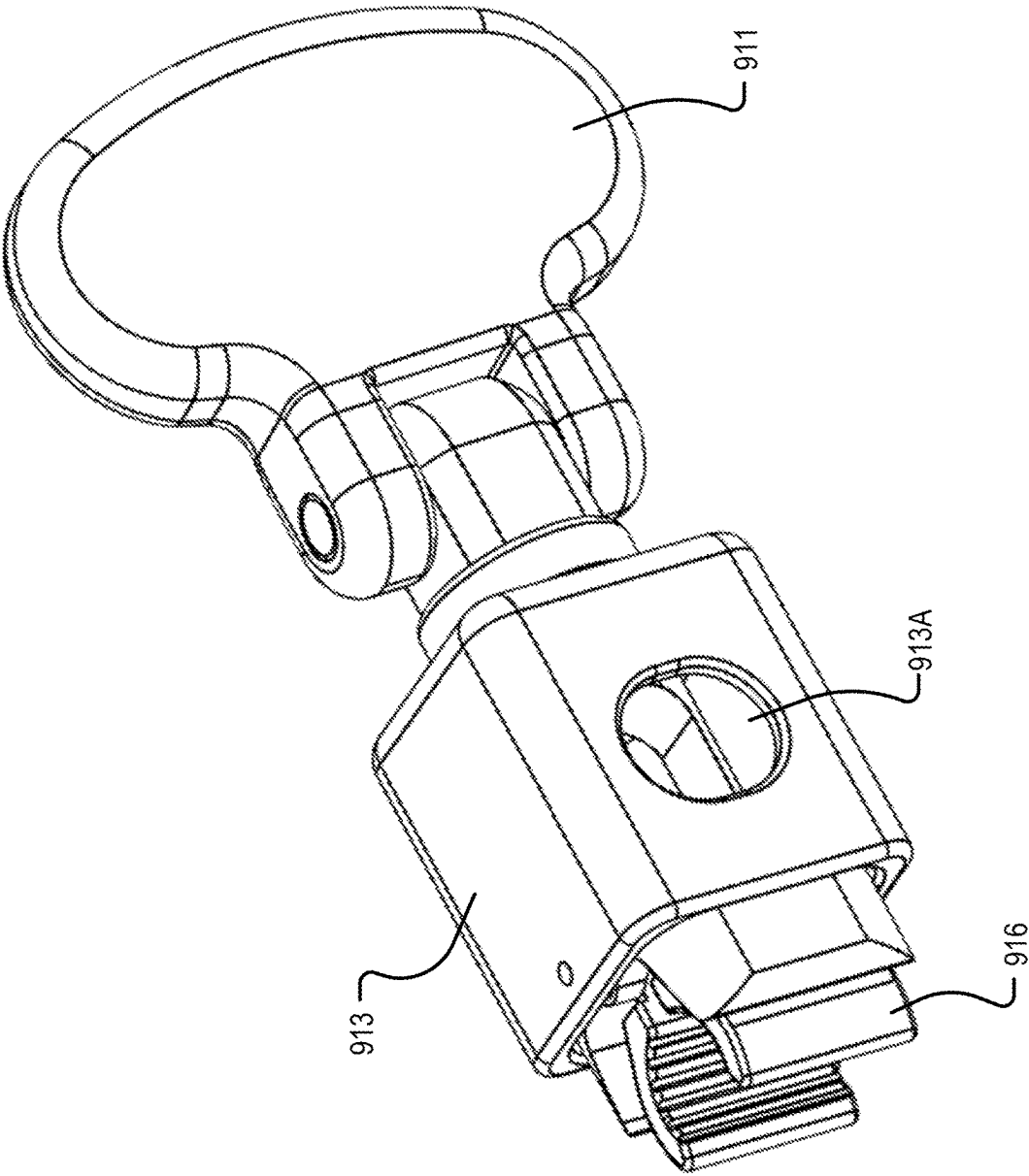


FIG. 118B



280
FIG. 118C

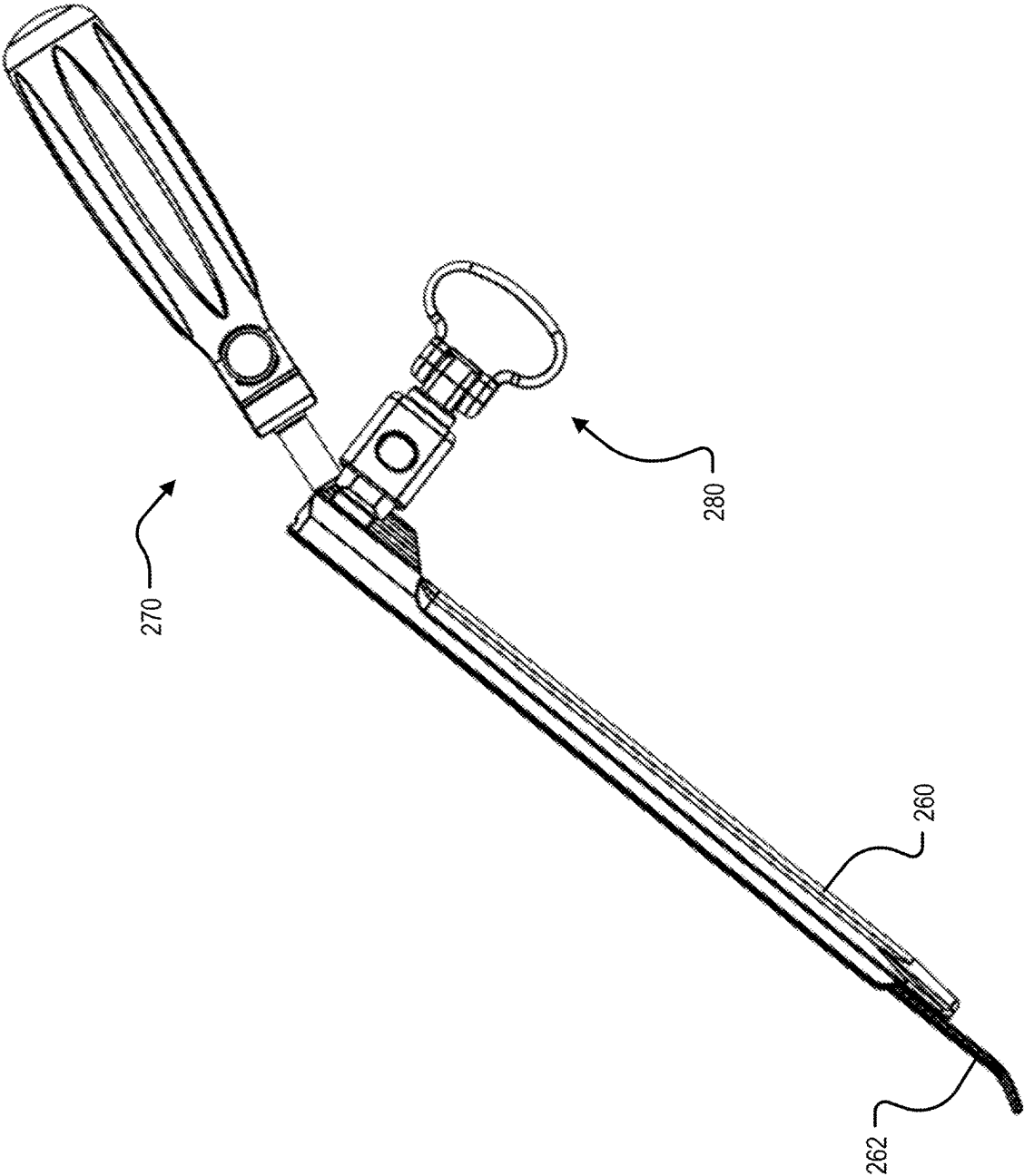


FIG. 118D

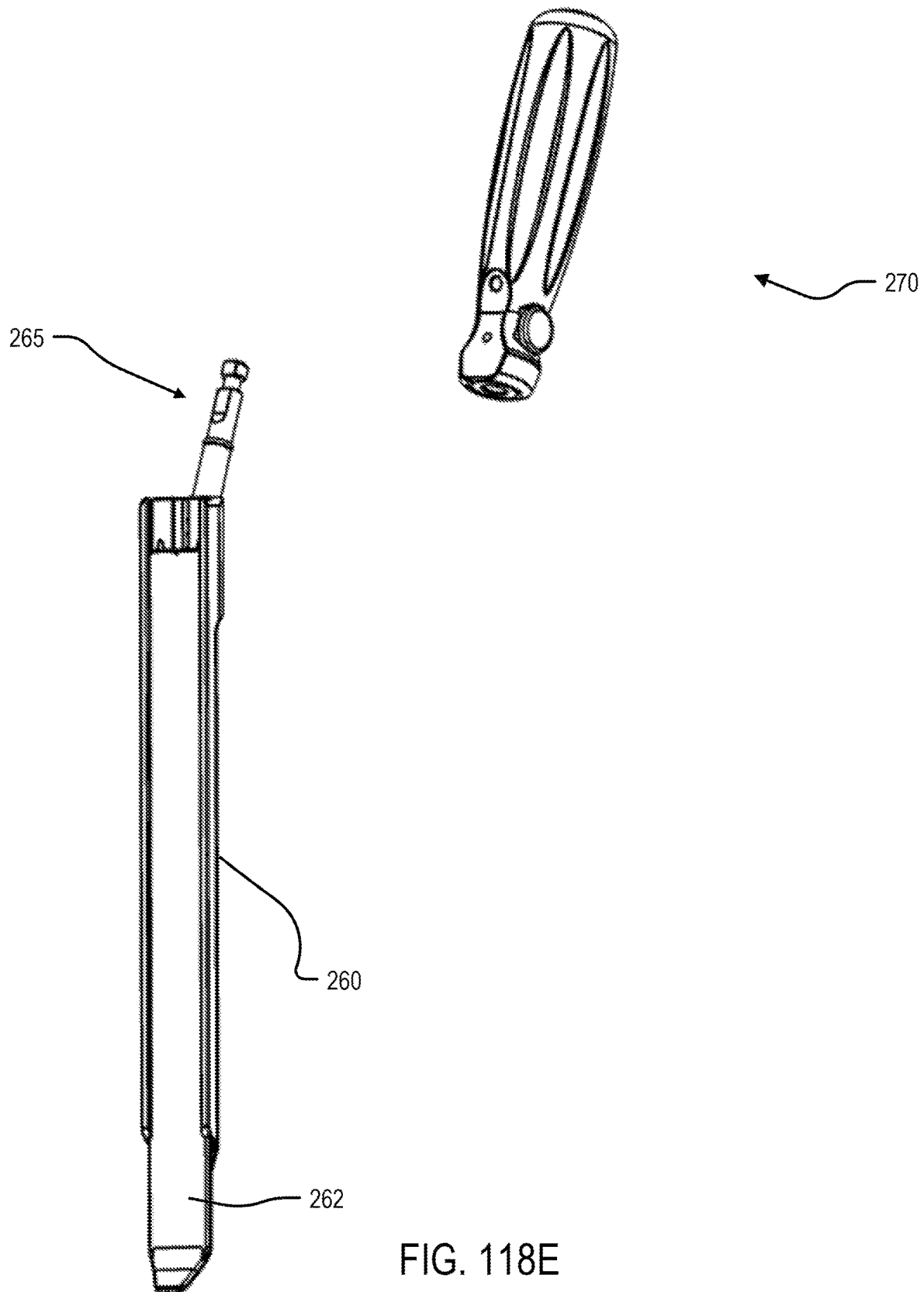
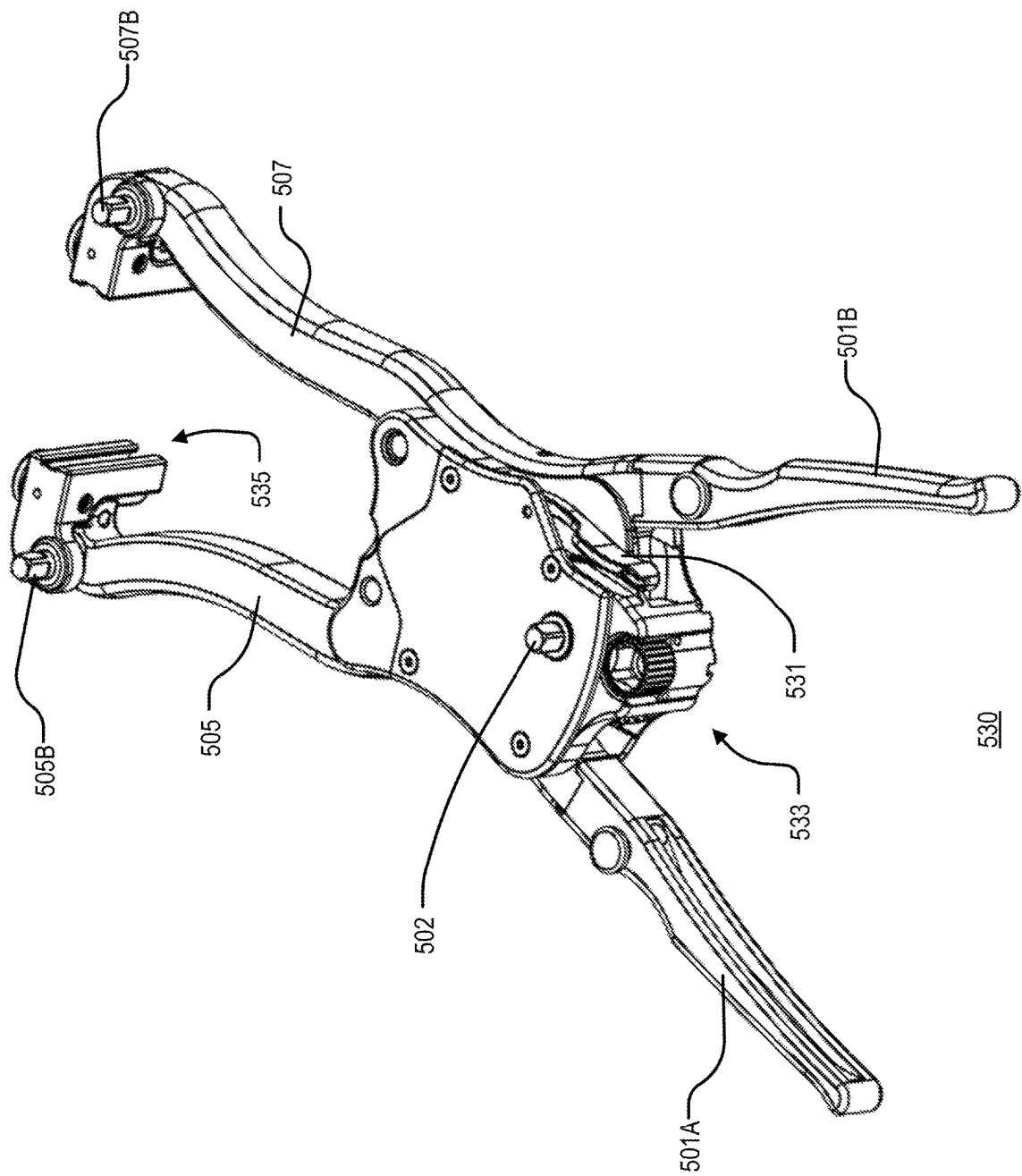


FIG. 118E



530
FIG. 119

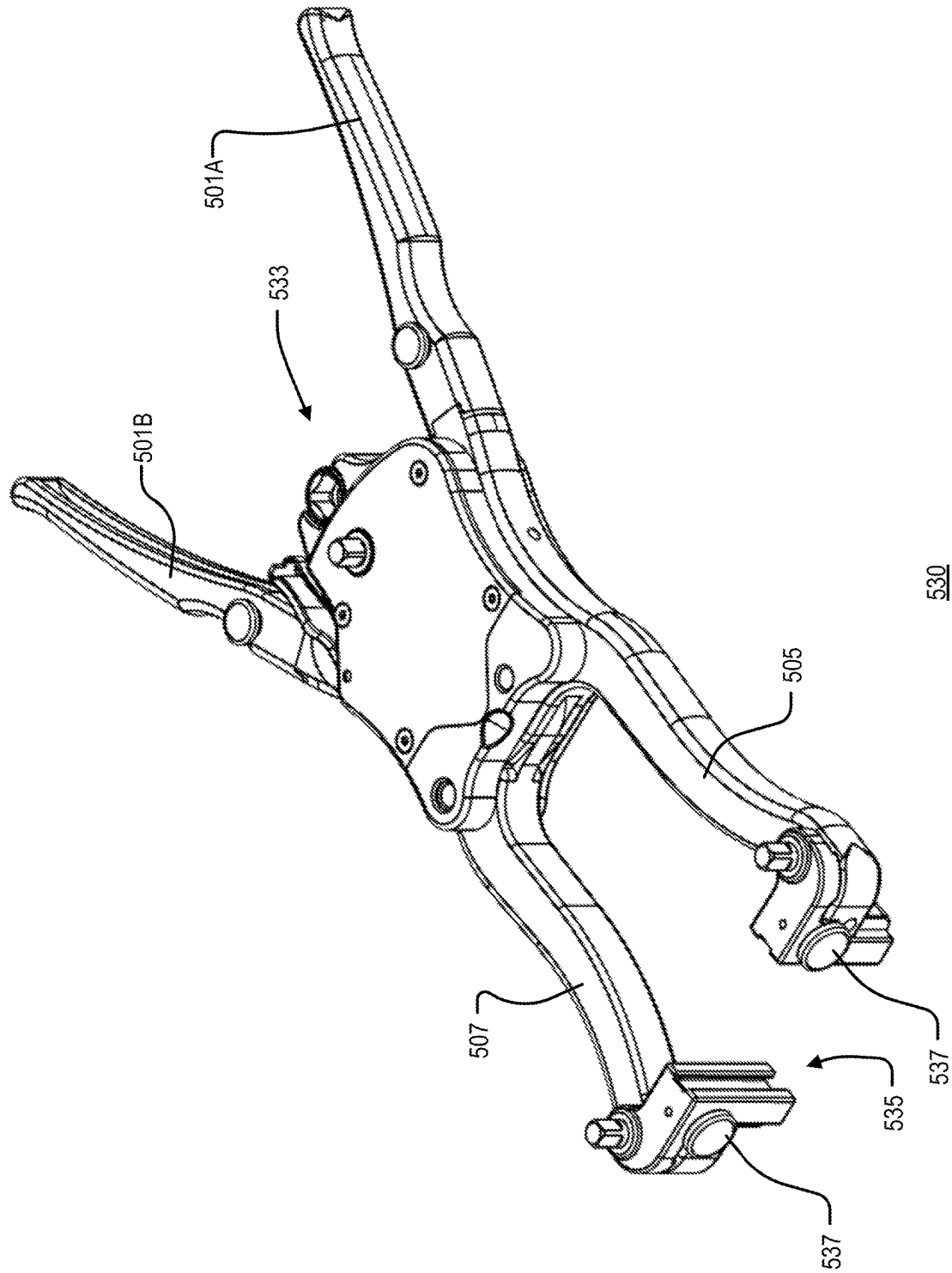
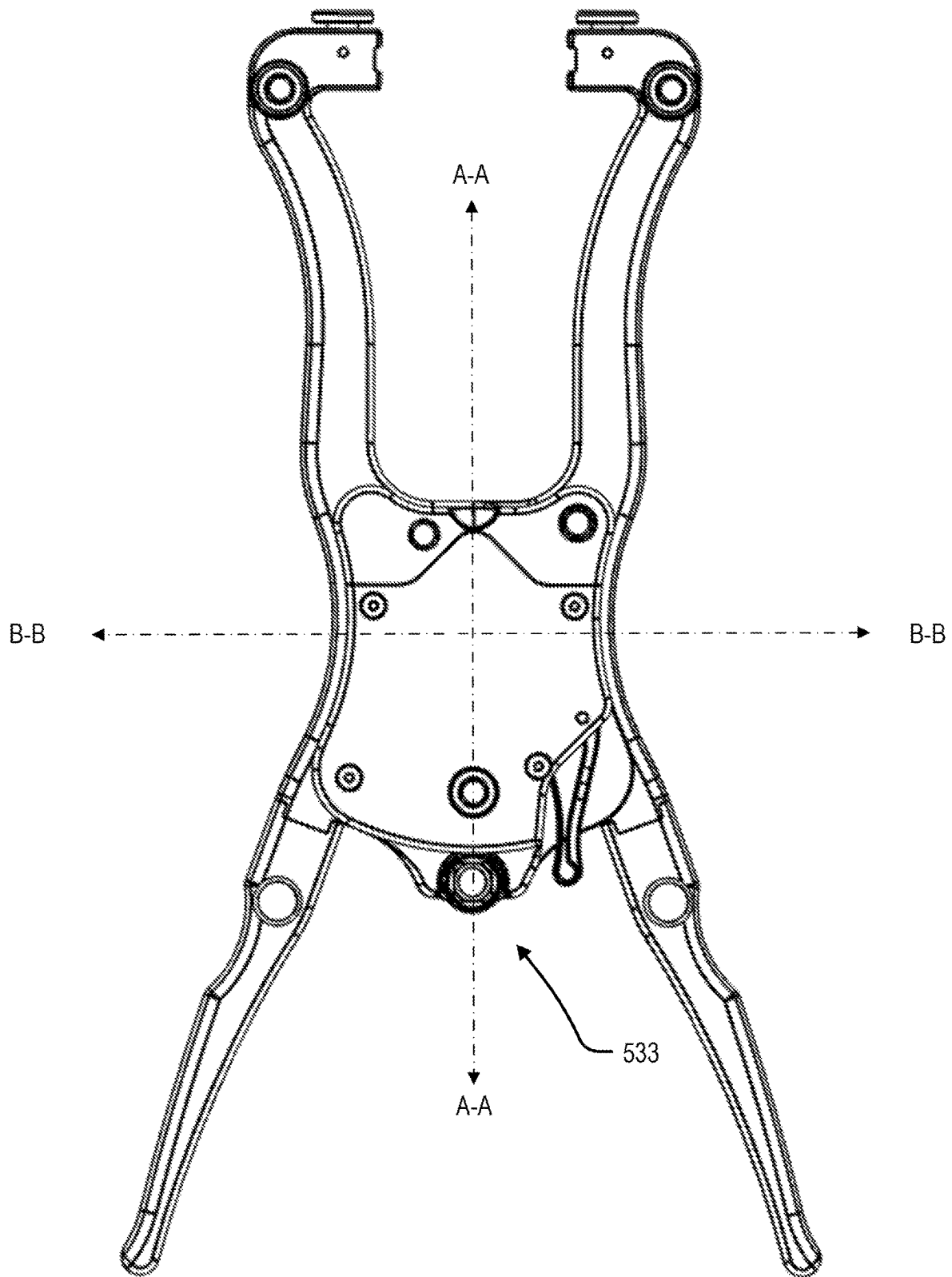


FIG. 120



530

FIG. 121A

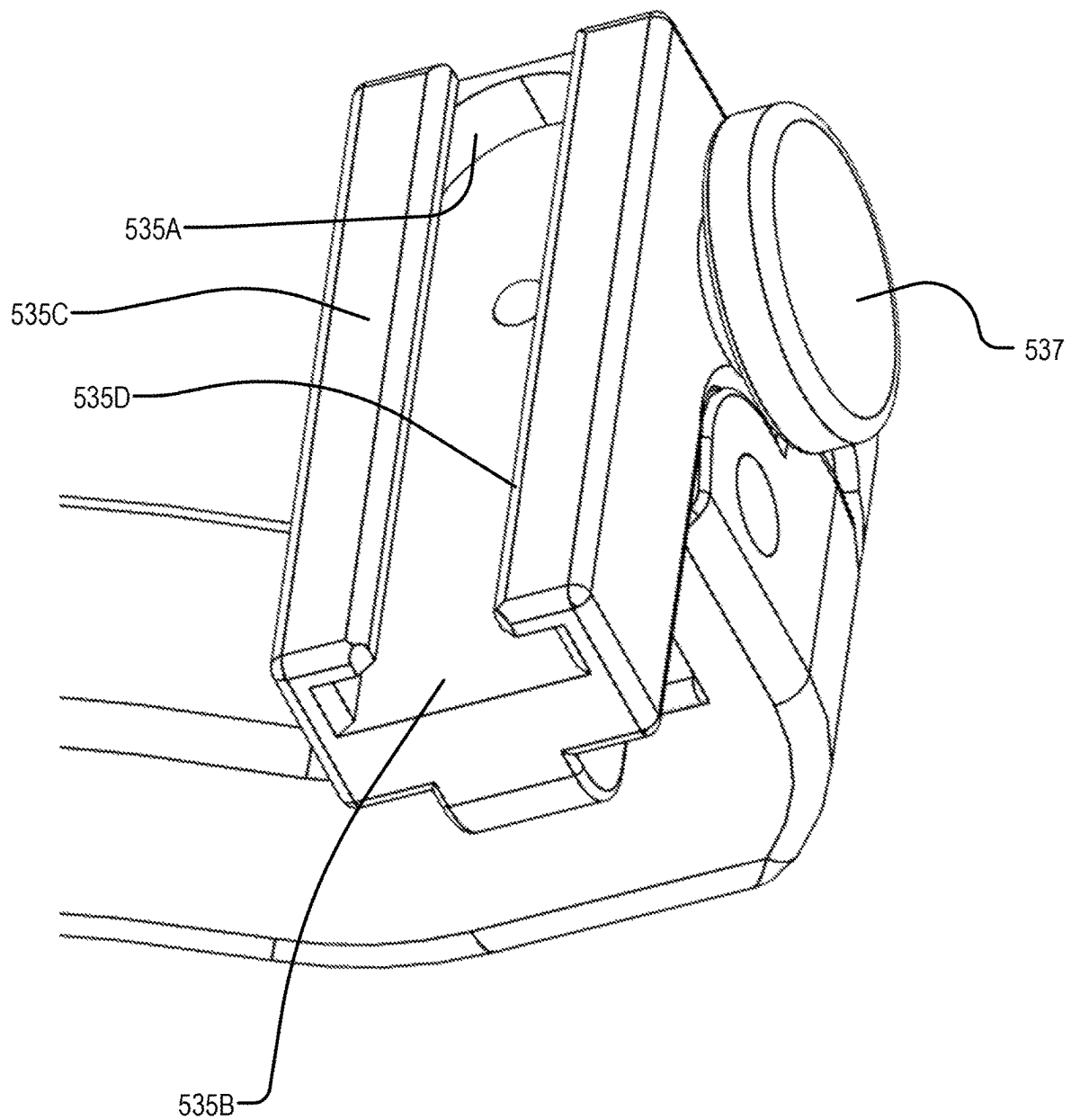
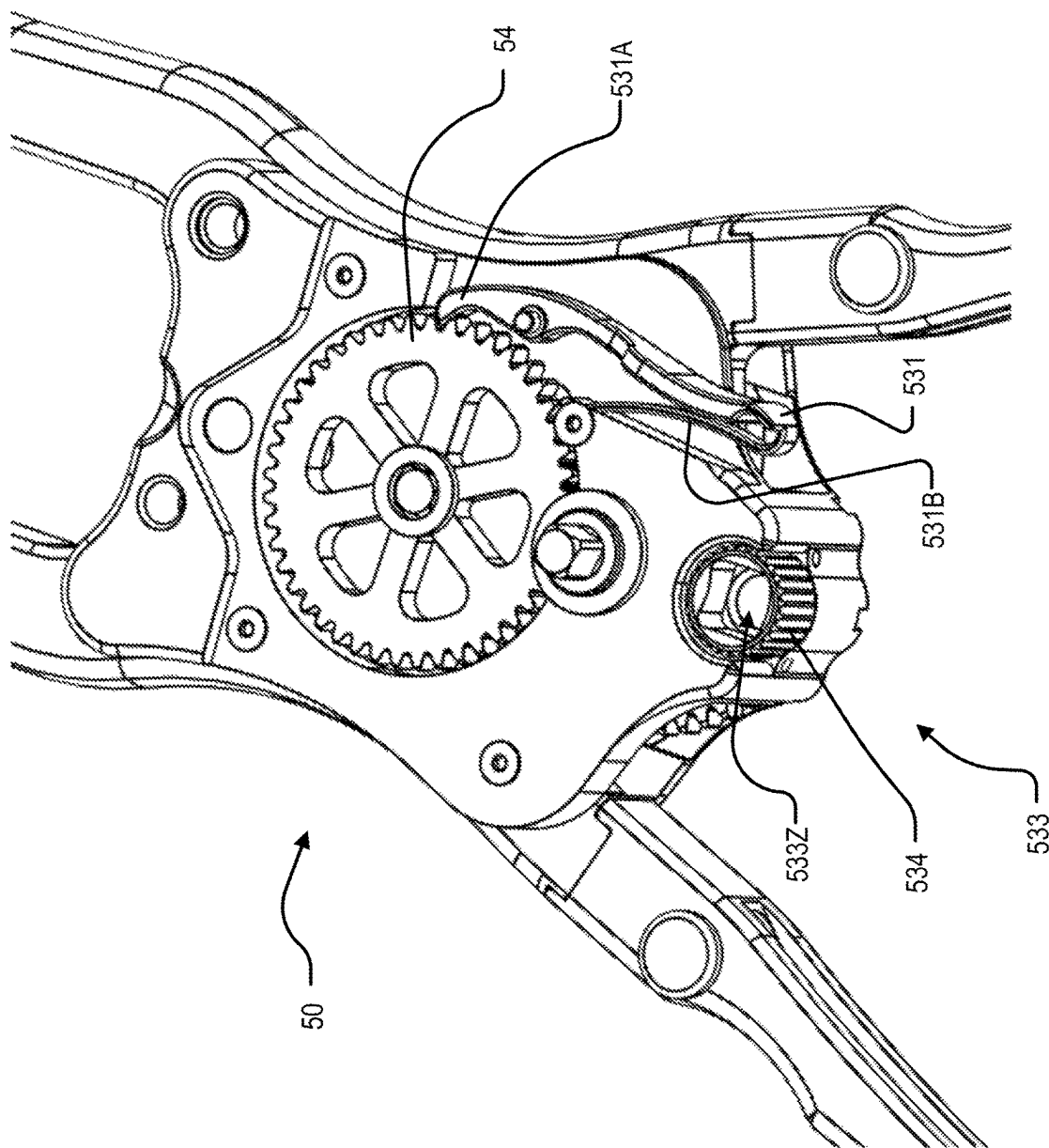


FIG. 121B



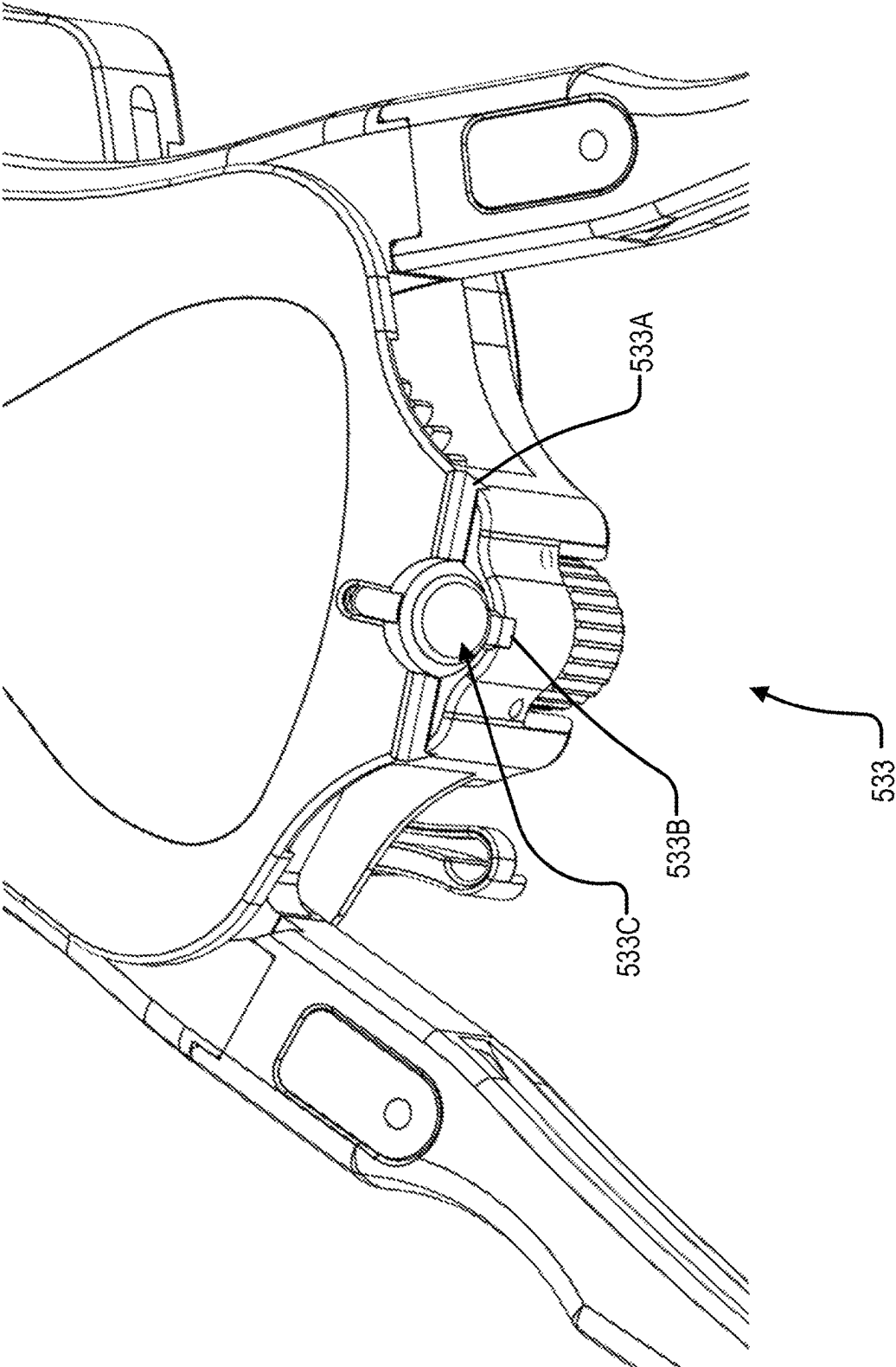
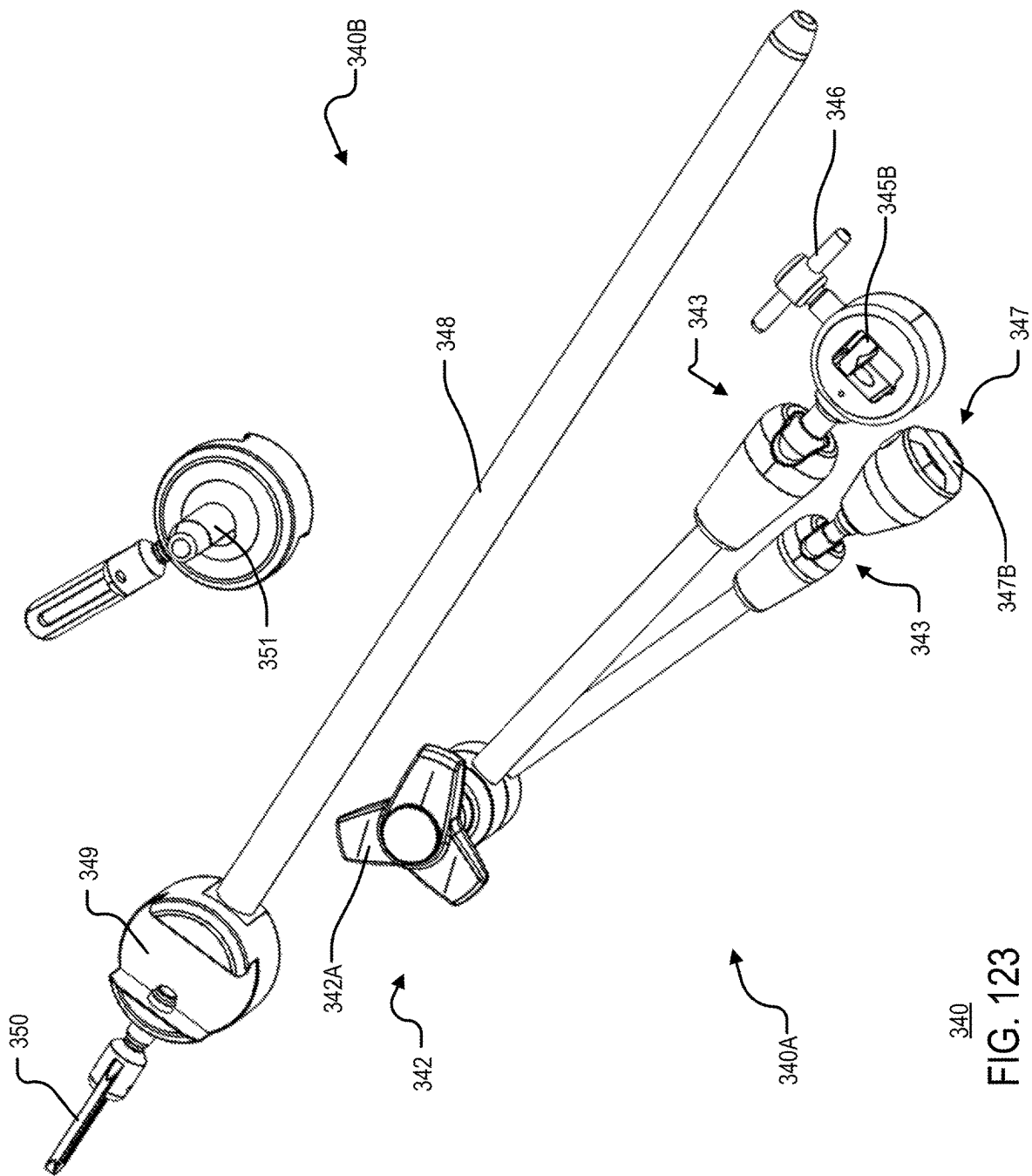
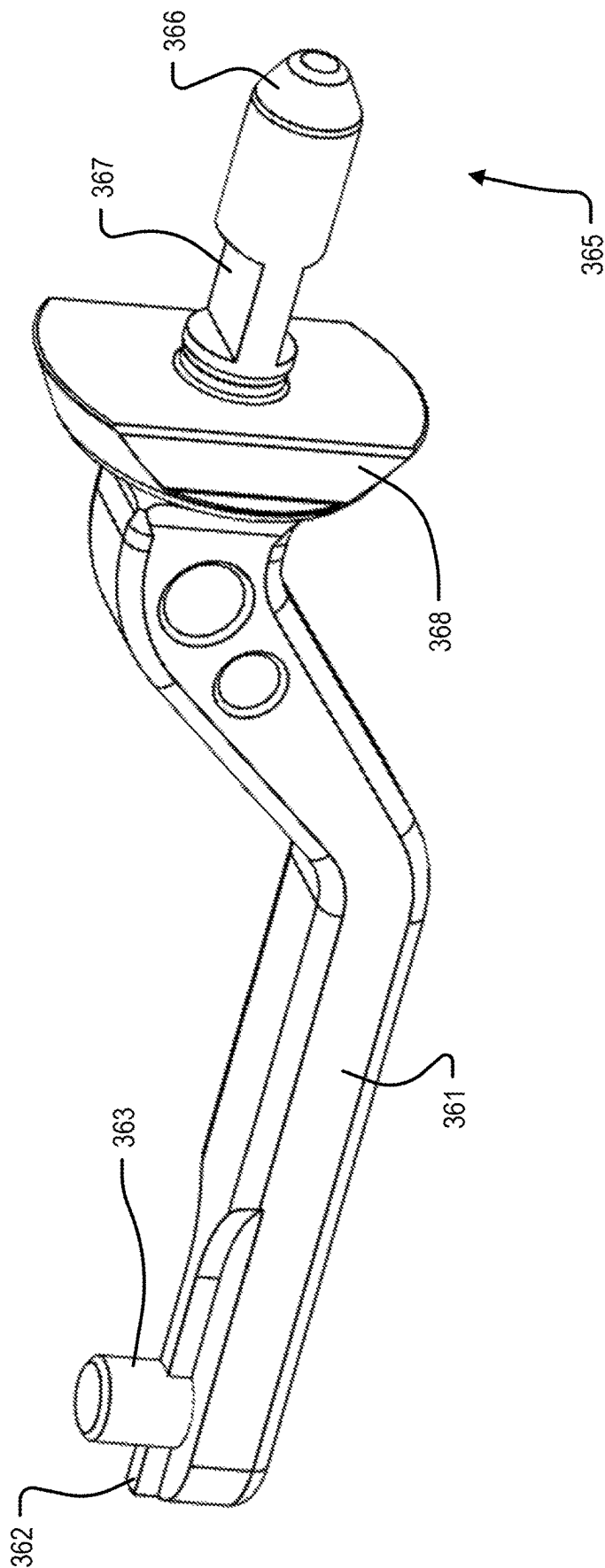
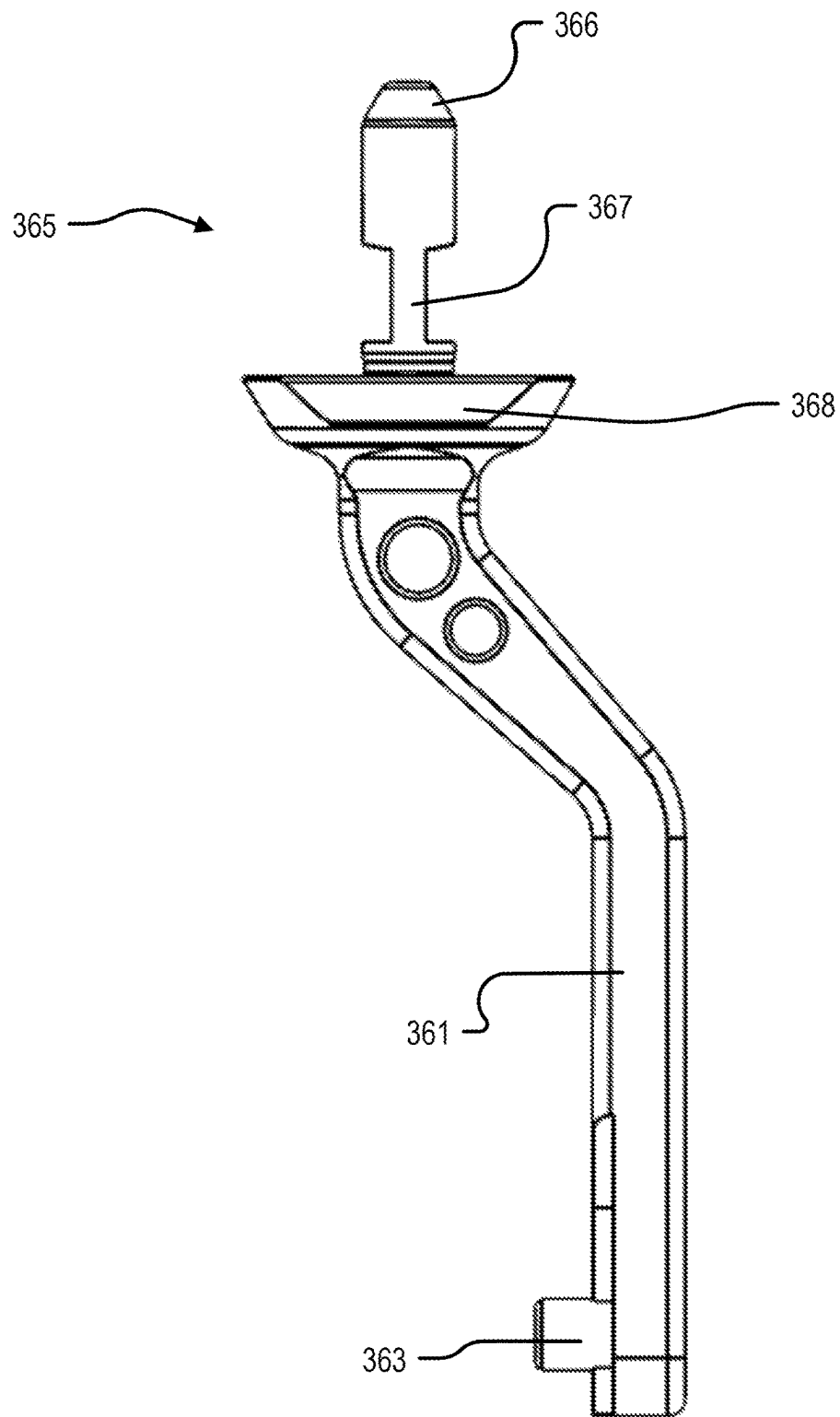


FIG. 122B





360
FIG. 124



360
FIG. 125

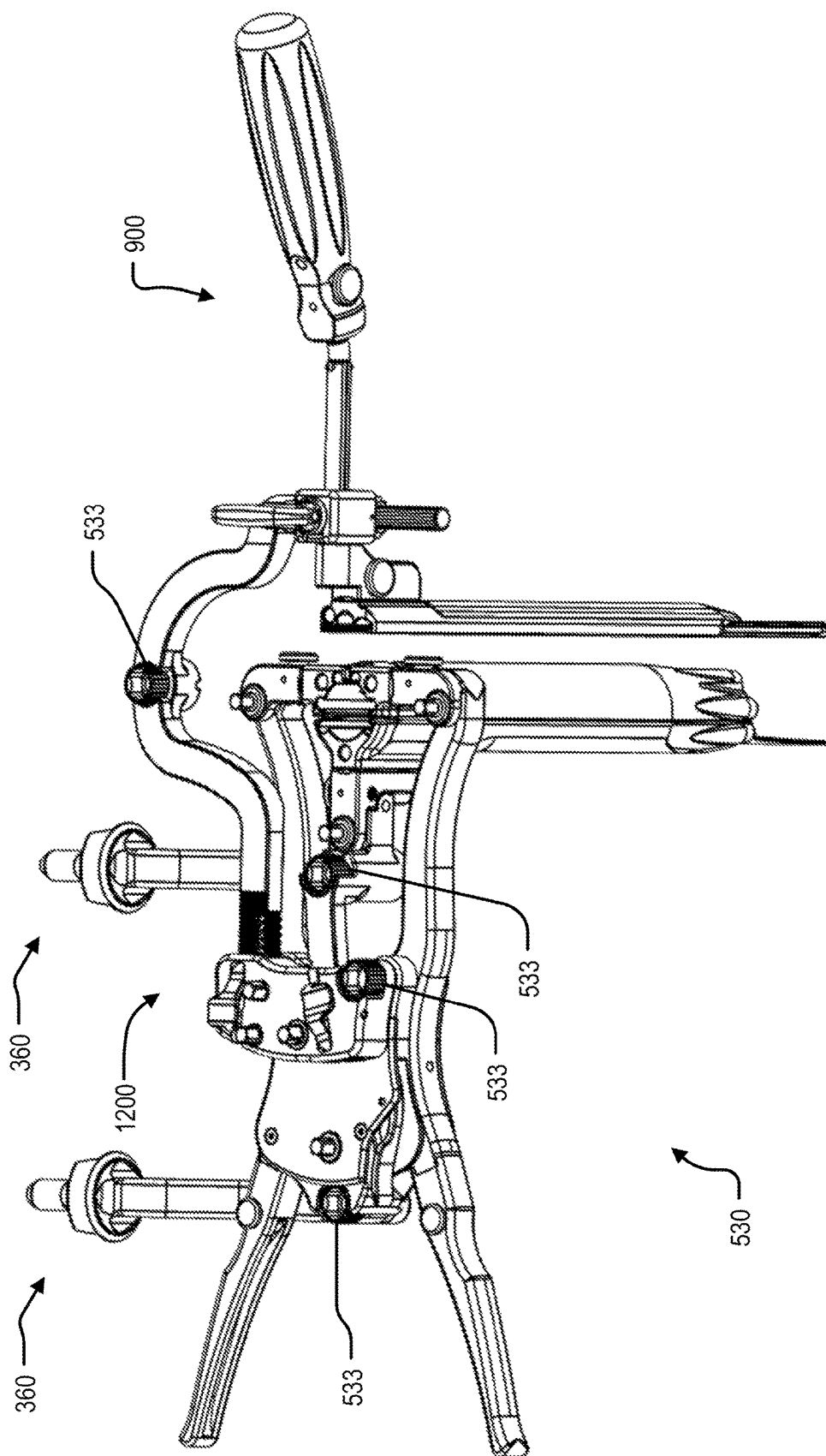


FIG. 126

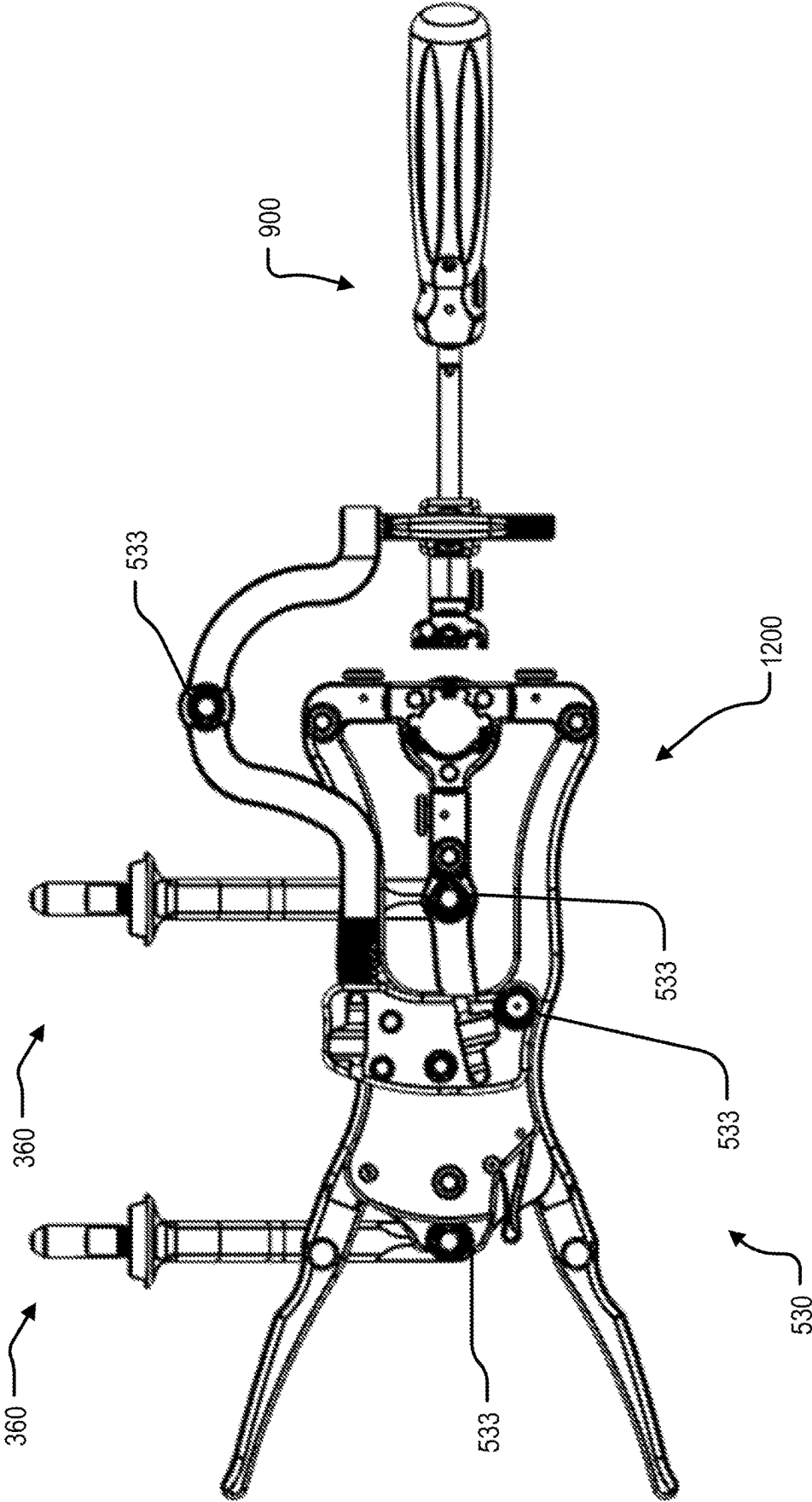


FIG. 127

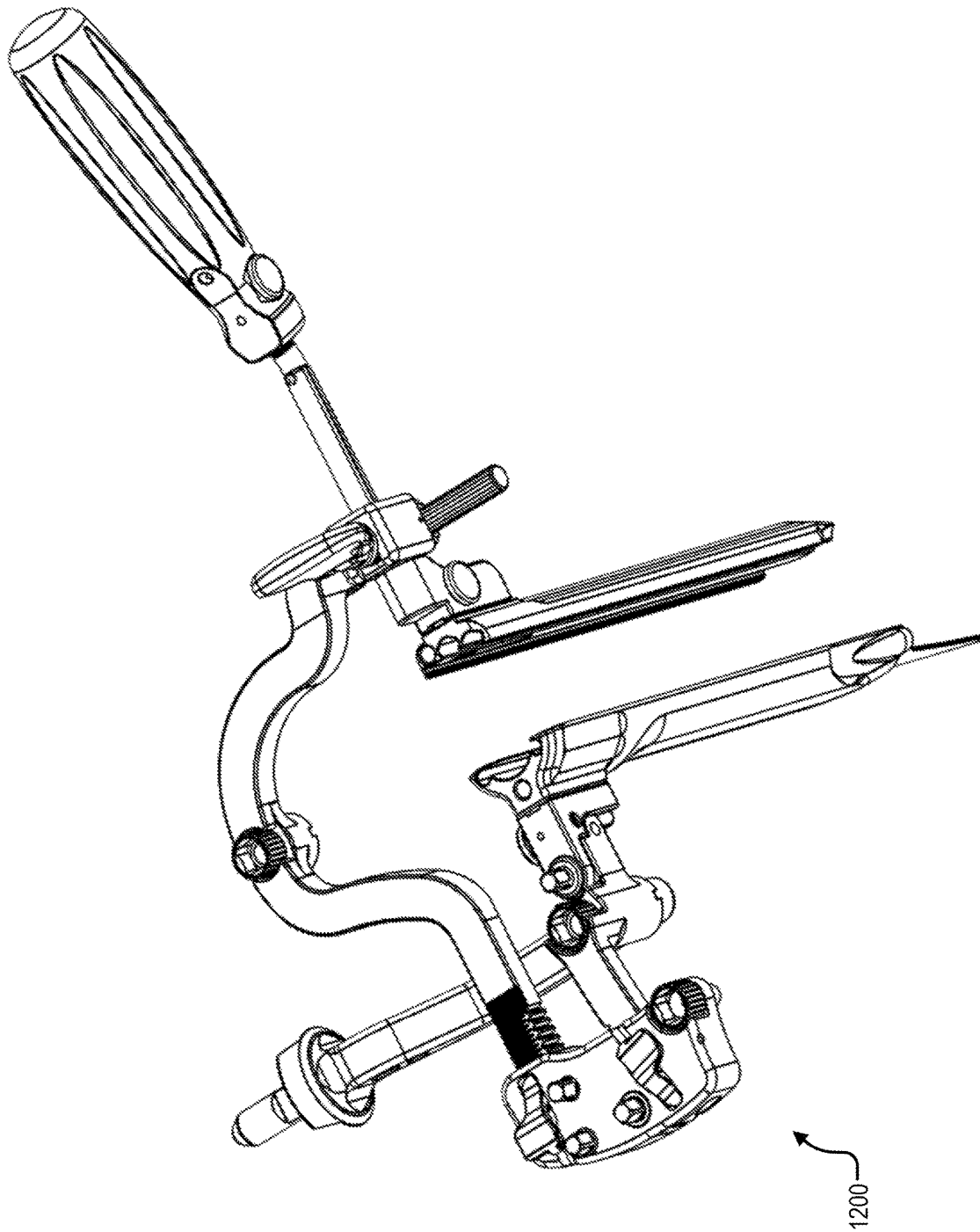


FIG. 128

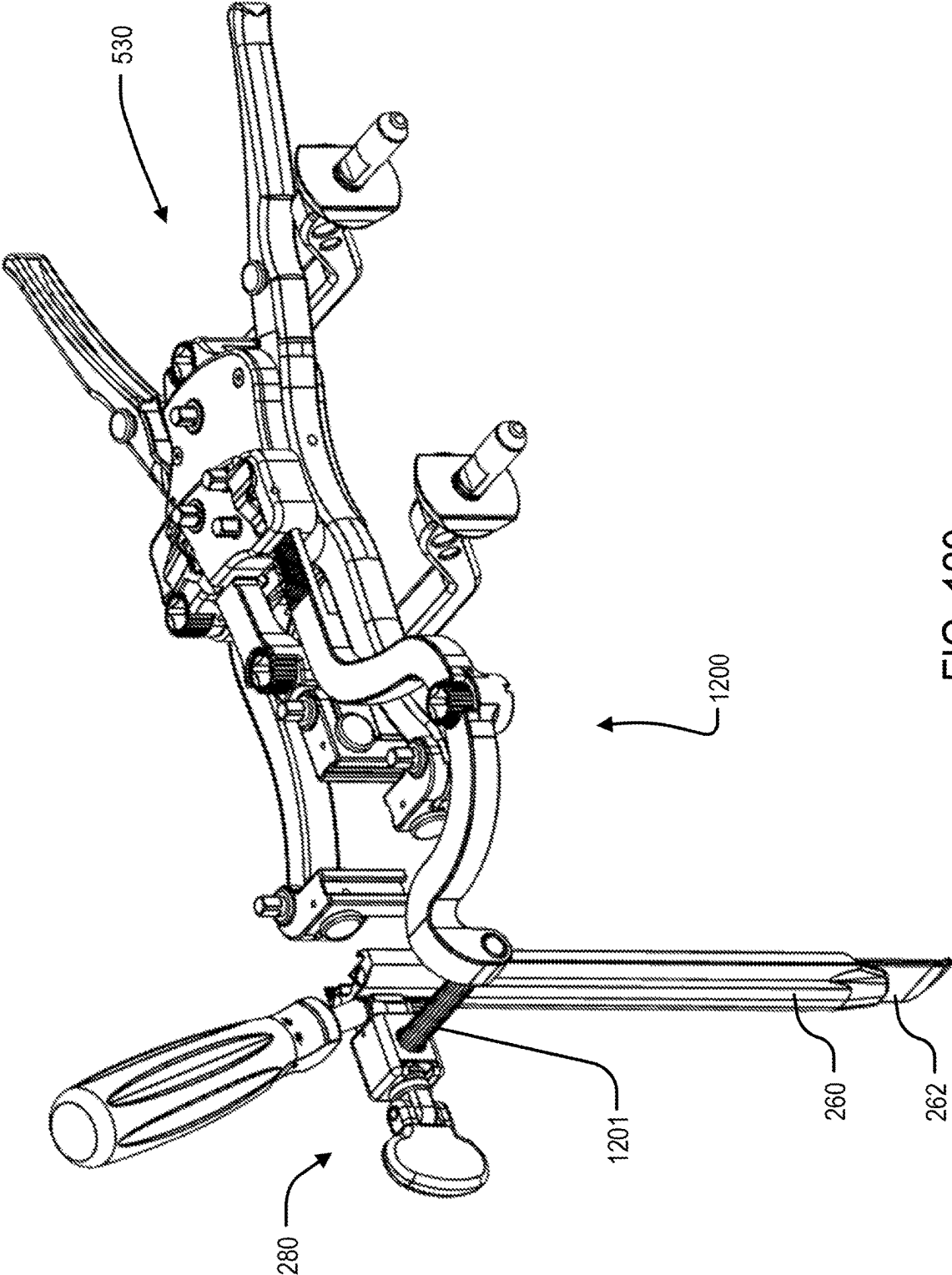


FIG. 129

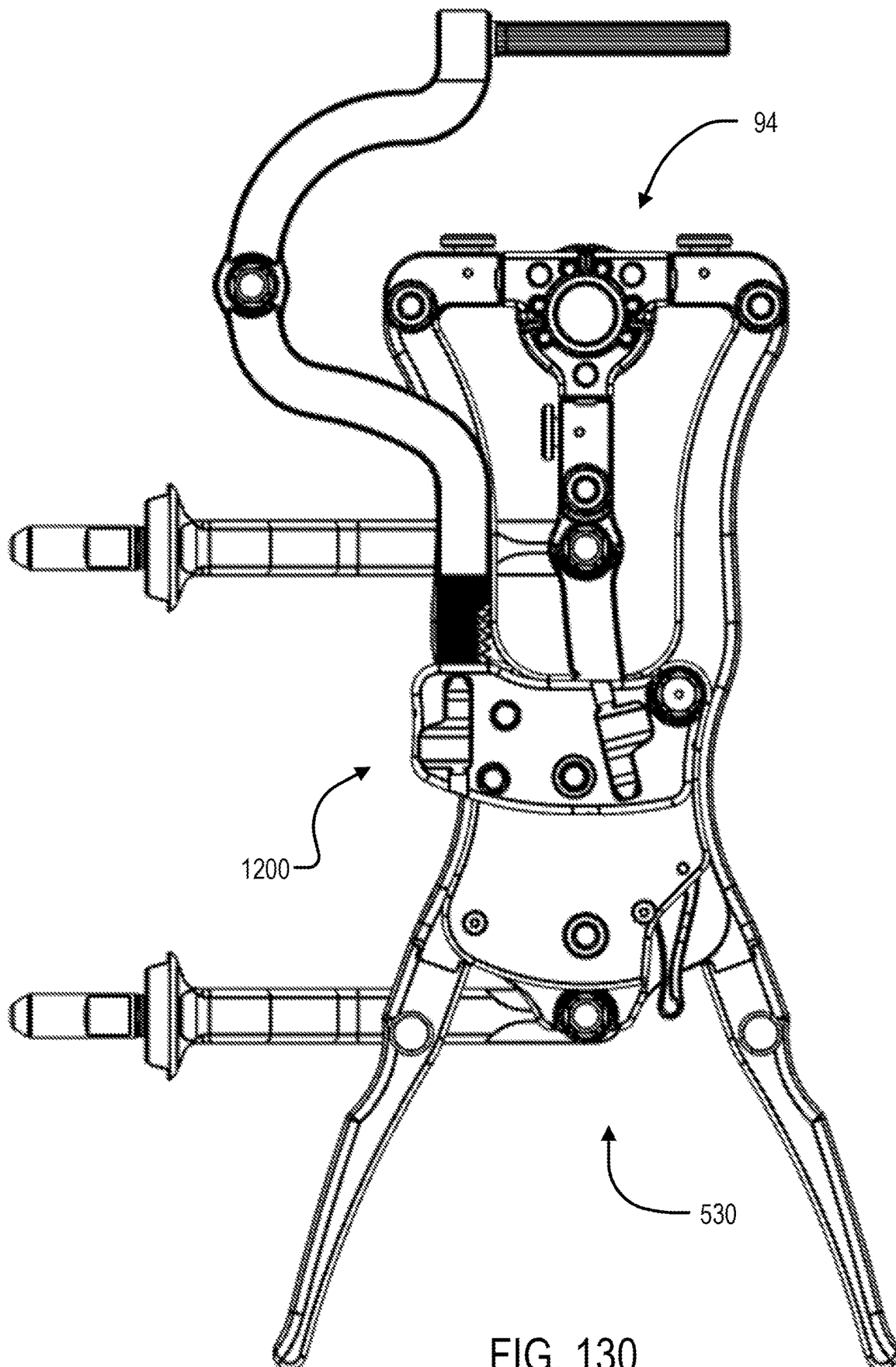


FIG. 130

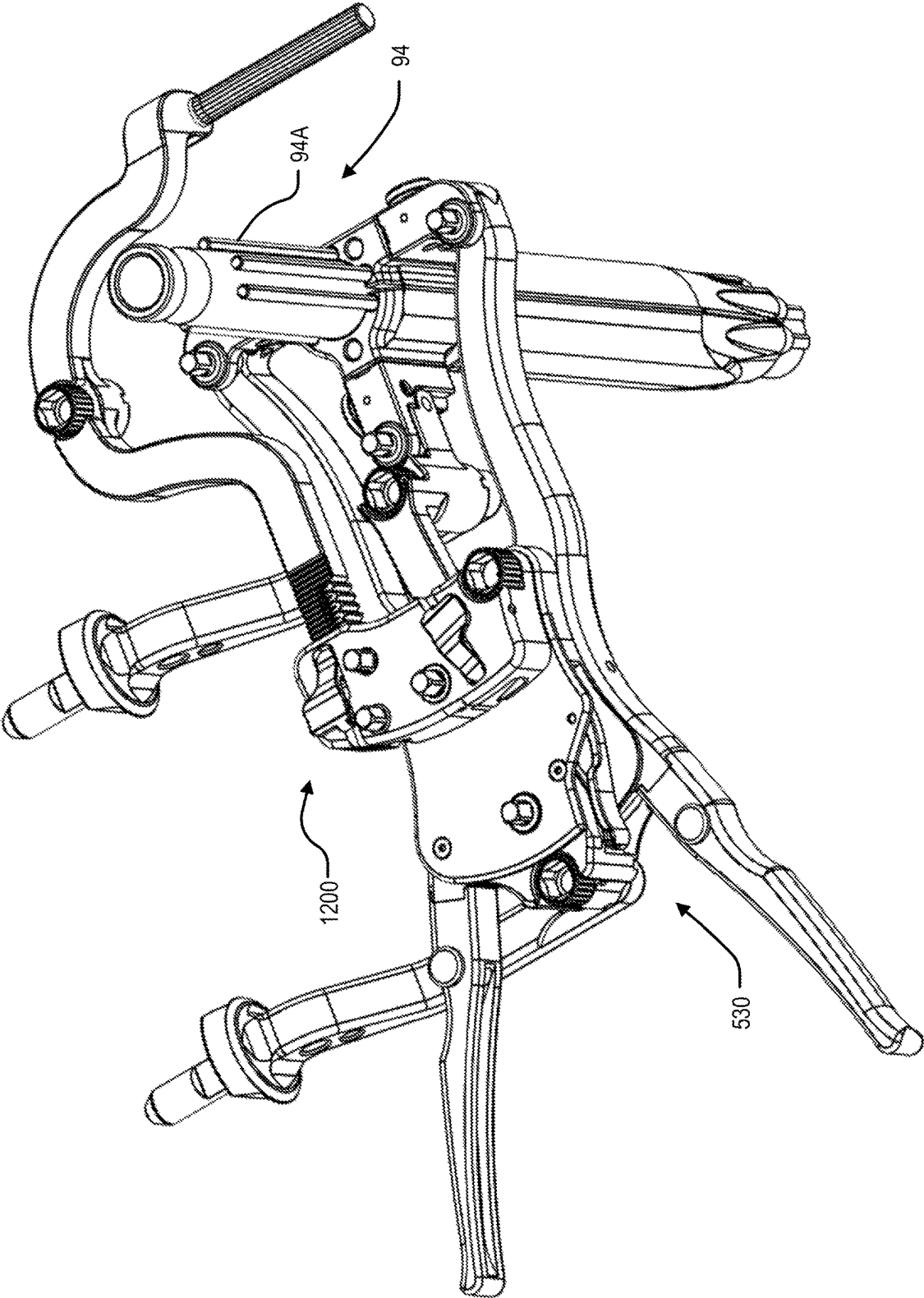


FIG. 131

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TISSUE RETRACTOR, RETRACTION MODULES, AND ASSOCIATED METHODS

CROSS-REFERENCE TO RELATED U.S. PATENT APPLICATION

This application is a continuation in part of: U.S. Non-provisional patent application Ser. No. 17/336,860 entitled "Tissue Retractor, Retraction Modules, and Associated Methods," filed Jun. 2, 2021 which is a continuation in part of U.S. Nonprovisional patent application Ser. No. 16/926,173 entitled "Tissue Retractor," filed Jul. 10, 2020. This application also claims priority to U.S. Provisional Application Ser. No. 63/254,929 filed Oct. 12, 2021. The entire disclosure of each of the above applications is incorporated by reference in its entirety.

FIELD

The present technology is generally related to medical devices to assist a surgeon during treatment of musculoskeletal disorders, and more particularly to a surgical system and method for accessing a surgical site to facilitate treatment. More particularly, the present disclosure is directed to a surgical retractor system including a primary retractor assembly and a secondary retractor assembly that are configured for various approaches to the spine, including for example, anterior, lateral, and oblique surgical techniques.

BACKGROUND

Spinal disorders such as degenerative disc disease, disc herniation, osteoporosis, spondylolisthesis, stenosis, scoliosis and other curvature abnormalities, kyphosis, tumor, and fracture may result from factors including trauma, disease and degenerative conditions caused by injury and aging. Spinal disorders typically result in symptoms including pain, nerve damage, and partial or complete loss of mobility.

Non-surgical treatments, such as medication, rehabilitation and exercise can be effective, however, may fail to relieve the symptoms associated with these disorders. Surgical treatment of these spinal disorders includes fusion, fixation, discectomy, laminectomy and implantable prosthetics. Surgical retractors may be employed during a surgical treatment to provide access and visualization of a surgical site. Such retractors space apart and support tissue and/or other anatomical structures to expose anatomical structures at the surgical site and/or provide a surgical pathway for the surgeon to the surgical site.

SUMMARY

This disclosure describes a plurality of different embodiments and modules for use as a modular retractor system. The system may use any of the variously disclosed blades, extendable blades, and dilators. Additionally, this disclosure describes a quick connect and release coupler for securing the modular retractor system to a table mount.

In an aspect, this disclosure describes a modular surgical retractor system including a modular retractor configured to couple and uncouple from a plurality of different add on modules. The retractor system may be designed for enabling access to a surgical site and/or forming an operative corridor, for example. In various embodiments, the retractor system may include a modular retractor having a longitudinal axis extending in a longitudinal direction and a lateral axis extending from a first lateral end to a second lateral end in

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a lateral direction, for example. In various embodiments, the modular retractor may include a first body portion that houses a distraction mechanism, a first arm and a second arm pivotally coupled together, and a first handle coupled to the first arm and a second handle coupled to the second arm, for example. In various embodiments, a first pivoting member may be coupled to a distal end of the first arm and a second pivoting member may be coupled to a distal end of the second arm, for example. In various embodiments, a first blade attachment mechanism coupled to the first pivoting member and a second blade attachment mechanism coupled to the second pivoting member, the first and second blade attachment mechanisms being configured to couple to first and second blades, respectively. In various embodiments, a first actuator may be operably coupled to the distraction mechanism for opening and closing the first arm and the second arm, for example. In various embodiments, a second actuator for adjusting the angulation of the first pivoting member, and a third actuator for adjusting the angulation of the second pivoting member may be provided, for example. In various embodiments, the modular retractor may include at least one table mount quick release coupler.

The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a perspective view of an exemplary embodiment of a retractor system including a primary retractor assembly and a secondary retractor assembly in accordance with the principles of the disclosure.

FIG. 2 is a perspective view of the primary retractor assembly of FIG. 1 in accordance with the principles of the disclosure.

FIG. 3 is a perspective view of the secondary retractor assembly of FIG. 1 in accordance with the principles of the disclosure.

FIG. 4 is a perspective view of the retractor system of FIG. 1 including a plurality of blades in accordance with the principles of the disclosure.

FIG. 5 is a top down view of the primary retractor assembly of FIG. 2 in accordance with the principles of the disclosure.

FIG. 6 is a top down view of the secondary retractor assembly of FIG. 3 in accordance with the principles of the disclosure.

FIG. 7 is a cutaway view of the retractor system of FIG. 1 in accordance with the principles of the disclosure.

FIG. 8 is an alternate cutaway view of the retractor system of FIG. 1 in accordance with the principles of the disclosure.

FIG. 9 is a perspective view of an exemplary blade for use with the retractor system of FIG. 1 in accordance with the principles of the disclosure.

FIG. 10 is an alternate perspective view of an exemplary blade and pin for use with the retractor system of FIG. 1 in accordance with the principles of the disclosure.

FIG. 11 is a perspective view of an exemplary blade and shim for use with the retractor system of FIG. 1 in accordance with the principles of the disclosure.

FIG. 12 is a perspective view of an exemplary set of nested dilators for coordinated use with the retractor system of FIG. 1 in accordance with the principles of the disclosure.

FIG. 13A is a top down view of the set of nested dilators of FIG. 12, and FIG. 13B is a top down view of a plurality

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of blades for use with the retractor system of FIG. 1 in accordance with the principles of the disclosure.

FIG. 14 is a top down view of an exemplary retractor system having a plurality of blades surrounding a set of nested dilators in accordance with the principles of the disclosure.

FIG. 15 is a top down view of an exemplary retractor system of FIG. 14 in a first partially expanded position after removal of the set of nested dilators in accordance with the principles of the disclosure.

FIG. 16 is a top down view of an exemplary retractor system of FIG. 14 in the first partially expanded position in accordance with the principles of the disclosure.

FIG. 17 is a side view of the exemplary retractor system of FIG. 14 having a plurality of blades surrounding a set of nested dilators in accordance with the principles of the disclosure.

FIG. 18 is a side view of the exemplary retractor system of FIG. 14 in a second expanded position in accordance with the principles of the disclosure.

FIG. 19 is a side view of the exemplary retractor system of FIG. 14 in the second expanded position in accordance with the principles of the disclosure.

FIG. 20 is a perspective view of a modular retractor.

FIG. 21 is a perspective view of a modular retractor.

FIG. 22 is a top down view of a modular retractor.

FIG. 23 is a perspective view of an adjustment tool for use with disclosed modular retractor embodiments.

FIG. 24 is a top down exploded parts view of a modular retractor.

FIG. 25 is a perspective exploded parts view of a modular retractor.

FIG. 26A is a perspective view of a distraction mechanism for use with disclosed modular retractor embodiments.

FIG. 26B is a top perspective view of a distraction mechanism.

FIG. 26C is an enlarged top perspective view of a distraction mechanism.

FIG. 26D is a bottom perspective view of a distraction mechanism.

FIG. 26E is an enlarged bottom perspective view of a distraction mechanism.

FIG. 27A is a top down view of a modular retractor coupled to a table mount.

FIG. 27B is a perspective view of a modular retractor coupled to a table mount.

FIG. 28A is a perspective view of a table mount rack.

FIG. 28B is a perspective view of a table mount rack.

FIG. 28C is an exploded parts view of a table mount rack.

FIG. 29 is a top perspective view of a first module for use with disclosed modular retractor embodiments.

FIG. 30 is a top perspective view of a first module for use with disclosed modular retractor embodiments.

FIG. 31A is a bottom perspective view of a first module for use with disclosed modular retractor embodiments.

FIG. 31B is a bottom perspective view of a first module for use with disclosed modular retractor embodiments.

FIG. 32 is an exploded parts view of a first module for use with disclosed modular retractor embodiments.

FIG. 33A is a perspective view of a first module coupled to a modular retractor.

FIG. 33B is a perspective view of a first module coupled to a modular retractor.

FIG. 34 is a top down view of a first module coupled to a modular retractor and a plurality of blades.

FIG. 35 is a top perspective view of a second module for use with disclosed modular retractor embodiments.

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FIG. 36 is a top perspective view of a second module for use with disclosed modular retractor embodiments.

FIG. 37 is a bottom perspective view of a second module for use with disclosed modular retractor embodiments.

FIG. 38 is a bottom perspective view of a second module for use with disclosed modular retractor embodiments.

FIG. 39 is an exploded parts view of a second module for use with disclosed modular retractor embodiments.

FIG. 40 is a top down view of a second module coupled to a modular retractor and a plurality of blades.

FIG. 41A is a top perspective view of a third module for use with disclosed modular retractor embodiments.

FIG. 41B is a bottom perspective view of a third module for use with disclosed modular retractor embodiments.

FIG. 42 is an exploded parts view of a third module for use with disclosed modular retractor embodiments.

FIG. 43 is a perspective view of a third module coupled to a modular retractor.

FIG. 44 is a perspective view of a third module coupled to a modular retractor and a plurality of blades.

FIG. 45A is a top perspective view of an alternate embodiment of a third module.

FIG. 45B is a bottom perspective view of an alternate embodiment of a third module.

FIG. 46 is a perspective view of a free hand module coupled to a third module for use with disclosed modular retractor embodiments.

FIG. 47A is a side view of a free hand module for use with disclosed modular retractor embodiments.

FIG. 47B is a side view of a free hand module for use with disclosed modular retractor embodiments.

FIG. 48A is an exploded parts view of a free hand module.

FIG. 48B is a perspective view with partially removed parts of a free hand module.

FIG. 49A is a perspective view of a free hand module in a sliding configuration.

FIG. 49B is a perspective view of a free hand module in a second position.

FIG. 50 is a perspective view of a free hand module and an telescoping blade system.

FIG. 51 is a perspective view of an telescoping blade system.

FIG. 52A is a perspective view of a blade connection channel.

FIG. 52B is a perspective view of a blade fastener.

FIG. 53 is a perspective view of a third module coupled to a modular retractor and a free hand module coupled to the third module.

FIG. 54 is a perspective view of a third module coupled to a modular retractor and a free hand module coupled to the third module.

FIG. 55 is a top perspective view of a fourth module for use with disclosed modular retractor embodiments.

FIG. 56 is a bottom perspective view of a fourth module for use with disclosed modular retractor embodiments.

FIG. 57 is an exploded parts view of a fourth module.

FIG. 58 is a top down view of a fourth module coupled to a modular retractor.

FIG. 59 is a perspective view of a fourth module coupled to a modular retractor.

FIG. 60 is a perspective view of a fourth module coupled to a modular retractor and first and second free hand modules coupled to the fourth module.

FIG. 61 is a perspective view of a fourth module coupled to a modular retractor and first and second free hand modules coupled to the fourth module.

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FIG. 62 is a top perspective view of a fifth module for use with disclosed modular retractor embodiments.

FIG. 63 is a bottom perspective view of a fifth module for use with disclosed modular retractor embodiments.

FIG. 64 is an exploded parts view of a fifth module.

FIG. 65 is a top perspective view of a fifth module.

FIG. 66 is a top view of a pair of blades for use with disclosed modular retractor embodiments.

FIG. 67 is a bottom view of a pair of blades for use with disclosed modular retractor embodiments.

FIG. 68 is a perspective view of a pair of blades for use with disclosed modular retractor embodiments.

FIG. 69 is an enlarged view of a top portion of a universal blade fastener.

FIG. 70 is a top view of three blades for use with disclosed modular retractor embodiments.

FIG. 71 is a bottom view of three blades for use with disclosed modular retractor embodiments.

FIG. 72 is a perspective view of three blades for use with disclosed modular retractor embodiments.

FIG. 73 is a top view of four blades for use with disclosed modular retractor embodiments.

FIG. 74 is a bottom view of four blades for use with disclosed modular retractor embodiments.

FIG. 75 is a perspective view of four blades for use with disclosed modular retractor embodiments.

FIG. 76 is a top view of a plurality of nested dilators.

FIG. 77A is a perspective view of a plurality of nesting dilators of FIG. 76 in a non-nested configuration.

FIG. 77B is a perspective view of a plurality of nesting dilators in a nested configuration.

FIG. 78 is a top view of a dilator.

FIG. 79 is a perspective view of the dilator of FIG. 78.

FIG. 80A is a top view of a dilator.

FIG. 80B is a perspective view of the dilator of FIG. 80A.

FIG. 80C is a perspective view of a set of nested and cylindrically shaped dilators.

FIG. 80D is an elevation view of the various dilators of embodiment of FIG. 80C.

FIG. 80E is a perspective view of the various dilators of embodiment of FIG. 80C.

FIG. 81 is a top down view of the various dilators of embodiment of FIG. 80C.

FIG. 82 is a perspective view of a modular blade.

FIG. 83 is a perspective view of a modular blade.

FIG. 84 is a perspective view of an extendable blade for coupling to the modular blade of FIGS. 82-83.

FIG. 85 is a perspective view of an extendable blade for coupling to the modular blade of FIGS. 82-83.

FIG. 86 is a front view of the modular blade of FIGS. 82-83 and the extendable blade of FIGS. 84-85.

FIG. 87 is a top down view of the modular blade of FIGS. 82-83 and the extendable blade of FIGS. 84-85.

FIG. 88 is a perspective view of a modular blade.

FIG. 89 is a perspective view of a modular blade.

FIG. 90 is a perspective view of an extendable blade for coupling to the modular blade of FIGS. 82-83.

FIG. 91 is a perspective view of an extendable blade for coupling to the modular blade of FIGS. 88-89.

FIG. 92 is a front view of the modular blade of FIGS. 88-89 and the extendable blade of FIGS. 90-91.

FIG. 93 is a top down view of the modular blade of FIGS. 88-89 and the extendable blade of FIGS. 90-91.

FIG. 94 is a perspective view of a modular blade.

FIG. 95 is a perspective view of a modular blade.

FIG. 96 is a perspective view of an extendable blade for coupling to the modular blade of FIGS. 94-95.

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FIG. 97 is a perspective view of an extendable blade for coupling to the modular blade of FIGS. 94-95.

FIG. 98 is a front view of the modular blade of FIGS. 94-95 and the extendable blade of FIGS. 96-97.

FIG. 99 is a top down view of the modular blade of FIGS. 94-95 and the extendable blade of FIGS. 96-97.

FIG. 100 is a perspective view of a modular blade.

FIG. 101 is a perspective view of a modular blade.

FIG. 102 is a perspective view of an extendable blade for coupling to the modular blade of FIGS. 100-101.

FIG. 103 is a perspective view of an extendable blade for coupling to the modular blade of FIGS. 100-101.

FIG. 104 is a perspective view of an extendable blade for coupling to the modular blade of FIGS. 100-101.

FIG. 105 is a front view of the extendable blades of FIGS. 103-105.

FIG. 106 is a perspective view of the outside surfaces of a modular blade and an extendable blade having a pointed end.

FIG. 107 is a perspective view of the inside surfaces of the modular blade and the extendable blade of FIG. 106.

FIG. 108A is a first exploded parts view of the modular blade and extendable blade of FIGS. 106-107.

FIG. 108B is a second exploded parts view of the modular blade and extendable blade of FIGS. 106-107.

FIG. 108C is a perspective view of the modular blade of FIGS. 106-107.

FIG. 108D is a top down view of the modular blade of FIGS. 106-107.

FIG. 108E is a perspective view of an impact driver for use with the modular blade and extendable blade of FIGS. 106-107.

FIG. 109 is a perspective view of a square shaped dilator.

FIG. 110 is a bottom perspective view of a pair of shims for coupling to various blades disclosed herein.

FIG. 111 is a perspective view of a relatively short shim having a pointed pin at a distal end thereof.

FIG. 112 is a perspective view of a relatively tall shim having a pointed pin at a distal end thereof.

FIG. 113 is a perspective view of a relatively tall shim having a blunted distal end.

FIG. 114 is a perspective view of a double-sided shim for coupling to various blades disclosed herein.

FIG. 115A is a perspective view of a blade adjustment and positioning tool.

FIG. 115B is an exploded parts view of a blade adjustment and positioning tool.

FIG. 116A is a perspective view of the blade adjustment and positioning tool engaged with a modular blade and an extendable blade.

FIG. 116B is a perspective view of the blade adjustment and positioning tool engaged with a modular blade and an extendable blade.

FIG. 117A is a perspective view of the inside surfaces of a modular blade and an extendable blade having a footed tip at the distal end thereof.

FIG. 117B is a perspective view of the modular blade and extendable blade of FIG. 117A.

FIG. 118A is a perspective view of a quick connect handle.

FIG. 118B is an exploded parts view of a quick connect handle.

FIG. 118C is a perspective view of a retractor mount coupler.

FIG. 118D is a perspective view of the modular and extendable blades of FIGS. 117A-117B coupled to the quick connect handle of FIGS. 118A-118B and the retractor mount coupler of FIG. 118C.

FIG. 118E is a perspective view of the modular and extendable blades of FIGS. 117A-117B before being coupled to the quick connect handle of FIGS. 118A-118B.

FIG. 119 is a first perspective view of an additional embodiment of a modular retractor.

FIG. 120 is a second perspective view of the additional embodiment of the modular retractor of FIG. 119.

FIG. 121A is a top down view of the embodiment of FIGS. 119-120.

FIG. 121B is a bottom perspective view of a blade coupling portion.

FIG. 122A is an enlarged view of the embodiment of FIGS. 119-121 from a top perspective with the top cover removed for ease of understanding of the internal gear system.

FIG. 122B is an enlarged view of the embodiment of FIGS. 119-121 from a bottom perspective showing various structural features of a table mount quick connect coupler.

FIG. 123 is a perspective view of various armatures of a quick connect table mount system for supporting various retractor embodiments disclosed herein.

FIG. 124 is a first perspective view of a quick connect coupler for connecting various retractor embodiments to various quick connect table mount systems disclosed herein.

FIG. 125 is a side view of the quick connect coupler of FIG. 124.

FIG. 126 is a perspective view of a modular retractor system including the quick connect couplers of FIGS. 124-125.

FIG. 127 is a top down view of the system of FIG. 126. FIG. 128 is a perspective view of a secondary module that may be coupled and uncoupled with various primary retractor embodiments.

FIG. 129 is a perspective view of the secondary module of FIG. 128 coupled to a primary retractor.

FIG. 130 is a top down view of a modular retractor supporting first and second blades to be slidably coupled to an outermost dilator.

FIG. 131 is a perspective view of the three blades being slidably coupled to an outermost dilator.

DETAILED DESCRIPTION

In one aspect, exemplary embodiments describe a retractor system 100 for use with anterior, lateral, and oblique surgical techniques. At least one use of retractor system 100 is to assist in the preparation of a surgical site to enable a surgeon to access a space between vertebrae of patient's spine. The retractor system 100 may assist a surgeon in accessing a space between vertebrae by enabling highly controlled dilation of the paraspinal muscles with a set of nested dilators and retraction of the various fibers and tissues at the surgical site with the use of a plurality of independently movable and inclinable blades.

Referring generally to FIGS. 1-8 exemplary retractor systems for enabling access to a surgical site are disclosed. FIG. 1 is a perspective view of an exemplary embodiment of a retractor system 100 including a primary retractor assembly 200 and a secondary retractor assembly 300 in accordance with the principles of the disclosure. Retractor system 100 is highly customizable and modular. For example, the primary retractor assembly 200 may be used as a standalone retractor system without the use of secondary retractor

assembly 300. Secondary retractor assembly 300 is configured to couple and uncouple on as needed basis with the primary retractor assembly 200 and secondary retractor assembly 300 can, for example, use one or two arms each having a corresponding blade.

Exemplary embodiments may include a primary retractor assembly 200 configured to open and close a first arm 105 and a second arm 107 along a first path of travel. The first path may be an arcuate path or segment defined by the length and geometry of the arms 105 and 107 and a handle pivoting mechanism 101c (see FIG. 8) configured to enable first handle 101a and second handle 101b to open and close. Other paths of travel are contemplated depending upon the geometry of the arms 105, 107 and the relative location of the handle pivoting mechanism 101c. The primary retractor assembly 200 may include a handle assembly having first and second handles 101a, 101b that are operably coupled to the first and second arms 105, 107 and configured to open and close the first and second arms 105, 107. For example, the first handle 101a may be coupled to the first arm 105 and the second handle 101b may be coupled to the second arm 107. The first and second arms 105, 107 may be operably coupled to first and second pivoting members 105a, 107a at a distal end thereof, respectively. The first and second pivoting members 105a, 107a may be configured to operably couple to first and second blades, 205, 207 (see FIG. 2), respectively, by a corresponding blade attachment mechanism as will be explained in more detail below during the discussion of FIGS. 9-13B.

In the exemplary embodiment, a first actuator 105b and a second actuator 107b are configured to adjust the angulation of first blade 205 and second blade 207, respectively. For example, the first actuator 105b may be configured to actuate the first pivoting member 105a to adjust the angulation of first blade 205 with respect to the first arm 105. Similarly, the second actuator 107b may be configured to actuate the second pivoting member 107a to adjust the angulation of second blade 207 with respect to second arm 107. In the exemplary embodiment, the first pivoting member 105a may be configured to independently adjust the angulation of first blade 205 with respect to the first arm 105 upon actuation of the first actuator 105b. Similarly, the second pivoting member 107a may be configured to independently adjust the angulation of the second blade 207 with respect to the second arm 107 upon actuation of the second actuator 107b. In disclosed embodiments, the first and second pivoting members 105a, 107a may each include a corresponding pin and socket mechanism enabling the pivoting members to pivot on a pin aperture 199 (see, e.g., FIG. 8). Additionally, the first and second pivoting members 105a, 107a may each include a corresponding blade attachment mechanism at a distal end thereof which will be explained in more detail below when discussing FIGS. 9-13.

In the exemplary embodiment, the primary retractor assembly 200 may include a primary actuator 102 that is configured to actuate a primary pinion gear mechanism 210 (see FIG. 7) to provide a precise and controlled mechanical advantage to open and close the first arm 105 and second arm 107. For example, the primary pinion gear mechanism 210 may include a primary pinion gear 210a fixedly coupled to the primary actuator 102 such that the primary actuator 102 may rotationally translate the primary pinion gear 210a. The primary pinion gear 210a may be engaged with the secondary pinion gear 210b, e.g., the primary pinion gear 210a and secondary pinion gear 210b may be toothed gears that are meshed with one another at a contact location (not illustrated). Furthermore, secondary pinion gear 210b may

be fixedly coupled to tertiary pinion gear **210c** which may be axially aligned with secondary pinion gear **210b** and disposed directly beneath secondary pinion gear **210b** (see FIG. 8). For example, secondary pinion gear **210b** may share an axis of rotation with tertiary pinion gear **210c** and secondary pinion gear **210b** may be relatively larger in diameter than tertiary pinion gear **210c**. This arrangement may resemble a two stage gear box or the like that allows for an increase in applied torque. In other embodiments, primary pinion gear mechanism **210** may be any other similar planetary gear system as would be understood by a person having ordinary skill in the relevant art. For example, those with skill in the relevant art will readily recognize that the particular diameter, tooth sizing, and tooth spacing of the primary pinion gear **210a** relative to the particular diameter, tooth sizing, and tooth spacing of the secondary pinion gear **210b** relative to tertiary pinion gear **210c** may control the amount of force (mechanical advantage or torque) that is applied to open and close the first and second arms **105, 107**.

In the exemplary embodiment of FIG. 8, tertiary pinion gear **210c** may be meshed with a first curved rack portion **210a-2** and a second curved rack portion **210b-2** disposed opposite the first curved rack portion **210a-2**. First curved rack portion **210a-2** may be fixedly coupled to second arm **101b** and second curved rack portion **210b-2** may be fixedly coupled to first arm **101a**. Each of curved rack portions **210a-2** and **210b-2** may feature a plurality of teeth extending along the curved body thereof and facing tertiary pinion gear **210c**. The first curved rack portion **210a-2** and second curved rack portion **210b-2** may be meshed with the teeth of tertiary pinion gear **210c** on opposite sides of tertiary pinion gear **210c**. In this way, when primary actuator **102** is rotated, primary pinion gear **210a** rotates which in turn rotates secondary pinion gear **210b** and tertiary pinion gear **210c**. In turn, tertiary pinion gear **210c** engages teeth on each of curved rack portions **210a-2** and **210b-2** and causes handles **101a, 101b** to open or close. In the disclosed embodiment, when tertiary pinion gear **210c** applies force to first curved rack portion **210a-2**, the first curved rack portion **210a-2** may extend through first handle **101a** at a corresponding first handle aperture **210a-1**. Similarly, when tertiary pinion gear **210c** applies force to second curved rack portion **210b-2**, the second curved rack portion **210b-2** may extend through second handle **101b** at a corresponding second handle aperture **210b-1**.

In disclosed embodiments, the primary pinion gear mechanism **210** may be operably coupled to the first and second handles **101a, 101b** and configured to simultaneously open and close the first and second arms **105, 107** along a first path of travel. For example, the primary actuator **102** may rotationally translate the primary pinion gear mechanism **210** in a clockwise direction which in turn rotationally translates the first arm **105** and second arm **107** such that they move away from one another, i.e., they open as explained above. Likewise, the primary actuator **102** may rotationally translate the primary pinion gear mechanism **210** in a counter clockwise direction which in turn rotationally translates the first arm **105** and second arm **107** such that they move towards one another, i.e., they close as explained above. Also as explained above, the particular diameter of primary, secondary, and tertiary pinion gears **210a, 210b, and 210c** may be adjusted to provide the desired amount of mechanical advantage or torque to open and close first and second arms **101a, 101b**.

In disclosed embodiments, primary retractor assembly **200** may include a primary retention lever **104** disposed between the first and second handles **101a, 101b** that is

configured to engage the primary retractor assembly **200** to control opening and closing of the first and second arms **105, 107** and thereby retain the first and second arms **105, 107** in a specific position. In the disclosed embodiment, primary retention lever **104** may frictionally engage curved rack portion **210b-2** to control opening and closing of the first and second arms. In other embodiments, the primary retention lever **104** may engage the primary pinion gear mechanism **210** at an outside portion of the circumference of the primary pinion gear **210a** (see FIG. 7) to thereby control and/or prevent rotation of the primary pinion gear **210a**. For example, the primary retention lever **104** may lock the primary pinion gear mechanism **210** in place to control opening and closing of the first and second arms. In some embodiments, the primary retention lever **104** may have a biasing element (not illustrated) that causes the primary retention lever **104** to naturally urge an angled tip portion of the body of the primary retention lever **104** against a portion of the primary pinion gear mechanism **210**. For example, a spring may naturally urge an angled tip portion of primary retention lever **104** to engage with a toothed portion of secondary pinion gear **210b**. Additionally, the primary retention lever **104** may be moved from an engagement position where primary retention lever **104** is in direct contact with the primary pinion gear mechanism **210** to a disengaged position where primary retention lever **104** is not engaged with the primary pinion gear mechanism **210**. For example, an end user such as a surgeon may depress primary retention lever **104** with their thumb to toggle primary retention lever **104** between the engaged position and the disengaged position. Furthermore, some embodiments may have a toggle feature (not illustrated) for maintaining the primary retention lever **104** in either of the engaged or disengaged positions.

In disclosed embodiments, the primary retractor assembly **200** may include a first table mount portion **106a** disposed adjacent the first handle **101a** and coupled to a body **200a** (see FIG. 5) or housing of the primary retractor assembly **200**. Similarly, the primary retractor assembly **200** may include a second table mount portion **106b** disposed adjacent the second handle **101b** and coupled to the body or housing of the primary retractor assembly **200**. The first and second table mount portions **106a, 106b** may each be attached to a surgical table (not illustrated) for fixing the primary retractor assembly **200** (and/or the retractor system **100**) in a fixed location in three dimensional space. In example embodiments, the primary retractor assembly **200** may be attached to a surgical table by at least one of the first and second table mount portions **106a, 106b** or by both.

At least one advantage of securing the primary retractor assembly **200** to a surgical table may be for enhanced stability and the even transfer of resultant forces from the primary actuator **102** through the first and second arms **105, 107** to the first and second blades **205, 207** and vice versa. For example, when the primary retractor assembly **200** is fixed to the surgical table and the primary actuator **102** is translated to open the first and second arms **105, 107** the primary pinion gear mechanism **210** may apply a precise controlled amount of force to open the first and second arms **105, 107** to thereby gently retract the tissue of a patient in a controlled manner. Additionally, when the primary retractor assembly **200** is fixed to the surgical table, it may be easier for an end user to independently move only one of the handles **101a, 101b** with respect to the surgical table. When moving only one of the handles **101a, 101b** the corresponding arm **105, 107** may move relative to the other. This scenario and functionality may assist a surgeon with precise surgical techniques where it may be desirable to indepen-

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dently move either of the first and second arms **105**, **107** along the first path of travel independently with respect to the other.

Disclosed embodiments described above may be configured to independently open and close the first arm **105** along the first path of travel by movement of the first handle **101a** relative to the second handle **101b** and independently open and close the second arm **107** along the first path of travel by movement of the second handle **101b** relative to the first handle **101a**. Additionally, because the primary pinion gear mechanism **210** includes a primary gear **210a** and a secondary gear **210b** operably coupled to the first and second handles **101a**, **101b** disclosed embodiments may be configured to provide a controlled mechanical advantage to open and close the first and second arms **105**, **107** along the first path upon actuation of the primary actuator **102**.

In accordance with disclosed embodiments, a secondary retractor assembly **300** may be configured to couple and uncouple from the primary retractor assembly **200** via a first recessed key portion **220a** disposed on the first arm **105** and a second recessed key portion **220b** disposed on the second arm **107** (see FIG. 2). Each of recessed key portions **220a**, **220b** may include a groove having a geometry that facilitates engagement of the primary retractor assembly **200** with the secondary retractor assembly **300** while also operably allowing the opening and closing of arms **105**, **107**. For example, the secondary retractor assembly **300** may have a corresponding outdent (e.g., dovetail) on an underside thereof configured to mate with an indent (e.g., dovetail groove) of the primary retractor assembly **200**. Additionally, secondary retractor assembly **300** may be fixed to primary retractor assembly **200** by turnkey **113**. Turnkey **113** may project from a central portion of the primary retractor assembly **200** through a central aperture **113a** (see FIG. 6) of the secondary retractor assembly **300**. In a first position, turnkey **113** may urge the primary retractor assembly **200** and secondary retractor assembly **300** towards each other and maintain direct contact to fixedly engage them to one another. Conversely, in a second position, turnkey **113** may be rotated such that turnkey **113** is aligned with central aperture **113a** and therefore has no bearing surface to urge the primary retractor assembly **200** and secondary retractor assembly **300** towards each other. Thus, in the second position the primary retractor assembly **200** and secondary retractor assembly **300** may be disengaged from one another. Other embodiments may use alternate means to securely engage the primary retractor assembly **200** with the secondary retractor assembly **300**, e.g., as fasteners, hexagonal grooves, channel locks, magnets, etc. provided that the primary retractor assembly **200** and the secondary retractor assembly **300** are securely engaged with one another such that resultant forces acting on the retractor system **100** may transfer between primary retractor assembly **200** and secondary retractor assembly **300** and also by extension to a surgical table via table mount portions **106a** and/or **106b**.

Secondary retractor assembly **300** may have a body portion **300a** generally defining a first channel **109d** and a second channel **111d**. Secondary retractor assembly **300** may be configured to independently extend and contract a third arm **109** and a fourth arm **111**, respectively. Although two channels **109d**, **111d** and two arms **109**, **111** are illustrated it is contemplated that secondary retractor assembly **300** may have any number of suitable channels and arms. Additionally, it is contemplated that only a single arm, e.g., third arm **109** or fourth arm **111** will be provided in some surgical settings.

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In disclosed embodiments, the secondary retractor assembly **300** may include a first channel **109d** having a curved or arcuate shape for operably retaining third arm **109** therein where third arm **109** has a corresponding curved or arcuate shape. The third arm **109** may be configured to extend outwards from first channel **109d** and contract within first channel **109d**. Similarly, secondary retractor assembly **300** may include a second channel **111d** having a curved or arcuate shape for operably retaining fourth arm **111** therein where fourth arm **111** has a corresponding curved or arcuate shape. The fourth arm **111** may be configured to extend outwards from first channel **111d** and contract within second channel **111d**. The geometry of the first channel **109d** and third arm **109** may define a second path of travel, e.g., an arcuate path of travel defined by the arcuate shapes of the first channel **109d** and third arm **109**. Similarly, the geometry of the second channel **111d** and fourth arm **111** may define a third path of travel, e.g., an arcuate path of travel defined by the arcuate shapes of the second channel **111d** and fourth arm **111**.

In disclosed embodiments, the secondary retractor assembly **300** may include a third actuator **109c** operably disposed adjacent the first channel **109d** and operably configured to extend and contract the third arm **109** via a pinion gear mechanism (not illustrated) having the same or similar components as primary pinion gear mechanism **210** of primary retractor assembly **200**. For example, a toothed pinion P1 (see FIG. 7) may be coupled to actuator **109c** and may operably engage a corresponding rack portion (not illustrated) on an adjacent surface of arm **109** to linearly translate, e.g., curvo-linear, third arm **109** forward and backward, i.e., extend and withdraw or translate away from the operative corridor. Similarly, the secondary retractor assembly **300** may include a fourth actuator **111c** operably disposed adjacent the second channel **111d** and operably configured to extend and contract the fourth arm **111** via a pinion gear mechanism (not illustrated) having the same or similar components as primary pinion gear mechanism **210** of primary retractor assembly **200**. For example, a toothed pinion P2 (see FIG. 7) may be coupled to actuator **111c** and may operably engage a corresponding rack portion (not illustrated) on an adjacent surface of arm **111** to linearly translate, e.g., curvo-linear, fourth arm **111** forward and backward, i.e., extend and withdraw or translate away from the operative corridor. For example, actuator **109c** may rotationally translate P1 in a clockwise direction which in turn linearly translates the third arm **109** arm such that it extends outward from channel **109d**. Similarly, actuator **109c** may rotationally translate P1 in a counter clockwise direction which in turn linearly translates the third arm **109** arm such that it contracts inward into channel **109d**. Likewise, actuator **111c** may rotationally translate P2 in a clockwise direction which in turn linearly translates the fourth arm **111** arm such that it extends outward from channel **111d**. Similarly, actuator **111c** may rotationally translate P2 in a counter clockwise direction which in turn linearly translates the fourth arm **111** such that it contracts inward into channel **109d**. Accordingly, in disclosed embodiments, the third arm **109** is configured to independently extend and contract along a second path of travel upon actuation of the third actuator **109c**, and the fourth arm **111** is configured to independently extend and contract along a third path of travel upon actuation of the fourth actuator **111c**.

In disclosed embodiments, the third and fourth arms **109**, **111** may be operably coupled to third and fourth pivoting members **109a**, **111a** at a distal end thereof, respectively. The third and fourth pivoting members **109a**, **111a** may be

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configured to operably couple to third and fourth blades **209**, **211**, respectively (see FIG. 3) by a corresponding blade attachment mechanism as will be explained in more detail below during the discussion of FIGS. 9-13B. In the exemplary embodiment, a fifth actuator **109b** and a sixth actuator **111b** are configured to adjust the angulation of third blade **209** and fourth blade **211**, respectively. For example, the fifth actuator **109b** may be configured to actuate the third pivoting member **109a** to adjust the angulation of third blade **209** with respect to the third arm **109**. Similarly, the sixth actuator **111b** may be configured to actuate the fourth pivoting member **111a** to adjust the angulation of fourth blade **211** with respect to fourth arm **111**. In the exemplary embodiment, the third pivoting member **109a** may be configured to independently adjust the angulation of third blade **209** with respect to third arm **109** upon actuation of the fifth actuator **109b**. Similarly, the fourth pivoting member **111a** may be configured to independently adjust the angulation of fourth blade **211** with respect to the fourth arm **111** upon actuation of the fourth actuator **111b**.

In disclosed embodiments, the third and fourth pivoting members **209a**, **211a** may each include a corresponding pin and socket mechanism enabling the pivoting members **209a**, **211a** to pivot on a pin disposed in a corresponding pin aperture **199** (see, e.g., FIG. 8). Additionally, the third and fourth pivoting members **209a**, **211a** may each include a corresponding blade attachment mechanism at a distal end thereof which will be explained in more detail below when discussing FIGS. 9-13.

In disclosed embodiments, the secondary retractor assembly **300** may include a first retention lever **109e** configured to engage the third arm **109** to control extension and contraction of the third arm **109** along the second path of travel and a second retention lever **111e** configured to engage the fourth arm **111** to control extension and contraction of the fourth arm **111** along the third path of travel. First and second retention levers **109e**, **111e** may have the same or similar components as described above with respect to primary retention lever **104**.

First retention lever **109e** and second retention lever **111e** may frictionally engage with the third arm **109** and fourth arm **111**, respectively, to control and/or prevent the extension and contraction of the third arm **109** and fourth arm **111**. For example, first retention lever **109e** and second retention lever **111e** may engage with a rack portion on an outside adjacent surface of the third arm **109** and fourth arm **111**, respectively, through an aperture **302** (see FIG. 8) projecting through a portion of channels **109d**, **111d**, respectively. In some embodiments, first and second retention levers **109e**, **111e** may include a biasing element having the same or similar components as explained above with respect to primary retention lever **104**. In some embodiments, first retention lever **109e** may engage a corresponding pinion gear mechanism operably associated with actuator **109c** to thereby control and/or prevent rotation of the corresponding pinion gear mechanism. Similarly, second retention lever **111e** may engage a corresponding pinion gear mechanism operably associated with actuator **109c** to thereby control and/or prevent rotation of the corresponding pinion gear mechanism.

Referring generally to FIGS. 1, 7, and 9-11 the pivoting members **105a**, **107a**, **109a**, and **111a** may each include the same or similar components and features. For example, pivoting members **105a**, **107a**, **109a**, and **111a** may each include a corresponding pin and socket mechanism. The pin and socket mechanism of pivoting members **105a**, **107a**, **109a**, and **111a** may be adjustable by way of actuators **105b**,

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107b, **109b**, and **111b** such that an inclination of pivoting members **105a**, **107a**, **109a**, and **111a** may be independently adjustable with respect to arms **105**, **107**, **109**, and **111**, respectively. In some embodiments, translation of actuators **105b**, **107b**, **109b**, and **111b** may cause a corresponding element, such as an internal pin, set screw or the like, to urge pivoting members **105a**, **107a**, **109a**, and **111a** to pivot outwards on a corresponding pin within a corresponding socket thereby enabling travel of pivoting members **105a**, **107a**, **109a**, and **111a** inwards and outwards with respect to arms **105**, **107**, **109**, and **111**, respectively. In some embodiments, pivoting members **105a**, **107a**, **109a**, and **111a** may pivot outwards, for example, within a range of 0-25 degrees, and more particularly within a range of 0-15 degrees with respect to arms **105**, **107**, **109**, and **111**.

Pivoting members **105a**, **107a**, **109a**, and **111a** may include corresponding blade attachment mechanisms **105f**, **107f**, **109f**, and **111f**, respectively (see FIG. 7). The blade attachment mechanisms **105f**, **107f**, **109f**, and **111f**, may each include a dovetail groove having a geometry that facilitates secure engagement with a corresponding one of blades **205**, **207**, **209**, and **211**. For example, blade attachment mechanisms **105f**, **107f**, **109f**, and **111f**, may have an indent portion on an inside surface thereof facilitating secure engagement with an outdent portion disposed on an outside surface of blades **205**, **207**, **209**, and **211** respectively. In some embodiments, the dovetail grooves of the blade attachment mechanisms **105f**, **107f**, **109f**, and **111f**, are tapered, and may for example be conically tapered, from one end to the other end to further securely retain blades **205**, **207**, **209**, and **211**. In other embodiments, the blade attachment mechanisms **105f**, **107f**, **109f**, and **111f**, may take alternate shapes, and have varying configurations provided that the shape thereof can securely engage with a corresponding one of blades **205**, **207**, **209**, and **211**. For example, an indent such as a square channel, hexagonal channel, or the like dimensioned to match to a corresponding outdent. Additionally, the blade attachment mechanisms **105f**, **107f**, **109f**, **111f** may have an outdent portion (rather than an indent portion as illustrated) and blades **205**, **207**, **209**, and **211** may have an indent portion (rather than an outdent portion as illustrated).

Referring generally to FIGS. 9-13B exemplary blades, shims, and dilators for use with, e.g., retractor system **100**, are disclosed. Referring to FIGS. 9-11, an exemplary blade, e.g., first blade **205** is illustrated. It shall be understood that characteristics of first blade **205** may be found throughout each of blades **205**, **207**, **209**, and **211** and the foregoing description is described with respect to first blade **205** solely for convenience of explanation. Moreover, although first blade **205** is illustrated as a relatively long and narrow curved blade **205** it can take any shape suitable for any particular type of surgery application. Indeed, it is contemplated that retractor system **100** is suitable for a multitude of different blades having different lengths, widths, and cross-sectional shapes thereof that can couple and uncouple to secondary blades, tools, and shims. For example, relatively shorter and wider blades having generally planar surfaces are contemplated. Furthermore, blade **205** may feature any number or type of secondary coupling members where shims, for example, may couple thereto. In at least one embodiment, blade **205** may have a relatively narrow portion at one end and fan out to a relatively wider portion at the opposite end, i.e., the blade **205** may have a width that increases along the length thereof from one end to the other end. Additionally, blade **205** may include channels, grooves, indents, outdents, etc. for fixation of secondary members

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such as shims, light fixtures other diagnostic tools such as endoscopes, electrodes, temperature sensors, suction devices, and etc.

In the exemplary embodiment, blade **205** has a proximate side **205a**, a distal side **205b** opposite the proximate side, an outside surface **205c** and an inside surface **205d** opposite the outside surface **205c**. The proximate side **205a** may be operably coupled to a distal end of pivoting member **105a** via an engagement feature **205e** disposed on the outside surface **205c** of blade **205**, for example. In some embodiments, blade **205** may include an elastic material allowing it to deflect at least partially. Additionally, in some embodiments a blade removal instrument may be required to install and/or remove blade **205** from a blade attachment mechanism.

In the disclosed embodiment, engagement feature **205e** is the outdent portion of a dovetail groove, i.e., the dovetail. In other embodiments, engagement feature **205e** may be a lap joint, tongue and groove type joint, a doweled butt joint, etc. In the exemplary embodiment, engagement feature **205e** features an indent portion **205f**. Indent portion **205f** may be a socketed portion facilitating secured engagement and retention with blade attachment mechanism **105f**. For example, indent portion **205f** may house a spring clip (not illustrated) to hold blade **205** in secure engagement with blade attachment mechanism **105f**. In embodiments that include a spring clip, a corresponding release tool or lever may be inserted into the indent portion **205f** to release the biasing force of the spring and thereby uncouple the blade **205** from blade attachment mechanism **105f**. In other embodiments, engagement feature **205e** may have an aperture for running a diagnostic tool such as an electrode or endoscope there through. In some embodiments, blade **205** may be conductive such that it may communicate with an external diagnostic tool (not illustrated). For example, blades may include a conductive material such as a metal like copper and be conductive and/or have terminals for electrical conduction between conductive pads placed external to retractor system **100**. In some embodiments, blade **205** may include partially conductive features, e.g., a semiconductor and/or other passive electrical devices such as resistors, diodes, and etc. In other embodiments, blade **205** may be an insulator such that it does not interfere with electrical signal processing of the aforementioned electrical devices.

In the exemplary embodiment, first blade **205** may include a longitudinal groove **205g** extending longitudinally along the inside surface **205d** that is sized accordingly to house and retain a corresponding pin **205p** therein. In at least one embodiment, pin **205p** may securely attach to a vertebra of a patient's spine by socketing in to the vertebrae or screwing into the vertebrae. In some embodiments, pin **205p** may be a conductive pin having a sensor at a distal end thereof or pin **205p** may be a hollow pin that houses electrical components and wiring therein. In other embodiments pin **205p** is purely mechanical in nature. In at least one embodiment, pin **205p** may be used to facilitate attachment of a shim **205s** to an inside surface **205d** of blade **205**. Shim **205s** may laterally extend from a side surface of the blade **205** and include a gripping portion at a proximate side thereof. Shim **205s** may also extend from the blade **205** to increase the operative length thereof and/or extend laterally to increase the operative width thereof. In some embodiments, the first, second, third, and fourth blades **105**, **107**, **109**, **111** are each configured to operably couple to a corresponding first, second, third, and fourth shim laterally projecting from a side portion thereof. In other embodiments, diagnostic tools such as an electrode, endoscope,

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fiber optic, light emitting diode or the like may extend along groove **205g**. In other embodiments still, a second groove (not illustrated) similar to groove **205g** may be provided so that a combination of the above described features may be used. For example, groove **205g** may house a corresponding pin **205p** and the second groove (not illustrated) may enable a diagnostic tool or the like to extend along the second groove (not illustrated).

Referring to FIG. **12** an exemplary set of nested dilators **400** is illustrated. Exemplary dilators **400** may include a neuro monitoring sensor or the like to help guide insertion of the dilators through muscle fibers. The set of nested dilators **400** may include a series of dilators having alternating circular and ellipsis (oval) cross sectional shapes or oblong cross-sectional shapes. For example, a first dilator **401** having a relatively small circular cross section is surrounded by a second dilator **403** having an ellipsis, or oval shaped cross section. The size and shape of the circular cross section of the first dilator **401** may be defined by a radius extending from a center point thereof and the shape of the ellipsis cross section may be defined by a major axis and a minor axis extending perpendicularly with respect to one another from a center point thereof.

In the exemplary embodiment, the second dilator **403** may, for example, have an ellipsis or elliptical cross section, or other cross sections, for example bi-convex or elongated and substantially flat sides with convex ends, and may have a curvature but may not be circular or elliptical, some such embodiments having a minor axis roughly corresponding to the radius of the circular cross section of first dilator **401**. For example, the minor axis of the ellipsis cross section of the second dilator **403** may only be slightly larger than the radius of the circular cross section of the first dilator **401**, and the major axis of the ellipsis cross section of the second dilator **403** may be relatively larger than the radius of the circular cross section of the first dilator **401** and the minor axis of the ellipsis cross section of the second dilator **403**. In some embodiments, the major axis of the ellipsis cross section of second dilator **403** may be roughly twice as large as the radius of the circular cross section of first dilator **401**. In some embodiments, the major axis of the ellipsis cross section of the second dilator **403** may be twice as large as the minor axis of the ellipsis cross section of the second dilator **403**. At least one advantage to this arrangement of alternating cross sections is that the second dilator **403** may be insert around the first dilator **401** between fibers of a muscle, e.g., the paraspinal muscle, such that the major axis of the second dilator **403** is initially arranged parallel with the fibers of the paraspinal muscle and can therefore be insert around the first dilator **401**. Once inserted around the first dilator **401**, second dilator **403** can be rotated such that the major axis of second dilator **403** is perpendicular to the orientation of the fibers of the paraspinal muscle thereby gently separating the fibers by orienting the second dilator **403** such that the major axis area of the second dilator **403** gently and controllably applies pressure to separate the fibers.

A third dilator **405** having a circular cross section may be insert around the second dilator **403**. The size and shape of the circular cross section of the third dilator **405** may be defined by a radius extending from a center point thereof. For example, the third dilator **405** may have a circular cross-sectional shape having a radius roughly corresponding to the major axis of the second dilator **403**. The third dilator **405** can freely rotate around the second dilator **403** and features a circular cross section having a radius that is only slightly larger than the cross-sectional major axis of the

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second dilator **403**. A fourth dilator **407** having an ellipsis cross section (oval) may be insert around the third dilator **405**. The fourth dilator **407** may be defined by an ellipsis cross section having a minor axis that is only marginally larger than the cross sectional radius of the third dilator **405**, i.e., the cross sectional minor axis of the fourth dilator roughly corresponds to the cross sectional radius of the third dilator **405**. Additionally, the cross-sectional major axis of the fourth dilator **407** is relatively larger than the cross sectional radius of the third dilator **405** and the cross sectional minor axis of the fourth dilator. In some embodiments, the major axis of the ellipsis cross section of fourth dilator **407** may be roughly twice as large as the radius of the circular cross section of third dilator **405**. In some embodiments, the major axis of the ellipsis cross section of the fourth dilator **407** may be twice as large as the minor axis of the ellipsis cross section of the fourth dilator **407**. At least one advantage to this arrangement of alternating cross sections is that the fourth dilator **407** may be insert around the third dilator **405** between fibers of a muscle, e.g., the paraspinous muscle, such that the major axis of the fourth dilator **407** is initially arranged parallel with the fibers of the paraspinous muscle and can therefore be insert around the third dilator **405**. Once inserted around the third dilator **405**, fourth dilator **407** can be rotated such that the major axis of fourth dilator **407** is perpendicular to the orientation of the fibers of the paraspinous muscle thereby gently separating the fibers by orienting the fourth dilator **407** such that the major axis area of the fourth dilator **407** gently and controllably applies pressure to separate the fibers.

FIG. **13A** is a top down view of the set of nested dilators **400** as explained above. As illustrated a set of nested dilators **400** that may sequentially gently separate fibers of a muscle are illustrated. The set of nested dilators **400** may be insert sequentially and rotated on an as needed basis to gently dilate an anatomical feature. FIG. **13B** is a top down view of blades **205**, **207**, **209**, and **211**. As illustrated blades **205**, **207** are relatively larger in width than blades **209**, and **211**.

FIGS. **14-19** illustrate various positions and modes of operation of retractor system **100** in use with the set of nested dilators **400**. For example, in FIG. **14**, retractor system **100** is shown in a closed position where arms **105**, **107** are closed and surround, at least partially, the set of nested dilators **400**. Additionally, arms **109**, **111** are fully extended and surround, at least partially, the set of nested dilators **400**. In FIG. **14**, the inside surfaces of blades **205**, **207**, **209**, and **211** (not labelled in FIG. **14**) together surround and contact an outside surface of a fourth dilator **407** (not labelled in FIG. **14**). For example, the blades **205**, **207**, **209**, and **211** surround and contact a set of nested dilators **400**. For example still, a side surface of each of blades **205**, **207**, **209**, and **211** contacts an adjoining side surface of a different adjacent blade of the blades **205**, **207**, **209**, and **211** thereby forming a closed shape. FIG. **17** is a side view of the arrangement of FIG. **14**.

In FIG. **15**, the set of nested dilators **400** is removed and the retractor system **100** is adjusted to a first partially opened position where arms **105**, **107** are partially opened and arms **109**, **111** are partially contracted. FIG. **18** is a side view of the first partially opened arrangement of FIG. **15**. In FIG. **16**, the retractor system is adjusted to a second partially opened position where arms **105**, **107** are further opened and arms **109**, **111** are further contracted. FIG. **18** is a side view of the second partially opened arrangement of FIG. **16**. FIG. **19**

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shows the angulation of each blade being adjusted outward approximately 15 degrees from the side view of FIG. **18**.

Additional Retractor Embodiments

Referring generally to FIGS. **20-81** an example modular retractor system including a modular retractor **500** and various add on retractor modules **600**, **700**, **800**, **900**, **1000**, and **1100** for use with modular retractor **500** are disclosed. In some embodiments, modular retractor **500** may include the same, substantially the same, and/or similar components and functionality as primary retractor **100** and the associated blades, dilators, and secondary retractor assembly **300**. Accordingly, those with skill in the art will understand the general principles, modes of operation, and associated methods of each example embodiment may be combined and/or modified in view of the skill of a person of ordinary skill in the art.

With reference to FIGS. **20-28C** a modular retractor **500** for enabling access to a surgical site, an adjustment tool **10**, a table mount **70**, and a table mount rack module **60** are disclosed. FIGS. **20-21** are perspective views of a modular retractor **500** and FIG. **22** is a top down view of the modular retractor **500** showing various axes and directions of operation. FIG. **23** is a perspective view of an adjustment tool **10** for use with disclosed modular retractor **500** embodiments. FIGS. **24-25** are exploded parts views of a modular retractor **500**. FIGS. **26A-26E** are various views of a distraction mechanism **50** for use with disclosed modular retractor **500** embodiments. FIGS. **27A** and **27B** are various views of a modular retractor **500** coupled to a table mount **70**. FIGS. **28A-28C** are various views of a table mount rack module **60**.

Modular retractor **500** is highly customizable and may be considered modular for reasons that will be readily apparent and explained in further detail below. For example, the modular retractor **500** may be used as a standalone retractor system without the use of additional add on modules or modular retractor **500** may be used with any of the disclosed modules discussed herein unless the context clearly suggests otherwise.

Modular retractor **500** may be configured to distract and retract a first arm **505** along a path of travel and a second arm **507** along a different path of travel. The various paths of travel may be an arcuate path or segment defined by the length and geometry of the arms **505** and **507**, respectively, and a handle pivoting mechanism **515** (see FIG. **26D**). Handle pivoting mechanism **515** may be configured to enable first handle **501a** and second handle **501b** to open and close, for example. Handle pivoting mechanism **515** may be a pin, screw, or the like, for example. Other paths of travel than those specifically shown are contemplated and those paths of travel may depend upon the geometry of the arms **505**, **507** and the relative location of the handle pivoting mechanism **515**. The modular retractor **500** may include a handle assembly having first and second handles **501a**, **501b** that are removably coupled to the first and second arms **505**, **507** and configured to open and close the first and second arms **505**, **507**. For example, the first handle **501a** may be coupled to the first arm **505** and the second handle **501b** may be coupled to the second arm **507**. Additionally, the arms **505**, **507** may extend through side channels of the body **503**, respectively, and/or be pivotable relative to body **503** and/or be operably coupled to body **503**. In various embodiments, the first handle **501a** and second handle **501b** may be removed and are held in place by first handle connection pin **508a** and second handle connection pin **508b**, for example. In various embodiments, the connection pins **508a**, **508b**

may be a pin, screw, knob, turnkey, and/or retaining fastener that a surgeon may quickly remove to uncouple the handle **501a**, **501b**, for example. Furthermore retractor **500** may include a table mount **506** extending in the lateral direction from body **503**. At least one advantage of having the first and second handles **501a**, **501b** be removable is greater freedom in performing a surgery due to the reduced structure adjacent a target surgical location, for example. For example still, after a surgeon has retracted a patient tissue, the surgeon may remove the handles **501a**, **501b** to prevent bumping into them.

In various embodiments, the first and second arms **505**, **507** may be coupled to first and second pivoting members **505a**, **507a** at a distal end thereof, respectively. The first and second pivoting members **505a**, **507a** may be configured to operably couple to first and second blades, **40** (see FIG. 22), respectively, by a corresponding blade attachment mechanism **505c**, **507c** as will be explained in more detail below. In the example embodiment, a first actuator **505b** and a second actuator **507b** are configured to adjust the angulation of first blade **40** and second blade **40**, respectively. For example, the first actuator **505b** may be configured to actuate the first pivoting member **505a** to adjust the angulation of first blade **40** with respect to the first arm **505**. Similarly, the second actuator **507b** may be configured to actuate the second pivoting member **507a** to adjust the angulation of second blade **40** with respect to second arm **507**. In the example embodiment, the first pivoting member **505a** may be configured to independently adjust the angulation of first blade **40** with respect to the first arm **505** upon actuation of the first actuator **505b**. Similarly, the second pivoting member **507a** may be configured to independently adjust the angulation of the second blade **40** with respect to the second arm **507** upon actuation of the second actuator **507b**. In disclosed embodiments, the first and second pivoting members **505a**, **507b** may each include a corresponding pin and socket mechanism enabling the pivoting members to pivot on a pin aperture **199** (see, e.g., FIG. 8).

As shown in FIG. 22, modular retractor **500** may extend in a longitudinal direction from a proximal end **500p** to a distal end **500d** in a longitudinal direction (or proximal-to-distal direction) parallel to longitudinal axis A-A. Additionally, modular retractor **500** may extend in a lateral direction (or widthwise direction) parallel to lateral axis B-B. The longitudinal axis A-A may be perpendicular to the lateral axis B-B and intersect at body **503** at a medial location of retractor **500**, for example. In various embodiments, and as shown by the Cartesian coordinate system in FIG. 22A, the longitudinal direction may be understood as the X direction and the lateral direction may be understood as the Y direction. Furthermore, a depth and/or thickness of modular retractor may be understood as the Z direction or vertical direction when viewed in a plan view.

FIG. 23 is a perspective view of an adjustment tool **10** for use with disclosed modular retractor **500** embodiments. In the example illustration, adjustment tool **10** may include a drive end **11** and a handle end **12**, for example. Drive end **11** may have a size and shape configured to rotate various actuators of modular retractor **500**, for example. In various embodiments, drive end **11** may take the shape of a hex-lobular drive end, a hex drive end, a torx drive end, a polygonal drive end, a square drive end, or the like. Similarly, actuators **502**, **507b**, **505b** may take any corresponding shape, for example.

FIG. 24 is a top down exploded parts view of a modular retractor **500** and FIG. 25 is a perspective exploded parts view of a modular retractor **500**. In the example embodi-

ment, arm **505** may include a rack portion **505d** at a distal end thereof and arm **507** may include a rack portion **507d** at a distal end thereof, for example. In various embodiments, rack portions **505d**, **507d** may be curved and be disposed at different relative distances from the distal end of the respective handle **505**, **507**, for example. Additionally rack portions **505d**, **507d** may be meshed with and movable by distraction mechanism **50**, for example. Distraction mechanism **50** may be operably drivable by actuator **502**, for example. Distraction mechanism **50** may include a plurality of gears to provide a mechanical advantage to open and close the arms **505**, **507** as will be explained in further detail below.

FIGS. 26A-26E are various views of a distraction mechanism **50** for providing a mechanical advantage to distract and retract arms **505**, **507**. Distraction mechanism **50** may principally be formed of a plurality of spur gears **51**, **52**, **54**, **55**, and a partial spur gear **57** that are meshed together and sized appropriately for providing a mechanical advantage to distract and/or retract arms **505**, **507**. For example, primary actuator **502** may be connected to first spur gear **51** and second spur gear **52** by shaft **53**, for example. In the example embodiment, primary actuator **502**, spur gears **51**, **52**, and shaft **53** are coaxially aligned in the vertical direction. Additionally, a partial spur gear **57** may be attached to shaft **53**. Partial spur gear **57** may be understood as a portion and/or slice of a relatively large spur gear having a central axis of rotation coincident with shaft **53**, for example. In the example embodiment, partial spur gear **57** may have an axis of rotation coincident with shaft **53**, for example. Additionally, first spur gear **51** may be meshed with third spur gear **54**. In turn, third spur gear **54** may be connected to fourth spur gear **55** by shaft **56**. Third spur gear **54**, shaft **56**, and fourth spur gear **55** may be coaxially aligned. In the example embodiment, third spur gear **54** is a relatively large spur gear and fourth spur gear **55** is a relatively small spur gear. Those with skill in the art will understand this arrangement may be advantageous for providing a relatively great mechanical advantage to perform distraction and/or retraction of arms **505**, **507**, for example. In the example embodiment, fourth spur gear **55** may be meshed with partial spur gear **57**. In this way, distraction mechanism **50** may comprise a plurality of spur gears having various teeth and recesses that are meshed and/or interconnected to one another.

FIG. 26B is a top perspective view of a distraction mechanism **50** and FIG. 26C is an enlarged top perspective view of distraction mechanism **50** with some parts removed for ease of understanding. In the example embodiment, third spur gear **54** may include teeth **54a** symmetrically radially disposed on a side surface around the circumference of third spur gear **54** and a rack **54b** may be radially disposed on a top surface of third spur gear **54** proximate the edge of spur gear **54**, for example. Primary pawl **504** may be configured to engage circular rack **54b** to allow spur gear **54** to rotate in a first direction (counter clockwise direction) and prevent third spur gear **54** from rotating in a second direction (clockwise direction). For example, primary pawl **504** may be disposed on a pivoting hinge and be biased such that a hook portion may be pushed downward against rack **54b** such that the hook portion is meshed within a valley between any pair of the teeth of rack portion **54b**, for example. In operation, an end user may rotate primary actuator **502** (via tool **10**, e.g.) counter clockwise such that primary pawl **504** moves in and out of the various valleys between teeth of rack portion **54b**. Notably, due to pawl **504** being biased against rack **54b**, pawl **504** may prevent third spur gear **54** from rotating in the clockwise direction. For example, as arms

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505, 507 are opened patient tissue may apply a closing force attempting to push arms 505, 507 back towards a closed position and pawl 504 may prevent and/or suppress arms 505, 507 from moving into a closed position. Additionally, in various embodiments pawl 504 may be depressible at a lateral end thereof opposite the hook portion that is engaged with rack 54b such that the hook portion of pawl 504 is moved upward in the vertical direction and prevented from engaging with rack 54b such that arms 505, 507 may be closed if and when desired. Furthermore, in FIG. 26C it is shown that spur gear 52 is meshed with rack portion 505d of arm 505 and rack portion 507d of arm 507. For example, rack portion 507d is meshed with a distal side of spur gear 52 and rack portion 505d is meshed with a proximate side of spur gear 52. Accordingly, rotation of spur gear 52 in a first direction will cause arms 505, 507 to distract outward by an equal amount and rotation of spur gear 52 in a second direction opposite the first direction will cause arms 505, 507 to retract inward by an equal amount. Alternatively, an end user may squeeze handles 501a, 501b to cause distraction and/or retraction by an equal amount which will also cause rotation of the various gears of distraction mechanism 50.

FIG. 26D is a bottom perspective view of distraction mechanism 50 and FIG. 26E is an enlarged bottom perspective view of distraction mechanism 50. In the example embodiment, the underside of partial spur gear 57 is shown as being meshed with spur gear 55 and being rotatably engaged with drive shaft 53. Additionally, suitable cutout portions 505z, 507z may be provided in the first handle 505 and second handle 507 that allow partial spur gear 57 to rotate a suitable distance when expanding arms 505, 507 such that partial spur gear 57 is fully contained within body 503 and does not clash with handles 505, 507, for example.

FIG. 27A is a top down view of a modular retractor 500 coupled to a table mount rack module 60 which is in turn coupled to a table mount 70. FIG. 27B is a perspective view of a modular retractor 500 coupled to a table mount rack module 60 which is in turn coupled to a table mount 70. In the example embodiment, the table mount 70 may be connected to and rigidly supported by a surgeons table via table mount portion 73, for example. Arms 72 and 71 may be adjustable by way of adjustment knob 74 to position table rack module 60 at a suitable location, for example.

FIGS. 28A and 28B are perspective views of a table mount rack module 60. FIG. 28C is an exploded parts view of table mount rack module 60. In the example embodiment, table mount rack module 60 may include an aperture 64 having a size and shape that corresponds to a size and shape of table mount arm 506 of modular retractor 500, for example. Depressible lever 65 may be used to lock table mount arm 506 when table mount arm 506 is insert inside of aperture 64, for example as shown in FIGS. 27A and 27B. Additionally, table mount arm 506 may slide in and out of aperture 64 to facilitate positioning modular retractor 500, for example. Table mount module 60 may include a connection arm 63, which may be insert into a corresponding aperture of table mount 70, for example as shown in FIGS. 27A and 27B to secure table mount rack module 60 to table mount 70. Connection arm 63 may be rigidly secured to body portion 66, for example. Additionally, extendable arm 67 may slide forward and backward through body 66 by a rack and pinion mechanism. For example, actuators 61, 62 may be securely coupled to body 66 and may each have pinion portions 61a, 62a having teeth that engage with rack portion 67a of extendable arm 67. Accordingly, rotation of actuator 61 and/or actuator 62 may rotate pinion portions

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61a and/or 62a such that teeth of pinion portions 61a, 62a cause extendable arm 67 to move forward and/or backward depending on the direction actuators 61 and/or 62 are rotated. Additionally, table mount rack module 60 may include a pawl 68 having a first hook portion 68a and/or a second hook portion 68b, for example. Pawl 68 may be pivotally coupled to body portion 66 at a pivot location 66b by a pin, for example. Pivot location 66b may enable pawl 68 to be toggled between a first position where pawl 68 allows extendable arm 67 to move forward but prevents extendable arm 67 from moving backward in the opposite direction. Similarly, in various embodiments, pawl 68 may be toggled to a second position where pawl 68 allows extendable arm 67 to move backward but prevents extendable arm 67 from moving forward. In some embodiments, pawl 68 may be moved to a third position, in the middle of the first position and second position, where pawl 68 prevents extendable arm 67 from moving forward and backwards. For example, in some embodiments, and in a third position pawl 68 may lock extendable arm 67 from relative motion in the forward and backwards direction. Other embodiments may utilize a locking element (not illustrated) to secure extendable arm 67 in an appropriate position. In this way, table rack module 60 may facilitate the relative motion of modular retractor 500 forward and backward in a direction defined by axis D-D of extendable arm 67. Additionally, table rack module 60 may facilitate the relative motion of modular retractor 500 from side to side in a direction defined by an extension direction C-C of table mount 506 (see FIG. 22), for example.

Referring generally to FIGS. 29-34 a first module 600 is disclosed. FIGS. 29 and 30 are top perspective views of a first module 600 and FIGS. 31A and 31B are bottom perspective views of a first module 600. FIG. 32 is an exploded parts view of a first module 600 and FIGS. 33A-34 are various perspective views of first module 600 coupled to modular retractor 500.

In accordance with disclosed embodiments, first module 600 may be configured to couple and uncouple from modular retractor 500 at connection points 503a, for example (see FIG. 20). In various embodiments, the first module 600 may have at least one corresponding connection point 603a on an underside thereof (see FIG. 31A) configured to couple, connect, and/or mate with a connection point 503a of the modular retractor 500. In the example embodiment, connection points 503a are indented apertures and connection points 603a are outdented posts having a corresponding size and shape to one another, respectively. In some embodiments, connection points 503a, 603a, may have slotted rails and/or grooves to facilitate a connection and/or prevent rotation of first module 600 relative to modular retractor 500, for example. Similarly, and in the example embodiment, one connection point of connection points 603a may be shaped like a circular post and the other connection point 603a may be shaped like an oval post to facilitate mating the first module 600 with modular retractor 500 in an appropriate orientation. Additionally, first module 600 may be locked to modular retractor 500 by lock 513 (see FIG. 20). Lock 513 may be pivotable such that in a locked position a flange portion of lock 513 may pivot into a locking aperture 603e of first module 600, for example. Similarly, in an unlocked position the flange portion of lock 513 may be unseated from aperture 603e. Other embodiments may use alternate means to securely engage the modular retractor 500 with the first module 600, e.g., as fasteners, hexagonal grooves, channel locks, magnets, etc. provided that the modular retractor 500 and the first module 600 are securely engaged with one

another such that resultant forces acting on the modular retractor **500** may transfer between modular retractor **500** and first module **600**.

First module **600** may include a first arm **605** and a second arm **607** that extend through body **603**. First arm **605** may extend through body **603** through a first contoured channel **603b** and second arm **607** may extend through body **603** through a second contoured channel **603c**, for example (see FIG. **32**). In various embodiments, contoured channels **603b**, **603c** may be L shaped channels. First module **600** may be configured to independently extend first arm **605** along a first path of travel and independently extend second arm **607** along a second path of travel by independent rack and pinion mechanisms, for example. The first path and second path may be an arcuate path or segment defined by the length and geometry of the arms **605** and **607**, for example. In various embodiments, the first path and second path may symmetrically fan out with respect to one another. Other paths of travel than those specifically shown are contemplated, e.g., a linear path.

First module **600** may include a table mount **606** extending laterally from a side surface thereof. Table mount **606** may facilitate the relative motion of first module **600** (and/or modular retractor **500** when coupled thereto) from side to side in a direction defined by an extension direction E-E of table mount **606** (see FIG. **34**), for example. Table mount **606** may be securely coupled to sliding frame **608**. Sliding frame **608** may be configured to slide forward and backward through sliding frame aperture **603d** of body **603**, for example (see FIG. **32**). Additionally, in various embodiments, sliding frame **608** may be configured to support first and second arms **605**, **607** at a bottom surface of first and second arms **605**, **607** proximate first pivoting member **605a** and second pivoting member **607a**, respectively (see FIGS. **31A** and **31B**). In various embodiments, support portion **609** may be pivotable relative to sliding frame **608** by pivot point **609a**, for example. In various embodiments, the first arm **605** may include first post **605f** and second arm **607** may include second post **607f** that extend through corresponding slotted apertures **609b**, respectively, of support portion **609**. In this way, and due in part to the size and geometry of the slotted apertures **609b**, support portion **609** may support both first arm **605** and second arm **607** while also enabling first and second arms **605**, **607** to be independently movable relative to one another, for example. In some embodiments, pivot point **609a** may be replaced by a non pivoting fastener such that first arm **605** and second arm **607** are not independently movable relative to one another (not illustrated) and distract and retract by equal amounts.

First module **600** may be configured to extend first arm **605** by activation of actuator **601**, e.g., by rotation of actuator **601**. Actuator **601** may be securely attached to body portion **603** and include a pinion portion **601a** (pinion gear and/or spur gear) having teeth that engage with and are meshed with curved rack portion **605d** disposed on a side surface of first arm **605**, for example. Accordingly, rotation of actuator **601** may rotate pinion portion **601a** such that teeth of pinion portion **601a** causes first arm **605** to move forward and/or backward depending on the direction actuator **601** is rotated. Additionally, first module **600** may include a first pawl **604a** that may be configured to engage the curved rack portion **605e** disposed on a top surface of first arm **605**, for example. First pawl **604a** may be configured to allow pinion portion **601a** to rotate in a first direction (counter clockwise direction) and prevent pinion portion **601a** from rotating in a second direction (clockwise direction). For example, first pawl **604a** may be disposed on a

pivoting hinge and be biased by a spring or the like such that a hook portion may be pushed downward against rack **605e** such that the hook portion is meshed within a valley between any pair of the teeth of rack portion **605e**, for example. In operation, an end user may rotate actuator **601** (via tool **10**, e.g.) counter clockwise such that pawl **604a** moves in and out of the various valleys between teeth of rack portion **605e** while first arm **605** extends outward away from body **603**. Notably, due to pawl **604e** being biased against rack portion **605e**, pawl **604a** may prevent first arm **605** from being pushed in an opposite direction. For example, as arm **605** is distracted outward patient tissue may apply a closing force attempting to push arm **605** back towards body **603** and pawl **604a** may prevent and or suppress this closing force. Additionally, in various embodiments pawl **604a** may be depressible at a lateral end thereof opposite the hook portion that is engaged with rack **605e** such that the hook portion of pawl **604a** is moved upward in the vertical direction and prevented from engaging with rack **605e** such that arm **605** may be closed if and when desired.

First module **600** may be configured to extend second arm **607** by activation of actuator **602**, e.g., by rotation of actuator **602**. Actuator **602** may be securely attached to body portion **603** and include a pinion portion **602a** (pinion gear and/or spur gear) having teeth that engage with and are meshed with curved rack portion **607d** disposed on a side surface of second arm **607**, for example. Accordingly, rotation of actuator **602** may rotate pinion portion **602a** such that teeth of pinion portion **602a** causes second arm **607** to move forward and/or backward depending on the direction actuator **602** is rotated. Additionally, first module **600** may include a second pawl **604b** that may be configured to engage the curved rack portion **607e** disposed on a top surface of second arm **607**, for example. Second pawl **604b** may operate in the same, substantially the same, and/or similar manner as explained above with respect to first pawl **604a**. Accordingly, duplicative description will be omitted.

In various embodiments, the first and second arms **605**, **607** may be coupled to first and second pivoting members **605a**, **607a** at a distal end thereof, respectively. The first and second pivoting members **605a**, **607a** may be configured to operably couple to third blade **45** and fourth blade **45**, respectively, by a corresponding blade attachment mechanism **605c**, **607c**. In the example embodiment, a first blade actuator **605b** and a second blade actuator **607b** are configured to adjust the angulation of blades **45** respectively (see FIG. **34**). For example, the first blade actuator **605b** may be configured to actuate the first pivoting member **605a** to adjust the angulation of blade **233** with respect to the first arm **605**. Similarly, the second actuator **607b** may be configured to actuate the second pivoting member **607a** to adjust the angulation of blade **234** disposed therein with respect to second arm **607**. In the example embodiment, the first pivoting member **605a** may be configured to independently adjust the angulation of a blade with respect to the first arm **605** upon actuation of the first actuator **605b**. Similarly, the second pivoting member **607a** may be configured to independently adjust the angulation of a second blade with respect to the second arm **607** upon actuation of the second actuator **607b**. In disclosed embodiments, the first and second pivoting members **605a**, **607a** may each include a corresponding pin and socket mechanism enabling the pivoting members to pivot, for example.

Referring generally to FIGS. **35-40** a second module **700** for use with modular retractor **500** is disclosed. FIGS. **35-36** are various top perspective views of a second module **700** and FIGS. **36-37** are various bottom perspective views of a

second module **700** for use with disclosed modular retractor **500** embodiments. FIG. **39** is an exploded parts view of a second module **700** and FIG. **40** is a top down view of a second module coupled to modular retractor **500** and a plurality of blades.

In accordance with disclosed embodiments, second module **700** may be configured to couple and uncouple from modular retractor **500** at connection points **503a**, for example (see FIG. **20**). For example, the second module **700** may have at least one corresponding connection point **703a** on an underside thereof (see FIG. **38**) configured to couple, connect, and/or mate with a connection point **503a** of the modular retractor **500** in the same, similar, and/or substantially the same manner as explained above. Accordingly, duplicative description will be omitted. Additionally, second module **700** may be locked to modular retractor **500** by lock **513** (see FIG. **20**). Lock **513** may be pivotable such that in a locked position a flange portion of lock **513** may pivot into a locking aperture **703e** of second module **700**, in the same, similar, and/or substantially the same manner as explained above. Accordingly, duplicative description will be omitted.

Second module **700** may include an arm **705** that extends through body **703**. Arm **705** may extend through body **703** through a first contoured channel **703b**. Second module **700** may be configured to extend arm **705** along a path of travel by a rack and pinion mechanism, for example. The path of travel may be an arcuate path or segment defined by the length and geometry of arms **705**, for example. Other paths of travel than those specifically shown are contemplated, e.g., a linear path.

Second module **700** may include a table mount **706** extending laterally from a side surface thereof. Table mount **706** may facilitate the relative motion of second module **700** (and/or modular retractor **500** when coupled thereto) from side to side in a direction defined by an extension direction F-F of table mount **706** (see FIG. **40**), for example. Table mount **706** may be securely coupled directly to arm **705** (see FIG. **39**), for example. Second module **700** may be configured to extend arm **705** by activation of actuator **701**, e.g., by rotation of actuator **701**. Actuator **701** may be securely attached to body portion **703** and include a pinion portion **701a** (pinion gear and/or spur gear) having teeth that engage with and are meshed with curved rack portion **705d** disposed on a side surface of arm **705**, for example. Accordingly, rotation of actuator **701** may rotate pinion portion **701a** such that teeth of pinion portion **701a** cause arm **705** to move forward and/or backward depending on the direction actuator **701** is rotated. Additionally, second module **700** may include a first pawl **704** that may be configured to engage the curved rack portion **705e** disposed on a top surface of arm **705**, for example. First pawl **704** may be configured to allow pinion portion **701a** to rotate in a first direction (counter clockwise direction) and prevent pinion portion **701a** from rotating in a second direction (clockwise direction) in the same, similar, and/or substantially the same manner as previously explained. Accordingly, duplicative description will be omitted.

In various embodiments, arm **705** may be coupled to pivoting member **705a** at a distal end thereof. Pivoting member **705a** may be configured to operably couple to a blade **35** by blade attachment mechanism **705c**. In the example embodiment, blade actuator **705b** may be configured to adjust the angulation of blade **35** (see FIG. **40**). For example, the blade actuator **705b** may be configured to actuate the first pivoting member **705a** to adjust the angulation of blade **35** with respect to arm **705**. In disclosed embodiments, the first pivoting member **705a** may include

a corresponding pin and socket mechanism enabling pivot member **705a** to pivot, for example.

Referring generally to FIGS. **41A-44** a third module **800** for use with the modular retractor **500** is disclosed. FIG. **41A** is a top perspective view of a third module **800** and FIG. **41B** is a bottom perspective view of a third module **800**. FIG. **42** is an exploded parts view of a third module **800**. FIG. **43** is a perspective view of a third module **800** coupled to a modular retractor **500** and FIG. **44** is a perspective view of a third module **800** coupled to a modular retractor **500** and a plurality of blades.

In accordance with disclosed embodiments, third module **800** may be configured to couple and uncouple from modular retractor **500** at connection points **503a**, for example (see FIG. **20**). For example, the third module **800** may have at least one corresponding connection point **803a** on an underside thereof (see FIG. **41B**) configured to couple, connect, and/or mate with a connection point **503a** of the modular retractor **500** in the same, similar, and/or substantially the same manner as previously explained. Accordingly, duplicative description will be omitted. Additionally, third module **800** may be locked to modular retractor **500** by lock **513** (see FIG. **20**). Lock **513** may be pivotable such that in a locked position a flange portion of lock **513** may pivot into a locking aperture **803e** of third module **800**, in the same, similar, and/or substantially the same manner as previously explained. Accordingly, duplicative description will be omitted.

Third module **800** may include an arm **805** that includes a straight portion **810** and a C shaped curved portion **811**. Straight portion **810** of arm **805** may extend through body **803** and move forward and backward in a longitudinal direction, for example. As seen best in FIG. **43**, when third module **800** is coupled to modular retractor **500** the C shaped curved portion **811** extends laterally outward in a lateral direction B-B farther than the farthest lateral edge of arm **807**. For example, the C shaped curved portion **811** does not obscure a surgeons viewing area and/or access to a surgical site. Furthermore, third module **800** may orient and/or support a blade **35** such that the blade faces the body portion **803** of third module **800**, the body portion of modular retractor **500**, and is also symmetrically disposed relative to the first arm **505** and second arm **507** of modular retractor **500**. For example, the C shaped curved portion **811** may support a blade **35** at a distal most position that is aligned in the longitudinal axis A-A of modular retractor **500** (see FIG. **43**). The straight portion **810** of arm **805** may extend through body **803** through a first contoured channel. In various embodiments, the contoured channel **803b** may be an L shaped channel, for example. Third module **800** may be configured to extend arm **805** along a path of travel by a rack and pinion mechanism, for example. The path of travel may be linear path, for example. Other paths of travel than those specifically shown are contemplated, e.g., an arcuate path.

Third module **800** may include a table mount **806** extending laterally from a side surface thereof in a direction defined by an extension direction G-G of table mount **806** (see FIG. **43**), for example. Table mount **806** may facilitate the secure placement of third module **800** such that third module **800** remains fixed in 3D space and/or facilitate the relative motion of third module **800** (and/or modular retractor **500** when coupled thereto) in any direction when moving table mount **70**, for example. Third module **800** may be configured to extend arm **805** by activation of actuator **801**, e.g., by rotation of actuator **801**. Actuator **801** may be securely attached to body portion **803** and include a pinion portion **801a** (pinion gear and/or spur gear) having teeth that engage

with and are meshed with straight rack portion **805d** disposed on a side surface of arm **805**, for example. Accordingly, rotation of actuator **801** may rotate pinion portion **801a** such that teeth of pinion portion **801a** cause arm **805** to move forward and/or backward depending on the direction actuator **801** is rotated in the same, similar, and/or substantially the same manner as previously explained. Accordingly, duplicative description will be omitted. Additionally, third module **800** may include a first pawl **804** that may be configured to engage rack portion **805e** disposed on a top surface of arm **805**, for example. First pawl **804** may be configured to allow pinion portion **801a** to rotate in a first direction (counter clockwise direction) and prevent pinion portion **801a** from rotating in a second direction (clockwise direction) in the same, similar, and/or substantially the same manner as previously explained. Accordingly, duplicative description will be omitted.

In various embodiments, curved arm portion **811** of arm **805** may be coupled to blade attachment mechanism **805c** at a distal most end. The curved arm portion **811** may support blade attachment mechanism **805c** such that it faces modular retractor **500** and is aligned with the longitudinal axis A-A of modular retractor **500** (see FIG. 43). In the example embodiment, blade attachment mechanism **805c** is fixed and a corresponding blade **35** does not pivot and/or angulate. However in other embodiments, third module **800** may include a first blade actuator (not illustrated) that is configured to adjust the angulation of a corresponding blade and a corresponding pivoting member with the same, substantially the same, and/or similar structural and characteristics as explained herein with respect to other embodiments.

Third module **800** may include a table mount **806** extending in a first lateral direction along axis G-G from arm **805** and a module mount **809** extending in a second lateral direction along axis H-H from arm **805**, i.e., in an opposite lateral direction (see FIG. 43). For example, table mount **806** may extend to the left direction and module mount **809** may extend to the right direction. Additionally, straight portion **810** of arm **805** may be supported by body **803** on the left side of the longitudinal axis A-A of modular retractor **500**. In this configuration, the module mount **809** may cross over the longitudinal axis A-A, for example. Module mount **809** may support a free hand module **900**, as will explained in further detail below.

Referring generally to FIGS. 45A-45B an alternative third module **800a** embodiment is disclosed. Third module **800a** may include the same, substantially the same, and/or similar components and functionality as third module **800**. Accordingly, duplicative description will be omitted. In the example embodiment, third module **800a** may be modified such that table mount **806** and module mount **809** are aligned. For example, table mount **806** and module mount **809** each extend from arm **805** in opposite directions and are aligned on the same common extension axis. For example still, axis G-G of table mount **806** and axis H-H of module mount **809** are aligned and extend in opposite directions.

Alternative third module **800a** may include a locking actuator **850**, for example. Locking actuator **850** may be rotatably secured within body portion **803** and be disposed above straight portion **810** of arm **805**, for example. In various embodiments, locking actuator **850** may include an outside thread pattern corresponding to an inside thread pattern of body **803** (not illustrated). In various embodiments, locking actuator **850** may be rotated in a first direction such that locking actuator **850** advances towards straight portion **810** of arm **805**. As locking actuator **850** advances, a bottom portion of locking actuator **850** may

contact an upper surface of straight portion **810** of arm **805** and apply a downward force to straight portion **810**. In this way, locking actuator **850** may provide a frictional force against straight portion **810** of arm **805** thereby preventing and/or suppressing arm **805** from moving forward and backward. For example, the greater the downward force applied to straight portion **810**, the greater the frictional force between the underside of locking actuator **850** and the upper surface of straight portion **810**. At least one advantage of locking actuator **850** may be that arm **805** may be locking in position such that it is fixed and is prevented from moving forward and backward, for example. In various embodiments, this may assist a surgeon in placement of third module **800** and/or modular retractor **500**. For example, a surgeon may lock arm **805** via locking actuator **850** and position modular retractor **500** and third module **800** as desired while arm **805** remains in place. Thereafter, the surgeon may release locking actuator **850** and extend arm **805** to distract patient tissue or retract arm **805**. Additionally, in some surgical settings, it may be advantageous to allow third module **800** to remain in a distracted position (while third module **800** is coupled to a table mount via table mount arm **806**) and remove modular retractor **500** while the surgical site remains distracted, or at least partially distracted, by third module **800**, for example. Additionally, any of the various disclosed modules may include a locking actuator **850** rotatably disposed in a corresponding body portion above a corresponding arm and work in the same, substantially the same, and/or similar manner as explained above.

In other embodiments, locking actuator **850** may be rotated between a locked position and an unlocked position. For example, in various embodiments, locking actuator **850** may include at least one locking tooth (not illustrated) that is disposed within locking cutout **850a** of arm **805**. For example, at least one locking tooth may jam with rack portion **805d** and prevent arm **805** from moving, for example. In other embodiments, locking actuator **850** may include at least one locking tooth that jams with pinion portion **801a**, for example. In other embodiments, locking actuator **850** may lock pawl **804** such that pawl **804** is engaged with rack portion **805e** and prevented from pivoting up and down relative to rack portion **805e**. For example, by locking pawl **804** in place such that pawl **804** is engaged with rack portion **805e**, arm **805** may be prevented from moving forward and backward. Moreover, the above described embodiments and functionality of locking actuator **850** are broadly applicable to all of the disclosed embodiments herein. For example, any of the various modules disclosed herein may include a locking actuator **850** having at least one locking tooth that jams with a corresponding rack portion of an arm, and/or a pinion portion of an actuator as explained above.

Referring generally to FIGS. 46-54 a free hand module **900** and a telescoping blade **20** for use with the modular retractor **500** and/or free hand module **900** is disclosed. FIG. 46 is a perspective view of a free hand module **900** and FIGS. 47A and 47B are side views of a free hand module **900** for use with disclosed modular retractor **500** embodiments. FIG. 48A is an exploded parts view of a free hand module **900** and FIG. 48B is a removed parts view of free hand module **900**. FIGS. 49A-49B are various perspective views of a free hand module **900** in various configurations. FIGS. 50-51 are various perspective views of a free hand module **900** and a telescoping blade system **20**. FIG. 52A is a perspective view of a blade connection channel **905d** and FIG. 52B is a perspective view of a blade fastener. FIGS.

53-54 are various perspective views of a third module 800 coupled to a modular retractor 500 and a free hand module 900 coupled to the third module 800.

In accordance with disclosed embodiments, free hand module 900 may be configured to couple and uncouple from third module 800 (see FIG. 46). For example, the free hand module 900 may be configured to couple, connect, and/or mate with module mount 809. In the example embodiment, gripping arms 916 may grip onto module mount 809, for example. Additionally, gripping arms 916 may include a plurality of rails and channels extending in the lateral direction on an inside surface thereof. The rails of gripping arms 916 may have a size and shape corresponding to rails and channels of module mount 809, for example. Accordingly, the gripping arms 916 may securely mate with module mount 809 by seating rails of gripping arms 916 in the channels of module mount 809 and seating the rails of module mount 809 in the channels of gripping arms 916. Furthermore, the gripping arms 916 may provide a clamping force against module mount 809 securely coupling the free hand module 900 to third module 800, for example.

In various embodiments, free hand module 900 may be configured to enable a surgeon to freely extend blade 20 forward and backward in the longitudinal direction along longitudinal axis A-A, for example. Free hand module 900 may not include a rack and pinion mechanism to extend the blade 20 and may rely on the manual operability of a surgeon, for example. In some surgical contexts, a free hand module 900 may afford a surgeon greater freedom in installation and facilitate the surgeon in retracting delicate patient tissue by hand. For example, when performing a retraction step with free hand module 900, patient tissue may resist the retraction and/or opening of a surgical access site. The degree of resistance of the patient tissue may be sensed by the surgeon as a form of haptic feedback informing the surgeon how much pressure has been applied to the patient tissue. In this way, the surgeon can sense and or prevent applying too much retraction force to a patient tissue and/or applying just the right amount of retraction force in delicate situations. Similarly, a free hand module 900 may be relatively easier for a surgeon to manipulate than a rack and pinion type of motion. This may allow the surgeon to quickly retract specific patient tissue with greater freedom in operation. Additionally, a length and/or height of telescoping blade 20 may be adjustable. Accordingly, a surgeon can retract various layers of patient tissue that are below (or above) the patient tissue which has been previously retracted by the other blades.

Free hand module 900 may include a handle 901 at a proximal end and a blade attachment mechanism 905c at a distal end, for example. Handle 901 may be rigidly secured to a shaft 905 and shaft 905 may define a longitudinal axis of free hand module 900, for example. Free hand module 900 may include a moving mechanism 910. As illustrated best in FIGS. 48A and 48B, moving mechanism 910 may include various components that enable an end user to toggle a lever 911 to enable the forward and backward movement of handle 901, shaft 905, and blade attachment mechanism 905c relative to moving mechanism 910 and module mount 809, for example. Moving mechanism 809 may include a body portion 913 having an aperture 913 extending there-through. Shaft 905 may extend through aperture 913a and slotted aperture 915b of gripper body 915, for example. An upper portion of lever support 912 may be disposed above body 913 and be operably coupled to lever 911 while a lower portion including an annular channel 912b may be disposed within body 913. For example, lever support 912 may be

securely attached to lever 911 by laterally extending posts 912a that extend through corresponding apertures 911a of lever 911. In this way, lever 911 may be pivotable about posts 912a and when depressed an upper surface of body 913 may act as a support surface such that depressing lever 911 pulls lever support 912 upward. For example, lever support 912 may be pivotable up and down in the vertical direction by depressing and/or rotating lever 911. For example still, pressing down on lever 911 may pull lever support 912 upward relative to body 913. Additionally, the lower portion of lever support 912 disposed within body 913 may prevent the over rotation of lever support 912 due to suitable retaining rails of body 913 being inset within annular ring 912b such that lever support 912 is fixedly retained by body 913, for example.

As shown best in FIGS. 48A-48B, an uppermost coupling portion 915a of gripper body 915 may be secured within a lower cavity of lever support 912. In this way, when lever 911 is actuated and pulls lever support 912 upwards, lever support 912 also pulls gripper body 915 upwards. In various embodiments, a stop block 914 may be disposed within body 913 at a bottom portion thereof. Stop block 914 may be disposed beneath shaft 905, for example. Additionally, stop block 914 may include inclined surfaces that may bias gripping arms 916 inwards (towards one another) to provide a gripping force against module mount 809, for example. In operation, an end user may actuate lever 911 such that gripper body 915 is pulled upwards and gripping arms 916 are biased inwards towards one another to securely couple to module mount 806 via clamping force.

In various embodiments, free hand module 900 via lever 911 may be adjustable and/or fixed in three modes of operation, for example. In a first mode of operation, and when lever 911 is in a first position, shaft 905 is extendable forward and backward through body 913 and gripping arms 916 are in an open position (see FIG. 49A). When gripping arms 916 are in an open position free hand module 900 may be positioned in place around and/or above module mount 809. In the first mode of operation, moving mechanism 910 and gripping arms 916 are both fully open, for example. In a second mode of operation, and when lever 911 is in a second position, shaft 905 is extendable forward and backward through body 913 and gripping arms 916 are in a closed position whereby gripping arms 916 provide a suitable clamping force to module mount 809. In the second mode of operation, moving mechanism 910 is movable in the longitudinal direction and free hand module 900 is securely coupled to module mount 809 due to gripping arms 916 being in the closed position, for example. In a third mode of operation, and when lever 911 is in a third position, shaft 905 is not extendable forward and backward through body 913 and gripping arms 916 are in a closed position (see FIG. 49B). In the third mode of operation, moving mechanism 910 is fixed relative to shaft 905 and free hand module 900 is fixed in 3D space due to gripping arms 916 being in the closed position and securely clamped on to module mount 809 (see FIGS. 53-54).

With reference to FIGS. 50-52B, a telescoping blade 20 is disclosed. Telescoping blade 20 may securely connect to blade attachment mechanism 905c of free blade module 900, for example. Additionally, telescoping blade 20 may securely connect to any of the other blade attachment mechanisms disclosed herein. Telescoping blade 20 may include a first blade 22 and a second blade 24 that is extending along axis Z-Z, for example. First blade 22 may include a channel 23 extending longitudinally down a length thereof from proximal end 20p to about the distal end 20d.

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Similarly, second blade **24** may include a rail **25** extending longitudinally down a length thereof from proximal end **20p** to about distal end **20d**. In various embodiments, the channel **23** and/or rail **25** may stop and/or terminate before the distal end **20d** to prevent the second blade **24** from extending too far. In various embodiments, the second blade **24** may slide upward and downward in a proximal-to-distal direction, shown by axis Z-Z in FIG. **50**. Additionally, an outside surface of blade **22** may include an engagement feature **26** for securely coupling to blade attachment mechanism **905c**, for example. Engagement feature **26** may include two spring loaded tabs **27** that are flexible towards one another and naturally biased away from one another, for example. In various embodiments, an end user may slide engagement feature **26** down into channel **905d** of blade attachment mechanism **905c** from above and the two spring loaded tabs **27** may push outward against side surfaces of channel **905d** to frictionally retain engagement feature **26** therein. Additionally, channel **905d** may include a stop feature **905e** adjacent a bottom surface thereof. In various embodiments, the stop feature **905e** may be a curved bottom surface corresponding to the geometry of the spring loaded tabs **27**, for example. In other embodiments, the two spring loaded tabs **27** may seat into corresponding channels or indents of blade attachment mechanism **905c** (not illustrated). Furthermore, in other embodiments the engagement feature **26** may be rotated about 180 degrees such that the blade **20** may be insert into blade engagement mechanism **905c** from below.

FIGS. **53-54** illustrate a modular retractor **500** with a third module **800** coupled thereto and a free hand module **900** coupled to the third module **800**. In the example embodiment, the telescoping blade **20** is attached to a distal end of free hand module **900** and blade **30** is attached to the distal end of third module **800**. Additionally, a centerline of telescoping blade **20** and a centerline of blade **30** are aligned with longitudinal axis A-A of modular retractor **500** (see FIG. **22**). However, it shall be appreciated that free hand module **900** is slidable along module arm **809** and can be positioned alternately than shown. Furthermore, the curved arm **811** curves out laterally farther than arm **507** of modular retractor **500**. As illustrated, blades **30**, **40**, and **20** form an opening for a surgical access location and/or a surgical access site.

FIGS. **55-56** are various perspective views of a fourth module **1000** for use with disclosed modular retractor **500** embodiments. FIG. **57** is an exploded parts view of a fourth module **1000**. FIGS. **58-59** are various views of a fourth module **1000** coupled to a modular retractor **500** and FIGS. **60-61** are various views of a fourth module **1000** coupled to a first free hand module **900** and a second free hand module **900**.

Fourth module **1000** may include the same, substantially the same, and or similar components as third module **800**. Accordingly, duplicative disclosure will be omitted and/or minimized. Fourth module **1000** may be configured to couple and uncouple from modular retractor **500** at connection points **503a**, for example (see FIG. **20**). For example, the fourth module **1000** may have at least one corresponding connection point **1003a** on an underside thereof (see FIG. **56**) that is configured to couple, connect, and/or mate with a connection point **503a** of the modular retractor **500** in the same, substantially the same, and or similar manner as explained above. Additionally, fourth module **1000** may be locked to modular retractor **500** by lock **513** (see FIG. **20**). Lock **513** may be pivotable such that in a locked position a flange portion of lock **513** may pivot into a locking aperture

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1003e of fourth module **1000**, in the same, substantially the same, and or similar manner as explained above.

Fourth module **1000** may include an arm **1005** that includes a straight portion **1010** and a C shaped curved portion **1011**. Straight portion **1010** of arm **1005** may extend through body **1003** and move forward and backward in a longitudinal direction, for example. As seen best in FIG. **58**, when fourth module **1000** is coupled to modular retractor **500** the C shaped curved portion **1011** extends laterally outward in a lateral direction farther than the farthest lateral edge of arm **507** of modular retractor **500**. For example, the C shaped curved portion **1011** does not obscure a surgeons viewing area and/or access to a surgical site. The straight portion **1010** of arm **1005** may extend through body **1003** through an L shaped contoured channel **1003b**, for example. Fourth module **1000** may be configured to extend arm **1005** along a path of travel by a rack and pinion mechanism, for example. The path of travel may be linear path, for example.

Fourth module **1000** may include a table mount **1006** extending laterally from a side surface thereof in a direction defined by an extension direction H-H of table mount **1006** (see FIG. **58**), for example. Table mount **1006** may facilitate the secure placement of fourth module **1000** such that fourth module **1000** remains fixed in 3D space and/or facilitate the relative motion of fourth module **1000** (and/or modular retractor **500** when coupled thereto) in any direction when moving table mount **70**, for example. Fourth module **1000** may be configured to extend arm **1005** by activation of actuator **1001**, e.g., by rotation of actuator **1001**. Actuator **1001** may be securely attached to body portion **1003** and include a pinion portion **1001a** (pinion gear and/or spur gear) having teeth that engage with and are meshed with straight rack portion **1005d** disposed on a side surface of arm **1005**, in the same, substantially the same, and or similar manner as explained above. Additionally, fourth module **1000** may include a first pawl **1004** that may be configured to engage the rack portion **1005e** disposed on a top surface of arm **1005**, for example. First pawl **1004** may be configured to allow pinion portion **1001a** to rotate in a first direction (counter clockwise direction) and prevent pinion portion **1001a** from rotating in a second direction (clockwise direction) in the same, substantially the same, and or similar manner as explained above.

Fourth module **1000** may include a table mount **1006** extending in a lateral direction along axis H-H away from arm **1005** and longitudinal axis A-A. Fourth module **1000** may include a proximal module mount **1009** extending along axis I-I in a lateral direction away from arm **1005** towards longitudinal axis A-A. For example, table mount **1006** may extend to the left direction and proximal module mount **1009p** may extend to the right direction. Additionally, in various embodiments, the C shaped curved portion **1011** may include a distal module mount **1009d** that extends along axis J-J from a side surface of curved arm portion **1011** such that it crosses over longitudinal axis A-A of modular retractor **500**. The distal module mount **1009d** and proximal module mount **1009p** may be symmetrically disposed relative to one another with respect to longitudinal axis A-A, for example (see FIG. **58**). Additionally, straight portion **1010** of arm **1005** may be supported by body **1003** on the left side of the longitudinal axis A-A of modular retractor **500**. In this configuration, module mounts **1009p**, **1009d** may cross over the longitudinal axis A-A, for example. Module mounts **1009p**, **1009d** may each independently support a free hand module **900**, in the same, similar, and/or substantially the same manner as explained previously. For example, as shown in FIGS. **60-61** proximal module mount **1009p** sup-

ports a free hand module **900** in a proximal position and distal module mount **1009d** supports a free hand module **900** in a distal position.

Referring generally to FIGS. **62-65** a fifth module **1100** for use with the modular retractor **500** is disclosed. FIGS. **62-63** are various perspective views of a fifth module **1100** for use with disclosed modular retractor **500** embodiments. FIG. **64** is an exploded parts view of a fifth module **1100** and FIG. **65** is a top perspective view of a fifth module **1100** coupled to a modular retractor **500**.

Fifth module **1100** may include the same, substantially the same, and or similar components as third module **800** and/or fourth module **1000**. Accordingly, duplicative disclosure will be omitted and/or minimized. Fifth module **1100** may be configured to couple and uncouple from modular retractor **500** at connection points **503a**, for example (see FIG. **20**). For example, the fifth module **1100** may have at least one corresponding connection point **1103a** on an underside thereof (see FIG. **63**) that is configured to couple, connect, and/or mate with a connection point **503a** of the modular retractor **500** in the same, substantially the same, and or similar manner as explained above. Additionally, fifth module **1100** may be locked to modular retractor **500** by lock **513** (see FIG. **20**). Lock **513** may be pivotable such that in a locked position a flange portion of lock **513** may pivot into a locking aperture **1103e** of fifth module **1100**, in the same, substantially the same, and or similar manner as explained above. Similarly, in an unlocked position the flange portion of lock **513** may be unseated from aperture **1103e**.

Fifth module **1100** may include an arm **1105** that includes a straight portion **1110** and a C shaped curved portion **1111**. Straight portion **1110** of arm **1105** may extend through body **1103** and move forward and backward in a longitudinal direction, for example. As seen best in FIG. **65**, when fifth module **1100** is coupled to modular retractor **500** the C shaped curved portion **1111** extends laterally outward in a lateral direction farther than the farthest lateral edge of arm **1105**. For example, the C shaped curved portion **1111** does not obscure a surgeon's viewing area and/or access to a surgical site. The straight portion **1110** of arm **1105** may extend through body **1103** through an L shaped contoured channel **1103b**. Fifth module **1100** may be configured to extend arm **1105** along a path of travel by a rack and pinion mechanism, for example. The path of travel may be linear path, for example.

Fifth module **1100** may include a table mount **1106** extending laterally from a side surface of arm **1105** adjacent a junction of curved portion **1111** and straight portion **1110**. Table mount **1106** may extend along axis H-H in a direction defined by an extension direction of table mount **1106** (see FIG. **65**), for example. In the example embodiment, table mount **1106** extends in a perpendicular direction to longitudinal axis A-A and/or a dominant extension direction of straight portion **1110**. Table mount **1106** may facilitate the secure placement of fifth module **1100** such that fifth module **1100** remains fixed in 3D space and/or facilitate the relative motion of fifth module **1100** (and/or modular retractor **500** when coupled thereto) in any direction when moving table mount **70**, for example. Fifth module **1100** may be configured to extend arm **1105** by activation of actuator **1101**, e.g., by rotation of actuator **1101**. Actuator **1101** may be securely attached to body portion **1103** and include a pinion portion **1101a** (pinion gear and/or spur gear) having teeth that engage with and are meshed with straight rack portion **1105d** disposed on a side surface of straight portion **1110** of arm **1105**, in the same, substantially the same, and or similar manner as explained above. Additionally, fifth module **1100**

may include a pawl **1104** that may be configured to engage the rack portion **1105e** disposed on a top surface of straight portion **1110** of arm **1105**, for example. Pawl **1104** may be configured to allow pinion portion **1101a** to rotate in a first direction (counter clockwise direction) and prevent pinion portion **1101a** from rotating in a second direction (clockwise direction) in the same, substantially the same, and or similar manner as explained above.

Fifth module **1100** may include a body **1103** having a curved body portion **1107** extending away from longitudinal axis A-A. In the example embodiment, curved body portion **1107** curves away in an opposite direction from arm **1105** and defines the distal most portion of body **1103**. Curved body portion **1107** may support and orient proximal modular mount **1109** such that it extends in a lateral direction towards arm **1105**, and crosses over longitudinal axis A-A. For example, table mount **1106** may extend to the left direction from a left side of arm **1105** and proximal module mount **1109p** may extend along axis K-K to the left direction from a left side of curved body portion **1107** and cross over longitudinal axis A-A. Additionally, in various embodiments, the C shaped curved portion **1111** may include a distal module mount **1109d** that extends from a side surface of curved arm portion **1111** such that it crosses over longitudinal axis A-A of modular retractor **500**. The distal module mount **1109d** and proximal module mount **1109p** may be disposed opposite one another and each cross over longitudinal axis A-A, for example. Additionally, straight portion **1110** of arm **1105** may be supported by body **1103** on the left side of the longitudinal axis A-A of modular retractor **500**. Module mounts **1109p**, **1109d** may each independently support a free hand module **900**, as explained previously. Furthermore, in various embodiments, module mount **1109p** may be relatively shorter than module mount **1109d**.

Referring generally to FIGS. **66-75** various blades for use with the modular retractor **500** and any of the various modules disclosed herein are disclosed. FIG. **66** is a top view of a pair of blades **30**, FIG. **67** is a bottom view of a pair of blades **30**, and FIG. **68** is a perspective view of a pair of blades **30** for use with disclosed modular retractor **500** embodiments. Blades **30** may be shaped like a half circle, for example. Accordingly, in various embodiments when blades **30** adjoin one another they may form a common circle in a fully closed position. Additionally, blades **30** may include an arcuate channel **31** extending along an inside surface thereof from a proximal end (see FIG. **66**) to a distal end (see FIG. **67**), for example. Arcuate channel **31** may have a size and shape corresponding to a size and shape of an arcuate outdent of a dilator, for example arcuate outdent **90a** of dilator **90** shown in FIGS. **80A**, **80B**. Alternatively, arcuate channel **31** may have a size and shape corresponding to a size and shape of an arcuate outdent of a shim (not illustrated). Additionally, blades **30** may include an aperture **32** extending through a top surface of blade **30** at the proximal end and penetrating the inside surface of blade **30** thereby providing access to the surgical access opening created by blades **30**, for example. Aperture **32** may provide access for light fixtures and other diagnostic tools such as endoscopes, electrodes, temperature sensors, suction devices, and etc. that may be insert therein and be protected while extending through blade **30**, for example.

FIG. **69** is an enlarged view of a top portion of a universal blade fastener **33**. Universal blade fastener **33** may be similar to blade fastener **26** of telescoping blade **20** (see FIG. **52B**) but in reverse. For example, blade fastener **33** may be configured for top loading blade **30** to a blade receiving mechanism. Blade fastener **33** may include a pair of spring

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loaded tabs **34** adjacent the upper surface of blade **30** and a curved support surface **35** therebelow. For example, spring loaded tabs **34** may be disposed at an upper region of blade fastener **33** and curved support surface **35** may be disposed at a lower portion of blade fastener. In this way, an end user can insert the support surface **35** within a blade receiving mechanism from above and the spring loaded tabs **34** can help retain blade **30** therein by a biasing force applied to sidewalls of a blade receiving mechanism. Furthermore, any other blade disclosed herein may include the same, similar, or substantially the same blade fastener **26**.

FIG. **70** is a top view of various blades **40** and **35** and FIG. **71** is a bottom view of the three blades **40** and **35** for use with disclosed modular retractor **500** embodiments. FIG. **72** is a perspective view of blades **35**, **40** for use with disclosed modular retractor **500** embodiments. In the example embodiment, a first blade **40**, second blade **40**, and a third blade **35** may form an oval shape. For example, when blades **40**, **35** are closed together such that they adjoin one another they may form an oval like shape. Blades **40** may include two arcuate channels **41** and an aperture **42**. Similarly, blade **35** may include two arcuate channels **36** and an arcuate channel **37**. Arcuate channels **41**, **37** may have a size and shape corresponding to a size and shape of an arcuate outdent of a dilator, for example arcuate outdent **81** of dilator **81** shown in FIG. **76**. Additionally, in the fully closed position where blades **40**, **35** adjoin one another, the six arcuate outdents of FIG. **76** may be disposed in a corresponding relative position and have a corresponding size and shape to the six arcuate channels **36**, **37** shown in the three blade configuration of FIG. **70**.

Furthermore, blades **40** may include an aperture **42** extending through a top surface of blade **40** at the proximal end and penetrating the inside surface of blade **40** and blade **35** may include an aperture **37** extending through a top surface of blade **35** at the proximal end and penetrating the inside surface of blade **35**. Apertures **37** and **42** may provide access for light fixtures and other diagnostic tools such as endoscopes, electrodes, temperature sensors, suction devices, and etc. that may be insert therein and be protected while extending through blades **35** and **40**, for example.

FIG. **73** is a top view of four blades **40**, **45** and FIG. **74** is a bottom view of the four blades for use with disclosed modular retractor **500** embodiments. FIG. **75** is a perspective view of the four blades **40**, **45** for use with disclosed modular retractor **500** embodiments. In the closed position, blades **40**, **45** may form an oval like shape. Blade **45** may include an arcuate channel **46** for securing to an arcuate outdent of a dilator having the same, similar, and or substantially the same attributes and purposes as explained above. Additionally blade **45** may include an aperture **47** extending through a top surface of blade **45** at the proximal end and penetrating the inside surface of blade having the same, similar, and or substantially the same attributes and purposes as explained above.

Referring generally to FIGS. **76-81** various dilators for use with the modular retractor **500** and the various blade embodiments disclosed herein are illustrated. FIG. **76** is a top view of a plurality of nested dilators **80** and FIG. **77A** is a perspective view of the plurality of nested dilators **80** in a non-nested configuration and FIG. **77B** is a perspective view of the plurality of nested dilators **80** in a nested configuration. In the example embodiment, five dilators are illustrated having progressively increasing sizing. A first dilator **85** may have a circular shape and a relatively narrow diameter for initiating a dilation process. An outside perimeter of the second dilator **84** may have an oval like shape and an inside

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diameter of dilator **84** may have a circular like shape corresponding to the outer diameter of first dilator **85**. An outside perimeter of the third dilator **83** may have an oval like shape and an inside perimeter of the third dilator **83** may have an oval like shape corresponding to the outer perimeter of second dilator **84**. Similarly, an outside perimeter of the fourth dilator **82** may have an oval like shape and an inside perimeter of the fourth dilator **82** may have an oval like shape corresponding to the outer perimeter of third dilator **83**. Similarly, an outside perimeter of the fifth dilator **81** may have an oval like shape and an inside perimeter of the fifth dilator **81** may have an oval like shape corresponding to the outer perimeter of fourth dilator **82**. In various embodiments, the dilators may be successively nested within one another to dilate a patient tissue before use of the various disclosed retractor embodiments. Additionally, fifth dilator **81** may include a plurality of arcuate outdents **81a** (e.g., an arcuate rail or the like) extending along an outside surface thereof. The arcuate outdent **81a** may mate with an arcuate channel of various blades as disclosed above.

FIG. **78** is a top view of a dilator **90** having an oval like outer perimeter and an oval like inner perimeter and FIG. **79** is a perspective view of dilator **90**. Dilator **90** may include a plurality of arcuate outdents **91a** (e.g., an arcuate rail or the like) extending along an outside surface thereof. In the example embodiment, arcuate outdents **91a** are disposed along roughly half of the available radial outer surface and extend in a proximal to distal direction, e.g., about half of the available perimeter includes arcuate outdents **91a** that extend from the proximal end to distal end. The arcuate outdents **91a** may mate with an arcuate channel of various blades in the same, similar, and/or substantially the same manner as explained above. FIG. **80A** is a top view of a dilator **95** and FIG. **80B** is a perspective view of dilator **95**. Dilator **95** may have a circular outer diameter and a circular inner diameter. Dilator **95** may include a plurality of arcuate outdents **95a** symmetrically radially disposed along the outer surface. The arcuate outdents **95a** may mate with an arcuate channel of various blades as disclosed above.

FIGS. **80C-80E** show various perspective and elevation views of a set of nested and cylindrically shaped dilators **99**. In the example embodiment, an innermost dilator **98** may be the thinnest and the longest dilator of the set, while the outermost dilator **94** may be the widest and the shortest dilator of the set. Of course the relative length and width of any dilator of the set of dilators **99** may be adjusted consistent with the particular surgery being performed. Additionally, there may be any number and size dilators in between the innermost dilator **98** and outermost dilator **94**, for example first inner dilator **97** and second inner dilator **96**. With reference to FIGS. **80D** and **80E**, it is seen that the outside circumferential surface of outermost dilator **94** has a plurality of rail portions **94A** that extend down its length in a proximal to distal direction. The rail portions **94A** generally have a size and shape corresponding to channel portions of various blades disclosed herein. For example, these rail portions **94A** may be disposed in particular locations along the outside circumferential surface of the outermost dilator **94** that takes into account the particular surgical approach employed by the surgeon and the location of the corresponding channel portions of the chosen blades, for example.

As seen best in the top down view of FIG. **81**, rail portions **94A** may be shaped like circular, oval, or arcuate outdents, for example. Additionally, pairs of adjacent rails **94A** may form coupling locations for any of the example blade embodiments disclosed herein. For example, as shown in FIG. **81** blade **1** and blade **2** may couple to respective pairs

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of rails **94A** in a first region (approximately upper half area of FIG. **81**) and blade **3** may couple to the remaining respective pair of rails **94A** in a second region (approximately lower half area of FIG. **81**). The illustrated spacing arrangement of rails **94A** takes into account a specific surgical approach chosen by the surgeon and the functionality of disclosed retractor embodiments, e.g., relative movement of retractor arms in a linear, arcuate, ratcheting, and/or pivoting motion, the types of blades and their relative locations, among other things. Further discussion regarding an example method of use of disclosed retractor and retractor module embodiments and surgical approaches utilizing the outer dilator **94** is shown in the top down view of FIG. **130** and the perspective view of FIG. **131**, among other places.

Referring generally to FIGS. **82-87** a modular blade **120** and an extendable blade **130** for coupling to modular blade **120** is disclosed. FIGS. **82** and **83** are various perspective views of a modular blade **120** and FIGS. **84** and **85** are various perspective views of an extendable blade **130** for coupling to modular blade **120**. FIG. **86** is a front view of modular blade **120** and extendable blade **130** side by side and FIG. **87** is a top down view of modular blade **120** and extendable blade **130**. In various embodiments, the modular blade **120** may be referred to as modular because it may couple to various extendable blades **130** such that extendable blades **130** may extend relative to modular blade **120**, e.g., blade **120** and blade **130** may be configured as a telescoping blade system.

Modular blade **120** may extend from a proximal end **120p** to a distal end **120d** in a proximal-to-distal direction (may also be referred to as longitudinal direction). The proximal end **120p** may include an engagement feature **126** having spring loaded tabs **127** for coupling to a blade engagement mechanism in the same, similar, and/or substantially the same way as explained above. The distal end **120d** may include a tip portion **121**. In the example embodiment, tip portion **121** comprises a substantially planar outer surface that tapers towards a centerline of modular blade **120** and terminates as a blunt chisel shaped end having a relatively smaller thickness than the remaining portions of modular blade **120**, for example. Modular blade **120** may include a pair of rails **124** that extend from proximal end **120p** towards distal end **120d**. For example, a first rail **124** may extend along a first side of blade **120** in the proximal-to-distal direction and a second rail **124**, opposite the first rail **124**, may extend along a second side of blade **120** in the proximal-to-distal direction. In various embodiments, rails **124** may define a receiving channel for receiving extendable blade **130** as will be explained in further detail below.

Additionally, modular blade **120** may include an aperture **122** extending through a top surface of blade **120** at the proximal end **120p** and penetrating through the inside surface of blade **120** at oval shaped opening **122a**, for example. In various embodiments, aperture **122** may comprise a passageway (in a cross section view) that is inclined away from the outside surface of blade **120** and towards the inside surface of blade **120** such that the passageway forms an oval shaped opening **122a** on the inside surface of blade **120**. In cross section, the passageway of aperture **122** may resemble a circle, oval, pentagon, square, rectangle, and/or any combination thereof. Aperture **122** may provide access for light fixtures and other diagnostic tools such as endoscopes, electrodes, temperature sensors, suction devices, and etc. that may be insert therein.

Modular blade **120** may include a contoured channel **123** for connecting with extendable blade **130** and facilitating the

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forward and backward relative motion of extendable blade **130** in the proximal-to-distal direction, for example. As shown best in FIG. **83**, contoured channel **123** may include a relatively large central arcuate channel portion **123a** having a pair of relatively smaller arcuate channels **123b** on opposite sides of channel portion **123a**, for example. Additionally, contoured channel **123** may include a plurality of indentations **125** extending in a proximal-to-distal direction, for example. In various embodiments, indentations **125** may be circular shaped indentations, oval shaped indentations, hexagonal shaped indentations, parallelogram shaped indentations, and/or any combination thereof. A distal end of contoured channel **123** may include a stop feature **129** for preventing extendable blade **130** from extending too far in the proximal-to-distal direction.

Extendable blade **130** may extend from a proximal end **130p** to a distal end **130d** in a proximal-to-distal direction (also referred to as a longitudinal direction). The distal end may include a tip portion **131** tapering towards a centerline of extendable blade **130** and terminating as a blunt chisel shaped end having a relatively smaller thickness than the remaining portion of extendable blade **130**, for example. In the example embodiment, an outside surface of extendable blade **130** may include an engagement feature **134** for connecting with contoured channel **123**, for example. Engagement feature **134** may include a proximal engagement rail **135** having a size and shape generally corresponding to a size and shape of contoured channel **123**. For example, proximal engagement rail **135** may have a size and shape generally corresponding to the relatively large central arcuate channel portion **123a** and the pair of relatively smaller arcuate channels **123b**, for example. Additionally, engagement feature **134** may include a medial engagement rail **136** having a width approximately corresponding to the relatively large central arcuate channel portion **123a** of modular blade **120**, for example. In various embodiments, an exposed surface of medial engagement rail **136** may be substantially planar although in other embodiments the exposed surface may be arcuately shaped to correspond and/or approximate the geometrical profile of contoured channel **123**, for example.

In various embodiments, engagement feature **134** may include at least one protrusion **133** having a size and shape generally corresponding to a size and shape of indentation **125**. For example, protrusion **133** may selectively be seated within any one of indentations **125** to secure extendable blade **130** in any one position of the plurality of positions defined by indentations **125**. In various embodiments, protrusion **133** may be a circular shaped protrusion, oval shaped protrusion, hexagonal shaped protrusion, parallelogram shaped protrusion, and/or any combination thereof. In various embodiments, protrusion **133** may extend away from extendable blade **130** in a direction perpendicular to the proximal-to-distal direction a distance that is relatively farther out than medial engagement rail **136** and/or proximal engagement rail **135**, for example. In some embodiments, protrusion **133** may be spring loaded and/or biased. In other embodiments, protrusion **133** may be a rigid non movable structure.

In various embodiments, engagement feature **134** may include a distal engagement rail **137** having a size and shape generally corresponding to a size and shape of contoured channel **123**. For example, distal engagement rail **137** may have a size and shape generally corresponding to the relatively large central arcuate channel portion **123a** and the pair of relatively smaller arcuate rails **123b**, for example. Additionally, engagement feature **134** may include a stop feature

138 that may abut against stop feature 129 of modular blade 120 to prevent extendable blade 130 from disengaging with modular blade 120, for example. For example, in a fully extended position, stop feature 138 of extendable blade may directly contact stop feature 129 of modular blade 120 and prevent extendable blade 130 from extending too far that engagement feature 134 becomes unseated from contoured channel 123.

With reference to FIG. 86, the inside surface of modular blade 120 and the outside surface of extendable blade 130 is illustrated. In various embodiments, extendable blade 130 may operably couple to modular blade 120 by inserting engagement feature 134 into channel 123. As explained above, extendable blade 130 may move forward and backward in a proximal-to-distal direction within contoured channel 123. For example, extendable blade 130 may extend forward and backward within contoured channel 123 and protrusion 133 may be seated within any one of indentations 125. For example, when modular blade 120 and extendable blade 130 are coupled together as a system, they may be referred to as a telescoping blade system.

With reference to FIG. 87, a top down view of modular blade 120 and extendable blade 130 is illustrated. In the example embodiment, it is shown that rails 124 define a cavity and/or channel for receiving extendable blade 130. For example, extendable blade 130 has a width in a lateral direction that corresponds to a distance between rails 124 and a thickness of extendable blade 130 corresponds to a depth of the cavity and/or channel between and defined by rails 124. In various embodiments, the outside lateral edges 1301 of extendable blade 130 may be inset within the receiving cavity defined by rails 124 such that they frictionally engage and slide across the interior side surfaces of modular blade 120, for example. In this way, rails 124 may provide a bearing surface for retaining extendable blade 130 therein while also allowing extendable blade 130 to move forward and backward in the proximal-to-distal direction. Additionally, it is shown that engagement feature 134 has a size and shape corresponding to contoured channel 123. For example, the curved surfaces of proximal engagement rail 135 may be inset within (mated within) contoured channel 123 and frictionally engage and slide across the interior surfaces defined by the relatively large central arcuate channel portion 123a and/or pair of relatively smaller arcuate channels 123b on opposite sides of channel portion 123a, for example.

Referring generally to FIGS. 88-93 a modular blade 320 and an extendable blade 330 for coupling to modular blade 320 is disclosed. FIGS. 88 and 89 are various perspective views of a modular blade 320 and FIGS. 90 and 91 are various perspective views of an extendable blade 330 for coupling to modular blade 320. FIG. 92 is a front view of modular blade 320 and extendable blade 330 side by side and FIG. 93 is a top down view of modular blade 320 and extendable blade 330. In various embodiments, modular blade 320 and extendable blade 330 may be configured as a telescoping blade system.

Modular blade 320 may extend from a proximal end 320p to a distal end 320d in a proximal-to-distal direction (may also be referred to as longitudinal direction). The proximal end 320p may include an engagement feature 326 having spring loaded tabs 327 for coupling to a blade engagement mechanism in the same, similar, and/or substantially the same way as explained above. The distal end 320d may include a tip portion 321. In the example embodiment, tip portion 321 comprises a substantially planar outer surface that tapers towards a centerline of modular blade 320 and

terminates as a blunt chisel shaped end having a relatively smaller thickness than the remaining portions of modular blade 320, for example. Modular blade 320 may include a pair of rails 324a and 324b that extend from proximal end 320p towards distal end 320d. For example, a first rail 324a may extend along a first lateral side of blade 320 in the proximal-to-distal direction and a second rail 324b, opposite first rail 324a, may extend along a second lateral side of blade 320 in the proximal-to-distal direction. First rail 324a may extend laterally away from extendable blade 320 farther than second rail 324b, for example. For example still, second rail 324b may be inset towards a center of modular blade 320 relative to first rail 324a and 324a may be outset relative to second rail 324b (see FIG. 93). First rail 324a may define a first receiving cavity 324y and second rail 324z may define a second receiving cavity 324z, for example. Additionally, modular blade may include a channel 319 extending along the outside surface of modular blade 320 in the proximal-to-distal direction and/or from a proximal end to a distal end. Channel 319 may have a size and shape generally corresponding to a size and shape of channel 339 of extendable blade 330, for example. In various embodiments, a stability pin may be positioned within channels 319 and 339, for example. In various embodiments, rails 324a, 324b, and channel 319 may define a contoured receiving channel for receiving extendable blade 330, as will be explained in further detail below.

Modular blade 320 may include an aperture 322 extending through a top surface of blade 320 at the proximal end 320p and penetrating through the inside surface of blade 320 at oval shaped opening 322a, for example. Aperture 322 may have the same, similar, and/or substantially the same features and functionality of aperture 122. Accordingly, duplicative description will be omitted. Modular blade 320 may include a contoured channel 323 for connecting with extendable blade 330 and facilitating the forward and backward relative motion of extendable blade 330 in the proximal-to-distal direction, for example. Contoured channel may include a relatively large central arcuate channel portion 323a having a pair of relatively smaller arcuate channels 323b on opposite sides of channel portion 323a, for example. Additionally, contoured channel 323 may include a plurality of indentations 325 extending in a proximal-to-distal direction, for example. In various embodiments, indentations 325 may be circular shaped indentations, oval shaped indentations, hexagonal shaped indentations, parallelogram shaped indentations, and/or any combination thereof. A distal end of contoured channel 323 may include a stop feature 329 for preventing extendable blade 330 from extending too far in the proximal-to-distal direction.

Extendable blade 330 may extend from a proximal end 330p to a distal end 330d in a proximal-to-distal direction (also referred to as a longitudinal direction). The distal end may include a tip portion 331 tapering towards a centerline of extendable blade 330 and terminating as a blunt chisel shaped end having a relatively smaller thickness than the remaining portion of extendable blade 330, for example. In the example embodiment, an outside surface of extendable blade 330 may include an engagement feature 334 for connecting with contoured channel 323, for example. Engagement feature 334 may include a proximal engagement rail 335 having a size and shape generally corresponding to a size and shape of contoured channel 323. For example, proximal engagement rail 335 may have a size and shape generally corresponding to the relatively large central arcuate channel portion 323a and the pair of relatively smaller arcuate channels 323b, for example. Additionally,

engagement feature 334 may include a medial engagement rail 336 having a width approximately corresponding to the relatively large central arcuate channel portion 323a of modular blade 320, for example. In various embodiments, an exposed surface of medial engagement rail 336 may be substantially planar although in other embodiments the exposed surface may be arcuately shaped to correspond and/or approximate the geometrical profile of contoured channel 323, for example.

In various embodiments, engagement feature 334 may include at least one protrusion 333 having a size and shape generally corresponding to a size and shape of indentation 325. For example, protrusion 333 may selectively be seated within any one of indentations 325 to secure extendable blade 330 in any one position of the plurality of positions defined by indentations 325. In various embodiments, protrusion 333 may be a circular shaped protrusion, oval shaped protrusion, hexagonal shaped protrusion, parallelogram shaped protrusion, and/or any combination thereof. In various embodiments, protrusion 333 may extend away from extendable blade 330 in a direction perpendicular to the proximal-to-distal direction a distance that is relatively farther out than medial engagement rail 336 and proximal rail 335, for example. In some embodiments, protrusion 333 may be spring loaded and/or biased. In other embodiments, protrusion 333 may be a rigid non movable structure.

In various embodiments, engagement feature 334 may include a distal engagement rail 337 having a size and shape generally corresponding to a size and shape of contoured channel 323. For example, distal engagement rail 337 may have a size and shape generally corresponding to the relatively large central arcuate channel portion 323a and the pair of relatively smaller arcuate rails 323b, for example. Additionally, engagement feature 334 may include a stop feature 338 that may abut against stop feature 329 of modular blade 320 to prevent extendable blade 330 from disengaging with modular blade 320 as explained above, for example.

With reference to FIG. 91, the inside surface of modular blade 320 and the outside surface of extendable blade 330 is illustrated. In various embodiments, extendable blade 330 may operably couple to modular blade 320 by inserting engagement feature 334 into channel 323. As explained above, extendable blade 330 may move forward and backward in a proximal-to-distal direction within contoured channel 323. For example, extendable blade 330 may extend forward and backward within contoured channel 323 and protrusion 333 may be seated within any one of indentations 325. For example, when modular blade 320 and extendable blade 330 are coupled together as a system, they may be referred to as a telescoping blade system.

With reference to FIG. 92, a top down view of modular blade 320 and extendable blade 330 is illustrated. In the example embodiment, it is shown that rails 324a, 324b and channel 319 define a cavity and/or channel for receiving extendable blade 330. For example, extendable blade 330 has a width in a lateral direction that corresponds to a distance between rails 324a, 324b and a thickness of extendable blade 330 corresponds to a depth of the cavity and/or channel between and defined by rails 324a and 324b. In various embodiments, the outside lateral edge 330l of extendable blade 330 may be mated within the receiving cavity 324y defined by rail 324a and an outside lateral rail 332 of extendable blade 330 may be mated within receiving cavity 324z, for example. In various embodiments, outside lateral rail 332 of extendable blade 330 may extend along the outside lateral edge of extendable blade 330 in the proximal-to-distal direction until about the tip portion 331, for

example. Additionally, channel 319 of extendable blade 330 may be mated within channel 319 of modular blade 320. In this way, rails 324a and 324b may provide a bearing surface for retaining extendable blade 330 therein while also allowing extendable blade 330 to move forward and backward in the proximal-to-distal direction. Additionally, it is shown that engagement feature 334 has a size and shape corresponding to contoured channel 323. For example, the curved surfaces of proximal engagement rail 335 may be inset within contoured channel 323 and frictionally engage and slide across the interior surfaces defined by the relatively large central arcuate channel portion 323a and/or pair of relatively smaller arcuate channels 323b on opposite sides of channel portion 323a, for example.

Referring generally to FIGS. 94-99 a modular blade 420 and an extendable blade 430 for coupling to modular blade 420 is disclosed. FIGS. 94 and 95 are various perspective views of a modular blade 420 and FIGS. 96 and 97 are various perspective views of an extendable blade 430 for coupling to modular blade 420. FIG. 98 is a front view of modular blade 420 and extendable blade 430 side by side and FIG. 99 is a top down view of modular blade 420 and extendable blade 430. In various embodiments, modular blade 420 and extendable blade 430 may be configured as a telescoping blade system.

Modular blade 420 may extend from a proximal end 420p to a distal end 420d in a proximal-to-distal direction (may also be referred to as longitudinal direction). The proximal end 420p may include an engagement feature 426 having spring loaded tabs 427 for coupling to a blade engagement mechanism in the same, similar, and/or substantially the same way as explained above. The distal end 420d may include a tip portion 421. In the example embodiment, tip portion 421 comprises a substantially planar outer surface that tapers towards a centerline of modular blade 420 and terminates as a blunt chisel shaped end having a relatively smaller thickness than the remaining portions of modular blade 420, for example. As best seen in FIG. 95, in some embodiments tip portion 421 may curve inward and/or arc inward in various embodiments, for example. Modular blade 420 may include a pair of rails 424a and 424b that extend from proximal end 420p towards distal end 420d. For example, a first rail 424a may extend along a first lateral side of blade 420 in the proximal-to-distal direction and a second rail 424b, opposite first rail 424a, may extend along a second lateral side of blade 420 in the proximal-to-distal direction. In various embodiments, modular blade 420 may be symmetrical on either side of a centerline extending in the proximal-to-distal direction, for example.

In various embodiments, first rail 424a may define a first receiving cavity 424y and second rail 424b may define a second receiving cavity 424z, for example (see FIG. 99). Additionally, modular blade 420 may include a first channel 419a and second channel 419b extending along the inside surface of modular blade 420 in the proximal-to-distal direction and/or from a proximal end to a distal end, for example. Channels 419a, 419b may have a size and shape generally corresponding to a size and shape of channels 439a and 439b of extendable blade 430, for example. In various embodiments, rails 424a, 424b, and channels 419a, 419b may define a contoured receiving channel for receiving extendable blade 430, as will be explained in further detail below.

Modular blade 420 may include at least one aperture 422 extending through a top surface of blade 420 at the proximal end 420p and penetrating through the inside surface of blade 420 at oval shaped opening 422a, for example. Apertures

422 may have the same, similar, and/or substantially the same features and functionality of aperture **122**. Accordingly, duplicative description will be omitted. Modular blade **420** may include a contoured channel **423** for connecting with extendable blade **430** and facilitating the forward and backward relative motion of extendable blade **430** in the proximal-to-distal direction, for example. Contoured channel may include a relatively large central arcuate channel portion **423a** having a pair of relatively smaller arcuate channels **423b** on opposite sides of channel portion **423a**, for example. Additionally, contoured channel **423** may include a plurality of indentations **425** extending in a proximal-to-distal direction, for example. In various embodiments, indentations **425** may be circular shaped indentations, oval shaped indentations, hexagonal shaped indentations, parallelogram shaped indentations, and/or any combination thereof. A distal end of contoured channel **423** may include a stop feature **429** for preventing extendable blade **430** from extending too far in the proximal-to-distal direction.

Extendable blade **430** may extend from a proximal end **430p** to a distal end **430d** in a proximal-to-distal direction (also referred to as a longitudinal direction). The distal end may include a tip portion **431** and extendable blade **430** may be generally shaped like a rectangle (in a plan view). In the example embodiment, an outside surface of extendable blade **430** may include an engagement feature **434** for connecting with contoured channel **423**, for example. Engagement feature **434** may include a proximal engagement rail **435** having a size and shape generally corresponding to a size and shape of contoured channel **423**. For example, proximal engagement rail **435** may have a size and shape generally corresponding to the relatively large central arcuate channel portion **423a** and the pair of relatively smaller arcuate channels **423b**, for example. Additionally, engagement feature **434** may include a medial engagement rail **436** having a width approximately corresponding to the relatively large central arcuate channel portion **423a** of modular blade **420**, for example. In various embodiments, an exposed surface of medial engagement rail **436** may be substantially planar although in other embodiments the exposed surface may be arcuately shaped to correspond and/or approximate the geometrical profile of contoured channel **423**, for example.

In various embodiments, engagement feature **434** may include at least one protrusion **433** having a size and shape generally corresponding to a size and shape of indentation **425**. For example, protrusion **433** may selectively be seated within any one of indentations **425** to secure extendable blade **430** in any one position of the plurality of positions defined by indentations **425**. In various embodiments, protrusion **433** may be a circular shaped protrusion, oval shaped protrusion, hexagonal shaped protrusion, parallelogram shaped protrusion, and/or any combination thereof. In various embodiments, protrusion **433** may extend away from extendable blade **430** in a direction perpendicular to the proximal-to-distal direction a distance that is relatively farther out than medial engagement rail **436** and proximal engagement rail **435**, for example. In some embodiments, protrusion **433** may be spring loaded and/or biased. In other embodiments, protrusion **433** may be a rigid non movable structure.

In various embodiments, engagement feature **434** may include a distal engagement rail **437** having a size and shape generally corresponding to a size and shape of contoured channel **423**. For example, distal engagement rail **437** may have a size and shape generally corresponding to the relatively large central arcuate channel portion **423a** and the pair

of relatively smaller arcuate channels **423b**, for example. Additionally, engagement feature **434** may include a stop feature **438** that may abut against stop feature **429** of modular blade **420** to prevent extendable blade **430** from disengaging with modular blade **420** as explained above, for example.

With reference to FIG. **98**, the inside surface of modular blade **420** and the outside surface of extendable blade **430** is illustrated. In various embodiments, extendable blade **430** may operably couple to modular blade **420** by inserting engagement feature **434** into channel **423**. As explained above, extendable blade **430** may move forward and backward in a proximal-to-distal direction within contoured channel **423**. For example, extendable blade **430** may extend forward and backward within contoured channel **423** and protrusion **433** may be seated within any one of indentations **425**.

With reference to FIG. **99**, a top down view of modular blade **420** and extendable blade **430** is illustrated. In the example embodiment, it is shown that rails **424a**, **424b** and channels **419a**, **419b** define a cavity and/or channel for receiving extendable blade **430**. For example, extendable blade **430** has a width in a lateral direction that corresponds to a distance between rails **424a**, **424b** and a thickness of extendable blade **430** corresponds to a depth of the cavity and/or channel between and defined by rails **424a** and **424b**. In various embodiments, the outside lateral rail **432a** of extendable blade **430** may be mated within the receiving cavity **424y** defined by rail **424a** and an outside lateral rail **432b** of extendable blade **430** may be mated within receiving cavity **424z**, for example. In various embodiments, outside lateral rail **432a**, **432b** of extendable blade **430** may extend along the outside lateral edge of extendable blade **430** in the proximal-to-distal direction until about the tip portion **431**, for example. Additionally, channels **439a**, **439b** of extendable blade **430** may be mated within channels **419a**, **419b** of modular blade **420**. In this way, rails **424a** and **424b** may provide a bearing surface for retaining extendable blade **430** therein while also allowing extendable blade **430** to move forward and backward in the proximal-to-distal direction. Additionally, it is shown that engagement feature **434** has a size and shape corresponding to contoured channel **423**. For example, the curved surfaces of proximal engagement rail **435** may be inset within contoured channel **423** and frictionally engage and slide across the interior surfaces defined by the relatively large central arcuate channel portion **423a** and/or pair of relatively smaller arcuate channels **423b** on opposite sides of channel portion **423a**, for example.

Referring generally to FIGS. **100-105** a modular blade **440** and various extendable blades **450a**, **450b**, and **450c** for coupling to modular blade **440** is disclosed. FIGS. **100-101** are various perspective views of a modular blade **440** and FIGS. **102-104** are various perspective views of extendable blades **450a**, **450b**, and **450c** for coupling to modular blade **440**. FIG. **105** is a front view of extendable blades **450a**, **450b**, and **450c**. In various embodiments, modular blade **440** and extendable blades **450a**, **450b**, and **450c** may be configured as a telescoping blade system. In various embodiments, the extendable blades **450a**, **450b**, and **450c** may have a relatively long and narrow tip section that may be advantageous for distracting soft tissues of a patient, for example.

Modular blade **440** may extend from a proximal end **440p** to a distal end **440d** in a proximal-to-distal direction (may also be referred to as longitudinal direction). The proximal end **440p** may include an engagement feature **446** having

spring loaded tabs **447** for coupling to a blade engagement mechanism in the same, similar, and/or substantially the same way as explained above. The distal end **440d** may include a tip portion **441**. In the example embodiment, tip portion **441** comprises a substantially planar outer surface that tapers towards a centerline of modular blade **440** and terminates as a blunt chisel shaped end having a relatively smaller thickness than the remaining portions of modular blade **440**, for example. Modular blade **440** may include a pair of rails **444** that extend from proximal end **440p** towards distal end **440d**. For example, a first rail **444** may extend along a first side of blade **440** in the proximal-to-distal direction and a second rail **444**, opposite the first rail **444**, may extend along a second side of blade **440** in the proximal-to-distal direction. In various embodiments, rails **444** may define a receiving channel for receiving any one of extendable blades **450a**, **450b**, and **450c**, for example.

Modular blade **440** may include a contoured channel **443** for connecting with extendable blades **450a**, **450b**, and **450c** and facilitating the forward and backward relative motion of extendable blades **450a**, **450b**, and **450c** in the proximal-to-distal direction, for example. As shown best in FIG. **83**, contoured channel **443** may include a relatively large central arcuate channel portion **443a** having a pair of relatively smaller arcuate channels **443b** on opposite sides of channel portion **443a**, for example. Additionally, contoured channel **443** may include a plurality of indentations **445** extending in a proximal-to-distal direction, for example. In various embodiments, indentations **445** may be circular shaped indentations, oval shaped indentations, hexagonal shaped indentations, parallelogram shaped indentations, and/or any combination thereof. A distal end of contoured channel **443** may include a stop feature **449** for preventing extendable blades **450a**, **450b**, and **450c** from extending too far in the proximal-to-distal direction.

Extendable blades **450a**, **450b**, and **450c** may extend from a proximal end **450p** to a distal end **450d** in a proximal-to-distal direction (also referred to as a longitudinal direction). The distal end may include a relatively long tip portion **451** that tapers near a medial portion of extendable blades **450a**, **450b**, and **450c** and then extends towards distal end **450d** at the same, similar, and/or substantially the same width. Relatively long tip portion **451** may terminate as an arcuate curved end with chamfered surfaces, for example. As seen best in FIG. **105**, extendable blades **450a**, **450b**, and **450c** are similar and have differently sized tip portions **451**. For example, extendable blade **450a** has a relatively wider tip portion **451** than extendable blades **450b** and **450c**, for example. Extendable blade **450b** has a relatively narrower tip portion **451** than extendable blade **450a** and a relatively wider tip portion **451** than extendable blade **450c**, for example. Extendable blade **450c** has a relatively narrow tip portion **451** than extendable blades **450a** and **450b**, for example. The other remaining features and components may be the same, substantially the same, and or similar.

In the example embodiment, an outside surface of extendable blades **450a**, **450b**, and **450c** may include an engagement feature **454** for connecting with contoured channel **443**, for example. Engagement feature **454** may include a proximal engagement rail **455** having a size and shape generally corresponding to a size and shape of contoured channel **443**. For example, proximal engagement rail **455** may have a size and shape generally corresponding to the relatively large central arcuate channel portion **443a** and the pair of relatively smaller arcuate channels **443b**, for example. Additionally, engagement feature **454** may include a medial engagement rail **456** having a width approximately

corresponding to the relatively large central arcuate channel portion **443a** of modular blade **440**, for example. In various embodiments, an exposed surface of medial engagement rail **456** may be substantially planar although in other embodiments the exposed surface may be arcuately shaped to correspond and/or approximate the geometrical profile of contoured channel **443**, for example.

In various embodiments, engagement feature **454** may include at least one protrusion **453** having a size and shape generally corresponding to a size and shape of indentation **445**. For example, protrusion **453** may selectively be seated within any one of indentations **445** to secure extendable blades **450a**, **450b**, and **450c** in any one position of the plurality of positions defined by indentations **445**. In various embodiments, protrusion **453** may be a circular shaped protrusion, oval shaped protrusion, hexagonal shaped protrusion, parallelogram shaped protrusion, and/or any combination thereof. In various embodiments, protrusion **453** may extend away from extendable blades **450a**, **450b**, and **450c** in a direction perpendicular to the proximal-to-distal direction a distance that is relatively farther out than medial engagement rail **456** and/or proximal engagement rail **455**, for example. In some embodiments, protrusion **453** may be spring loaded and/or biased. In other embodiments, protrusion **453** may be a rigid non movable structure.

In various embodiments, engagement feature **454** may include a distal engagement rail **457** having a size and shape generally corresponding to a size and shape of contoured channel **443**. For example, distal engagement rail **457** may have a size and shape generally corresponding to the relatively large central arcuate channel portion **443a** and the pair of relatively smaller arcuate channels **443b**, for example. Additionally, engagement feature **454** may include a stop feature **458** that may abut against stop feature **449** of modular blade **440** to prevent extendable blades **450a**, **450b**, and **450c** from disengaging with modular blade **440**, for example. For example, in a fully extended position, stop feature **449** of extendable blade may directly contact stop feature **458** of extendable blades **450a**, **450b**, and **450c** and prevent extendable blades **450a**, **450b**, and **450c** from extending too far that engagement feature **454** becomes unseated from contoured channel **443**.

In various embodiments, extendable blades **450a**, **450b**, and **450c** may operably couple to modular blade **440** by inserting engagement feature **454** into channel **443**. As explained above, extendable blades **450a**, **450b**, and **450c** may move forward and backward in a proximal-to-distal direction within contoured channel **443**. For example, extendable blades **450a**, **450b**, and **450c** may extend forward and backward within contoured channel **443** and protrusion **453** may be seated within any one of indentations **445**. For example, when modular blade **440** and any one of extendable blades **450a**, **450b**, and **450c** are coupled together as a system, they may be referred to as a telescoping blade system and such system may be particularly advantageous for distracting and retracting various soft patient tissue, for example.

In the example embodiment, rails **444** may define a cavity and/or channel for receiving any one of extendable blades **450a**, **450b**, and **450c**. For example, extendable blades **450a**, **450b**, and **450c** may have a width in a lateral direction that corresponds to a distance between rails **444** and a thickness of each extendable blades **450a**, **450b**, and **450c** may correspond to a depth of the cavity and/or channel between and defined by rails **444**, for example. In various embodiments, any one corresponding pair of outside lateral edges of extendable blades **450a**, **450b**, and **450c** may be inset within

the receiving cavity defined by rails **444** such that a pair of lateral edges frictionally engages and slides across the interior side surfaces of modular blade **440**, for example. In this way, rails **444** may provide a bearing surface for retaining any one of extendable blades **450a**, **450b**, and **450c** therein while also allowing any inserted blade to move forward and backward in the proximal-to-distal direction. Additionally, in various embodiments engagement feature **454** has a size and shape corresponding to contoured channel **443**. For example, the curved surfaces of proximal engagement rail **455** may be inset within (mated within) contoured channel **443** and frictionally engage and slide across the interior surfaces defined by the relatively large central arcuate channel portion **443a** and/or pair of relatively smaller arcuate channels **443b** on opposite sides of channel portion **443a**, for example.

Additional Retractor Embodiments

Referring generally to FIGS. **106-108D** a modular blade **140** and an extendable blade **150** having a pointed end **151** is disclosed. In some embodiments, extendable blade **150** may be referring to as an “impact blade” on account of being configured with a pointed end **151** that may be driven into a disc space, for example. FIG. **106** is a front view of the modular blade **140** and extendable blade **150** slidably coupled together and FIG. **107** is a rear view of the modular blade **140** and extendable blade **150** slidably coupled together. FIGS. **108A** and **108B** are various exploded parts views of the modular blade **140** and extendable blade **150**. FIG. **108C** is a perspective view of the modular blade **140** and FIG. **108D** is a top down view of the modular blade **140**.

In the example embodiment, modular blade **140** may include an engagement feature **146** having similar functional attributes to blade engagement feature **126** explained above with respect to blade **120**. However, in this embodiment, the engagement feature **146** of modular blade **140** does not include spring loaded tabs **127**, for example. Rather, as best seen in FIGS. **108B**, **108C**, and **108D**, engagement feature **146** comprises a raised rail **146B** having an arcuately shaped and/or curved shaped uppermost surface **146A**, for example. The raised rail portion **146B** may be offset from the outside surface of modular blade **140** by platform **146D**. In the example embodiment, a thickness or dimension of platform **146D** in a lateral direction is less than a thickness or dimension of raised rail **146B** in the lateral direction. In the example embodiment, channel portions **146C** are formed on opposite sides of platform **146D** and are located between the outside surface of modular blade **140** and the adjacent inside surface of raised rail **146B**, for example. In this way, engagement feature **146** may be configured to slidably connect to a corresponding blade coupling portion of an arm of the variously disclosed blade coupling portions having corresponding male/female features. In at least one embodiment, modular blade **140** may be securely coupled to blade coupling portion **535** shown in FIGS. **119-121B** by sliding the modular blade **140** from beneath blade coupling portion **535** upwards into blade coupling portion **535**. For example, modular blade **140** may be configured for bottom loading as will be explained in further detail below. Additionally, any blade disclosed herein may be configured with the same, similar, or substantially the same type of engagement feature **146**. However, consistent with the disclosure herein, modular blade and engagement feature **146** may also be reversed for top loading.

As seen best in FIGS. **108B**, **108C**, and **108D**, modular blade **140** may comprise a pair of channels **143** extending

along the interior surface thereof from a proximal end **140P** to a distal end **140D**. In various embodiments, the interior surface of modular blade **140** may be a curved surface **148**. Additionally, the extendable blade **150** may comprise a pair of outdented rail portions **153** extending along the side surfaces thereof from a proximal end **150P** to a distal end **150D** thereof. In the example embodiment, the blade surface of extendable blade **150** has a curved sidewall surface **158** having a size and shape generally corresponding to the interior curved surface **148** of modular blade **140**, for example. In use, the extendable blade **150** may be configured to couple to the modular blade **140** by sliding the outdented rails **153** into the interior channels **143** of modular blade **140**, for example. In this way, extendable blade **150** may be configured to slidably couple to modular blade **140** in an operable way, for example. However, it shall be understood that other embodiments may rely on a single rail **153** and a single channel **143** and that channels **143** and rails **153** may take various shapes, for example oblong, square, trapezoidal, dovetail, tongue and groove, etc.

FIG. **108E** is a perspective view of a driver **149** for use with the modular blade **140** and extendable blade **150**, for example. In some embodiments, driver **149** may be a referred to as an impact driver on account of being suitable for sustaining an impact force to drive extendable blade **150** forward, for example. In the example embodiment, driver **149** may extend in a longitudinal direction from a proximal end to a distal end. For example, driver **149** may include a proximal end comprising a striking end or surface **149A** having a relatively flat or smooth top surface and a circumferential side surface **149B** having various texturing. In the example embodiment, the texturing of side surface **149B** extends in a proximal to distal direction along the outside side surface of the proximal end of driver **149** as raised rails and indented valleys therebetween. The texturing may assist a surgeon in rotating driver **149** within channel **143** and in some embodiments, rotating driver **149** may be used to rotate a threaded pin (not illustrated). In some embodiments, the contouring and/or texturing may correspond to a torx head or a similar driver end, for example. In various embodiments, a shaft **149C** may couple to or be integrally formed together with the proximal end and distal end, for example. In the example embodiment, shaft **149C** may have a diameter substantially corresponding in size and shape to a diameter of channel **143**, for example. Additionally, a distal end of shaft **149C** may comprise a blunt distal end **149D**. In operation, the blunt end **149D** may contact the proximal end of extendable blade **150** for pushing extendable blade **150** forward. Additionally, a first and second driver **149** may be used together to push extendable blade **150** forward within both of channels **143**, for example.

FIG. **109** is a perspective view of a rectangular shaped dilator **160**. In the example embodiment, dilator **160** extends in a longitudinal direction from a proximal end **160P** to a distal end **160D**. The proximal end **160P** may include a pair of opposing curvilinear indents **165** for gripping dilator **160**, for example. In various embodiments, dilator **160** may comprise an aperture **163** or opening extending from proximal end **160P** to distal end **160D** through dilator **160**, for example. In various embodiments, dilator **160** may have planar side surfaces **161** extending in a proximal-to-distal direction. In the example embodiment, dilator **160** may further include a chiseled end or inclined end defining the distal end **160D** surfaces. For example, planar side surfaces **161** may terminate into inclined surfaces **162** which may facilitate insertion of dilator **160** into an operative corridor for example. In the example embodiment, dilator **160** is

rectangular shaped and in other embodiments dilator **160** may be square shaped. Dilator **160** may be used with any type of blade disclosed herein. Dilator **160** may be particularly advantageous for use with a relatively flat planar blade such as blade **130** shown in FIG. **84** and/or the footed tip blades disclosed below. At least one surgical configuration may comprise the utilization of a four-blade configuration comprising four substantially planar blades that generally surround the rectangular shaped dilator **160**, for example.

Referring generally to FIGS. **110-114** various example shims **170**, **180** are disclosed. FIG. **110** is a bottom perspective view of a pair of shims **170** for coupling to various blades disclosed herein. FIG. **111** is a perspective view of a relatively short shim **170** having a pointed pin **173** at a distal end thereof and FIG. **112** is a perspective view of a relatively tall shim **170** having a pointed pin **173** at a distal end thereof. FIG. **113** is a perspective view of a relatively tall shim **170** having a blunted distal end. FIG. **114** is a perspective view of a double-sided shim **180** for coupling to various blades disclosed herein. In the example embodiments, shim **170** may extend in a longitudinal direction from a proximal end **170P** to a distal end **170D**. The proximal end may include a tab **172** for gripping shim **170** and pushing shim **170** downward or pulling shim **170** upward, for example. As seen best in FIG. **110**, shim **170** may comprise an arcuate rail **171** having a size and shape generally corresponding to a channel of a blade, for example channel **143** of modular blade **140**. Additionally, in various embodiments, tab **172** may comprise a relatively smooth planar upper surface that is strong enough to sustain an impact for driving or tapping of shim **170**. For example, a surgeon may provide an impact force to tab **172** by a mallet or hammer which drives shim **170** forward while remaining partially constrained within a corresponding channel of a blade. In striking tab **172**, the shim **170** may be thrust forward or in a distal direction thereby inserting pin **173** into patient anatomy such as a bone or disc space, for example. Each shim **170** may have various dimensioned side surfaces for abutting against patient tissue. As seen in FIG. **111**, some shim **170** embodiments may include a working surface **175** extending for a relatively short distance in the longitudinal direction and for a relatively great distance in a lateral direction substantially perpendicular to the longitudinal direction of shim **170**, for example. As seen in the example embodiment of FIG. **112**, some shim **170** embodiments may include a working surface **176** extending for a relatively long distance in a longitudinal direction and for a relatively short distance in a lateral direction substantially perpendicular to the longitudinal direction of shim **170**, for example. In the example embodiment of FIG. **112**, working surface **176** extends in the longitudinal direction for slightly less than half the length of the shim **170** in the longitudinal direction. As seen in the example embodiment of FIG. **113**, some shim **170** embodiments may include a working surface **177** extending for a relatively long distance in a longitudinal direction and for a relatively short distance in a lateral direction substantially perpendicular to the longitudinal direction of shim **170**, for example. In the example embodiment of FIG. **113**, working surface **177** extends in the longitudinal direction for more than half the length of the shim **170** in the longitudinal direction.

FIG. **114** is an example embodiment of a double-sided shim **180**. In this embodiment, shim **180** includes a pair of tabs **182**, and a pair of rails **181**. In this way, shim **180** may slidably couple to a pair of corresponding channels of a blade, for example channels **143** of modular blade **140**. In the example embodiment of FIG. **114**, shim **180** comprises

a first working surface **184** extending away from shim **180** in a first lateral direction perpendicular to the longitudinal direction and a second working surface **185** extending away from shim **180** in a second lateral direction perpendicular to the longitudinal direction. In this embodiment, working surfaces **184**, **185** are disposed on opposite sides of a centerline **189** of shim **180**.

Referring generally to FIGS. **115A-116B** a blade adjustment and positioning tool **250** is disclosed. FIG. **115A** is a perspective view of a blade adjustment and positioning tool **250** and FIG. **115B** is an exploded parts view of the blade adjustment and positioning tool **250**. FIG. **116A** is an outside surface perspective view of the blade adjustment and positioning tool engaged with a modular blade and an extendable blade and FIG. **116B** is an inside surface perspective view of the blade adjustment and positioning tool engaged with a modular blade and an extendable blade, for example. In the example embodiment, tool **250** may extend in a longitudinal direction from a proximal end **250P** to a longitudinal end **250D**, for example. In the example embodiment, the proximal end may be defined by a rotatable turnkey **251** and the distal end may be defined by a tab **256**.

In the example embodiment, the first portion **250A** may comprise a gripping portion **255** having a plurality of gripping indentations extending along a length thereof, for example. Additionally, the gripping portion may include a centrally disposed shaft extending therethrough in the longitudinal direction between proximal aperture **257A** and medial aperture **257B**, for example. Additionally, the second portion may include a matting rail **258** having a size and shape that corresponds to a size and shape of a channel of modular blade **120**, for example channel **123** shown in FIG. **83**. However, it shall be understood that reference to channel **123** is by example only, and that matting rail **258** and the corresponding channel of modular blade **120** does not necessarily need indentations **125** and arcuate channel portions **123B**, for example. In various embodiments, the distal end of first portion **250A** may be defined by tab **256**. In the example embodiment, tab **256** is offset laterally from an extension axis extending through proximal aperture **257A** and medial aperture **257B**, for example.

Second portion **250B** may include a turnkey **251** and/or a knob at a proximal end thereof. Turnkey **251** may be coupled to or monolithically formed with primary shaft **252** and extension shaft **253**. In various embodiments, extension shaft **243** may comprise a drive feature or driving head **254** at a distal end thereof. In operation, second portion **250B** may be insert inside of first portion **250A** by inserting the driving head **254**, extension shaft **253**, and primary shaft **252** through proximal aperture **257A**. Due to the particular design of this embodiment, the primary shaft **252** may be rotatably disposed within the central shaft of the first portion and primary shaft **252** may have a size and shape generally corresponding to a size and shape of the central shaft of the first portion, e.g., substantially the same diameter and a length substantially the same as a distance between proximal aperture **257A** and medial aperture **257B**. Extension shaft **253** may be partially mated to and/or disposed within an open channel **258C** of matting rail **258** such that it may freely rotate and so can driving end **254**.

With reference to FIGS. **116A** and **116B**, tool **250** may couple to a modular blade **120** by inserting matting rail **258** with channel **123**. Additionally, as seen best in FIG. **116B**, due to the offset nature of tab **256** an inside surface of tab **256** may be directly adjacent to and/or directly contact an inside surface of extendable blade **130**. In this way, tool **250** may provide a fulcrum or handle to manipulate the modular

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blade 120 and extendable blade 130. At least one particularly advantageous use of tool 250 may be when modular blade 120 and extendable blade 130 are coupled to a free hand module, as explained above, which may not have actuators to cause pivoting and/or angulation, e.g., free hand module 900 as shown in FIG. 49B. Additionally, in some embodiments, an end user may initially mate first portion 250A to modular blade 120 and extendable blade 130, then insert second portion into first portion, and slide second portion forward in a distal direction such that it pushes extendable blade 130 forward. In some embodiments, a chiseled end of second portion, e.g., drive feature 254, may unseat a protrusion of extendable blade 130 from a corresponding indentation of modular blade 120, e.g., protrusion 133 and indentations 125 (see FIGS. 83-84). In this way, an end user may utilize tool 250 to extend a position of extendable blade 130 via second portion 250B and may use tool 250 as a fulcrum or handle for applying a mechanical advantage as a fulcrum to modular blade 250 and extendable blade 230, for example. Additionally, in at least some embodiments, because drive end 254 may be rotatable, it may facilitate the unseating of the protrusion of the extendable blade 130 from the corresponding indentation of modular blade 120.

Referring generally to FIGS. 117A-118E, various views of a modular blade 260 and an extendable blade 262 having a footed tip 263, a quick connect handle 270, and a retractor mount coupler 280 are disclosed. FIG. 117A is a perspective view of the inside surfaces of the modular blade 260 and extendable blade 262 and FIG. 117B is a perspective view of the outside surfaces of the modular blade 260 and extendable blade 262. In the example embodiment, the extendable blade 262 has a footed tip 263 and is slidably coupled to the modular blade 260 similarly as explained above. Accordingly, duplicative description is omitted or only briefly described again. In this embodiment, modular blade 260 comprises an attachment rail 264 on the outside surface thereof adjacent the proximal end. Attachment rail 264 may include various surface texturing on the outside surface thereof, e.g., rail like peaks and channel like valleys therebetween extending in a proximal to distal direction around the outside curved surface of attachment rail 264. In at least some embodiments, attachment rail 264 is integrally formed with modular blade 260 and in others it may be removably coupled thereto. In various embodiments, attachment rail 264 may include attachment shaft 265 extending from the proximal end of modular blade 260, for example. The proximal most portion of attachment shaft 265 may comprise a generally cylindrical extension shaft having a planar indent 267, a necked down portion 268, and an end 266 that is wider than the necked down portion 268, for example. In various embodiments, the attachment shaft 265 may quickly couple to and uncouple from a quick connect handle 270, for example.

FIG. 118A is a perspective view of a quick connect handle 270 and FIG. 118B is an exploded parts view of the quick connect handle 270. In the example embodiment, quick connect handle 270 extends in a longitudinal direction from a proximal end 270P to a distal end 270D. The distal end 270D may comprise a coupling aperture 269 for connecting to an attachment shaft 265 of a modular blade 260 when attachment shaft 265 is inserted therein, for example. Quick connect handle 270 may include a main body portion 275 or handle and the distal end 270D may comprise a coupling mechanism having various mating features comprising a size and shape generally corresponding to a size and shape attachment shaft 265, for example. In the example embodiment, the pin 272 and mating features 274 are actuated by

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actuator 271. Quick connect handle 270 may couple to modular blade 260 by depressing actuator 271 and sliding attachment shaft 265 into aperture 269 such that sliding barrel 276 lockingly engages attachment shaft 265, for example. In various embodiments, barrel 276 may be biased towards the proximal end 270D by set pins 279 and springs 278. In various embodiments, upon activation of actuator 271, e.g., by depressing actuator 271 button, barrel 276 may linearly translate forward to securely couple to modular blade 260.

FIG. 118C is a perspective view of a retractor mount coupler 280. Retractor mount coupler 280 may have many of the same, similar, and/or substantially the same components and functionality as free hand module 900 of FIGS. 47B and 48A, for example. Accordingly, duplicative description will be omitted and/or only briefly described. In this embodiment, retractor mount coupler 280 may include a pair of gripping arms 916 for gripping on to the outside textured surface of rail 264, for example. Additionally, retractor mount coupler 280 may include a lever 911 and a body 913 having an aperture 913A. Lever 911 may function in the same or substantially the same way as previously explained with respect to free hand module 900. As seen best in FIG. 129, retractor mount coupler 280 may couple and couple to a rod, pole, table mount extension, and/or lateral arm 1201 of a secondary module 1200, for example. FIG. 118D is a perspective view of the modular blade 260 and extendable blade 262 of FIGS. 117A-117B coupled to the quick connect handle 270 of FIGS. 118A-118B and the retractor mount coupler 280 of FIG. 118C. FIG. 118E is a perspective view showing the quick connect handle 270 uncoupled from the attachment shaft 265 of modular blade 260.

Referring generally to FIGS. 119-122 an additional embodiment of a modular retractor 530 is disclosed. Modular retractor 530 may have the same, similar, and or substantially the same components and functionality as explained above with respect to modular retractor 500 and modular retractor 530 may be used with any of the various add on retractor modules 600, 700, 800, 900, 1000, and 1100 that are disclosed herein and shown in the various views of FIGS. 20-81, for example. Accordingly, significant duplicative description will be omitted although some aspects will be repeated for ease of explanation. FIGS. 119 and 120 are various perspective views of modular retractor 530. FIG. 121A is a top down view of modular retractor 530 and FIG. 121B is a bottom perspective view of a blade attachment mechanism 535. FIG. 122A is an enlarged view of the embodiment of FIGS. 119-121 with the top cover removed for ease of understanding of the internal gear system. FIG. 122B is an enlarged view of the embodiment of FIGS. 119-121 from a bottom perspective showing various structural features of a table mount quick release coupler 533.

In the example embodiment, modular retractor 530 may include handles 501a, 501b and arms 505 and 507. The handles and/or actuator 502 may serve to open the retractor 530 by spreading the arms 505, 507 apart from one another as explained previously. Modular retractor 530 may differ from retractor 500 in a few ways. For example, the blade attachment mechanism 535 may be configured for bottom loading rather than top loading, the blade release actuator 537 may be disposed on a side surface of blade attachment mechanism 535 rather than a top surface, a release mechanism 531 may be relied upon rather than pawl 504, and a table mount quick release connection 533 may be provided rather than table mount arm 506, for example.

As seen best in FIGS. 120 and 121B, blade attachment mechanism 535 may be configured for bottom loading of

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various blade engagement features, for example blade engagement feature **146** shown in FIGS. **108B**, **108C**, and **108D**. Generally, blade attachment mechanism **535** may have a size and shape generally corresponding to a size and shape of a corresponding blade engagement feature **146**. In the example embodiment, blade attachment mechanism **535** includes a channel **535B** that is open to the bottom and closed at the top by a curved top surface **535A**, for example. The curved top surface may have a curvature generally corresponding to a curvature of the curved uppermost surface **146A** and the channel **535B** may have a width, length, and height generally corresponding to a width, length, and height of raised rail portion **146B**, for example. In some embodiments, the curved top surface **535A** may be referred to as a stop surface and/or stopping wall. In various embodiments, channel **535B** may be flanked by supports **535C**, for example. In use, when blade engagement feature **146** is securely coupled to blade attachment mechanism **535** the side surfaces of platform **146D** may contact the side surfaces **535D** of supports **535C** such that the channel portions **146C** are mated with supports **535C**, for example. Additionally, in various embodiments the side surfaces **535D** may be chamfered and/or inclined, for example as seen best in FIG. **121B**. This arrangement may facilitate insertion of a blade thereon, for example. Furthermore, in various embodiments an outside surface of a corresponding blade and/or platform **146D** may contact the outermost surface of supports **535C**. Further still, blade release mechanism **537** may securely couple and uncouple a blade to blade attachment mechanism **535** when actuated.

As seen best in FIG. **122A**, modular retractor **530** may include a distraction mechanism **50**. In this embodiment, spur gear **54** includes a plurality of teeth on a side surface thereof but may not include a plurality of teeth on a top surface thereof like the example modular retractor **500** embodiment. Additionally, a release mechanism **531** may include a tooth **531A** or tip at an end thereof having a size and shape generally corresponding to a valley between adjacent teeth of spur gear **54**. Release mechanism **531** may be biased by a spring tab or leaf spring **531B** which naturally urges tooth **531A** into a meshed arrangement with the teeth of spur gear **54** such that the spur gear **54** is prevented from rotating in a direction which would cause the arms of modular retractor **530** to collapse or close. Additionally, because release mechanism **531** may pivot in and out of a meshed arrangement with spur gear **54** an end user may cause expansion or distraction between arms **505**, **507** of modular retractor **530** without needing to actuate release mechanism **531**. In this way, release mechanism **531** functions similarly to a pawl preventing the collapse of arms **505**, **507** while simultaneously allowing, for example, an uninhibited expansion of arms **505**, **507**.

As also seen best in FIGS. **122A** and **122B**, modular retractor **530** may include a quick connect table mount **533**. Quick connect table mount **533** may include an aperture **533Z** and a tightening knob **534**. Aperture **533Z** may have a size and shape corresponding to a square or polygonal driver, for example a drive end of a wrench such as the egg wrench **10** illustrated in FIG. **23**. As seen best in FIG. **121A**, quick connect table mount **533** may be generally disposed in a central position of the main retractor body and when viewed in a plan view may be aligned along a longitudinal axis bisecting the retractor body. This arrangement may facilitate a symmetrical load distribution, for example. However, it shall be understood that quick connect table mount **533** may be alternately disposed, for example on a side surface on the left side, medial or central area, and/or right

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side of the retractor body (with respect to plan view of FIG. **121A**). Additionally, a plurality of quick connect table mounts **533** may be disposed in any viable region of modular retractor **530** and the various add on modules disclosed herein. As seen best in FIG. **122B**, quick connect table mount **533** may be configured to receive a corresponding post **363** and/or rail **362** of a quick connect arm **360** (see FIGS. **124-125**). In the example embodiment, quick connect table mount **533** may include a centrally disposed mating aperture **533C** accessible from a bottom side of modular retractor **530**, for example. In some embodiments, mating aperture **533C** is coaxially aligned with aperture **533Z** although this is not a requirement. In various embodiments, mating aperture **533C** may include a circumferential ring surface having a relatively greater diameter than the central portion of aperture **533C** to facilitate seating of post **363** (see FIGS. **124-125**) in an arrangement similar to concentric circles of varying depths. Additionally, various counter torque mating features may be disposed around and/or surround aperture **533C**, for example. In the example embodiment, a first groove **533A** extends in a direction that is substantially parallel with the longitudinal axis of modular retractor **530** and a second groove **533B** extends in a direction that is substantially perpendicular with the longitudinal axis of modular retractor **530**, e.g. second groove **533B** extends in a lateral direction with respect to modular retractor **530**. In the example embodiment, the first and second grooves **533A**, **533B** may resemble a cross shape and/or an X shape. Additionally, the ends of grooves **533A**, **533B** may be open or closed, for example one side of groove **533B** is closed and has an arcuate end surface which ensures proper alignment quick connect arm **360**, for example. In various embodiments, the rail **363** of quick connect arm **360** may nest within at least one of grooves **533A**, **533B**. Accordingly, in this embodiment an orientation of quick connect arm **360** is adjustable between a direct head on orientation type and a side or lateral orientation type, for example. At least one advantage of this configuration may be providing flexibility in orientation to a surgeon depending on different types of procedures being performed and or changes in orientation mid-procedure.

FIG. **123** is a perspective view of a table mount system **340** adapted for use with various retractor components disclosed herein. For example, various armatures of a quick connect table mount system may be used for supporting and manipulating various retractor embodiments disclosed herein. In the example embodiment, table mount system **340** may include a first armature **340A** and a second armature **340B**. The first armature **340A** may be slidably connected to the second armature **340B** by armature connection mechanism **345**. In at least one embodiment, second armature **340B** may include a table mount channel **349** and a table mount clamp **350**. In use the second armature **340B** may be rigidly and removably secured to an operating table by tightening clamp **350** such that table mount channel **349** is tightened to the table and arm **348** extends in a vertical direction with respect to a horizontal surface or plane of a table (not illustrated). Thereafter, the first armature **340A** may be coupled to arm **348** by positioning arm **348** within the aperture of armature connection mechanism **345** and tightening turnkey **346** such that a movable platform **345B** clamps down on to armature **348** by closing and/or reducing the size of the aperture, for example. In the example embodiment, movable platform **345B** has a channel for seating the curved surfaces of arm **348** and in some embodiments may have grooving or other texturing to facilitate a relatively strong connection.

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First armature portion **340A** may include a first arm **341A** and a second arm **341B** that are hingedly connected together by hinge mechanism **342**, for example. In various embodiments, hinge mechanism **342** may allow for a full 360 degree rotation, or a subset thereof. At least one embodiment may include corresponding teeth that may mesh together when tightened or clamped together by a tightening knob **342A** that urges the corresponding teeth into corresponding valleys, for example. In various embodiments, second arm **341B** may be movably coupled to armature connection mechanism **345** by a ball and socket joint **343**, for example. Additionally, first armature **341** may be coupled to a snap on connector **347** by a ball and socket joint **343**, for example. Consistent with the disclosure herein, it shall be understood that any connection between the various armatures of disclosed table mount systems may be a rotatable hinge like connection, a sliding connection, and/or a ball and joint connection. Additionally, these connection types may be readily swapped and/or substituted. For example, post **351** may be inset within a hollow interior of any armature end to change the connection type and/or functionality depending solely on the particular needs of an end user.

FIG. **124** is a first perspective view of a quick connect coupler **360** for connecting various retractor embodiments to various quick connect table mount systems disclosed herein. FIG. **125** is a side view of the quick connect coupler **360**. Quick connect coupler **360** may include an arm **361** supporting a post **363** and rail **362** on a first end thereof. In various embodiments, the arm **361** may follow a diagonal, straight, and/or curved profile, for example. On an opposite end, quick connect coupler **360** may include an armature coupler **365**, for example. Armature coupler **365** may include a post having an inclined, chamfered, and/or dimpled end **366**, for example. Additionally, armature coupler **365** may include a grooved portion **367** to facilitate a rigid and secure engagement with a quick connect coupler of a table mount system, for example connector **347** shown in FIG. **123**. Furthermore, a base portion of quick connect coupler **360** may include a counter torque surface **368** for resisting a rotation of quick connect coupler **360**, for example surface **368** may directly contact a corresponding counter torque surface **347B** of connector **347** (see FIG. **123**). It shall be appreciated that armature coupler **365** may take any shape and have any form and type of various indentations, outdents, apertures, posts, slots, and etc. to facilitate attachment to a table mount arm whether in a snap on quick connect style as illustrated in the corresponding FIGS. or by, for example, a clamp on ratcheting style or even a mushroom expansion style.

FIG. **126** is a perspective view of a modular retractor system including the quick connect couplers of FIGS. **124-125**. FIG. **127** is a top down view of the system of FIG. **126**. Consistent with the disclosure herein, modular retractor **530** is securely coupled to a secondary module, e.g., module **1200**. Module **1200** may have the same, similar, and/or substantially the same features and functionality as the various other secondary modules discussed above. However, in this embodiment, secondary module **1200** is capable of linearly extending a centrally disposed first arm and a C shaped second arm, for example. In this embodiment, the C shaped second arm is supporting a free hand module **900** and the first arm is securely connected to a quick connect arm **360** via a table mount quick release coupler **533**, for example. Additionally, a body portion of module **1200** includes a table mount quick release coupler **533** and the second C shaped arm includes a table mount quick release

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530 is securely connected to a quick connect coupler **360** at the table mount quick release coupler **533**. Accordingly, various modular retractor systems may comprise a plurality of quick release couplers **533** whether they be on the primary retractor **530** or secondary module **1200**, for example. In this way, a surgeon has maximum flexibility in attaching the modular retractor system to a table mount. FIG. **128** is a perspective view of module **1200** in an uncoupled position with respect to modular retractor **530**. FIG. **129** is a perspective view of module **1200** coupled to modular retractor **530**. In the example embodiment, retractor mount coupler **280** is secured to attachment rail **264** of the modular blade **260** and extendable blade **262**. Additionally, the retractor mount coupler **280** is coupled to lateral arm **1201** of retractor module **1200**.

FIG. **130** is a top down view of modular retractor **530** supporting first and second blades and module **1200** supporting a third blade. FIG. **131** is a perspective view of three blades being slidably coupled to a dilator **94**. With reference back to the set of dilators **99** disclosed in FIGS. **80C**, **80B**, **80E**, and FIG. **81** an example method of use will be disclosed. In a first step, a surgeon may insert an initial dilator or pin, e.g., innermost dilator **98**. In some embodiments, and depending on the particular surgical approach dilator **98** may be insert into a patient from a lateral, anterior, or trans psoas approach, e.g. Once the initial dilator **98** is insert in the patient, a dilator having a relatively wider size may be insert over the initial dilator **98**, e.g., dilator **97**. After dilator **97** is slipped over innermost dilator **98**, a dilator having a relatively wider size than dilator **97** may be slipped over dilator **97**, e.g., dilator **96**. Any number of successive and iteratively increasing in size dilators may be slipped over one another in this process. Thereafter, an outermost dilator **94** having a plurality of rail portions **94A** may be slipped over the immediately prior dilator, e.g., dilator **96**.

Next, a surgeon may position modular retractor **530** and retractor module **1200** over the outermost dilator **94**. For example, a surgeon may install first and second blades to the first and second arms of modular retractor **530** and a third blade may be installed on the proximal arm of retractor module **1200**. The three blades may be collapsed such that edge portions contact one another and a circular void space is formed by the interior surfaces of the three blades. Next, the surgeon may slip the three blades over the outermost dilator **94** such that the corresponding channel portions of the blades slidably couples to the rail portions **94A** of the outermost dilator, see FIG. **81**. Thereafter, the surgeon may move the modular retractor **530** and retractor module **1200** such that the blades slide down along the length of outermost dilator and into the operative corridor. Once the surgeon has moved the three blades into the operative corridor, and the blades are supporting adjacent tissue, the surgeon may remove the outermost dilator **94**. After the outermost dilator **94** has been removed, the surgeon can freely manipulate any one of the three blades in any manner or relative movement as previously explained to enlarge the operative corridor.

It should be understood that various aspects disclosed herein may be combined in different combinations than the combinations specifically presented in the description and accompanying drawings. For example, features, functionality, and components from one embodiment may be combined with another embodiment and vice versa unless the context clearly indicates otherwise. Similarly, features, functionality, and components may be omitted unless the context clearly indicates otherwise. It should also be understood that, depending on the example, certain acts or events of any of the processes or methods described herein may be per-

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formed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques).

Unless otherwise specifically defined herein, all terms are to be given their broadest possible interpretation including meanings implied from the specification as well as meanings understood by those skilled in the art and/or as defined in dictionaries, treatises, etc. It must also be noted that, as used in the specification and the appended claims, the singular forms “a,” “an” and “the” include plural referents unless otherwise specified, and that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

What is claimed is:

1. A retractor system for enabling access to a surgical site, comprising:

a modular retractor having a longitudinal axis extending in a longitudinal direction and a lateral axis extending from a first lateral end to a second lateral end in a lateral direction, the modular retractor including:

a first body portion that houses a distraction mechanism; a first arm and a second arm pivotally coupled together, a first handle coupled to the first arm and a second handle coupled to the second arm;

a first pivoting member coupled to a distal end of the first arm and a second pivoting member coupled to a distal end of the second arm;

a first blade attachment mechanism coupled to the first pivoting member and a second blade attachment mechanism coupled to the second pivoting member, the first and second blade attachment mechanisms being configured to couple to first and second blades, respectively;

a first actuator operably coupled to the distraction mechanism for opening and closing the first arm and the second arm;

a second actuator for adjusting an angulation of the first pivoting member; and

a third actuator for adjusting an angulation of the second pivoting member, wherein the first body portion comprises at least one table mount quick release coupler.

2. The retractor system of claim 1, wherein the first body portion further comprises a connection point for connecting to at least one retractor module.

3. The retractor system of claim 1, wherein the at least one table mount quick release coupler is disposed on the first body portion such that the longitudinal axis crosses over the at least one table mount quick release coupler.

4. The retractor system of claim 1, wherein the at least one table mount quick release coupler comprises a centrally disposed mating aperture and a tightening knob.

5. The retractor system of claim 1, wherein the first body portion further comprises at least one channel adjacent the at least one table mount quick release coupler.

6. The retractor system of claim 1, wherein the first body portion further comprises a first channel adjacent the at least one table mount quick release coupler and a second channel adjacent the at least one table mount quick release coupler.

7. The retractor system of claim 6, wherein the first channel extends substantially in the longitudinal direction and the second channel extends substantially in the lateral direction.

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8. The retractor system of claim 1, further comprising at least one table mount arm configured to releasably couple to the at least one table mount quick release coupler.

9. The retractor system of claim 8, wherein the at least one table mount arm comprises a post.

10. The retractor system of claim 8, wherein the at least one table mount arm comprises at least one rail.

11. The retractor system of claim 1, further comprising at least one table mount arm configured to releasably couple to the at least one table mount quick release coupler,

wherein the table mount arm comprises a post and a rail.

12. The retractor system of claim 1, further comprising a third blade,

wherein at least one of the first blade and second blade is a modular blade and the third blade is slidably coupled to the modular blade.

13. The retractor system of claim 1, further comprising an extendable blade,

wherein the extendable blade comprises at least one rail that is slidably disposed within at least one channel of the first blade.

14. The retractor system of claim 13, wherein the extendable blade comprises a footed tip.

15. The retractor system of claim 14, further comprising a rectangular dilator.

16. The retractor system of claim 14, wherein the first and second blades comprise substantially planar working surfaces.

17. The retractor system of claim 1, further comprising an adjustable armature for supporting the retractor system, wherein:

the adjustable armature comprises a first armature portion coupled to a second armature portion,

the first armature portion is configured to couple to an operating table having a horizontal surface, the first armature portion having a first armature extending in a vertical direction substantially opposite a plane defined by the horizontal surface,

the second armature portion is slidably coupled to the first armature,

at least one table mount arm is coupled to the second armature, and

the at least one table mount arm is releasably coupled to the at least one table mount quick release coupler.

18. The retractor system of claim 17, wherein the first armature portion comprises a ball and socket joint.

19. The retractor system of claim 17, wherein the second armature portion comprises a second armature and a third armature that are hingedly connected.

20. The retractor system of claim 17, wherein the at least one table mount arm is coupled to the second armature at a ball and socket joint.

21. A method of forming an operative corridor in a patient, comprising:

providing a modular retractor and at least one retractor module for coupling to the modular retractor,

the modular retractor comprising:

a first body portion that houses a distraction mechanism;

a first arm and a second arm pivotally coupled together, a first handle coupled to the first arm and a second handle coupled to the second arm;

a first pivoting member coupled to a distal end of the first arm and a second pivoting member coupled to a distal end of the second arm;

a first blade attachment mechanism coupled to the first pivoting member and a second blade attachment

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mechanism coupled to the second pivoting member, the first and second blade attachment mechanisms being configured to couple to first and second blades, respectively;

a first actuator operably coupled to the distraction mechanism for opening and closing the first arm and the second arm;

a second actuator for adjusting an angulation of the first pivoting member;

a third actuator for adjusting an angulation of the second pivoting member;

wherein the first body portion comprises at least one table mount quick release coupler;

the at least one retractor module comprising:

a third arm supporting a third pivoting member and a third blade attachment mechanism at a distal end thereof, the third blade attachment mechanism being configured to couple to a third blade;

a fourth actuator operably coupled to the at least one retractor module for extending and contracting the third arm;

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a fifth actuator for for adjusting an angulation of the third pivoting member;

attaching the first blade to the first arm, the second blade to the second arm, and the third blade to the third arm;

sequentially inserting at least two dilators into a patient, the at least two dilators including an outermost dilator and an innermost dilator;

inserting a first dilator into a patient, wherein the outermost dilator forms an operative corridor;

positioning the first blade, second blade, and third blade directly above the outermost dilator;

sliding the first blade, second blade, and third blade over the outermost dilator and into the operative corridor.

22. The method of claim **21**, wherein sliding the first blade, second blade, and third blade over the at least one second dilator further comprises mating respective channel portions of the first blade, second blade, and third blade with corresponding rail portions of the outermost dilator.

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